GAMING THE IRON CURTAIN

How Teenagers and Amateurs in Communist Czechoslovakia Claimed the Medium of Computer Games

JAROSLAV ŠVELCH

Gaming the Iron Curtain

Game Histories

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Gaming the Iron Curtain: How Teenagers and Amateurs in Communist Czechoslovakia Claimed the Medium of Computer Games, Jaroslav Švelch, 2018

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To Lenka

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Series Foreword

What might histories of games tell us not only about the games themselves but also about the people who play and design them? We think that the most interesting answers to this question will have two characteristics. First, the authors of game histories who tell us the most about games will ask big questions. For example, how do gameplay and design change? In what ways is such change inflected by societal, cultural, and other factors? How do games change when they move from one cultural or historical context to another? These kinds of questions forge connections to other areas of game studies, as well as to history, cultural studies, and technology studies.

The second characteristic we seek in "game-changing" histories is a wide-ranging mix of qualities partially described by terms such as *diversity*, *inclusiveness*, and *irony*. Histories with these qualities deliver interplay of intentions, users, technologies, materials, places, and markets. Asking big questions and answering them in creative and astute ways strikes us as the best way to reach the goal not of an isolated, general history of games but rather of a body of game histories that will connect game studies to scholarship in a wide array of fields. The first step, of course, is producing those histories.

Game Histories is a series of books that we hope will provide a home—or maybe a launchpad—for the growing international research community whose interest in game history rightly exceeds the celebratory and descriptive. In a line, the aim of the series is to help actualize critical historical study of games. Books in this series will exhibit acute attention to historiography and historical methodologies, whereas the series as a whole will encompass the wide-ranging subject matter we consider crucial for the relevance of historical game studies. We envisage an active series with output that will reshape how electronic and other kinds of games are understood, taught, and researched, as well as broaden the appeal of games for the allied fields such as history of computing, history of science and technology, design history, design culture, material culture studies, cultural and social history, media history, new media studies, and science and technology studies.

The Game Histories series will welcome but not be limited to contributions in the following areas:

- Multidisciplinary methodological and theoretical approaches to the historical study of games.
- Social and cultural histories of play, people, places, and institutions of gaming.
- Epochal and contextual studies of significant periods influential to and formative of games and game history.
- Historical biography of key actors instrumental in game design, development, technology, and industry.
- Games and legal history.
- Global political economy and the games industry (including indie games).
- Histories of technologies pertinent to the study of games.
- Histories of the intersections of games and other media, including such topics as game art, games and cinema, and games and literature.
- Game preservation, exhibition, and documentation, including the place of museums, libraries, and collectors in preparing game history.
- Material histories of game artifacts and ephemera.

Henry Lowood, Stanford University
Raiford Guins, Stony Brook University

Preface

> Už jsi zase proháněl pimprlíky? (Have you been chasing puppets again?)
>
> —My parents

Before you start reading this book, I must admit it. I have been chasing puppets.

In the 1980s, my dad would take me with him to work, at the computing department of the Solo factory in the small Czech town of Sušice. Solo was a match factory. Its surroundings smelled of lumber, but the IT room smelled of computers and adventure. Men and women sat by computers, programming and entering data about matches, people, and other things. I would sit at an East German computer, the Robotron 1715, and play games on a green-screen text-only monitor. Only much later did I find out that these games were made in the United States by a company named Yahoo Software (unrelated to the Yahoo search engine). One of them was a clone of *Pac-Man* and the other a clone of *Donkey Kong*.

In 1990, when I was nine, I got a computer of my own. My dad and I went to Prague and met a bunch of scruffy engineers in a makeshift office—a typical example of the computer importing businesses that sprang up after the fall of the Iron Curtain. The machine was a Sam Coupé—an obscure British product that was one of the last 8-bit computers ever launched in Western markets. I had long been enthralled by computers, but having my own was simply amazing. I learned to program in BASIC, and played the few games that were already released for the machine. But I wanted more. Luckily, the machine was compatible with the Sinclair ZX Spectrum, the country's most widespread 8-bit computer, also made in the United Kingdom. I made friends with other kids who had Spectrums (or their less cool, but otherwise compatible Czechoslovak clones). We visited each other, played together, and carried around plastic bags full of cultural capital—cassette tapes full of games.

The Coupé had a 3.5″ floppy disk drive, but to load Spectrum games, I had to connect a cassette player (made in Asia, but marketed by the German company Elta) and load them up from a tape. There were many tapes to choose from and about twenty games on each tape. Some older, some newer; most in English, but also a few in Spanish; and some even in Czech or Slovak! There came my first peek into computer game history. New pixelated worlds beckoned me, and new player characters—or puppets, as my parents would jokingly call them—would compete for my attention. But from the very beginning, I was just as inclined to catalog, collect, and preserve games as I was to play them. Perhaps my career as a game scholar had already started.

When I began my dissertation research on games in Czechoslovakia and the Czech Republic, without any clear leads or plans, I was visiting the Comparative Media Studies department at the Massachusetts Institute of Technology. While there, I met colleagues not only from the United States but also from Spain and Singapore. We compared our personal gaming histories and found many differences and many similarities—and my experience was clearly more similar to the Spanish and Singaporean ones than the American one.

I lived in Somerville, Massachusetts, and one day, my landlady told me that as someone interested in computer games, I should go to an upcoming house party across the street. The house belonged to a senior programmer for a large local game development studio. “You should meet this guy,” the programmer said, and introduced me to Michal Hlaváč. As it soon turned out, Michal had made several Spectrum games in Czechoslovakia in the 1980s and 1990s—including a minor hit, *Pouch the Beetle*—before moving to the United States to study at Harvard and MIT. Needless to say, he was surprised to meet somebody who wanted to discuss *Pouch the Beetle*.

This chance meeting in 2008, followed by a proper interview, started the collection of material for my project, which has slowly taken on the shape of this book. This is, to a large extent, a book about movements and flows of computers and games. Its making also involved movements and travels, some international and many within the Czech Republic and Slovakia. Since that moment in 2008, I have not only been chasing puppets in computer games but also chasing interviewees, old magazines, design notes, and photographs. I keep chasing more. This volume should not be considered the final word on the history of games in Communist Czechoslovakia, but a snapshot from a longer journey.

Read More on the Book's Website

Visit the book's website at ironcurtain.svelch.com to find updates, extra content, additional images, and links to playable versions of games discussed in this book.

Acknowledgments

I worked on the research for this book on and off for about ten years. It went through several iterations and benefited from help, advice, and inspiration from many people. Initial seeds were sown during my 2007–2008 visit with MIT's Comparative Media Studies program. Clara Fernández-Vara, Matthew Weise, and Philip Tan, in particular, gave me the initial encouragement to pursue a Czechoslovak history of games. Upon my return to Prague, my PhD supervisor Professor Jiří Kraus generously accepted this choice and handed me troves of inspiration from multiple fields. The breadth of his knowledge kept astounding me.

My wife, Lenka Švelchová, selflessly offered continual advice and support, and helped me identify the important themes and events in my stories. I had forgotten to warn her that by marrying me, she was by extension also marrying this monograph. My parents, Jaroslav Švelch and Jana Švelchová, provided help and asylum countless times. Without their support and encouragement, I would never have become an academic. My bandmates in Rest In Haste provided me with many mind-opening musical experiences that reinvigorated my creative thinking.

I did most of the research and drafting work while at Charles University in Prague. I would like to thank my colleagues Roman Hájek, Radim Hladík, Jernej Prodnik, Irena Reifová, Jiřina Šmejkalová, Markéta Štechová, Václav Štětka, and Lenka Vochocová for their unwavering dedication to good and honest scholarship and for creating a friendly and creative atmosphere.

My brother and fellow game scholar Jan Švelch has always been the one with whom I discussed my ideas first, over beer and during lunches and long phone calls. With his (often cautious) approval, I knew I could move further. Along with Jan, Maria B. Garda and Alex Wade kindly read all chapters, and I am very grateful for their commitment and their spot-on comments. Three anonymous reviewers also read the whole manuscript, and their thoughtful suggestions helped me tighten and strengthen my arguments. I received insightful feedback on various parts and versions of the text from a wonderful interdisciplinary group of colleagues, including Pavel Mücke, Irena Reifová, Radim Hladík, Lucie Česálková, Kateřina Svatoňová, Jakub Macek, Vít Šisler, Ksenia Tatarchenko, Melanie Swalwell, Gleb J. Albert, Alison Gazzard, Lies van Roessel, Carly Kocurek, Bobby Schweizer, Vítězslav Sommer, Zbigniew Stachniak, and Kieran Nolan. I received additional valuable tips and material from Martin Bach, Helena Durnová, Patryk Wasiak, Ivana Čapková, Denisa Nečasová, Martin Husák, Milan Tuček, Riccardo Fassone, and Martin Mana. Tereza Krobová enthusiastically provided invaluable help and support in the final years of the project. I would also like to thank Robert Jameson, who kindly shared his then unpublished research on the history of computing in Czechoslovakia, and Petr Vnouček, who helped me gather material in the project's early stages.

This book would never have existed without the kind help of my informants, who delved deep into their memories and personal archives to share the material that forms the backbone of my research. I am most grateful for their help and their time. My gratitude also goes to fan archivists, whose diligent and enthusiastic efforts help preserve Czechoslovakia's 8-bit culture.

Speaking of hobbyists, the book's snapshots of life in 1980s Czechoslovakia were taken by amateur street photographer and professional mathematician Karel Bucháček. A treasure trove of his photographs was recently discovered by his nephew Roman Bucháček, who has—as the collection's heir and curator—kindly granted permission to reprint the photographs for free.

When I started my research on the topic, I wondered who would ever want to read a book about obscure games from Communist-era Czechoslovakia. I was delighted when series editors Henry Lowood and Raiford Guins told me they would. Along with MIT Press acquisitions editor Doug Sery, they believed in this project and gave me much-needed confidence. I am grateful for the opportunity. The book has also benefited from the care of the ever-helpful MIT Press staff, and from the editing skills of staff at Westchester Publishing Services, who read the manuscript more thoroughly than I ever could have.

A Note on Translations and Pronunciation

For the sake of readability, I use English translations to refer to games, magazines, and book publications. The original Czech or Slovak titles can be found in the references and bibliography. All quotes from material are presented in my own translations from Czech or Slovak into English. Many of these quotes are taken from some of the earliest Czechoslovak writings on microcomputers and computer games, before a discourse about these topics was even established. Others feature the unwieldy jargon of Communist-era bureaucracy. If they sounded stilted in the original, I strove to preserve their stiltedness.

To correctly pronounce the Czech and Slovak names and terms, a few basic rules need to be observed: There is a strong correlation between writing and pronunciation, meaning that one letter generally corresponds to a single sound and no vowels are silent. The stress almost always falls on the first syllable of a word.

The following graphemes are read in a significantly different way from English:

a	"u" in *run*
i, y	"i" in *pin* (always)
u	"oo" in *book*
c	"ts" in *bits*
ch	"ch" in *loch*
j	"y" in *young*
r	rolled as in Italian

The most frequent graphemes with diacritics are pronounced as follows:

á	"a" in *car*
é	a long "e," similar to the "a" in *rare*

í, ý	“ee” in *week*
ů, ú	“oo” in *bloom*
ě	“ye” in *yell*; softens the preceding consonant
č	“ch” in *child*
l’	“li” in *Liu* (the Chinese last name)
ř	similar to “rs” in pronunciation of *version*, but rolled and trill
š	“sh” in *shoe*
t’	“ti” in *Tian Shan*
ž	“s” in *measure*

Introduction

In March 1988, the underground magazine *Illegal* published "The European Crackers Map, Anno 1988." The "zine" was a flagship of a youth subculture that engaged in hacking and pirating commercial game software, and listed contributors from countries including Germany, Belgium, Denmark, Sweden, Holland, France, the United States, England, and Turkey. The map was supposed to show all the major groups that were engaging in these clandestine activities on the European continent. At first glance, one thing is clear—according to the map, no groups were operating to the east of the Iron Curtain. Instead, the whole region was labeled with the words "Communist in sight."[1]

For many in the West in the 1980s, the Soviet bloc was a blank space, an uncharted territory that they could fill with fantasy—or prejudice. The only 1980s Soviet bloc game to become part of mainstream gaming culture was *Tetris*. Styling its title as "TETЯIS" or "ТЕТРИС," Western publishers emphasized the elements of Soviet mystique to market the game as unique and different. Played against background images of Soviet landscapes and landmarks, its Western versions were stylized as souvenirs from a strange, distant country. The packaging of one of them even inaccurately placed its origin in the "blasted plains beyond the Urals."[2]

Soviet bloc computer games might have seemed like a fantastic, contradictory idea. After all, games appeared to stand for everything that was Western. They were a culmination of the postwar technological and consumer cultures; they have evolved to "epitomize an ideal type of global postindustrial neo-liberal cultural product."[3] If they seem to be so intertwined with the capitalist mode of production, then what role did they play in the East?

The Czechoslovak Socialist Republic—the setting of our story—was a country run by one of the region's most dogmatic and conservative regimes, which crushed civil liberties, eliminated private enterprise, and mismanaged the economy, resulting in recurrent shortages of consumer goods. The country should not, however, be imagined as just a gloomy and backward outpost of state socialism. In a number of ways, it was a country that was ready for computer games. It had a fairly developed industry and large urban centers; it offered a decent standard of living, as well as solid and free education and healthcare. The eradication of public life made people retreat into the domains of hobbies and leisure. Czechoslovakia—similar to neighboring socialist countries such as Hungary—was thus a home to an intriguing kind of industrialized consumerist society with a very narrow selection of things to consume, but a strong do-it-yourself movement. At the same time, the country's authorities had long touted technological progress and invested into research and technical education, creating thousands of potential microcomputer enthusiasts.

Given the power that communist or socialist parties held in the Soviet bloc, a Westerner might think that the Communist Party of Czechoslovakia had a *plan* for games—either that they were banned, or that they were used as tools of ideological propaganda. But in fact, the party was not interested. Communist technocrats saw computers as industrial machines and overlooked their potential impact on culture and entertainment; they could not and did not provide enough machines for the people that craved them.

In the 1980s—despite the scarcity of home computers—a remarkably active enthusiast scene emerged, which revered Western games but also created hundreds of its own. The amateur scene developed its own genre conventions, its own canon, and even its own programmer stars. It discovered that games were a medium, and used them not only for entertainment but also as a means of self-expression. In the final years of Communist rule, Czechoslovak amateurs became some of the first in the world to continuously make activist games about current political events, predating trends observed later in independent or experimental titles. This book will chronicle how Czechoslovak users "gamed" the Iron Curtain, imported computers and software from the West, and created their own world of games.

At the same time, the book aims to cover more than just games. I will branch out to political and economic history, to histories of technology, foreign trade, education, labor, leisure, gender relations, interior design, and even agriculture. Rather than serving as an inanimate backdrop, these topics will often animate my narrative and show how people and institutions in late socialist societies approached the first mass-market digital technologies, and technologies in general. I will tell stories of fandom, hacking, and activism, and shed light on leisure and hobbies in the former Soviet bloc, as well as the ways in which people could self-organize and work for a common goal under Communist rule.

The Histories So Far

In her 2006 book on game industry, Aphra Kerr noted that "a good social history of digital games remains to be written. Such a history would focus less on dates and inventors and more on struggles and uncertainties."[4] Ten years later, in the introduction to *Debugging Game History*, Henry Lowood and Raiford Guins pointed out that "there is a shocking lack of social and cultural histories of games."[5] This book wants to offer a remedy to this long-lasting and alarming deficit. But why should we kick off with a book that covers 1980s Czechoslovakia—a relatively short and peripheral slice that cannot be connected to any canonical or even internationally recognizable game hardware or software? What could be the benefit of such an obscure outlier?

I believe that a history of games written from the margins can be a valuable intervention into the field. In fact, recent developments in three related streams of historical research—digital game history, history of technology, and regional history—point toward the need for seemingly peripheral perspectives. The first of the three, history of digital games, has been developing in the shadow of industry narratives. Many books have been published about the industry at large and about individual companies such as Nintendo or id Software, mostly written by journalists and enthusiasts.[6] As critics of these histories have pointed out, these works were descriptive rather than interpretative, celebratory rather than critical, and they reproduced a dominant, hegemonic narrative of games from a Western perspective.[7] Huhtamo has notably called them artifacts of a "chronicle" era of video game history, and predecessors to future, more nuanced histories.[8] The establishment of game history as a field of scholarship progressed through a number of steps, including the 2009 special issue of the *IEEE Annals of the History of Computing* on the history of computer games, the 2013 History of Games conference and the resulting special issue of the *Game Studies* journal, and the 2016 launch of MIT Press's Game Histories series. Today, the subfield is buzzing with activity, and game histories are becoming more critical and diverse.

Despite this current surge in game history, little academic work has been published on the history of microcomputers and games in the Soviet bloc, save for insightful but short accounts of 1980s Polish gaming culture by Patryk Wasiak, one article by Zbigniew Stachniak on hobby computing in the Soviet Union, and the entries on Poland, Hungary, (East) Germany, and Russia in the *Video Games Around the World* anthology, which vary wildly in focus and analytical depth.[9] The book's informative but cursory entry on the Czech Republic unfortunately suffers from multiple factual inaccuracies.[10] Yet despite the lack of scholarship on games behind the Iron Curtain per se, game history has been opening up to research that questions and subverts hegemonic narratives, and welcomes marginal and peripheral points of view—resulting in histories that media historian Jaakko Suominen has called "emancipatory."[11] Melanie Swalwell's work on game histories in Australia and New Zealand brought attention to *the local*, and rightfully recovered the narratives of players, enthusiasts, amateurs, and homebrewers.[12] Works by Alison Gazzard, Graeme Kirkpatrick, and Alex Wade, among others, have shown that the UK game industry and culture produced many extraordinary achievements and had a lasting influence on many other European countries (including Eastern European ones), complicating the major industry narrative of American and Japanese dominance.

The movement toward marginal perspectives need not be interpreted only geographically. Although they focus on the United States—a default backdrop of many game histories—Carly Kocurek's monograph on arcades in the 1970s–1980s United States, and Laine Nooney's work on the games by Roberta Williams, manage to "other" the West by tracing the formation of masculine player and developer cultures that we too often take as a given.[13] Ongoing research on computer subcultures, specifically cracking, piracy, and demoscenes, pursued by Markku Reunanen, Patryk Wasiak, and others, has opened doors to another peripheral narrative—that of young people, mostly in Western Europe, breaking the rules of the game industry, and often even the law.[14] Last but not least, crucial progress has been made regarding the materiality of games. For a long time, games and their hardware *as things* had been silenced. But scholars of game preservation, platform studies, and the media archaeology of games, such as Raiford Guins, Nick

Montfort, Ian Bogost, and James Newman, have started listening in to their stories. As I will explain below, considering the computer's materiality may bridge the analytical gap between the machine as an abstract standard and the everyday life reality in which it is embedded.

Beyond contributing a mosaic piece to the general sum of knowledge of game histories, I hope this book can refresh the ways these histories are written. Shifting the focus to a marginal location with no established grand narrative can liberate us from canon and focus on the "struggles and uncertainties" anticipated by Aphra Kerr. In the 1980s, Czechoslovakia was a state socialist country without a functioning hardware or software market, where much of the microcomputer activity took place within paramilitary clubs and youth organizations. When faced with such an environment, a historian must forget the stereotypical arcs of game history narratives—the rises and falls of commercial companies, the "behind the scenes" accounts of famous titles, the development of franchises and intellectual properties. We no longer have to (or can) follow templates established by those who have had the power to write themselves into history, or pay lip service to manufacturers and publishers. The contextual features that Western game histories have taken for granted—the existence of distribution infrastructures, the availability of goods and services, the business logic of game industries, and the discourse-defining role of marketing and specialist press—will not apply in this case. To understand and analyze this context, we must forge new methods and approaches and populate our narratives with new actors; we must invest seemingly trivial and banal activities with a renewed sense of drama.

Besides its most evident compatriots in the subfield of game history, this book is grounded in the more general areas of history of technology and history of computing—which have also been welcoming to research coming from geographical and other margins. Kristen Haring's research on ham radio enthusiasts invited the amateur to the history of the medium and can be read as a prequel to any work on hobby computing and gaming communities.[15] Eden Medina's *Cybernetic Revolutionaries*—a celebrated book about government application of cybernetics in 1970s Chile—has sparked interest in regional histories of technology that had thus far been obscured.[16] Benjamin Peters's monograph *How Not to Network a Nation* follows the attempts of Soviet engineers to build a national computer network.[17] In the introduction to the latter book, Sandra Braman welcomes the "disorientation that results when the familiar is encountered in an unfamiliar context, broadening and deepening what we believe that we know about the familiar."[18] In both Medina's and Peters's work, as well as in Ksenia Tatarchenko's research on exchanges between Soviet and Western computer scientists, this disorientation offers a fresh opportunity for deeper discussions of politics, technology, and the way things and ideas travel around the world.[19] My intention is also to disorient—at least temporarily—and to show the familiar medium of computer games in the unfamiliar environment of the Soviet bloc.

This book also draws from—and hopefully contributes to—regional history, namely the history of Czechoslovakia in the late socialist period. The lingering issue with many of the lay and academic histories of the country has been their focus on the bipolar dynamics of state repression and dissent, which left little space for the investigation of technology and everyday life, the two subjects most relevant to my project. The recent overview by Kevin McDermott, *Communist Czechoslovakia 1945–89*, challenges the "totalitarian" paradigm that sees state socialism as a foreign-imposed and monolithic top-down authoritative system, instead treating it "as a living dynamic organism in which individuals struggled to empower themselves and thereby shape and make sense of the world around them."[20] This shift is exemplary of wider efforts to rejuvenate the period's historiography by focusing on the day-to-day functioning of institutions and the ordinary lives of individuals. I am indebted to the pioneering research on the country's science and education policies by Vítězslav Sommer, Jiří Hoppe, and Pavel Urbášek; the work on the perestroika discourse by Michal Pullmann; and oral history projects by Miroslav Vaněk and Pavel Mücke.[21] When writing about Czechoslovak history, international scholars often allow themselves more liberty to choose topics that might seem too recent or too banal to local historians. The domestication of computing and games, for example, fits into the histories of do-it-yourself practices under state socialism, convincingly recounted in several works by Paulina Bren.[22] My work also resonates with Kimberly Zarecor's monograph on the country's socialist-era prefabricated housing projects.[23] Overall, this book will connect the histories of technology and everyday life, claiming that although Czechoslovak economic and technology policies largely failed, education efforts and research funding generated anticipation and enthusiasm about computing, and opened up possibilities for creative work that unfolded in hobbyist environments rather than the state-controlled economy.

Finding History

A peculiar feature of digital game history as an academic endeavor is that it has rarely been practiced by trained historians.[24] Consequently, it often favors cultural and sociological analysis over more traditional historiographical work. This may be, in part, a reaction to the criticism of "chronicle" era histories and the perceived devaluation of historiographical work. This project, too, started out by looking at the *discourse* about games in 1980s Czechoslovakia, a direction that seemed more feasible to a graduate of media studies and linguistics such as myself. However, as I delved deeper into the material, it became clear that I could not make any statements about the discourse without reconstructing its mise-en-scène. At that point, textual analysis gave way to historiographical research.

As noted in the preface, I am a borderline insider in the history I will be telling, having been continuously involved in computer gaming since 1990. That gives me the benefit of familiarity with hardware, software, and the vernacular of computer users around that time. At the same time, it carries the risk of overreliance on personal experience. In part, I decided to avoid this bias by limiting the focus to the 1980s. The main thrust of the story starts when the first microcomputers began appearing in the country around 1982, and is bookended by the events of 1989, which led to the dismantling of both the Soviet bloc and the Iron Curtain. The focus on a limited spatiotemporal chunk of history calls for thick descriptions and chunky analyses.

Another part of that risk was prevented by this book's orientation to an international audience. When I started writing this book, I dreaded the daunting task of explaining the workings of Communist-era Czechoslovakia to foreign readers.[25] But this obligation turned out to be a blessing in disguise. Putting myself in the shoes of a person unfamiliar with the minutiae of everyday life under state socialism—and getting feedback from international scholars—called for a renewed and meticulous engagement with the local context, which generated many new revelations and interpretations.

I have drawn from oral history interviews, archival material, published material, and preserved games themselves. Over a ten-year period, I interviewed forty-one participants in the story. Most of them gave detailed and candid answers that proved to be invaluable sources of insight into the player and maker experience. At the same time, the method has its apparent drawbacks, well documented in oral history methodologies.[26] People's memories of events are often blurry and biased; some interviewees colored their narratives with nostalgia for the 1980s, contempt for the Communist regime, or both. I believe that through comparisons among the interviews, and collating them with information from other sources, I have nonetheless reached an acceptable approximation of the experience of historical subjects. Throughout the project, I have engaged with the (arguably tiny) community of local gaming history enthusiasts via popularization efforts. Although an overwhelmingly rewarding and productive experience, this had its downsides, too. A couple of eager community members had read my popular and academic articles before I interviewed them, and referred to my interpretations of past events, thus creating an unwelcome feedback loop. I discarded those passages in these interviews that may have been informed by my own writing.

Archival material included primarily a diverse mix of computer club newsletters, magazines, and ephemera. Thanks to the tireless efforts of fans, many of these have been scanned and digitally preserved in unofficial online archives. The games themselves have also been preserved by the community, usually by individual users digitizing their collections and contributing to platform-specific databases. To examine the games, I played all the individual titles mentioned in this book, originally written for a variety of 8-bit platforms. Although I have access to genuine 1980s machines, I performed these play-throughs using emulators, or "virtual machines, programs that implement earlier computers and can run software for them."[27] These do not capture the physical experience of interacting with the games on original hardware, but make exploration and archival of the games' content more convenient and efficient.[28]

Overall, the material I have been working with is extremely rich, but not without its blind spots. Many of the relevant archival collections, such as those of the late Communist-era education or industry ministries, have not yet been made available to scholars. When documenting the institutional and policy perspectives, I am therefore more reliant on secondary sources. Another blind spot concerns statistics. The anecdotal, idiosyncratic nature of the historical evidence I am working with makes it difficult to estimate the scope and impact of various computing practices. Although my focus will be mostly on microhistories, I believe that the quantitative prevalence of a practice profoundly affects the qualitative experience of such practice—we can imagine how different it may feel to play computer games when it is a fringe practice versus when it is a commonplace pastime. However, there were next to no statistics on computers and games in private homes during this period, and I will have to cite estimates that have their own biases.

Yet another blind spot is gameplay itself. We have a reasonable amount of information about the life around games and the games that were produced in the country. However, we are on shakier ground when reconstructing the moment-to-moment interactions with computers and games—and especially their affective dimensions—as these were rarely photographed, written up, or otherwise documented at the time. This also explains why there is no individual chapter dedicated to the issue of interaction with games, and the topic is instead embedded in other passages.

What Is "Social" in a Social History

I have mentioned several approaches to game history—including chronicles as well as critical, emancipatory works. None of them could have been written without an implicit or explicit stance on the place of games in a society. My general approach is that of "social history." This term has been associated with attention to "groups of people, particularly those far from the summits of power," as well as "social behaviors and ideas."[29] Accordingly, mine will be primarily a history of users.

My ambition is to approximate an ethnography of the past by studying users and their everyday practice. On this journey, I will follow the lead of the *alltagsgeschichte* (everyday life histories) movement in German historiography.[30] Rather than focusing on "big structures" and "large processes," the proponents of *alltagsgeschichte* called for a "systemic decentering of analysis and interpretation" of history.[31] Aiming for a qualitative understanding of everyday life in families, households, or schools, they focused primarily on *microhistories*, case studies based on individual biographies or local contexts. Instead of trying to grasp an overarching structure that governed people's lives, they sought to reconstruct the linkages and gaps between localized meanings, practices, and contexts. Their method included "careful observation" of material, followed by "reconstructive linking together of individual elements in a network of interrelations," resulting in patchworks rather than linear chain-of-event narratives.[32]

In *alltagsgeschichte*, a social history is also a *local* history, in which spaces and places are an indispensable element of analysis on a variety of levels. On the geopolitical level, we cannot begin to understand the specifics of Czechoslovak microcomputing and game communities until we learn about trade embargoes, currency reserves, and limitations on personal travel, all of which shaped the flows of hardware, software, and information into (and out of) the country. Distribution of games and microcomputers depended on national education and research policies and the regional network of youth and paramilitary clubs. But most importantly, interactions with computers and games unfolded in everyday settings and were subject to everyday interactions within club workshops, living rooms, and bedrooms. With local histories, there is a certain temptation to present these spaces as exotic locales. I, instead, will look for the quotidian as much as for the novel and the extraordinary. Many of the events and artifacts in this narrative—such as the Czechoslovak game creators' fixation on the character of Indiana Jones—may initially seem peculiar, but will start making sense when embedded into a larger narrative meshwork.

This book presents a local history delimited by a *national* focus; it presents a Czechoslovak history as a case of a Soviet bloc history, and more generally as a case of peripheral history. Comparisons with existing research about other Eastern European countries will show both similarities and differences. Many of the similarities—such as the unavailability of hardware and state control over hobby activities—result from the shared state socialist policy doctrine and the geopolitical situation. At the same time, the Soviet bloc was by no means a monolithic territory with "communists in sight," and each country interpreted the Soviet model in its own distinct way. In the Soviet bloc, and in the preglobalization world in general, international mobility of the population was low, and foreign trade limited. Media markets, communities, and hobby organization infrastructures tended to be national. A small group of cosmopolitan politicians, managers, academics, artists, and tourism industry workers traveled to other countries on a regular basis, but much of the population was spending their lives in their home towns or cities. People were aware of the developments in the West and the rest of the Soviet bloc, but the reference frame for their everyday life and hobby activities was primarily national.[33] Although Polish magazines, for example, had considerable influence on the Czech hobby computing scene, there was little to no actual collaboration with Polish computer hobbyists.

The *national* focus is therefore not just an arbitrary methodological choice, but reflects the lived experience of the actors in my history.

Beyond the focus on people, there is one more aspect of the "social" that needs to be discussed—the social as a system of associations between people or artifacts. Bruno Latour has made the distinction between the "standard sociology of the social," and the "sociology of associations." The social, according to Latour, has traditionally been used to denote "that which has already been assembled and acts as a whole."[34] Standard sociology of the social therefore mostly works with established, durable actors that do not seem to require further disassembly. Latour gives IBM and France as examples; for our story, we may add COMECON, the Warsaw Pact, or the Czechoslovak secret police. On the other hand, sociology of associations questions the links between people and things, "reassembling the social" from its constituent parts. Latour himself has deployed his approach in both historical and contemporary studies of scientific and technological projects, in which new actors, connections, and conflicts of interests emerged.[35]

Although I will not fully subscribe to the actor-network approach Latour advocates, some of its considerations are in line with my aim to forget what we might take for granted. The history of computer games in Czechoslovakia revolves around the proliferation of a new type of technology—the 8-bit microcomputer, a phenomenon that necessitated new connections and infrastructures. To understand how it entered people's lives, we must pay close attention to how new networks were being assembled, how positions of actors were being negotiated and new alliances forged. Whereas in the West, games were at the inception of an industry, in Communist Czechoslovakia they became part of hobby practices within paramilitary and youth organizations. We would not get very far in this history if we made blanket statements about the effects of "Communist ideology" on gaming on Czechoslovakia, which was indirect at best, or about "state power," which was distributed into many agencies, groups, and individuals. Instead we need to look at the nitty-gritty details of individual interactions.

This book will combine the two approaches. In the background, I will often reap the benefits of the "sociology of the social" and work with established, well-understood actors, while in the foreground, I will use the lessons of the "sociology of associations." My basic heuristic is tracing the interplay of four basic categories of elements that, hand in hand, shaped the history of computer games in the country: (1) the individual users and creators who had to overcome numerous obstacles in order to participate in hobby scenes, form communities, and establish software distribution networks; (2) institutions that put forward various policies and norms that alternately inhibited and encouraged the growth of the microcomputer community; (3) the hardware and software, whose limitations and functionalities structured the users' activities; and (4) the discourses in which the officials, hobbyists, and players were all engaged, struggling over the place of computers and games in society. Instead of ascribing primary agency to any of these elements, I investigate the multiple ways in which they influenced each other.

The next three sections elucidate some of the specific concepts I will be using, including *vnye* (which pertains to the relationship between users and institutions), *bricolage* (concerning the relationship between users and technology), and *coding acts* (the means of connecting users with their peers via computer software).

"Gaming" and Being *vnye*

The book's title prominently features the word "gaming," which begs a comprehensive clarification. Besides referring to computer games, the title builds on the analogy to the widely used phrase "gaming the system," the purposeful bending, manipulation, or misuse of rules, usually against their original purpose and to one's advantage. As this book will show, the country's relative isolation behind the Iron Curtain, as well as its rigid economy and out-of-touch bureaucratic leadership, created obstacles and constraints for people who wanted to engage in hobby computing. But at the same time, 1980s Czechoslovakia was not a full-on dystopian totalitarian society. People could and did take advantage of gaps in the Iron Curtain and state authority. According to McDermott, the citizens learned to "play the system."[36]

This dynamic has been famously conceptualized in *The Practice of Everyday Life* by Michel de Certeau, who distinguishes two types of action—*strategies* and *tactics*. The "technocratic" top-down *strategies* are ordained by those who hold power over spaces or things, whereas "transversal" bottom-up *tactics* "do not obey the law of the place, for they are not defined or identified by it." A tactic is related to the art of "making do" with limited resources. De Certeau continues:

> It operates in isolated actions, blow by blow. It takes advantage of "opportunities" and depends on them, being without any base where it could stockpile its winnings, build up its own position, and plan raids. What it wins it cannot keep. This *nowhere* gives a tactic mobility, to be sure, but a mobility that must accept the chance offerings of the moment, and seize on the wing the possibilities that offer themselves at any given moment. It must vigilantly make use of the cracks that particular conjunctions open in the surveillance of the proprietary powers. It poaches in them. It creates surprises in them. It can be where it is least expected. It is a guileful ruse.... In short, a tactic is an art of the weak.[37]

De Certeau's language, teeming with military metaphors, might suggest an open conflict between those who employ strategies and those who engage in tactics. But tactics tend to feed off strategies; "the generalization and expansion of technocratic rationality have created, between the links of the system, a fragmentation and explosive growth of these practices."[38] Grand strategies often lead to the emergence of efficient tactical practices and networks. Let us now focus on two sets of these tactics, the first associated with the movement of objects from the West, and the second with the building of domestic networks and communities.[39]

The existence of microcomputer culture in Czechoslovakia was enabled by tactical movements of computer hardware and software across the Iron Curtain and within the country. Practices such as the smuggling of electronics and computers from abroad were commonplace, considered ethically justified, and—to some extent—even tolerated by the authorities. Tracing these movements is key to understanding Czechoslovak computer and gaming cultures. Here, we can follow the lead of John Urry's sociology of *mobilities*. Urry introduced the concept of mobilities to describe how movements of people, objects, and ideas shape identities and societies. For our purposes, three basic types of mobility will play a key role—*corporeal* mobility (travels of individual people), *object* mobility (movement of hardware or software products), and *imaginative* mobility (engagement with things and texts from other places). As chapter 2 will show, despite the relatively firm boundary of the Iron Curtain, individual travelers and smugglers were bringing Western products to the East. An important aspect of these movements is the overlap between corporeal and object mobilities. Given the state of Czechoslovak foreign trade channels, one often had to personally and individually import goods, or have someone do it. This made people personally invested in their machines and their hobby.

Paramilitary computer clubs, ostensibly training future military cadres, became important hubs of leisure hobby computing and gaming. These activities may have contained elements of resistance and nonconformism, but users and players usually did not challenge the "regime." In fact, a regime (as well as the title's "Iron Curtain") should be understood not as a monolith, but as a collection of numerous rules that the users circumvented or subverted, as well as institutional and material infrastructures whose resources they used to their own advantage, creating an informal network that became the foundation for a vibrant and prolific game culture. My use of the term "gaming" therefore does not imply confrontational, but rather tactical behavior, the two-way process of *adaptation to* the institutional infrastructures, and *adaptation of* these infrastructures to the hobbyists' own needs.

The relationship of the Communist leadership to hobby computing and computer games was far from straightforward. Although the state repeatedly failed to procure the microcomputers that hobbyists and the public were asking for, its educational policies supported hobby endeavors. Party authorities rarely interfered in the day-to-day activities within hobby groups, letting them emerge as relatively autonomous spaces. To explain the nature of these spaces, I will use Alexei Yurchak's concept of *vnye*, which has proved inspiring and influential in the historiography of Czechoslovakia.[40]

In his 2006 book *Everything Was Forever, until It Was No More*, Yurchak singles out deterritorialized "contexts that were in a peculiar relationship to the authoritative discursive regime—they were 'suspended' simultaneously inside and outside of it, occupying the border zones between here and elsewhere." To describe these creative, dynamic, and open-minded milieus, Yurchak uses the term *vnye*. Often translated from Russian as "outside," this word can also denote "a condition of being simultaneously inside and outside of some context—such as, being within a context while remaining oblivious of it, imagining yourself elsewhere, or being inside your own mind."[41] The *vnye* groups were neither aligned with nor poised against the system. As Yurchak notes: "Although uninterested in the Soviet system, these milieus heavily drew on that system's possibilities, financial subsidies, cultural values, collectivist ethics, forms of prestige."[42] He gives the example of Moscow's Pioneer Palace, the headquarters of the youth organization whose overt goal was to bring up young people "as well-educated and devoted followers of the party." But some of the palace's clubs, although in line with socialist values, "actively promoted the types of knowledge, critical judgment, and independent thinking that taught children to question authority and ideological pronouncements."[43] Yurchak's examples include literature, archaeology, and theoretical physics groups, but Czechoslovak computer clubs served similar functions.

Computers and games were well suited to become focal points of *vnye* milieus. In the Soviet bloc, computing, along with other technical pursuits, tended to be portrayed as ideologically neutral and economically beneficial. The widely discussed economic and educational contribution of computers was an important bargaining chip when seeking and justifying support for hobby computing. Although related to mathematics or ham radio, the microcomputer represented a new category of technology, which generated a need for new support groups. The fact that computers were quite rare contributed to the emergence of unexpected connections; many of the people interviewed for this book recall friendships brought about by computer ownership. At the same time, these environments were often rather exclusive, and I will try to understand why women, specifically, were underrepresented.

There is one key point at which my understanding of *vnye* diverges from Yurchak's original conceptualization. In his view, these spaces were constant, paradigmatic manifestations of late socialist regimes, and an "indivisible, if somewhat paradoxical, element of the Soviet state's cultural project."[44] He is quick to warn readers not to consider them "spaces of authenticity and freedom that were clandestinely 'carved out' from the spatial and temporal regimes imposed by the state."[45] Existing research on Czechoslovak history, including mine, confirms that these places were indeed plentiful under Communist rule. But Yurchak's account paints them—somewhat ahistorically—as places that had always already existed: a part of the "forever" he alludes to in the title of his book. But by accepting such a proposition, we would deny credit to the individual activists who started these clubs, sometimes after frustrating negotiations with authorities. Chapter 3 will show that establishing and sustaining *vnye* environments required a considerable amount of organizational work.

Tiny Interventions

Microcomputer users had been prototypical "active users" of digital technology engaging in the creation of "user generated content" long before mainstream media studies noticed.[46] To flesh out the relationship between users and technological artifacts and elucidate the practices of hobbyists and homebrewers, I want to revisit the concepts of *bricoleur* and *bricolage*, originally inspired by analog do-it-yourself hobbyist activities, but loosely adapted to many other areas of practice.[47] Sherry Turkle has used them in relation to computing, to describe the tinkerer attitude of programmers who favored *in situ* manipulations of virtual objects over careful planning and analysis.[48] The terms have also been adopted by cultural and media studies to describe subcultural fashion trends and fan work.[49] But whereas Turkle emphasizes the conceptual and mental dimensions of tinkering, cultural studies tend to connect bricolage to the tactical, bottom-up use of salvaged objects. The latter becomes especially relevant under the conditions of shortage typical of 1980s Czechoslovakia.

In the original conceptualization of *bricoleur* by Claude Lévi-Strauss, we can observe an extremely relevant dichotomy—lost in many subsequent iterations—between a *bricoleur* and an *engineer* as two distinct models of working with objects and symbols: "The bricoleur is adept at performing a large number of diverse tasks; but unlike the engineer, he does not subordinate each of them to the availability of raw materials conceived and procured for the purpose of the project. His universe of instruments is closed and the rules of his game are always to make do with 'whatever is at hand.'"[50] Whereas the engineer sets out a plan or a project in advance and sets aside the required resources, the bricoleur works with whatever material is available, making do and employing the tactics of the weak. Claudio Ciborra further develops the idea for the field of software, linking bricolage to the practice of *hacking*:

> The power of bricolage, improvisation, and hacking is that these activities are highly situated; they exploit, in full, the local context and resources at hand, while often pre-planned ways of operating appear to be derooted, and less effective because they do not fit the contingencies of the moment. Also, almost by definition, these activities are highly idiosyncratic, and tend to be invisible both because they are marginalized and because they unfold in a way that is small in scope.[51]

Being at odds with the modernist project, bricolage tends to be marginalized and "only tolerated as a hobby or pastime."[52] But as Lévi-Strauss notes, the difference between the engineer and the bricoleur is "less absolute than it might appear"[53] and users can find themselves in between these poles or routinely switch from one to the other, being engineers-turned-bricoleurs or bricoleurs-turned-engineers. Due to limited access to hardware, tools, and resources, Czechoslovak microcomputer enthusiasts often became bricoleurs not by choice, but by necessity, improvising and reusing existing bits and pieces to build hardware and software. As we will see, the engineer approach was often employed by hobbyists, many of whom had technical education, and bricolage by professional hardware designers. Throughout the book, I highlight the situated and localized nature of hardware and software bricolage. It took place within everyday life, often within the household, and was deeply influenced and limited by the concrete conditions of the users. As Ciborra puts it, bricolage and hacking operate on the level of "small forces, tiny interventions and on-the-fly add-ons."[54] Rather than construct a history of large projects, this book will trace such small forces and tiny interventions.

Working with unsystematically collected sets of resources, a bricoleur's approach is inextricably connected to the nature of those resources. Here, I draw more inspiration from Latour, and specifically his soft spot for non-human actors. Although I have the utmost interest in people, the stories I am going to tell will be of people *and* machines. The relationship between the two has been the subject of dramatic discussions in several disciplines, oscillating on the spectrum between technological determinism (the belief that technological change drives social change) and social constructivism (the belief that technology is an incidental result of social processes).[55] But Latour argues that if we trace associations on the micro level, we can see a much more diverse picture: "[There] might exist many metaphysical shades between full causality and sheer inexistence. In addition to 'determining' and serving as a 'backdrop for human action', things might authorize, allow, afford, encourage, permit, suggest, influence, block, render possible, forbid, and so on."[56]

We commonly say that computers "do" something or "behave" in a certain way. The fact that computers perform a wide variety of actions was one of the reasons that hobbyists found them so attractive. As Turkle notes, kids often considered computers "sort of alive."[57] Such expressions should not be interpreted as naïve animism, but as faithful accounts of computer user experience. Of course, the computer could not act without human input. At the same time, a specific piece of hardware encourages certain types of actions—it makes one thing easier and another difficult. Let us consider the example of the cassette tape, the most common data storage medium among Czechoslovak users. Its notorious slowness and its need for rewinding structured many of the temporal features of gaming practice. On some platforms, local enthusiasts came up with faster data encoding schemes, but eventually reached the physical limits of the magnetic tape technology and available computer hardware. Hardware and software can be stretched and adjusted, but not indefinitely.

Much of this book will deal with 8-bit home computers, whose interface design strongly encouraged programming. Most machines came with an interpreter of the BASIC programming language stored in their ROM, straight from the factory. When switched on, they booted into the machine's version of BASIC, making it the default way of interacting with the computer. This certainly contributed to the popularity of homebrew, amateur programming efforts. On the other hand, the ubiquitous BASIC interpreters could not hide the simultaneous existence of multiple incompatible hardware platforms.[58] A program for the Sinclair ZX Spectrum would not run on a Commodore 64 and vice versa. Platform differences have tremendous emic significance for our story, and structured local gaming culture, including user practices and community network building. People did not join Atari clubs or Sinclair clubs because of their aesthetic preferences or exposure to persuasive discourses, but for pragmatic reasons: they benefited from information and software for their particular platform.[59]

Coding Acts

Although this book crosses over into the fields of the history of technology (and computing) and regional history, I consider myself primarily a scholar of digital games, and therefore feel obliged to clarify the role of games in my story. Since its beginnings in the 1990s and institutionalization in the 2000s, the field of game studies has been rocked by several ontological and methodological debates. Some scholars adopt a more "formalist" approach and try to describe digital games as structures of rules and fictional audiovisual or textual content, but others stress that games acquire meaning when played by real people in social contexts, and bring attention to the embeddedness of games in the social world.[60] These views are not necessarily mutually exclusive, but tend to lead to different methodologies and arguments.

To properly situate games within my narrative, I have had to attune my sensibilities to the ways Czechoslovaks in the 1980s understood computer games. To a much larger extent than today, games were treated as pieces of code that were programmed, copied, cracked, dissected, and passed around. Their rulesets and stories were relevant, of course, but often considered less significant than technical execution and audiovisual spectacle (however modest these may seem in retrospect). In the research on this period, making and playing games cannot be isolated from other hobby computing activities such as hardware tinkering, cracking, or programming of utility and demo software. For most of the decade, using computer technology only to play games was considered an aberration rather than a norm, and most people in my story took part in more than one type of projects. Grouping some of these activities together, Swalwell has introduced the concept of *vernacular digitality*, which stresses their popular, amateur (or *homebrew*) dimension.[61] In a similar manner, Montfort speaks of *creative computing*, a category that includes writing games, interactive fiction, poetry generators, and other artwork in both “revered and popular” forms.[62]

Much of users’ creative efforts remained in the closed circuit between them and their machines, as results of private experimentation. These are mostly lost to history, leaving us with only those pieces of software that were presented and distributed to others. One of my missions in this book is to understand the reasons why they were made, presented, and distributed. Following recent discussions about code as speech and hacking as communication, I want to draw inspiration from pragmatics, namely speech act theory.[63] One of its founders, J. L. Austin, pointed out that speech communication amounts to much more than making verifiable statements about the world. In natural language communication, speaking is also doing. We may utter a sentence with a sum of semantic content, and therefore perform a *locutionary* act. Simultaneously, the very sentence may be a performance of our intention to change the state of the world, and therefore we are engaging in an *illocutionary* act.[64] The intention or “force” of the illocutionary act cannot be deduced from the semantic content itself. A usual example is the sentence “Can you pass me the salt?”—styled as a question, but conventionally interpreted as a *requestive*. Another example is a promise—which is both an utterance and an action of committing to the said promise. The locutionary and illocutionary acts are considered components of a *speech act*.[65]

Analogously, I will argue that a program or a game is more than a piece of code, a computational artifact, a system of rules, or a narrative. The programmer does not merely hand over his program, but does so with intentions that can be interpreted based on the norms and conventions of a community. A game may be more than just an offer of entertainment.[66] It also serves as a statement about the author's skills and willingness to participate in a community; it may summon recognition and respect. As Reunanen and his colleagues have shown, demos made by young members of computer subcultures were a means of intergroup communication.[67] Similarly, creating a game about an antiregime protest may raise awareness about that fact, or even invite people to the next demonstration. I thus propose to treat the making of games and other software, as well as hacking and cracking, as *coding acts*—acts of writing and releasing computer code to attain a goal in a given communication context.

This approach has three basic benefits for my analysis. First, like *vernacular digitality* and *creative computing*, it lets us group game making with other related activities such as demo making and cracking. Second, it allows us to connect the artifact to the people around it. Engaging in a coding act entails communication with a group of other users with an intention and a goal. It allows us to analyze not only the meanings within the semiotic material of the game but also the meaning of *having made* the game. In this respect, my notion of the coding act is closely related to the concept of *technicity*, defined by Dovey and Kennedy as the expression of identities "formed around and through … technological differentiation."[68] Through coding acts—but also through hardware tinkering—users demonstrated their skills and affirmed their affiliation to technophile circles. Third, the coding act approach overcomes the methodological issue of object demarcation. In the period I am studying, games circulated in many cracked, hacked, and otherwise mangled versions. Focusing on the act of making or modifying games frees us from the thankless task of isolating an idealized version of a game as a discrete artifact. Instead, we can see the creation and release of its code as situated action.

The concept of a *coding act* is not meant to replace existing game or software ontologies, or to extend speech act theory to the field of software. Moreover, it may not fit current industrial settings, in which software is made by large teams within complex organizations. But in a homebrew, amateur environment, it can be a helpful heuristic tool. For a social historian, the question "What social effect was this author trying to achieve with this piece of code?" is more productive than "What is the content of this game?" Although the latter question also needs to be asked, it is but a stepping stone to answering the former. A coding act is a vector rather than a point, a verb rather than a noun. As such, it allows a historian to breathe life into software artifacts. Methodologically speaking, the illocution cannot always be deduced from the rules or fictions of the games, or from the runtime behavior of a program. Quite often, we have to arrive at it by interviewing the author or investigating auxiliary, paratextual material such as credits, loading screens, in-line comments in the code, and so on. And because computer games and other programs are complex multimodal artifacts, they can "do" more than just one thing.

What Lies Ahead

This book's narrative both is and is not chronological. The order of the chapters follows a loosely temporal logic—it starts with a general overview of technology policies from the 1960s onward, then continues with the acquisition of technology and self-organization, to the distribution and making of games. But each chapter concerns itself with one theme, follows its own internal chronology, and builds around a set of key concepts.

To stress the interconnectedness of games to the more general context, the first three chapters of this book are not even about games per se, but about the microcomputer hobby, of which games were an integral part. Chapter 1 identifies the place and meaning of the home computer in late Communist-period Czechoslovakia. It analyzes the political, economic, and cultural context of 1980s Czechoslovakia, and explains the elusive ideals of the so-called *scientific-technological revolution*. Following the stories of Czechoslovak efforts at producing home computers, it highlights a fundamental disagreement between state authorities and computer enthusiasts over the definition of what a computer was good for. The Communist leadership failed to acknowledge and act upon the people's desire to own and play with computers. Due to this disagreement—which I call a "clash of teleologies"—home computing became almost exclusively a domain of amateur enthusiasts. Chapter 2 will trace the journeys of computer hardware into socialist-era homes under the conditions of a *shortage economy*. Throughout the 1980s, computers were mostly unavailable for retail sale, and people had to resort to hunting them down through the channels of individual imports and smuggling. Thanks to its low price and versatility, the British Sinclair ZX Spectrum computer ended up being the default platform for many domestic users and remained so long after it became obsolete in the United Kingdom. This chapter will also investigate how computers fit into people's everyday lives and households. Chapter 3 will show how hobby computing enthusiasts built their communities and networks. Because any kind of gathering and collective ownership required the patronage of a socialist organization, local hobbyists moved into the *vnye* spaces of paramilitary and youth organizations and formed clubs that became hubs of hobby computing. We will look at the clubs' gender makeup and the activities their members engaged in, including hardware bricolage—which was widespread in Czechoslovakia's shortage economy. The end of the chapter will compare the apolitical *vnye* spaces with the spaces of dissent, where computers were used (although only marginally) to assist in the publication of samizdat.

Chapter 4 will explore the discussion about the dangers and benefits of computer games. Although playful experimentation had long been an integral part of both professional and hobby computing, commercial titles differed from the older type-in games and "diversions." Prominent members of the amateur community feared games would turn users into passive recipients of prefabricated content. Club leaders promoted the idea that users should instead treat these games as material for bricolage. Unlike in some Western countries, where games became a stand-alone category of commercial products, Czechoslovak discourse saw them as pieces of code ready for hacking and modification. Chapter 5 will shed light on the informal distribution networks that supplied local users with games and other software, mainly by reconstructing the journey of one particular title, the 1987 British game *Exolon*. It will trace the national "sneakernet," as well as its transnational ties to the West. Although Czechoslovak users played Western titles, they depended on unauthorized copies without packaging and instructions, and therefore had little knowledge about how particular games were intended to be played, leading to both misunderstandings and experimentation. While examining distribution networks and mobilities, the chapter also engages with the material culture that they helped create—hand-drawn cassette covers, serendipitous compilations, and personal walkthroughs. It concludes with an account of the local arcade machine culture.

The final two chapters tell the stories of Czechoslovak game creators. Chapter 6 makes a case for studying conversions and clones, which are often left by the wayside in favor of presumably more original, canonical titles. Although many domestic programmers started out imitating

Western or Japanese titles, in the process they created unique artifacts very much tied to the local context. Through a process of creative imitations, two subgenres became emblematic of 1980s Czechoslovak production—text adventures (starting with a series of unlicensed games about Indiana Jones) and hacking games. In the latter, the player takes on the role of a computer hacker and cracks puzzles using a simulated operating system. Chapter 7 will explore the use of games for personal expression and activism, using the notion of *tactical media*. From short personal messages to titles protesting the Communist regime, people were using games to communicate their experiences and opinions. Such early use of games for these purposes is quite rare in worldwide comparison, and serves as a testament to the extent to which games were integrated in the everyday communication practices of their fans. And finally, after a conclusion, the epilogue will sketch out the influence of the 1980s developments on the later stages of Czech and Slovak gaming cultures.

1 Micros in the Margins: Computer Technology in the State Socialist Society

> They cost a lot of money, they are faulty—that's about all I know about computers.
> —An older man interviewed in a vox pop for Czechoslovak Television, 1988[1]

In 1988, Czechoslovak state television broadcast its first educational series dedicated solely to microcomputers, entitled *Computer Dilemmas*. Its scripts were based on the interplay between two moderators: a thirty-something male expert, played by the program's author David Gruber, and a skeptic, played by a woman in her early twenties. The skeptic challenged the expert to convince her, as well as the viewers, of the benefits of home computing in everyday life. Accordingly, the program was set in a studio decorated to resemble the homes and offices of late socialist Czechoslovakia, using various tones of homely beiges and browns, complemented by yellow desk lamps. In many ways, the program's goals and structure resembled the influential 1982 British series *The Computer Programme*—only six years later.[2]

In the first installment, the expert optimistically claimed that the program would "focus on computers, which you already have at home, or soon will have."[3] However, the subsequent vox pops with people on the street indicated that they did *not* usually own home computers and had little idea what benefits these machines could bring them. Although certainly not representative of the whole population, the answers seem to be authentic and unscripted, and offer a rare glimpse into ordinary citizens' thinking about computers.

Nineteen out of twenty people said they did not have a computer, roughly matching the 1989 official statistics, according to which 1.8 percent of households owned a machine, compared with 18.6 percent in the United Kingdom, 18 percent in the Netherlands, 15 percent in the United States, and 13 percent in West Germany.[4] Asked if they would like to have a computer at home, a majority of respondents questioned its utility or said it was unaffordable or unavailable.[5] As a blue-collar worker put it: "I don't personally have one, although I naturally have nothing against having one—but I don't expect to have one in a foreseeable future, because neither financial circumstances nor availability are favorable."[6] The only interviewee who did own a computer was a male engineer in his thirties, a typical representative of the country's younger generation of technical intelligentsia, interviewed in his lab. Another vox pop in a later episode suggested that if it were affordable, ten out of twenty people would buy a computer for themselves, and sixteen out of twenty would buy one for their children. As the moderator concluded in the final episode: "People would like to have computers, but these are inaccessible; there is quite a lot of information, but all of it seems to be so technical that it tends to drive people away."[7]

In the West, the personal or home computer was an intensely advertised piece of consumer electronics, its concept originating in a "rich interplay of cultural forces and commercial interests."[8] The former included the hobbyist tradition and the capitalist ideal of personalized consumption. The proliferation of computers was driven by the interests of traditional semiconductor, electronics, and computer companies, as well as new entrepreneurs such as Apple. In Communist-era Czechoslovakia, the hobbyist tradition was also strong, but computing did not become a customer-oriented business. Instead, it was a part of broader technological policies of Communist leadership, originating in the idea of *scientific-technological revolution* (STR), a visionary project that promised to improve the country's economy and society by purposeful implementation of technology. Originally a techno-socialist utopian concept that informed reformist policies in the 1960s, it later mutated into a technocratic approach that stressed the deployment of computer technology in the industry but offered no visions for its use in everyday life. Although the rhetoric of STR incited interest in computers, actual machines remained out of reach for most ordinary citizens, continually suspended in an elusive near future, or locked behind the doors of factories, universities, and state institutions. Microcomputers for the home or personal use never made it into the five-year plans of the state socialist economy. As a renowned Czechoslovak computer engineer put it in 1989: "In our country, no one is responsible for the production of personal microcomputers. No one was assigned that task, no one had to complete it, and therefore no microcomputers are available."[9] But as the country's authorities detached themselves from the idea of a personal microcomputer, it was eagerly pursued by the country's enthusiastic hobbyists.

This chapter will show how microcomputers were sidelined into the margins of the Czechoslovak economy and society. I will start by laying out the basic points of the political and social history of postwar Czechoslovakia and painting a picture of the so-called *normalization* era. I will examine the Czechoslovak computer industry, and follow the country's technology policies, retracing the story of STR from its inception toward a point I call a *clash of teleologies*—a mismatch of views between technocratic state officials and technophile computer enthusiasts about the ultimate place and mission of computing in society. Finally, I will explicate this mismatch by documenting the Sisyphean struggles to produce microcomputers in the country, and—more specifically—by analyzing the debate about the failure of the Ondra platform, an 8-bit microcomputer designed to be used by kids at their homes. This debate was one of the crucial moments at which relevant social actors defined the uses and values of microcomputers. At that point, the hobby community clearly emerged as an autonomous, vocal, and resourceful group critical of the state's technological policies.

Toward Normalization

In Czechoslovakia, the pioneering age of home microcomputing coincided with the final decade of Communist rule. Local computer enthusiasts' memories of excitement over new technologies, and of hours spent playing and making games, are thus inextricably linked to memories of the last years of the stagnant regime.

Czechoslovakia became part of the Soviet bloc in February 1948, when the Communist Party of Czechoslovakia (Komunistická strana Československa or KSČ), which had already secured a government majority and considerable popular support, seized power in a Soviet-assisted coup. For the Soviets, the small country in the heart of Europe was an important asset both geopolitically and economically. With its heritage of intermittently prosperous periods during the late Austro-Hungarian Empire and the interwar democratic republic, Czechoslovakia had one of the most developed industrial sectors in the region, especially when compared with the Soviet bloc's other, more agricultural economies. It produced high-quality armaments, automobiles, heavy machinery, and consumer items such as shoes and glass, as well as punch-card and calculation devices.

Immediately after the coup, thereafter dubbed “Victorious February,” Czechoslovak Communists imposed a hard-line Stalinist rule. They disposed of tens of thousands of actual and potential political opponents through a combination of show trials, purges, and labor camp internment.[10] Industry was completely nationalized (although this process had already started in 1946), agriculture was collectivized, and central planning replaced the free market; strict censorship was imposed.

The regime started to loosen up slowly after Stalin’s death in 1953. This was most palpable in arts and design, where the dogmatic insistence on socialist realism gave way to the pursuit of elegant modernism and formal experimentation. One of the most pivotal and fondly remembered moments of Communist-era Czechoslovak culture is the triumph of Czechoslovak designers and artists at the 1958 Expo in Brussels. As censorship started to lift around the mid-1960s, the newfound hope and artistic freedom brought a powerful surge in cultural activity. It was during this period that the most influential movement within Czechoslovak film, the Czechoslovak New Wave, produced its best works, both joyful and socially critical. A less visible but even more impactful process was taking place in the country’s power structures, where de-Stalinization resulted in a shift of power from apparatchiks to experts. A strong expert culture was shaping policies, particularly those related to science and technology.[11]

Around the mid-1960s, progressive members of the ruling Communist Party, joined by students and prominent members of the intelligentsia, started to question the Stalinist paradigm of Communist rule. They reasoned that the total dominance of the party over all matters stifled social and economic progress, stunted civil liberties, and stood in the way of the proclaimed goal of creating an equal and just society. In January 1968, the reformists, led by Alexander Dubček, secured control of the party and set out to build a more democratic socialism, or a “socialism with a human face.”[12] But the Prague Spring, as the brief reformist-led period came to be called, was not to survive the summer.

The Soviets—concerned about these developments, which threatened the balance of power in Europe—came up with a swift and resolute response. With the support of some Czechoslovak conservatives and the forced compliance of the leading reformists, Warsaw Pact armies invaded the country in August 1968 to widespread popular condemnation, but no organized resistance. Among the conditions of capitulation set by Moscow, there was a call for a normalization of affairs. As a result, party liberals were removed, demoted, or co-opted; the plans for reforms were scrapped. In 1969, the country became a federation of Czech and Slovak republics to appease the Slovak branch of the party, adding another layer to the mushrooming bureaucracy. The new Soviet-approved party leadership commenced purges and screenings that were less brutal but more extensive than the ones carried out in the 1950s.[13] The humanities-based intelligentsia, seen as the primary culprit for the pre-invasion "crisis," took a big hit. Purges were "less intense in the technical intelligentsia, which was thought to be shielded from corrupting ideas by their scientific rationality,"[14] although even here, they led to a significant reshuffling of cadres. As I will show, the softer stance against technical workers reflected the apparat's belief that technology was not an ideological instrument.

Normalization did not stop at screenings. The goal of the normalizers was to pacify the population and prevent future crises. Gustáv Husák, the aging apparatchik who was to become the party leader and president for most of the 1970s and 1980s—whose omnipresent but expressionless portraits became a symbol of normalization—put it this way at a 1968 party meeting:

> A normal person wants to live quietly, without certain groups turning us into a jungle, and therefore we must appeal to people so that they condemn this. This party wants to safeguard the quiet life. These [reformist] groups want to terrorize not only the party leadership, but citizens as well. We must keep order in this state with a firm hand.[15]

In 1970, the then president Ludvík Svoboda described the normal life as depending on three certainties: "an assured opportunity to work calmly and systematically 'for a better tomorrow,' the 'leading role' of the party, and hermetic alliance with the Soviet Union."[16] These statements marked the beginning of an era in which conformity was valued above all else. Under normalization, Czechoslovak leadership was among the most politically, economically, and culturally conservative of all Soviet bloc countries.

Beyond the Quiet Life

On the surface, normalization was a remarkably successful project. The citizens seemed to adapt to the situation and resume their quiet lives. They more or less voluntarily participated in the ritualistic manifestations of support, such as the parades celebrating Red October and Victorious February. There were almost no signs of social unrest during the two decades of normalization, unlike in neighboring Poland, where the Solidarity trade union became a formidable opposition force.[17] In Czechoslovakia, small groups of dissidents were relegated to the margins of society without much recognition or support from the general population.[18] Explaining this acquiescence is one of the key challenges of the historiography of Communist Czechoslovakia. There are at least two major complementary explanations.

The first and perhaps more conservative explanation attributes this to fear of reprisal.[19] Soviet troops remained stationed in the country after the 1968 invasion as a powerful reminder of the reforms' fate. The screenings, along with zealous surveillance by the State Security (Státní bezpečnost or StB), sowed the seeds of paranoia and distrust among citizens. As one of my informants put it, as soon as there were three people around the table, it was likely that one them was an StB agent.[20] A Czechoslovak counterpart of East Germany's infamous Stasi, the StB recruited thousands of informants, often through blackmail, and traded information about anti-regime activities for favors and promises of career promotion. Suspicion of having engaged in such activity, especially when combined with a history of reformist sympathies, could ruin one's—and one's children's—career prospects or ban one from traveling abroad. Still, some avenues of life were watched more closely than others. The arts, literature, music, and humanities were considered more liable to ideological subversion than the natural sciences, technology, or electronics. The stereotypical ideological enemy, as seen in propagandistic TV serials, was a failed reformist poet collaborating with Western spies, rather than a rogue engineer.[21] Experts invaluable to the running of the economy were approached with more lenience, allowing elements of the expert culture to carry over to the normalization period.

The second explanation of the acquiescence is a general "shift of the society from the public to the private."[22] According to a tacit but well-understood social contract, the socialist state guaranteed housing, healthcare, job security, social benefits, and a satisfactory living standard in exchange for "public political compliance."[23] Fatigued by the turbulent political events of 1968, people found comfort in private spaces, among family and friends, enjoying leisure activities and consumption of the goods offered by the rusty but more or less functional economy. The party encouraged this drive toward individualist consumerism despite its apparent incompatibility with the Marxist-Leninist doctrine. In October 1969, General Secretary Husák requested financial assistance from the Soviets. It is quite telling that instead of investing the money in the country's aging industrial infrastructure, he used it to freeze the prices of food. As Bren notes, these freezes remained largely intact for another twenty years and, as a result, consumer consumption "could, and did, increase."[24] Between 1980 and 1989, the average salary rose steadily from 2,637 to 3,123 Czechoslovak crowns.[25]

Socialist consumerism mirrored its Western equivalent in many aspects, but it had its own idiosyncratic aesthetics. Although many people yearned for Western products, the stores were mostly supplied with locally designed and manufactured goods, and the domestic market, including electronics, was a reserve for a limited range of national brands of varying quality. At the same time, state-run Czechoslovak television became an important tool for social cohesion, offering not only straightforward propaganda but also popular dramatic serials about working and middle-class men and women who overcame personal challenges to be able to enjoy their quiet and normal socialist lives.[26]

Socialist consumerism had its spatial and material implications. According to Bren, who calls this phenomenon *private citizenship*, "[People] came to understand their rights (and obligations) as citizens as existing not within a political collective but within individualized spaces of self-realization."[27] Do-it-yourself and hobby activities were thriving, often partially supported by state or military funding. Hiking and tourism were popular among all generations; people from urban areas were building cottages (*chata*) in the countryside as their weekend getaways, "do-it-yourself paradises," and opportunities for individualist self-fulfillment.[28] In larger cities, almost half the population across professions would leave for the weekends, leaving the streets abandoned.[29] But eventually, socialist consumerism turned out to be a double-edged sword. Leisure-time bricoleurs pilfered resources from their state-owned employers or spent many of their working hours moonlighting on other people's cottages and hobby projects, with adverse effect on workforce productivity.

Despite the major roles played by the apparat and the secret police, normalization-era Czechoslovakia was not a totalitarian state in the traditional sense associated with the Stalinist Soviet Union or the Nazi dictatorship. The party was never fully in charge, being subjected to external constraints and "composed of concrete fallible human beings, who … at the higher levels tended to scrabble around pretending to be in control over processes that were often beyond them."[30] It did have a path in mind for each "socialist citizen"—the path of a normal, quiet life that did not threaten the status quo.[31] However, people learned to "play" or "live" the system in order to satisfy their consumer demands, and simply to do the things they enjoyed.[32] They approached socialism pragmatically, even cynically, engaging in "moonlighting, bartering, bribery, hoarding, speculating, pilfering and smuggling."[33] These tactics had become so engrained in the daily lives of Czechoslovak citizens that they also became "normal."[34] By the 1980s, many, and possibly most, no longer believed in the state socialist project and privately disagreed with the official ideology. One of my informants recalls, "We were very well accustomed to this kind of doublespeak. At home, we all knew that Dubček was a good guy, but I knew I couldn't say that at school."[35] Behind the surface of parades and slogans, the legitimacy of the regime was irreversibly eroding.

Isolation behind the Iron Curtain was never hermetic, either. Some citizens—although invariably monitored—could travel to the West and bring home luxury items, technology, or magazines. Hollywood and other "capitalist" films were regularly screened in Czechoslovak cinemas—although they could not make up more than 30 percent of the titles shown.[36] Some people watched black market VHS tapes with films that did not make it into official distribution; others bought imported vinyl records at illegal bazaars. Those who lived close enough to the borders could listen to German or Austrian radio or even watch Western TV stations.

Those who did could clearly see that the ailing Czechoslovak economy was falling further and further behind the West. Throughout the 1980s, growth and productivity stagnated.[37] Workforce productivity was at about 53 percent of the level achieved by developed Western European countries, and the Czechoslovak economy was burning through far more resources.[38] The overwhelming majority of goods and services were provided by nationalized industries, which progressed along five-year plans that outlined the goals and strategies for the whole period. Until 1988, private enterprise was technically illegal—unlike in Poland or Hungary, where the collectivist doctrine was applied with less fervor and small businesses or private farms were allowed to survive.[39] Decision-making in the economy was often driven by personal bias or bureaucratic interests rather than the proclaimed socialist ideals; just as ordinary citizens learned to game the system in their everyday pursuits, so did industry managers and workers, bending the rules to fulfill quotas they were not personally invested in.

However, party elites did not find the courage to implement reforms akin to Mikhail Gorbachev's 1985–1988 perestroika. The country's leadership faced a difficult dilemma. Clearly, they were expected to follow the Soviet example. Some officials saw perestroika as a career opportunity and pushed for reforms, whereas many others feared that the reforms—which bore eerie resemblance to the Prague Spring's "socialism with a human face"—would compromise the normalization project, and hence their power. The conservative clique prevailed and admitted nothing more than a series of cautious, mostly economic measures, accompanied by signs of liberalization in the cultural life. But rather than appeasing the populace, these concessions revealed that the regime was stuck in repetitive monotony and unprepared for substantial change.[40]

A Revolution That Was Normalized

I have already mentioned the bad shape of the Czechoslovak economy in the 1980s. Besides external causes such as the 1970s oil crises, this state of affairs was a heritage of failed technology policies, whose story will help us understand the peculiar fate of microcomputers in Czechoslovakia. In the 1950s, the country's command economy had achieved extensive growth through juggernaut industrialization, but its top-down, centralized nature made it difficult to introduce technological innovation and respond to customer demand.[41] The 1962–1963 national economic crisis urged experts to rethink the country's policies.

One of the suggested remedies was the introduction of *cybernetics* into the management of the economy, traced on the Soviet example in the works by Slava Gerovitch and Benjamin Peters.[42] Initially criticized as a reactionary pseudoscience, cybernetics was reinterpreted in the late 1950s by leading Soviet bloc academics as a scientific, rational solution to complex problems of governance, and became part of research agendas and reform proposals. The interest in computers as tools of cybernetic management sparked projects such as the OGAS computer network in the Soviet Union, and the boom of industry automation projects. Czechoslovak interest in cybernetics closely followed the Soviet example. Informal groups of cybernetics enthusiasts started forming in the 1950s. A translation of Norbert Wiener's *Cybernetics* was published in 1960, two years after the Russian one; in 1965, the Czechoslovak Academy of Sciences started publishing the *Cybernetics* (*Kybernetika*) scholarly journal.[43]

On a more general philosophical level, the deployment of cybernetics can be considered a component of STR. The term *scientific-technological revolution* was probably coined in 1955 by the chairman of the Soviet Council of Ministers, Nikolai A. Bulganin, and inspired by the work of the Marxist historian of science John Desmond Bernal. Conceived, according to Stefan Guth, as a "consultative process in which political leadership was to be joined by technocratic expertise," it was a response to de-Stalinization and the decline of charismatic leadership.[44] By the early 1960s, it had become a buzzword in the Soviet bloc's expert cultures. In Czechoslovakia, it found its expression in the work of Radovan Richta, a prominent philosopher of considerable influence in the Communist Party. When the reformist tendencies within the party were peaking, Richta and his multidisciplinary team were tasked with searching for ways to restructure and improve socialism. The resulting 1966 book, *Civilization at the Crossroads*, became a domestic hit and gained international renown, thanks to numerous translations.[45]

In their work, Richta and his think tank went even further than their Soviet colleagues, calling for profound political changes along with economic and managerial ones. The authors criticized the state of the Czechoslovak economy, a result of a one-sided industrialization that favored quantity over quality, as well as the ineffective "administratively directive system of management."[46] In order to compete with the West—they argued—socialist countries had to systematically employ cybernetics, automation, and advanced technologies. They suggested that this could not be achieved without a deeper political and social reform, describing the then current "hypertrophy of directives" as something that damages the human factor, "dulls initiative, destroys the feeling of responsibility, cultivates mediocrity, teaches one to go with the flow, not to stand out and not to push for the new, but rather to sail through the pores of directives."[47] They stressed the importance of creativity and lifelong education, and called for participatory management, interdisciplinary research, and widely accessible information systems that would allow for both top-down and bottom-up flow of information—a prerequisite for efficient "collective reasoning" with the participation of individual workers and citizens.[48] Such changes would have to go hand in hand with creating "free space for self-determination and self-development of the socialist citizen."[49] If all these conditions were met, the socialist society would eventually leapfrog capitalist ones, because technological progress would benefit not just capitalists but all citizens. Revisiting the book as historians of microcomputers, we can easily read many of these passages as endorsements of mass access to computer technology. The authors' vision was participatory, not technocratic. In fact, they explicitly warned against "technocratic tendencies," which resulted not from science and technology alone but from the partial interests of powerful groups within the society.[50]

Richta's audacious vision of technosocialism was part and parcel of the Prague Spring's "socialism with a human face" project, which was squashed after the Soviet invasion. One might have therefore expected Richta to be punished and the book banned. Instead, both ended up "normalized." The original diagnosis of the economy's problems continued to hold throughout the normalization era, but the proposed remedy was far too radical for a regime preoccupied with maintaining the status quo. In a 1972 follow-up book, Richta and his coauthor denounced reformist ideas, stating that attempts at modernized models of the socialist society "objectively lead to refusal and denial of socialism," and even to "attempts at counterrevolution."[51] The term *scientific-technological revolution* remained in the official vocabulary, but was stripped of its reformist undertones. In later normalization-era publications, the word *revolution* was gradually replaced by *development* or *progress*.[52] Like *cybernetics* in the Soviet Union, STR seemed to become a buzzword that gave legitimacy to the very technocratic tendencies that Richta was warning against.[53] As science historian Vítězslav Sommer observes, the party was unwilling to give up control over science and technology: "Science was not an active actor of development but a policy instrument constructed by experts for the purposes of centralized governance."[54]

Although the importance of cybernetics, automation, and technological progress continued to be stressed throughout the 1970s and 1980s, a deeper structural reform was never realized.[55] The measures taken to stimulate and modernize the economy were piecemeal and largely ineffective "improvements" through "rationalization" or "intensification" of the existing directive system.[56] Vast amounts of money were invested into applied research with meager results. Science and research fields employed twelve times as many people as in Denmark or Finland.[57] (However, this number included not only scientists but also manual and skilled laborers in research branches of large industrial concerns.) In addition, individual efforts in industrial research and development could be rewarded by financial bonuses under a broad "improvement proposal" scheme, which remunerated innovation in a distant echo of Richta's reform plans.

But rampant mismanagement in central planning prevented innovation from making significant impact. According to education historian Pavel Urbášek, "research and development plans existed in complete isolation from production plans."[58] In a system that kept incentivizing quantity of production over quality, factory managers avoided technological innovation in fears of increasing production costs. This misalignment of research and production generated a disproportionate number of prototypes with little economic or social impact, and continually frustrated ambitious designers and engineers. This was not the scientific-technological revolution they had been hoping for.

The State of the Computer Industry

Central planning required gathering and evaluating massive amounts of statistics about industry and commerce, and Soviet bloc countries were therefore well aware of the potential benefits of computing.[59] Mainframe computers in particular seemed to be an epitome of centralized decision-making and planning—these massive wonders of engineering could be localized in a particular controlled environment and fed vital information from all corners of the country.[60]

In Communist-era Czechoslovakia, numerous state and military institutions, factories, universities, and research centers had been purchasing and running computers since the 1950s. In 1974—not too long before the advent of microcomputers—358 domestically produced and 709 foreign-made computers were running throughout the country. Out of the foreign ones, 372 were made outside of COMECON, the Soviet bloc's trade treaty.[61] These included IBM, Hewlett-Packard, UNIVAC, and Control Data Systems mainframes, as well as DEC and HP minicomputers (or "minis"). However, importing computers was far from straightforward. The Czechoslovak crown was not convertible to Western currencies, and machines therefore had to be purchased with funds drawn from convertible currency reserves controlled by the State Bank. In addition, computers were subject to embargoes—each larger purchase had to be approved by CoCom, a Cold War–era institution founded by Western countries to oversee embargoes on foreign trade, especially involving technology with potential military applications.[62] These limitations made domestic production a desirable project.

Czechoslovakia had promising prerequisites for building a computer industry. It had the Aritma factory, one of the most advanced punch-card machine manufacturers in the region, founded in 1940.[63] And it had Antonín Svoboda, a world-renowned mathematician and computing pioneer who headed the Research Institute for Mathematical Machines. During World War II, Svoboda made an adventurous escape from Prague through France into the United States, where he worked at the MIT Radiation Laboratory and met, among others, the computing pioneer Howard Aiken. In Czechoslovakia, he became an early proponent of cybernetics. At the Institute, he created innovative machines such as SAPO, the first Czechoslovak automatic digital computer (launched in 1956, see figure 1.1), and 1963's EPOS, a decimal computer with sophisticated fault-tolerance features, motivated by error-prone relays the team had to use.[64] Throughout his career, he struggled—and often failed—to secure political support and resources, and, disillusioned, left the country in 1964 for the United States.

The Cold War era tends to be associated with military funding of technological advancements such as the Soviet and American space programs or the ARPANET.[65] However, the role of the Czechoslovak People's Army in the development of computer technology was that of a receiver rather than a leader. It did have privileged access to resources, and therefore used computers relatively early, starting with punch-card machines in the 1950s, mainly for administrative tasks. Historical research on the topic is woefully lacking, but post-1989 accounts by military officers indicate that the army acquired its first digital computer, the Svoboda team's EPOS, in 1967. Starting in the 1970s, it used portable minis (and later, micros) for field calculations and mission command automation. Due to embargoes on military technology, the army did not use Western machines and relied on domestic or COMECON ones.[66] But despite running three research centers and gaining renown for high-quality airspace reconnaissance technology, the army's computer equipment was no more cutting edge than that of other branches

Figure 1.1
The SAPO computer, designed at the Research Institute for Mathematical Machines by a team headed by Antonín Svoboda. Photo by Karel Mevald. © 2017 ČTK/Karel Mevald.

of the state.[67] Nevertheless, the army did play an important indirect role in supporting hobby computing through Svazarm, an independent paramilitary organization that was largely managed by army officials.

Soon after Svoboda's departure to the United States, Czechoslovakia gave up on original development of mainframes. Around the same time, the local computer industry started to clone (or, less frequently, license) Western hardware.[68] In 1968, the country joined the COMECON-wide Uniform System of Electronic Computers program, which focused on the manufacturing of "functional equivalents" of IBM 360 mainframes, sometimes with the help of industrial espionage.[69] This was followed by the System of Small Electronic Computers initiative, under which several socialist countries, including Czechoslovakia, jointly manufactured clones of DEC minis.[70]

These initiatives alleviated the dependency on Western hardware by providing hundreds of mainframes and minis. By 1982, domestic models made up more than half of the country's state-owned computers (1,029 out of 2,036).[71] However, reliance on the copying of Western technology, along with mismanagement of resources, brain drain, and other factors, resulted in the "computer gap" between the West and the Soviet bloc.[72] According to a contemporary CIA report, this gap ranged between two and eight years depending on the particular type of computing equipment.[73] Importantly for our story, the initiatives cemented the Soviet bloc's orientation toward the West as a center of innovation in computer technology, a sentiment that would be shared by computer hobbyists.

Electronization Programs of the 1980s

According to normalization-era technology policies, the role of computers was to become tools or "helpers," applied in particular locations and processes, and subservient to grander goals, such as "increasing workforce productivity" or "saving labor."[74] This is clearly visible in the resolution produced by the 1981 convention of the Communist Party of Czechoslovakia, written in the typically cumbersome and circuitous party jargon:

> For the fulfillment of the decisive tasks of the national economy, fast development of electrotechnical industry, namely microelectronics and automation tools, is critical. For that, it is necessary to create cadres and material conditions for expeditious application of electronics and microelectronics in all branches of the national economy.[75]

This application-oriented teleology of computing created a vicious circle. Electronics and computers that were supposed to improve the economy could not be produced by that very economy in sufficient volume or quality, and neither could they be imported.

To support and manage the electronics industry, the Czechoslovak Federal Assembly instated the Federal Ministry of Electrotechnical Industry in 1979. Its first and only head was Minister Milan Kubát, a university professor with ample experience in semiconductor design and production. The task of explaining the ministry's policies was often up to his adviser Ivan Malec (born 1930), a seasoned manager and industry insider. His story provides an important counterpoint to the stories of computer hobbyists. Having worked in industry automation since 1952, Malec cofounded INORGA, the Institute for Automation and Management of Industry, which had access to an IBM mainframe.[76] After 1968, he successfully "pulled the Institute through the trouble of screenings," but his wife lost party membership, which made him ineligible for high-ranking positions. He became a manager at the Central Databank of the Federal Statistical Office, which he equipped with a Control Data Systems mainframe (costing $3 million) that processed the results of the 1970 census. Besides his consultancy job at the new federal ministry, he also served as the editor in chief of *Elektronika Electronics* magazine. Launched in 1987, it became the ministry's mouthpiece and set out to explain its policies to a largely hobbyist and enthusiast readership.[77]

Among other things, the ministry controlled the two largest manufacturers of electronic technology—the Industry Automation Works concern (known by the acronym ZPA), focused mostly on industrial machinery and mainframes, and TESLA, the many-armed electronics behemoth with over thirty branches that produced everything from transistors and microprocessors to radios and TVs and ran several research centers. The latter's brand name was initially inspired by the Serbian inventor, who briefly studied in Prague. After Tito's Yugoslavia fell out of Soviet favor, the name was reinterpreted as "TEchnika SLAboproudá" (low-current technology, the alternative Czech term for electronics).[78] However, Czechoslovak people jokingly interpreted it as "TEchnicky SLAbé," or "technologically weak," hinting at the company's reputation among end users.[79]

Despite there being a whole new ministry for electronics led by established experts, there was an important political obstacle to its efficient operation: bureaucratic infighting within the power structures of Communist Czechoslovakia. As in many other Soviet bloc countries, the country was nominally run by the government (or, more precisely, federal and republic governments), but much of the political power was concentrated in the Communist Party's Central Committee, whose competing departments mirrored some, but not all ministries.[80] When interviewed for this book, Malec admitted that ministries and departments did not always see eye to eye:

> The organizational structure of the Party's Central Committee was not too friendly toward the development of computer technology.... Its many departments profited from having sovereignty over a certain resort, and did not like each other. There was for example the department for automotive industry, and they weren't happy when somebody was telling them what to do. But computer technology in a way connected it all; it would have needed a center.[81]

It is quite telling that the only time Malec saw universal agreement within the Central Committee on the use of computers was when he was asked to create an electronic database of party membership.

Despite these difficulties, two major programs of "electronization," with a prominent focus on computers, were introduced in the 1980s, both likely inspired by corresponding measures in the Soviet Union.[82] The Long-Term Complex Program of Electronization of Czechoslovak National Economy, launched in September 1984, focused mainly on the automation of existing industrial production and management.[83] The following Long-Term Complex Program of Electronization in Training and Education, ratified in November 1985, started with a premise similar to the abovementioned party resolution, but indicated greater incorporation of microcomputers:

> The basic prerequisite of a successful construction of a developed socialist society is the utilization of scientific-technological progress. This requires the widest possible application of science in manufacturing and its transformation into means of production. A necessary condition of this process is the wide utilization of electronics in the entire national economy. This puts extraordinary requirements on the training of cadres. The resort of education then has the important task to prepare not only qualified experts for electronization of individual branches of the national economy, but also to prepare today's young generation for the everyday use of electronics.[84]

Indeed, most of the document focuses on the education of expert "cadres" for the various branches of the national economy. Only in the sections about elementary schools does the program specify the aim to bring pupils up toward a "positive attitude toward technology" and to make work with electronic and computer systems "a natural part of their daily lives." To achieve that, the document recommends that educators "support interest in electronic and computer games." At higher levels, games disappear from the curriculum, to be replaced by algorithmization; the goal is for every high school student to be able to "program basic tasks for personal computers."[85]

Compared with similar education policy programs in the United Kingdom, or even the Polish Meritum school computer project, the Czechoslovak one was relatively late and hardly groundbreaking.[86] Nevertheless, the federal ministry at least appeared dedicated to the program. In 1987, it reported that education was and would remain "the biggest consumer" of the ministry's computer production.[87] A year later, the program shipped seven thousand machines to high schools.[88] Its deployment nonetheless foundered on the continuing lack of hardware, disinterest of teachers, and lack of skilled personnel. In 1988, it was projected that its goals would only be met in 1995, five years after the original deadline.[89]

Despite these attempts to quantitatively increase the penetration of hardware, the Communist leadership never fully embraced the "revolution" component of the scientific-technological revolution and treated computers mainly as a means of improving the output of existing industrial infrastructure. This is a marked difference from, for instance, Allende's socialist government in Chile, which made an enthusiastic attempt at integrating cybernetics into the very central management of the economy, or the failed OGAS project proposed by Victor Glushkov in the Soviet Union, which would have allowed workers to participate in industry management.[90] On the other hand, the Czechoslovak political establishment at least rhetorically promoted electronics, computers, and—to some extent—even computer games, as progressive and positive developments. Although this rhetoric did not always translate to actual availability of devices, the official acceptance of computers could be used to justify various decentralized bottom-up activities in the amateur spheres. At the same time, computing was not tainted by being fully "owned" by the regime, and citizens could therefore interpret it independently of the party's goals and ideologies. As demonstrated throughout the book, the work of revolution was in many ways taken up by hobbyists.

Men, Women, and Machines

The rise of expert culture and the policies of electronization expanded the ranks of the *technical intelligentsia*, a social group of technology experts and academics within what are now known as the STEM fields.[91] Many members of this group would become influential microcomputer enthusiasts, and understanding the group's values and demographics is therefore crucial for analysis of the computer hobby.

The country's education policies promoted technical schooling. The admission quotas for technical study programs in higher education were set so high that schools were desperate for students, actively sought out female applicants, and even accepted students with politically problematic backgrounds.[92] Students of technical subjects were the second-fastest-growing group after students of economics. Their number almost doubled between 1961 and 1981, to about 85,000, whereas humanities only grew by one-third, to 69,000. Technically oriented secondary education experienced similar growth.[93] Computing, specifically, was an established segment of higher education by 1980s. University lectures on computers had already started in 1951 when Antonín Svoboda taught the mathematical machines course at the Czech Technical University (CTU).[94] By the late 1950s, programming was taught at the CTU, Charles University, and the Brno University of Technology. In 1973, several full-fledged programs dedicated to computing opened at Czechoslovak universities, later complemented by programs on automation.[95]

Belonging to the technical intelligentsia was an important identity marker.[96] Graduates from technical programs boasted the academic title of *inženýr* (engineer) and used the corresponding abbreviation "Ing." to mark their occupational identity and status, even in semiformal settings such as hobby groups. Because of the state's wage-leveling policies, technical experts—save for management positions reserved for party members—did not earn significantly more than manual laborers.[97] On the other hand, the group enjoyed professional prestige and respect due to its critical role in the state economy, emphasized in the discourse of STR. As demonstrated in the case of Poland, members of the technical intelligentsia were often self-confident torchbearers of scientific and technological progress, and extolled scientific rationality as a solution to both small- and large-scale problems.[98] As Lipovetsky argues in the case of the Soviet Union, the technical intelligentsia tended to define itself in opposition to the party apparat on the one hand and the masses on the other.[99] Its members struggled for autonomy and control over their own work and careers.[100] Despite some degree of success, they were nonetheless largely subjected to the party's policies and managerial decisions. Due to the lack of career development opportunities, many of them thus partook in hobby activities.

The technical intelligentsia was predominantly, although not exclusively, a masculine domain. Despite some progressive policies, Czechoslovakia was still a deeply patriarchal society, and women rarely reached the salaries or career opportunities of their male counterparts.[101] As the example of the *Computer Dilemmas* television program suggested, women were stereotypically portrayed as the ones to whom technology needed to be explained. Nevertheless, several factors contributed to an increasing number of women joining technical professions. Czechoslovakia, like most Soviet bloc countries, followed the doctrine of full, compulsory employment of both men and women. Education mobility, although generally rather low, was higher among women.[102] Because men were incentivized to leave the school system early to enter heavy industry, many women continued toward university diplomas, sometimes in technical subjects.[103] Between 1961 and 1981, the proportion of female students in technical higher education rose from 11 percent to 19 percent; in 1978, women made up 20.2 percent of expert technical personnel within the socialist economy.[104]

Women had an especially strong presence in professional computing and data processing. According to 1986 data, the total percentage of women in the field was 59.4 percent. As in the West, women clustered in less lucrative clerical jobs, making up 97.2 percent of data-entry personnel, and only 29.2 percent of management. But women also accounted for 58.3 percent of the country's 7,338 professional programmers.[105] In comparison, when the proportion of women programmers peaked in the United States in the late 1980s, it reached around 35 percent.[106] Such a high percentage of female programmers supports (and even surpasses) the findings from United States and United Kingdom, which have shown that computing as a profession was more gender diverse in its early decades than it is today.[107]

Despite women accounting for almost 60 percent of the professional computing workforce, they were much less likely to participate in computing as a hobby. As discussed in more detail in chapter 3, amateur computer clubs had as few as 3 percent female members. The dramatic disparity between the two figures will seem less surprising when we realize that, at the time, professional and hobby computing were two largely separate fields of practice. One was a job, and the other a voluntary leisure activity. The latter therefore structurally excluded women, who had less free time than men (see chapter 3). Furthermore, women often entered computing because of practical circumstances rather than previous personal interest in technology.[108] In a 1989 interview in one of the hobbyist newsletters, the female professional programmer Inka Jelínková was asked what had set her on the journey to programming, and she answered quite explicitly: "The fact that I didn't get into the school I wanted to go to. I was persuaded to enroll at the Czech Technical University, where they were opening the new degree in programming."[109] At the time, the urgent demand for labor in institutional computing overrode cultural stereotypes about women and technology, opening new, fast-growing specializations in computing and automation to women.

The professional and hobby domains each revolved around a distinct type of interaction with computers, derived from the type of hardware they tended to employ. Since the 1950s, Czechoslovak professionals had traditionally worked in large teams on powerful mainframes stationed in institutions and factories, and their routines were embedded in managerial and administrative processes. Because mainframe machine time was expensive, programmers were used to writing code on paper, without real-time access to the computer. Hobbyists, on the other hand, enjoyed individual interaction with micros, even at the time when these were little more than toys. A significant part of the hobby activities revolved around tinkering and gadgets associated with stereotypically masculine engineering professions. Even within the professional circles, we observe that the closer a computing job position was to hardware, the fewer women were present. According to the 1986 report, women accounted for 29.5 percent operating system programmers (compared to 58.3 percent of regular programmers), 11.7 percent computer technology engineering specialists, and 4.0 percent service technicians.[110]

Overall, the hobby scene was often driven by the general ideals of the technical intelligentsia, such as the belief in technological progress and the desire to be in control of one's work. However, the overlap between *professional* computing and the enthusiast sphere was limited, despite both using computers. Hobby computing emerged as a separate endeavor, more likely to be taken up by male electrical engineers than female programmers.

Side Roads to Micros

By the early 1980s, the country's technical intelligentsia became aware of the utility of microcomputers in both the industrial and personal realms, and a demand for these machines was building up. The COMECON-wide computer manufacturing programs, however, focused on mainframes and minis and notably neglected production of microcomputers for the mass market.[111] Compared to their bigger brothers, micros were more difficult to plan; they required a close interplay between design, production, and marketing, and flexible responses to fleeting consumer preferences—exactly the things that the country's centrally directed concerns tended to lack.[112] In addition, the Czechoslovak electronics industry was short on material resources, including microprocessors—the chips that served as the microcomputer CPUs—as well as seemingly more trivial components, such as quality laminate and chemical compounds required for precise manufacturing of printed circuit boards.[113]

Some of the demand could be covered by imports, but schools, hobby clubs, and households did not possess enough convertible currency to routinely purchase foreign machines.[114] With no microcomputer plan underway, the Federal Ministry of Electrotechnical Industry greenlit bottom-up initiatives to design domestic microcomputers. As the ministry's Ivan Malec retrospectively explained in 1987, "While trying to offer a creative outlet for young inventors, we temporarily allowed initiative within individual factories."[115] The plan had one important caveat: the computers were to be assembled solely from COMECON-made components. Several industrious engineers rose to the challenge, driven by curiosity and ambition, as well as the potential "improvement proposal" bonuses.

Although the first documented Czechoslovak micro was 1982's SAPI-1,[116] designed mainly for industrial applications, its impact pales in comparison to three mid-1980s machines usually grouped under the heading of "school computers" because of their use in education. The three machines—the PMD 85, the IQ 151, and the Ondra—defied the principles of central planning by being a motley crew of incompatible devices, none of which adhered to an established Western or domestic standard.[117] These machines grew in the cracks and margins of the plans, rather than according to them.

The PMD 85 was built by Roman Kišš, a young and ambitious engineer who joined one of the flagship factories of the TESLA concern, based in the Slovak spa town of Piešťany (see figure 1.2). Upon taking the job, he was immediately disappointed by pervasive waste of labor and the low morale that the STR was supposed to fix. In his view, the employees worked for the company just 20 percent of the allotted work hours, spending the rest of the time building black market devices for personal profit, namely the sought-after convertors and amplifiers that allowed people in Southern Slovakia to receive Austrian TV. He pondered what he would do with his remaining 80 percent.[118]

At the time, the factory manufactured the MHB8080A 8-bit microprocessor, a clone of the Intel 8080, and its director encouraged Kišš to come up with possible applications. Inspired by Hewlett-Packard's HP-85, he started designing the PMD 85 on his own, unsupervised by the management.[119] Like

Figure 1.2
The PMD 85 has become a political project. Roman Kišš shows his work to Jozef Lenárt, first secretary of the Communist Party of Slovakia's Central Committee (standing behind Kišš). Peter Pfliegel, director of TESLA Piešťany, stands in the back with folded arms. Photo courtesy of Roman Kišš.

many other Soviet bloc computer designers, he had to work with a limited range of generic components and without "custom chips," which enabled the graphic or sound capabilities of most contemporary 8-bit micros, such as the Sinclair ZX Spectrum or the Commodore 64. The biggest challenge was to provide responsive graphics at a reasonable resolution. After some time, Kišš devised a work-around—a system in which the video processor shared dynamic memory with the CPU. This allowed for fast graphics display, acknowledged in the machine's name—the Piešťany Microcomputer with Display (Piešťanský mikropočítač displejový). The fact that the display was monochrome was hardly seen as a downside; color TVs were so scarce that most domestic computers were not designed with color in mind.

After demonstrating a prototype at an industry conference, Kišš received an audience from Minister Kubát, who proceeded to greenlight the production. As with all domestic computers, demand vastly exceeded supply, and various institutions and companies fought to secure their PMDs. As Kišš recounts, "This director, that director, everybody wanted it. They didn't know what for, but they wanted it." Although the computer was designed to work with substandard components and faulty memory chips, production soon started running out of even the faulty chips. Kišš remembers being told by his boss: "You better step aside, this has become a political project."[120] From that moment on, the project was co-opted by party authorities. Kišš was discouraged from improving the architecture of the computer and shifted focus to different projects. Meanwhile, at least 14,200 PMD machines were manufactured by 1989, making it the most influential domestically designed microcomputer platform.[121] It spawned several clones and became popular in school and youth club settings. In high schools, it competed with the slightly older and more obscure IQ 151 machine, manufactured by Industry Automation Works in runs of about 1,000 per year, or 4,000 pieces in total.[122]

Neither the PMD 85 nor the IQ 151 were sold to end users. They would not even have been affordable. According to a ministry official, a set of components for one domestic microcomputer before assembly "was many times more expensive than prices of completed devices on international markets" due to inefficient low-volume production.[123] All the while, Czechoslovak machines were inferior to their Western or Japanese counterparts, generally outperformed by the 1982 British Sinclair ZX Spectrum.

The Ondra project was supposed to fix the situation. Before the Ondra's author, Eduard Smutný, started designing his machine, he had already created the SAPI-1 and become a "hardware guru" of the Czechoslovak hobby scene, as well as a frequent commentator on the state of the industry.[124] Inspired by the ZX Spectrum, he wanted to create an affordable computer that would be widely available for home use to "educate and entertain our youth."[125] Moreover, he dreamed of a machine that would be sturdy enough for use by his son Ondra, who suffered from cerebral palsy and who gave the machine its name. In a 1986 interview, Smutný recounted:

> I was waiting for an idea, or a miracle. Then one day in early 1985, the miracle happened. TESLA Jihlava started manufacturing a simple and cheap keyboard for a foreign customer. The keyboard was not suitable for professional systems, because it didn't have enough keys, but for kids … it was absolutely ideal. Now I only needed a microcomputer to match the keyboard![126]

He took advantage of a laid-back summer at his job in the Prague-based research branch of TESLA, inventing and patenting his own display work-around enabling satisfactory monochromatic Spectrum-grade graphics. Instead of the power-hungry MHB8080A, he employed the U880 CPU, an East German clone of the Zilog Z80 that allowed for a sleeker and more portable case than those of the bulky IQ 151 and PMD 85.

Smutný had big plans for the computer. He appeared in the November 1985 TV broadcast from the Electronization and Automation Exhibition, uncharacteristically dressed up in a three-piece suit, and announced that the Ondra would be available in retail for 3,200 crowns. The price would make it a reasonably affordable machine. The fact that Minister Kubát appeared in the same broadcast, praising developments in the fields of mini- and microcomputers, gave weight to the promise.[127] However, after a test run of about two thousand machines, production stopped and never resumed. As Smutný admitted a year later: "The Ondra computer was manufactured … in TESLA Liberec, under—let me say—completely ridiculous conditions; it had to go down various side roads, not the normal way of planning and production."[128]

According to Smutný, no manufacturing plant considered production of the Ondra profitable. Apart from the aforementioned avoidance of costly innovation, this was due to a policy that valued labor-intensive work over precision manufacturing. The cancellation triggered a notably open public debate about the use of microcomputers in Czechoslovakia, which will be discussed in the next section.

This short-lived experiment in bottom-up hardware projects offers several insights. First, they were not entirely bottom-up, because the projects' success strongly depended on political patronage. It was impossible for individual engineers to penetrate the plan, secure components, allocate resources, and gain access to retail distribution. This privilege was given to the PMD 85, but not to the Ondra. Kišš's boss was right. What had started out as a hobby project necessarily became a political one.

Despite the admirable ingenuity of their designers, the school computers' quality was poorer than that of Western imports. An education expert even noted that the presence of these low-quality computers in schools caused "political damage" by revealing the backwardness of domestic technology.[129] They were incompatible with Western standards and with one another. Unlike the Polish Meritum or Hungarian Primo, which were partial clones of the TRS-80 computer, Czech machines were not designed with software compatibility in mind. They were brainchildren of hardware engineers, and showed little concern for real-life use scenarios. Aside from a very limited catalog of programs for the IQ 151, production of software was no more than an afterthought, and left mostly to amateurs.[130] The fact that none of these Czechoslovak machines included Czech or Slovak diacritics suggested that they were not designed for text work, but primarily for programming education (see figure 1.3).

The machines also created problems for the ministry, because they competed for scarce resources.[131] In a 1987 interview for his own magazine, Malec announced that the ministry was committed to narrowing down the range of microcomputer models and warned against "the praise in the press and in television for various microcomputers constructed on a shoestring and manufactured in three months," which undermined "serious efforts at progress in technology and organization of production."[132] At that point, he finally disowned the bottom-up attempts to produce micros. Instead, he proposed pooling the resources of the electrotechnical industry and joining

Figure 1.3
Kids programming on PMD 85s at a youth computing retreat, 1987. Photo by Vlastimil Veselý.

up with other COMECON countries to overcome the deficiencies of a small isolated market. His plan, although reasonable, never came to fruition.

Despite their subpar performance and restricted availability, domestic microcomputers played an important part in Czechoslovak computer culture. They were well known due to their deployment in education and provided a point of comparison to the shinier Western machines. Although they were criticized at the time of their release, the current wave of retro computing has rediscovered them as imperfect but truly local originals, and they have become coveted objects of technonostalgia. As of 2018, retro computing enthusiasts continue making games for both the PMD and the Ondra.

Who Needs a Home Computer?

In the mid-1980s, the unavailability of micros for home use was becoming painfully apparent, as was the shortage of other electronics such as VHS players. Popularization and educational efforts inadvertently contributed to people's awareness of the problem. Youth and hobby magazines convinced their readers that computers were the future, but that future kept being postponed. When the Ondra project—which promised to offer the Czechoslovak citizens a domestic alternative—was canceled, it sparked a heated and prolonged debate, eventually revealing that microcomputers for the home were simply not on the government's agenda.

Before we analyze the debate, it is important to reiterate that normalization-era media discourse consisted of much more than plain propaganda. It was possible to publicly discuss particular problems, including quality or accessibility of goods.[133] All critique was subject to editorial control and censorship, but some mainstream outlets identified as more progressive and offered a conduit for people's discontents. Among these outlets were the *Young World* (*Mladý svět*) magazine and the *Youth Television Club* (*Televizní klub mladých*) TV program, both targeted at older teenagers and young adults. In January 1987, Eduard Smutný, by then a bona fide hero of local computer enthusiasts, appeared on *Youth Television Club* for a studio interview (see figure 1.4). Apparently upset and unusually outspoken given the tame standards of the show, he expressed his disappointment over the cancellation of the Ondra platform:

Figure 1.4
Eduard Smutný interviewed on the *Youth Television Club*, his hand on the Ondra computer. Photo courtesy of Czech Television.

> Unless we manufacture twenty thousand machines, it makes no sense to write programs for it, it makes no sense to write manuals for it, it makes no sense to teach people how to use it, to put it into schools. Clearly, we can't have a different computer in each town. And we need twenty thousand computers in schools. A thousand won't make a difference.[134]

He continued with a critique of the national electrotechnical industry, pointing out the lack of incentives for the nationalized industry to produce micros:

> We're talking here about electronization. We've been talking about electronization for quite a few years and haven't really started yet. … The situation is absolutely critical. We are lagging ten to fifteen years behind Western computers. Even comrade minister will admit that. … It really must be our goal to have these computers—the goal to give cheap Czechoslovak computers to people. … There must be such stimuli that would make us want to do it, so that it's profitable.[135]

For Smutný, the failure to deliver hardware to the people was not just a practical problem, but a policy failure—a lack of political support for accessible computing.

The *Young World* magazine supported Smutný's fight. In an article titled "We Want Computers!"—a report from a computer hobbyist meeting that featured a round table with Smutný—the journalist praised him for not mincing words. The dramatic and emotional article concludes with the author's provocative manifesto of discontent:

> Dear comrades, ministers of electrotechnical industry, education, domestic and foreign trade, but also both ministers of culture (because computers are becoming a part of national culture), I would like to tell you on behalf of a few enthusiastic beginners that we are tremendously dissatisfied. And if any of your subordinates say something else, they are liars.[136]

On top of its confrontational tone, the article is also notable for being one of the first instances of computers being dubbed a part of "culture," a claim absolutely foreign to official policies, but very perceptive given the inventive use of games by Czechoslovak amateurs.

Soon after *Young World*, Smutný also received support from *Mikrobáze*, the country's most influential computer hobby newsletter. Referring to the *Youth Television Club* interview, it portrayed Smutný and his Ondra as victims of "stupid bureaucracy, driven by ignorance and social indifference."[137] Three years later, the academic Bohuslav Blažek even indirectly called Smutný "a martyr of computer culture."[138]

Given the lack of nonstate actors involved in the computer industry, the criticism over the lack of home computers was directed at the government, specifically the Ministry of Electrotechnical Industry. The ministry swiftly responded to the "campaign" in the February 1987 editorial of *Elektronika*. Taking up the thankless task of defending the ministry's position, Ivan Malec admitted that the criticism was partially justified:

> The rational core of the criticism is quite clear: computers for personal use are not available in the required volumes, and if they are, they are not at all affordable. However, this undeniable fact has become entangled in too much emotion, and in subjective interests and opinions that lead to disinformation.[139]

He went on to attribute the problem to a shortage of components that could only be surmounted by massive investment in the industry. In subsequent articles and interviews, he delved deeper into the issue, questioning the people's desire to own microcomputers:

> Furthermore, I have been asking myself: if computers, then what for and what kind? I have to admit that in regard to the demand for "computers in every household," I have not found a satisfactory answer to the first part of the question. I can imagine the necessity of this demand once we can realistically think of automated ingredient management in meal preparation, automated environment-dependent diet control, ordering goods from home, remote bill payments, etc.[140]

He went even further in a controversial 1988 interview, entitled "I Am a Skeptic," for the Polish magazine *Bajtek*, saying that "as a society we are not mature enough for the general use of microcomputers—except for games, of course."[141] Although these statements drew the ire of computer hobbyists, the core arguments were substantiated. Most 8-bit micros were not powerful enough for serious work, and were mostly incapable of the networking applications that would eventually become an essential application of home computers. Games—certainly one of the most profitable and massive applications of mass-market micros—were nonessential to Malec, as was hobbyist tinkering. In the interview for this book, he recalled:

> I kept being accused by hobbyists that I was preventing the development of computer technology, because I did not want the 8-bit machines. … I could be quite blunt, too. I couldn't just let people walk all over me and say that the IQ 151 was genius. That just wasn't true. I had seen something better, I have seen how much further things could go. Those [8-bit] machines were only for calculation, but not for industry management and other applications.[142]

Overall, he does not remember the squabbles with hobbyists as particularly important for the ministry, although he admits that some of the hobbyists were clever.[143]

It would be too easy to interpret this debate as a conflict between a "martyr" and an "oppressor." Presumably both parties—enthusiasts, represented by Smutný, and the ministerial cadres, represented by Malec—were dedicated to scientific-technological progress. Rather than a clash of ideologies, we are witnessing a *clash of teleologies*—two distinct ways of thinking about the application of computer technology, and two distinct ways of interconnecting them with human and nonhuman actors. As a seasoned manager and government bureaucrat who had overseen large mainframe operations, Malec saw computers through the prism of industrial and institutional application. He was proud of having acquired the big and expensive computer that helped him process the census data in 1970, but he appreciated it mainly as a powerful tool, harnessed in a chain of managerial processes. To him, 8-bit machines such as the Spectrum or Ondra were mere toys. Although he supported their adoption at schools and computer clubs, he dismissed the importance of having one's personal machine. The latter was not the ministry's priority—and even if it was, the government was unlikely to deliver due to the generally bad shape of the national economy.

Smutný, on the other hand, saw the value in hands-on contact with technology. His premises resonated with the tenets of Richta's take on STR. He envisioned a society in which citizens—and especially young people—had the right to participate in technological progress. For him and other hobbyists, playing with computers at home was a joyful and stimulating activity, and an essential prerequisite for further creative work with computers—including "useful" applications. The benefits of this work could not be planned and technocratically controlled, but would instead emerge from playful experimentation and feedback loops in which the computer was not just a tool but also a partner.

This clash of teleologies and the ministry's indifference to the desires of individual enthusiasts hint at a conceptual disconnect between the state and hobbyists. On the one hand, hobbyists felt neglected by the state-controlled industry. On the other hand, this allowed home computing to emerge as a field that neither the party nor the government had stakes in, allowing it to remain relatively uncontrolled and autonomous.

Farm Computers and the Courageous Clone

Not too long after Smutný's TV appearance, the first domestic computer that was available in retail—and, moreover, compatible with the Sinclair ZX Spectrum standard—finally arrived. It was not the Ondra. It was not even manufactured by a TESLA plant. Instead, it was produced by Didaktik, a factory based in the small Slovak town of Skalica. Unlike TESLA, Didaktik was formally a manufacturing cooperative (*výrobní družstvo*), not a state-owned enterprise. Therefore, it did not fall under the control of the Ministry of Electrotechnical Industry and enjoyed greater autonomy. As the cooperative's director admitted in a 1988 interview, "being a cooperative probably gives us more space and independence, but also responsibility."[144]

Following subtle changes in economic policies inspired by the Soviet perestroika reforms, many—especially agricultural—cooperatives engaged in so-called *auxiliary production* (*přidružená výroba*), using surplus resources to manufacture nonagricultural products that were currently in demand. The farms could sell their agricultural production in the West and use the convertible currency to purchase, for instance, electronic components on the capitalist market.[145] The most notable of these was the Unified Cooperative Farm Slušovice, whose ambitious and politically adept management transformed it into a conglomerate of gigantic proportions, producing not only grain, fertilizer, and vegetables but also 8-bit and 16-bit microcomputers—besides running its own tourist agency and a horse track.[146] Its line of mostly business-oriented machines was called TNS, or Ten Náš Systém (That System of Ours).[147] Farm computers were not a Czechoslovak specialty, but rather an anomaly of late socialism—the first Hungarian computer available on the consumer market, 1984's Primo, was also manufactured by an agricultural cooperative "under crude circumstances" and "on kitchen tables."[148] These efforts became a target of ridicule and satire, notably in the 1987 song "Auxiliary Production" by the Bratislava-based folk-rock band Lojzo. In the video for the song, the musicians, dressed as farmers, walk through a cowshed, drop their rakes, change into lab coats, and enter a high-tech IT department where they pretend to work on computers.[149]

Didaktik Skalica was not a farm, but a cooperative manufacturer of school equipment, including laboratory tools for physical experiments. Compared with its peers, Didaktik was technologically advanced and therefore considered a flagship of Slovak cooperatives. As Didaktik's then chief engineer Ľudovít Barát notes, its reputation and partial autonomy enabled it to bypass the rules of the ministry. While the engineers at TESLA scrambled to build computers exclusively from COMECON-made parts, Didaktik became the first Czechoslovak factory that managed to import the Spectrum's original ULA custom chips, made in the United Kingdom by Ferranti, avoiding the need for improvised work-arounds. Its Didaktik Gama was, in effect, the first Czechoslovak-made global machine, combining British chips and Soviet RAM with domestic components. Like the PMD 85 and the Ondra, the Gama was also a partially bottom-up project. Barát refuses the credit for "designing" the machine, which was basically a clone: "I wouldn't call it 'design,' but I would say that I was its father. A person who was perhaps a bit more courageous that the rest."[150] In his view, Didaktik's competitive advantage was its vertical integration and material self-sufficiency, which alleviated the supply problems rampant in domestic manufacturing. The factory melded plastic, printed its own circuit boards, and even produced its own packaging.[151]

Unlike the IQ 151 and the PMD 85, Didaktik Gama was sold primarily in retail, despite a name that suggested educational use. It initially sold for 6,200 crowns, which was the equivalent of about two average monthly salaries.[152] Tens of thousands of machines were manufactured in the following years, but as shown in the next chapter, production could not satisfy demand.[153]

Despite the success of Didaktik Gama, micros struggled on the fringes of the command economy. The idea of a home computer for the masses was alien to the country's official policies, and computers were mostly unavailable to the general public. The educational and employment policies produced large crowds of people with solid technical education who were ready to use micros, but the machines were not coming.

The reasons for this development were both material and political. First, microcomputers were scarce due to import limitations and a shortage of resources and facilities for domestic production. Second, there was little political support for affordable 8-bit microcomputers. Whereas Radovan Richta's version of STR envisioned technology as a cornerstone of political life, normalization-era officials saw it as an apolitical tool that could be used for quantitative—rather than qualitative—improvements of existing industrial processes. Ivan Malec's remark—cited earlier in this chapter—that support for computing would have required agreement between different departments seems very relevant. Although the planning was centralized, it was shaped by the vested interests of distinct arms of the bureaucracy. Power over computer production was mostly exercised by the domain of electrotechnical industry, with additional input from the domain of education. Whereas the former portrayed the benefits of computers in terms of industry automation, the latter considered micros little more than school equipment. There was no tendency, for instance, to use computers or computer games for the purposes of propaganda. They were viewed as devices for calculation, not culture.

This technocratic view clashed with the technophile approach of computer hobbyists, as seen in the example of the Ondra controversy. The ensuing debate exemplified a mismatch of teleologies—it was not just a debate about how to produce computers, but also about their potential uses. The ministry considered computers a means to an end, whereas hobbyists saw them as ends in themselves—as more autonomous, independent objects.

The stories of domestic micro production show the drive of some engineers to produce computers for ordinary people, as well as the impediments to those efforts. When it came to microcomputers, the computer industry was doubly paralyzed—there was neither enough space for bottom-up processes, nor any centralized plan. The attempts to squeeze the micros through the sieve of central planning led to stillborn projects such as the Ondra. Only with substantial political support or autonomy was it possibly to "game" central planning and manufacture micros. Although the PMD 85 and Didaktiks made a dent in the pre-1989 computer culture, they were easily outnumbered by individually imported machines, a phenomenon discussed in the next chapter.

If the idea of a home computer was an ill fit for the structures of the Czechoslovak economy and the official policies, then the video game console was even worse off. The late 1970s seemed promising. Hobby magazines published several schematics for transistor-based "TV games," and around 1980, TESLA Piešt'any was even producing *Pong*-style consoles featuring specially designed integrated circuits.[154] But given how rarely they appear in fan collections, these were probably never made in large batches. Czechoslovak production never moved past the so-called "first generation" unprogrammable consoles, and more advanced, microprocessor-based devices were not manufactured in the country. Given the worrisome shortage of microprocessors in the context of a political discourse that favored their industrial and educational uses, placing these chips into entertainment machines would have seemed frivolous and wasteful. However, handheld electronic games by the Soviet company *Elektronika*, and especially *Nu Pogodi*, were a popular import. These were blatant clones of the Nintendo Game & Watch line, and did not use computer-grade chips.[155] The Soviets also designed and manufactured dozens of original electro-mechanical and digital arcade games, but this was not the case in Czechoslovakia.[156]

Overall, Czechoslovakia had neither a policy nor a market that would provide machines for the home. Microcomputers were in a power vacuum—they were not a part of the plan, their use was not regulated, and they were not considered ideological tools by the authorities or by citizens. Surrounded by the context of normalization, they were neither normal nor abnormal. They were left out of the state agenda and available for appropriation by prospective users. People seized the opportunity and made computers their own, bringing them into homes and computer clubs.

2 Hunting Down the Machine: Trajectories of Microcomputer Domestication

> "What do you currently offer in the category of home computing equipment?"
> "Currently, nothing."
>
> —From an interview with František Nevrlý, deputy manager of the only specialized home computer store in Prague, 1988[1]

In Communist Czechoslovakia, one could not simply go to a store and buy a computer. As seen in the previous chapter, the country's disorganized computer industry failed to produce enough machines to sate the demand of state institutions and schools, let alone the general public. Bulk imports of computers for home use were severely constrained by the limitations on foreign trade, namely export embargoes and the country's limited convertible currency reserves.[2]

Czechoslovakia's was a *shortage economy*, described by the Hungarian economist János Kornai as an economy with "general, frequent, intensive and chronic" shortages in all avenues of economic life.[3] The root cause of these shortages was centralized bureaucratic control, which gave sellers and manufacturers incentives to receive recognition from their superiors by fulfilling arbitrary quotas rather than to satisfy their customers. Although the state-controlled command economy managed to provide for people's basic needs, shortages were common enough to become a part of everyday life. In popular discourse, the image most frequently used to represent scarcity was a line of people queueing up in front of a grocery store, usually to get hold of bananas—a notoriously scarce grocery item. But in place of bananas, we could easily picture consumer electronics, including microcomputers (see figure 2.1).

Although the availability of computers was improving little by little throughout the 1980s, there was still only *one* specialized computer store

Figure 2.1
People waiting in an overnight queue for television sets in a Prague shopping arcade, early 1989. Photo by Karel Bucháček.

in Prague as late as 1988, a small place situated on a busy downtown street close to the city's historical center. This 1988 interview with its deputy manager illustrates the thankless job of running a computer store in a shortage economy. Having admitted he had no computers to offer, he continued:

> So far, we have imported Sinclair, Delta,[4] and Sharp computers. Last Christmas, we were selling the first batch of the Czech version of the Sinclair machine—the Didaktik Gama. …
>
> *And will there be more of these?*
>
> We don't know when, but there will definitely be some. Despite the fact that I can only promise to deliver what we currently have in stock. …
>
> *Let me ask a mean question. What do you have if you don't have anything?*
>
> We have just started selling data storage media, cassettes and floppy disks. We try to find and sell cassette players that would work well with home computers. But none of those are being imported.[5]

To a reader unfamiliar with socialist retail, the interview might seem like an exercise in Kafkaesque absurdity—the deputy manages a computer store that is currently not selling any computers, and keeps waiting for machines that may or may not arrive. But the tone of the interview, published in a computer club newsletter, is informative rather than satirical; the empty shelves were very real. Luckily, another batch of Didaktik Gama machines arrived just before Christmas 1988. In a later issue, the newsletter reported:

> When a delivery of 50 pieces of the Didaktik Gama computer was to arrive at [the store], a line started forming already at 6 A.M. At 9 A.M., around 35 enthusiastic customers stood in the line. The last, 45th one joined at around 12.30 P.M. … Due to security reasons, only fifteen people at a time could wait inside the store. As in other extraordinary circumstances, the power of human solidarity prevailed. Those waiting in the store were switching places with those on the street (it was not freezing, but an unpleasant cold wind was blowing outside). And the store's staff made coffee for their exhausted customers! At 4.10 P.M., the longed-for computers were finally delivered.[6]

As this story shows, people were willing to brave chilly weather to secure a computer that was already outdated by Western standards. Although today's fans and enthusiasts also wait in lines to be the *first* ones to lay their hands on a new gadget or video game, the forty-five Czechoslovak customers knew that this might be their *only* chance, for months to come, to secure a home computer.

This chapter will show how microcomputers and computer games entered the everyday lives of people in Czechoslovakia. The United Kingdom, arguably the center of game industry closest to Czechoslovakia, will serve as a useful point of comparison. As Alex Wade sums it up, in the United Kingdom, "[The] movement of videogames' technologies into the domestic environment was predicated on the formation of an industry that took individuals' convictions of their products and projects, and, through sharp use of advertising and public relations, moved computers and videogames into the home."[7] Wade's analysis, informed by the fast-growing historiography of the British game industry, ascribes decisive agency to the industry and advertising. In de Certeau's terms, we could say that these actors deployed *strategies*. But in a Czechoslovak shortage economy, no such strategies applied. It was up to the users themselves to procure machines, using a variety of grassroots and black market logistics. Instead of investigating large-scale processes, we will follow the lead of *alltagsgeschichte* scholarship and make a descent into microhistories to see how acquisition and domestication of machines played out in everyday life contexts.

In media and cultural studies, adoption of new technologies by end users has been the subject of *domestication* scholarship, kick-started by Roger Silverstone. He explains: "Domestication does, perhaps literally, involve bringing objects in from the wild: from the public spaces of shops, arcades and working environments; from factories, farms and quarries. The transition … of objects across the boundary that separates public and private spaces is at the heart of what I mean by domestication."[8] Following in the footsteps of British cultural studies, domestication research digs into the ways in which technologies are appropriated into the cultures of individual households, families, and other social groups. It situates technologies within the sites of everyday struggles between actors (such as family members), and connects these struggles to wider social issues such as class and gender. Although valuable and inspiring for my analysis, domestication is but a part of the whole story. As a research tradition originating in a developed capitalist economy, it presumes that acquiring hardware is a trivial matter, which was not the case in a shortage economy.

Returning to Silverstone's metaphor of *the wild*, I believe that *hunting* can serve as an adequate metaphor for the process that precedes and complements domestication—a metaphor that captures the drama of the shortage economy. While hunting after computers, users had to learn about the terrain, develop and adopt tactics, coordinate with their compatriots, and exercise patience. For extended periods of time, they may have seen the computer as an exclusive and elusive object of dreams and desires. Many of them had to travel (or have somebody travel) to the West to poach a machine from the capitalist retail network. John Urry has argued that these unsanctioned mobilities of Western products allowed Eastern Europeans to build their identities independently of the socialist ideology. Upon their arrival, those products were "circling free from the conditions under which they had been produced, and moved and located within new patterns of social life."[9] Computers entered socialist homes, but offered ample opportunities for imaginative mobility. Through playing Western games and interacting with Western (or Japanese) technology, one could establish a tentative connection with a distant cultural context.

This chapter will follow the succession of steps one had to take when getting hold of a computer, from the initial seeds of interest and first encounters to the acquisition of a machine abroad, on the black market, or in specialized stores. In the latter part of the chapter, I will move to domestication in the strict sense and show how microcomputers—though Western or Japanese—were embedded in the local material culture and household dynamics. But first, we should mention two basic prerequisites for obtaining a machine–interest and opportunity. Regarding interest, it bears repeating that an 8-bit micro had very limited practical utility; to use Melanie Swalwell's words, home computers at the time were "a technology in search of a use."[10] Their use for word processing and office tasks was inconvenient because of low memory and limited connectivity to peripherals. Despite a widespread expectation that computer technology would play a decisive role in the future, it was not *necessary* to have a personal machine, and the role of a microcomputer in one's personal life remained undefined by the official teleology. Two main groups made the micro their own: electronics enthusiasts, for whom it represented the latest and most exciting installment in a long history of electronics that could be tinkered with and mastered; and children and young people who saw it as an object that could produce captivating, colorful, and interactive experiences unmatched in their striking audiovisual presence. As for opportunity, individual importation of micros was largely dependent on a small group of more "cosmopolitan" citizens permitted to travel to the West or at least correspond with someone who lived there; these people, and their families and friends, often counted among the early adopters, especially in the early to mid-1980s.[11]

A Machine That Obeys

Many of the earliest home computer enthusiasts had pursued technical interests long before microcomputers first appeared in Czechoslovakia in the early 1980s, often alongside their jobs as mechanical and electrical engineers, automation experts, or research scientists. Twin brothers Eduard and Tomáš Smutný, both of whom play important parts in our story, were in many ways typical representatives of the Czechoslovak technical intelligentsia. They were born in 1944, first built their own crystal radios in the 1950s, and in the 1960s graduated from a military high school that gave them a good foundation in electronics (see figure 2.2). Tomáš went on to study at a military college and developed his career at the Army Research Institute, working on radar-based speed measurement technology. Eduard earned an electrical engineering degree from the CTU and worked as a design engineer at a research and development branch of the TESLA electronics concern. Here, he created several computers, including the TESLA JPR 12, a 12-bit mini; the SAPI-1, arguably the country's first microcomputer; and the

Figure 2.2
The Smutný twins in the 1950s as students of a military high school in Bratislava, in today's Slovakia. Eduard on the left, Tomáš on the right. Image courtesy of Tomáš Smutný.

ill-fated Ondra, discussed in the previous chapter. For the Smutný brothers, moving to microcomputers was a natural progression, the next chapter of their personal narratives. As soon as the first microcomputers became available in the country in the early 1980s, they started writing about them for *Amateur Radio* (*Amatérské radio*) and joined the hobby computing circles.[12]

The *Amateur Radio* monthly had a status on the hobby scene similar to that of *Popular Electronics* in the United States or *Radio* in the Soviet Union. Unlike the commercial American magazine, however, *Amateur Radio* was funded by the state and followed an overt policy goal: to support and promote the growth of hobbies with military or industrial application. But although the magazine did contain a portion of obligatory ideological material, regular hobbyists bought it for the news, practical information, and schematics, often contributed by other readers. *Amateur Radio* started covering microcomputers irregularly in the late 1970s, and introduced a regular section called *Microelectronics* (*Mikroelektronika*) in 1982. Its austere and unattractive design notwithstanding, the magazine was massively popular, selling 114,000 copies in 1982 (rising to 150,000 copies in 1989) in a country of 15 million.[13]

To understand what drove people toward technical hobbies, we can draw from Haring's account of ham radio enthusiasts. In her view, technical hobbies are attractive because of "the sense of power that [comes] from having full knowledge of the system."[14] They provide materials and blueprints for creative work and tinkering, and access to communities in which one's expertise and mastery are recognized and appreciated. The engagement with computers can also be understood as a matter of personal politics. Sherry Turkle observed that many early US hobbyists used computers to mitigate their political frustration, brought about by living in a world they could not fully grasp or control—that is, they used computers as "a compensation for dissatisfactions in the world of politics and the world of work."[15] Decades before networked computers started to be associated with fear of surveillance, 1980s micros could be truly personal machines that granted control and autonomy. Presumably, the appeal of control and autonomy was even stronger in late socialist Czechoslovakia, where consumerism and individualism were thriving as they were in the West, but where much of the control over one's life was usurped by a nondemocratic regime. As an informant remembers, "Opening a Spectrum in the evening and writing a program was a terrific respite" from the politically motivated harassment she experienced at work.[16]

Western histories of hobby computing usually start with 1970s kit-based machines such as the MITS Altair or the Acorn System 1.[17] However, these were too niche to get into the hands of enthusiasts in the Soviet bloc in any substantial numbers. In Czechoslovakia, the beginnings were connected to 1970s programmable calculators from Hewlett-Packard and Texas Instruments, as well as the first European mass-market micros, namely the 1981 Sinclair ZX81. Local early adopters, too, were amazed by the fact that computers did what they were told, and by the power they granted to their users. This is evidenced by a story told in a 1991 interview with electronic scoreboard engineer Jiří Pobříslo—a member of the technical intelligentsia, a reader of *Amateur Radio*, and a well-known member of the ZX Spectrum community:

> Sometime in early 1983 … I was reading up in Amateur Radio about how Englishmen invented the [Sinclair ZX] Spectrum and what it's capable of. At the same time, I got hold of a BASIC course … and I eventually learned the language. I wrote a couple of programs as an exercise, but then I grew impatient, took one of those programs—a lowest common denominator calculation—and

> brought it with me to the Nisa factory, where my old friend … had a ZX81, the Spectrum's predecessor. Ing. Mixánek typed the program into the ZX81 and we ran it. And it worked—it worked exactly as I wrote it; it computed what it was supposed to compute, and displayed what it was supposed to display.[18]

Microcomputers were empowering even to those who had already had access to mainframes and minis. The experience of working with micros was qualitatively different—more convenient, personal, and intimate. Vít Libovický, the author of several 1980s games, recounted his memories of visiting the workplace of his father, a researcher at the Physics Institute of the Czechoslovak Academy of Science. By the early 1980s, the Institute had purchased a Video Genie 8-bit microcomputer (TRS-80 compatible) in addition to their mainframe. The young Vít preferred the Video Genie over the massive—and more powerful—mainframe. Whereas the mainframe took fifteen minutes to boot, one could just turn on the Video Genie and a user-friendly BASIC interpreter was instantly ready to take orders.[19]

Libovický was one of the lucky kids and teenagers who gained access to computers through parents who worked as engineers, programmers, scientists, or computer operators. Throughout the 1980s, many Czechoslovak microcomputers worked such double shifts, operated by parents during workdays, and by their children during evenings or weekends. The kids who were less lucky could read about micros in magazines such as *ABC of Young Technicians and Scientists* (*ABC mladých techniků a přírodovědců*) or *Science and Technology for Youth* (*Věda a technika mládeži*). Although they rarely mentioned games, their articles on computing sowed the seeds of interest in the young minds of their readers, who started dreaming of having their own machines.

Wandering Programmers

Luckily, engagement with microcomputers did not necessarily require ownership of a machine. Several notable Czechoslovak programmers learned BASIC long before they had their own computer, "tracing programs on paper to understand how they worked."[20] Programming on paper was not unusual at a time when mainframes were shared among multiple people or institutions—and it is still done casually during prototyping and design. Among young Czechoslovak programming adepts of the 1980s, though, emulating computers on paper was a temporary substitute for a physical machine.[21]

In 1981, *ABC of Young Technicians and Scientists* published the Computer Game System, a paper model of a simple computer (see figure 2.3). The system had little in common with the outward appearance of a computer, but it simulated a computer's functionality. With a handful of instructions and ten memory registers, it allowed users to run simple programs—if the users followed the commands and carried out the calculations themselves. Named to appeal to the magazine's young audience, the Computer Game System included two program cards with simple games, one of them being a version of *Lunar Landing Game*, made popular on 1970s minicomputers.[22]

Figure 2.3
The Computer Game System, a popular paper model published in *ABC of Young Technicians and Scientists*, year 1981, issue 7. The user would insert the program card into the wider opening on the left-hand side of the model and follow the program by vertically sliding the card and executing the commands. The right-hand side of the model was reserved for memory registers. By popular demand, the model was later reprinted by *Science and Technology for Youth*. Illustration by Jaroslav Velc.

Similar paper models had been used in the United States starting in the 1960s, including Bell Telephone Laboratories' CARDIAC (Cardboard Illustrative Aid to Computation).[23] In Czechoslovakia, paper models were an integral part of the 1980s youth club practice, and have been cited alongside programmable calculators as important and foundational stepping stones in personal computing narratives.[24] Although a paper computer could evoke a sense of control and accomplishment by itself, part of its appeal was the anticipation of real machines.

Before one could have regular access to a computer in a club, at school, or at home, there was yet another option—going to specialized trade fairs and exhibitions that displayed computers. These ranged from specialized automation or education events to more general electronics fairs, some of which even hosted Western or Japanese manufacturers in addition to the ones based in the Soviet bloc. The exhibitions were heavily attended by the public, who found in them a chance to see—or even try out—otherwise unavailable products. Though mostly targeted at delegations from the industry and state institutions, computer stands were often swarmed by groups of young people who wanted to play with the machines, or at least grab promotional materials.[25]

Vlastimil Veselý was one of them. He first saw a computer in 1979 at an exhibition in his home city of Ostrava. Enchanted by the new technology, he started traveling around the country to attend exhibitions. A self-professed "wandering programmer," he would hijack exhibited machines and occupy them for as long as he was allowed. He soon learned BASIC from a course published by *Amateur Radio* and wrote programs that he then typed in and executed at exhibitions. This way, he created his own version of the strategy game *Hamurabi*: "There was no listing available, so I programmed it in my own way, based on what the game looked like on another computer. I wrote it at home, then I went to the exhibition and typed in all 1,400 lines of code. That took about an hour and a half. Then I entered the magical word RUN. And sometimes there was a syntax error."[26]

At various exhibitions, he connected with other enthusiasts who also used exhibitions as transient spaces for coding, experimenting, and gathering.[27] A group of young people would flock around the stand of TESLA Eltos (TESLA's distribution and service branch), operated by Sylva Prokšová (née Vošahlíková), a junior Eltos employee tasked with promoting TESLA's microcomputers (see figure 2.4).[28] She enjoyed interacting with the

Figure 2.4
A snapshot from the ZENIT 86 exhibition in Prague, 1986. Sylva Prokšová is on the right. Photo by Vlastimil Veselý.

group: "When one got bored standing around, there was nothing better than a group of dedicated fans."[29] Along with some of these amateurs, most of them younger than herself, she later started one of Prague's most notable computer clubs. Veselý—whose story we will revisit in chapter 6—joined a club in Ostrava, but occasionally visited and collaborated with the Prague group.

Spectacle from the West

In the West, advertising played an important role in convincing the public that microcomputers were worth purchasing—either as an investment in the future or as a taste of what was to come. Behind the Iron Curtain, there was little to advertise. But despite the limited availability of microcomputers, word about them was spreading not only among technical hobbyists but also among the lay public. Young people, in particular, seemed to be instantly awestruck by computer games for machines such as the Sinclair ZX Spectrum.

Those who first played in the 1980s feel a unique nostalgia for the early computer games, for they had next to no prior experience with digital interactive media.[30] Their recollections of their first encounters with computers are vivid and specific, and often resort to words such as "unreal," "unbelievable," or "insane." As Michal Hlaváč, later an author of several text adventure games, remembers:

> One day, we had a call from our dad's friend, who said: "Boys, you have to come over." We immediately drove to their place. There was a ZX Spectrum on their desk, the original rubber one. So we plugged it into the TV set and played the five games that came with it, *Jetpac* and *Pssst* among them. And we were blown away—it was insane. We had seen a PMD 85 before, but you could only draw white lines on it. This was in color, animated, simply unreal. We stayed there until midnight, which was unheard-of on a night before school. It made such a deep impression that I will never forget it.[31]

Clearly the hobbyist urge to understand and control the machine, discussed above, was not the only reason to enjoy a computer. Kids such as the Hlaváč brothers were in it for the spectacle, at least initially. There was a striking difference between a black-and-white text display and the color animations found in action games for the Spectrum. Inside, there was still a computer, and a small one at that. But playing a fast-paced action game such as *Jetpac* was a qualitatively different experience.[32] Sprites flying around a luminescent CRT at the player's whim looked much more alive, and demonstrated the computer's potential in a far more striking and accessible fashion. Although this appeal was likewise felt in the West, the sensory overload of such games was an even starker contrast to the slow-paced and visually austere television programming of 1980s Czechoslovakia.[33]

The result of this fascination was that many kids dreamed of having a computer. For most families, micros were simply too expensive to be bought as toys. But there was another reason for parents to buy computers for their kids—to give them a head start in learning how to operate computers, and thus prepare them for the future. In the West, 8-bit machines were sometimes advertised as learning tools, but in Czechoslovakia this message was more likely to spread by word of mouth. Hlaváč says his father was one of the few "visionaries" who believed in computers and would do anything to secure one for his sons. Other young people had to spend years convincing their parents that their interest in computers was genuine, and that they would indeed benefit from the sizable investment.[34]

Another notable motivation to buy a computer was its status symbol value, derived from both its scarcity and its aura of being a Western product. In a 1988 opinion piece for a computer club newsletter, the publication's editor situates the ZX Spectrum on the timeline of "symbols of public recognition." First was chewing gum, which had been famously scarce. Then came the transistor radio, the car, the black-and-white television set, the cassette player, the color TV, the Sinclair ZX81, the VHS player, the ZX Spectrum, and satellite television. By 1988, the ZX Spectrum had already been pushed out of the wish list by an IBM PC-grade machine. The author went on to complain that, for IBM PC owners, he was "but a poor wretch, left to rot in the withering Spectrumland."[35]

Lamenting that the never-ending chase after the next exclusive gadget eroded social ties in hobbyist communities, this piece attests that conspicuous consumption of technology and Western goods was very much at home even in Soviet bloc socialist countries, which, though ostensibly egalitarian, had manufactured inequalities of their own.[36] Yet keep in mind that, compared with stereos, TVs, and VHS players, the computer was still a relatively niche and expensive product that would hardly be purchased with the sole purpose of showing off—and that an IBM PC was out of reach of the vast majority of the population.[37] In fact, some of the "wandering programmers" could not afford an 8-bit machine until the 1990s.

Importing the Standard

Deciding to purchase a machine was just the start of the adventure. Hunting down a computer required a combination of both financial and social capital, and often the added bonus of pure luck. For most Czechoslovak citizens, computers were prohibitively expensive. In the mid-1980s, the price of a Sinclair ZX Spectrum on the informal market was about twelve to sixteen thousand crowns—four to five times the average monthly salary in 1980s Czechoslovakia. Even when it dropped to about five thousand in 1989, it was still a hefty sum.[38] Price was far from the only obstacle. Another was the import barrier. To get a computer, one had to look for a gap in the Iron Curtain and employ one of the tactics Czechoslovak citizens had developed to acquire unavailable goods. These included self-importing from abroad, shopping in imported goods stores, and tapping into the gray and black markets.

According to period reports, the majority of home computer users—especially in the early to mid-1980s—had individually imported their machines from abroad.[39] This tactic could only work under two conditions. First, one had to obtain permission to travel to the West, or find friends or acquaintances who could. This permission was by no means a common thing, but rather a privilege of a lucky few. Second, one had to have enough disposable hard currency. Again, this was quite rare. Western money could not be freely exchanged within Czechoslovakia, and was extremely difficult to acquire legally.

A significant group of citizens with such opportunities were the ones hired for temporary jobs in the "friendly" socialist countries of Africa and the Middle East. This experience was common enough to become a theme in popular music—the novelty song "White Daddy" by the pop-bluegrass musician Ivan Mládek tells a story of a Czechoslovak man who had fathered a son in Mali while working on the country's construction sites, but left before his son could ever meet him.[40]

During layovers in Western Europe, temporary workers could spend the money they had earned in developing countries on luxury items and electronics. In this way, one of my interviewees got his Sinclair ZX Spectrum from his father, who had worked as an electrical engineer in North Africa. Another dad was a chemistry teacher in Uganda and bought a Spectrum in Paris; yet another worked his stint as a geologist in Syria.[41] Besides temporary workers, there were particular professions that were also likely to obtain travel permits—scientists, foreign trade officials, tourist agency employees, and creative professionals. That last category includes Michal Hlaváč's "visionary" father, who worked in the film industry. In the mid-1980s, he went on a work trip to Spain, sacrificing his own comfort to secure the precious machine: "He was eating canned food the whole time, so that he could save up his per diems, and then he took all that money and bought us a Spectrum."[42] Those who traveled to the West on a more regular basis functioned as so-called "trade tourists" or "shopping tourists" and could supply Western goods to their friends and acquaintances, or to black market resellers.[43]

Before a micro could enter Czechoslovakia, another major obstacle awaited—the exorbitant customs fees on home electronics, only suspended in January 1988.[44] In 1987, the fee was 150 crowns per 1 kB of memory—bumping up the cost by 7,200 crowns (more than two average monthly salaries) in the case of a 48 kB ZX Spectrum.[45] It is no wonder that many opted to smuggle the micros instead. As Jiří Pobříslo, the engineer quoted earlier in this chapter, remembered in an interview:

> I paid a visit to an acquaintance who was about to travel to England to visit his wife, and asked him to bring—or I should rather say smuggle in—the computer for me. He agreed, but he obviously didn't have the money. We had to get the sum by the next day—and my wife actually managed to get those 500 German marks in one day. He left for England and smuggled the Spectrum back, wrapped in paper as a sandwich and hidden in a basket among other sandwiches.[46]

The practice of self-importing was so widespread that even research institutes and youth clubs—which were short on hard currency and unlicensed to trade with the West—resorted to individually importing computers and then "legalizing" them by selling them at pawn shops and immediately buying them back as "used" items. Although the police initially frowned upon these practices, they soon stopped bothering. In the end, "it was all for the good of socialism."[47]

By the mid-1980s, the Sinclair ZX Spectrum had emerged as the most widespread platform in Czechoslovakia, although it was never advertised or continuously distributed in the country. According to 1988 estimates, there were one hundred thousand Spectrums in the country, out of which only 16 percent were purchased domestically.[48] Early on, the Spectrum's price and versatility were the decisive factors. It was cheaper than competitors such as the Commodore 64 and the 8-bit Ataris. It did not require any special peripherals and could easily be connected to a Czechoslovak TESLA TV set and any cassette player. Its size probably also played a role. Being light, compact, and smaller than a letter-sized sheet of paper, it could easily be smuggled into the country once unboxed.

As soon as a sizable user base was established in the country, the Spectrum community snowballed, as other prospective users sought to benefit from the existing support networks and extensive collections of pirated software. Because there were no authorized services or software distributors in the country, going with the flow and getting a Spectrum was clearly the most practical solution. The considerable British influence on Czechoslovak gaming, through the Spectrum platform, was therefore an unplanned outcome of Czechoslovak users' improvised tactics rather than a product of an intentional strategy on the British side. The Spectrum upheld its status as the most widespread 8-bit platform until the 1990s.

In the latter half of the decade, the Atari 8-bit platform (including the Atari 800, XE, and XL models), originally designed in the United States, started to compete with the Spectrum. Nevertheless, it is telling that one of the first Czechoslovak Atari users was at first disappointed when his friend—whom he had instructed to buy a Spectrum in West Germany—brought him an Atari instead, because the Sinclair machine happened to be sold out.[49]

The Shiny Side of Retail

Regular shops had empty shelves, but a whole other world of retail glistened on the edges of socialist economies. In many Czechoslovak cities, one could find Tuzex stores, stocked with a tempting selection of Western luxuries: American jeans, Japanese hi-fi sets, Danish Legos, West German deodorants, and occasionally microcomputers. There was just one catch: the shops did not accept Czechoslovak crowns. The only way one could pay for these wonders was with hard currency or special coupons.

Hard-currency shopping chains were common in most Soviet bloc countries—Czechoslovakia had Tuzex, East Germany had Intershop, Poland had Pewex, and so on. The original intention behind them was to offer Western tourists luxury goods, trinkets, and souvenirs in order to collect their precious hard currency.[50] In the 1970s and 1980s, Tuzex opened up to a local clientele and transformed into a chain of luxury stores for Czechoslovak citizens. People who worked abroad could not keep Western currency, but had to exchange it for "consumption coupons" (colloquially called *bon*, plural *bony*), which could be spent in a Tuzex store.

The new system seemed to have killed two birds with one stone. As Paulina Bren notes in her analysis of the Tuzex phenomenon, "Expanding the pool of Tuzex customers meant more foreign currency in the state's coffers as well as a more satisfied consuming public."[51] But Tuzex also generated problems of its own, namely a burgeoning black market embodied in the figures of jean-clad, digital watch–wearing hustlers. Called *veksláci* in Czech (singular *vekslák*, derived from the German word *wechseln*—to exchange), they stood about on their spots in the vicinity of Tuzex stores, offering to sell *bony*, hard currency, or Western goods. For many prospective computer owners, they were the necessary middlemen on the way to the desired machine. The nominal value of a *bon* corresponded to one crown—and was not tied to a specific category of goods—but one *bon* cost five crowns on the black market. This meant that a microcomputer bought in Tuzex for black market *bony* cost five times its nominal price.[52] Sometimes, one could buy a micro directly on the black market, usually from a *vekslák*. However, this practice seems to have been more common in Poland, where pawn shops and flea markets ranked among the most substantial sources of computers.[53]

Official bulk imports started to play a significant role in the latter half of the decade, likely as a result of the relaxation of CoCom embargoes in July 1984.[54] Tuzex did on several occasions offer various quantities of microcomputers, and its erratic deliveries significantly shaped the local user bases. In the West, regional distribution of microcomputer platforms was, to a large extent, a product of corporate marketing strategies. In socialist Czechoslovakia, the decisions about importing particular microcomputer models were made not by businessmen, but by officials working for a limited number of "foreign trade enterprises," which were granted state monopoly over foreign trade. As Kornai would remind us, the rationale behind their actions was not to make profit, but to fulfill the instructions of their superiors with as little effort as possible.[55] Most likely they were tasked with importing computers to alleviate public demand. They did not, however, necessarily listen to the customers who wanted more Spectrums, and instead struck whichever deal was advantageous for them, even if that meant bringing discounted stock of machines that had failed in other markets. While intermittently stocking Sinclair machines, this import system also generated at least two platforms that became "big in Czechoslovakia," but nowhere else in Europe.

In 1984, Tuzex sold a thousand Sord M5 computers, creating a lively but isolated user base for this obscure Japanese-designed and Irish-manufactured 8-bit micro.[56] A couple of years later, Tuzex and several regular retailers stocked Japanese Sharp MZ 800 machines. According to a widely circulated but unconfirmed story, Czechoslovak officials—always short on Western hard currency—exchanged shipments of Sharp machines with the Japanese for shipments of wooden spoons.[57] Details of the deal notwithstanding, the Sharp MZ 800 became one of the country's top five 8-bit platforms, despite being virtually unknown in neighboring countries.[58] In Hungary, a similar honor belonged to the UK-designed 8-bit platform called Enterprise.[59] To a socialist consumer, the end results of these machinations could seem arbitrary and uncoordinated, very much like domestic computer production.

But foreign trade enterprises could be gamed, too. Oldřich Burger (born 1946), a cofounder of the country's first Atari club, reportedly used his contacts in the Socialist Union of Youth—a major Czechoslovak youth organization known for its vast resources and political clout—to lobby for importing Atari computers instead of the ones made by Commodore. Around Christmas 1987, the first shipments of Ataris arrived at Tuzex stores, and Atari clubs all over the country started to collapse under the weight of new members.[60] This shows how a single decision within the bureaucracy could have a major impact on the local computing and gaming communities. (For period estimates of machines by platform, see table 2.1. Note that their accuracy is up for debate, and it is possible that hobbyists overstated the estimates to grant legitimacy to their hobby or their platform.)

Let us conclude our excursion into computer hunting. It involved a diverse range of tactics that often took the hunters on crooked paths through space and time. Many of these tactics had already been established as a response to the shortage economy, and were tolerated by the authorities. But at the same time, computers were different from jeans, bananas,

Table 2.1
Estimated numbers of machines in the country per platform, compiled from various contemporary sources, mainly written by hobbyists.

Year	Estimates of penetration per platform
1985	20,000 ZX81s[a]
1986	40,000 Spectrums[b]
1987	80,000–100,000 Spectrums, up to 50,000 8-bit Ataris, 10,000 ZX81s, 1,000 Commodore 64s, hundreds of others[c]
1988	100,000–150,000 Spectrums[d]
1989	The above plus 14,200 PMD 85s, and about 4,000 IQ 151s[e]

[a] AR, "AR výpočetní technice '85."
[b] Myslík, "mikro PF 86."
[c] Libovický and Dočekal, "Domácí počítače, s nimiž se (možná) setkáte"; Ladislav Zajíček, "Kopisté versus Mikrobáze," *Mikrobáze* 3, no. 6 (1987): 2–4; Jan Hlaváček, "Zeptali jsme se za vás … Ondřeje Šebesty, vedoucího technicko-poradenského střediska v.d. STYL," *Zpravodaj Atari klubu (487. ZO Svazarmu)* 1, no. 6 (1987): 54–56.
[d] L. Kalousek, "Náš interview se zástupci 666. ZO Svazarmu." *Amatérské radio* 37, no. A6 (1988): 201–202; zav [pseud.], "Celní úleva a počítače."
[e] (VV), "PMD-85: Verzia 85-1 & 85-2"; Malec and (r), "Mikropočítače z 'druhé strany.' "

perfumes, or even cassette players. Buying a computer was a significant financial sacrifice and a major decision that determined which software one could use and which user community one would join. As Kirkpatrick has shown in his interviews with early Polish computer users, the situation was very similar in Poland, where people were saving up for years before they could buy a machine.[61] Accounts of computer hunting therefore became an integral part of one's personal computing narrative; they were stories to be told.

Narratives of computer hunting shaped the meanings people ascribed to Western, as opposed to domestic, computers. In Poland, the domestic Meritum machines were often dismissed as unattractive, outdated, and fault-ridden, whereas their Western counterparts represented affluence and sophistication.[62] In Czechoslovakia, Spectrums and Ataris evoked ideas of adventure, achievement, and freedom, whereas local models such as the PMD 85 and IQ 151 were associated with the quotidian realities of the normalization era. Paradoxically, imported machines dominated in domestic, private spaces, whereas domestically produced models—generally unavailable in retail—were situated in public spaces. This changed with the launch of the Spectrum-compatible Didaktik Gama in 1987, which problematized the distinction between the domestic and the Western options by offering a middle ground—a decent, if basic, domestic copy of the British machine. Nevertheless, not even the Gama eliminated computer hunting, because it could not by itself satiate the ever-growing demand for micros.

A Room of Its Own

Whether it was imported or bought domestically, the microcomputer's arrival in a socialist household was an event whose significance was felt by the whole family. In a 1988 interview, the Czechoslovak interdisciplinary scholar and microcomputer enthusiast Bohuslav Blažek compared the introduction of the computer into one's home to a deus ex machina:

> The computer makes a thunderous entrance into the family scene, like a deus ex machina. It sweeps the family savings and requires extensive reconstruction of the family's leisure time. Significant changes are made regarding the layout of objects within the apartment, and ownership of everything in the apartment is redistributed, including all spaces and passages.[63]

Although his statement is clearly hyperbolic, there were certain objective reasons that could make domestication of a computer in Czechoslovakia a dramatic affair. Some of them are related to the country's housing and interior design. In the postwar decades, Czechoslovakia suffered from a prolonged housing shortage. It was not uncommon for two generations to share a two-bedroom apartment. The crisis was partially alleviated by the massive state-funded construction of prefabricated panel high-rises, called *paneláky* (singular *panelák*) in Czech, starting in the late 1960s. These were usually built in large and condensed housing projects that loomed over the outskirts of towns and cities.[64] Simultaneously, another type of housing was changing the Czechoslovak landscape—the *bytovka*, a smaller, more communal standardized apartment building, sometimes built from brick, other times from panels. These developments led to large masses of people moving to new, unfamiliar places; even today, about one-third of Czechoslovak citizens live in *paneláky*.[65] But despite the magnitude of construction, lack of living space continued to be an issue in urban areas, in part because of the 1970s baby boom.[66]

When it came to furniture, the inhabitants of these new apartments did not have much choice—a common refrain in a shortage economy. Desirable pieces were produced in very small quantities, forcing the majority of citizens to choose from a narrow selection of mass-produced sets. An extensive 1979 housing study bemoaned the unergonomic and aesthetically misguided interior design solutions, many of which were due to lack of consumer choice. Most families (87 percent of those living in the new housing projects) fitted their living rooms with large wall-to-wall, floor-to-ceiling wall units, which were imposing but impractical (see figures 2.5 and 2.6).[67] The wall unit served as "an altar" for the TV set, which was the centerpiece of the whole living room.

Despite the limited space, the average socialist household was full of things. People compensated for the anonymous, impersonal nature of their new apartments and their mass-produced furniture by decorating their rooms with colorful wallpapers, tablecloths, and assorted trinkets that evoked petit bourgeois coziness rather than modernism; rustic and vintage elements were freely combined with plastics and synthetic fiber carpets.[68] Given the size of apartments, it was unusual to have a study. All children in a family typically shared one bedroom, furnished with leftover pieces from other rooms.[69] The situation was slightly different in older buildings, where

Figure 2.5
A wall unit arrangement, as advertised in *Domov* (*Home*) magazine, year 1974, issue 2.

people soldiered on with aging furniture that did not meet the parameters of modern housing.

Usually, 8-bit computers such as the Spectrum or the Atari connected to regular TV sets. Television was widespread in Czechoslovakia in 1980, when 88.7 percent of households owned at least one set—but only 5.5 percent of these were color TVs.[70] Especially in the early 1980s, a large portion of households only had one set. Therefore, microcomputer users often had to share the TV set with the rest of the household, and the computer's position in the living room was only temporary. Because most TVs at the time did not have a spare antenna socket, this meant switching the cable input manually every single time, a banal but tiresome task, especially when the TV was embedded in the wall unit. The price of the micros contributed to the care with which they were manipulated. The Hlaváč brothers, for example, were forbidden from eating at the computer to avoid damaging it.[71]

Figure 2.6
A drawing of "usual living room furnishing," based on the 1979 housing study. The illustration generalizes photographic evidence collected during research. Illustration by Ivana Čapková.

Another informant remembers storing the micro in the original packaging after each use.[72] All this suggests that finding a place for the computer was not a trivial matter.

In the interview about computers in the home, Blažek recommended using the micro as an "interface" that would facilitate a partnership between generations, or more specifically, between the father and children.[73] In some families, computers did fulfill that function. The Hlaváč brothers used to sit for long hours in front of a shared TV set.[74] Another informant, a mathematics professor and mother of three, used to cook dinner or mend socks at the same time as she instructed her children, who were programming or playing games on a ZX Spectrum in the kitchen.[75] Nevertheless, most recollections of dedicated computer enthusiasts point in the opposite direction—there was a tendency to move the micro from the shared living room to the private space of the machine's primary user. At some point, the novelty that had brought the family together wore off. The fact that microcomputer games—unlike video game consoles—offered limited multiplayer options also did not help. The Spectrum's tiny keyboard and lack of a standard joystick interface made it particularly difficult to share. Beyond play and simple experiments, making one's own software involved long, concentrated, and unspectacular effort that could hardly be enjoyed by the whole family.

Placing one's computer in a separate room required suitable furniture.[76] Given the chronic scarcity of such furniture, computer club newsletters published instructions on how to build do-it-yourself computer desks. One of these articles described the motivations behind the micro's journey into more autonomous spaces:

> The interests of everyone who wants to dedicate themselves to working with a computer, will sooner or later clash with the requirements of other family members, which are—though justified—still a bit of an impediment to this activity. It is therefore advisable that every user create an autonomous workplace of sorts, where one can work without limiting others.[77]

The movement of machines away from shared spaces was further enabled by the introduction and proliferation of a new line of TESLA-manufactured portable black-and-white TVs, which included the Satelit, Pluto, and Merkur models. These were commonly used as monitors, not only at home but also in computer clubs. Many players and programmers in the 1980s therefore experienced the decade in monochrome and only used color TVs on special occasions. As a respected author of homebrew actions games admitted: "All the colors in my games were made on a black-and-white set, based on my reading of grayscale and the mental image I had of Spectrum's colors. Those games were totally color-blind."[78]

In his genealogy of the UK game industry, Alex Wade rightly points out that for all their interest in 1980s "bedroom coding," game historians writing about this phenomenon have been more interested in "coding" than "bedrooms."[79] As a countermeasure, let us take a couple of peeks into the Czechoslovak bedrooms. Figure 2.7 shows the room of the prominent homebrew programmer František Fuka, in which a computer and a TV set share space with an old painting and old-fashioned wardrobes and carpets. Fuka was lucky enough to live in a spacious art nouveau–period apartment that provided enough space for a sturdy office desk.

We can make another, interactive, visit to a Czechoslovak bedroom thanks to the game *Demon in Danger*, written in 1988 by then fifteen-year-old

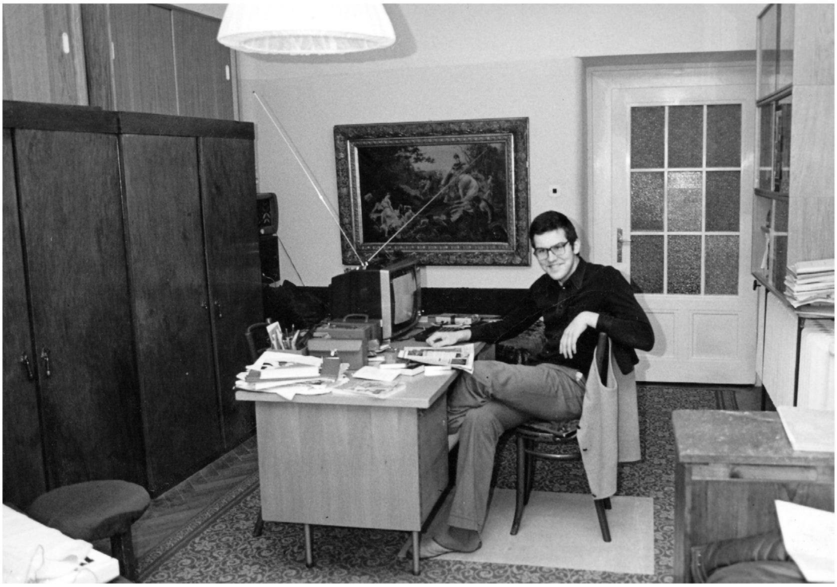


Figure 2.7
František Fuka in his bedroom. The antenna indicates that the portable TV set was used to watch television in addition to serving as a display for the ZX Spectrum. Photo courtesy of František Fuka.

Martin Malý (see figure 2.8). This text adventure takes place in his family's home, a *bytovka* apartment in the village of Chleby, about 70 km from Prague. The game is a reminder that domestication not only brings computers into the home but—as explored in chapter 7—also brings the home into the computer.[80] Within the game, Malý, who used the Demonsoft label, describes his bedroom:

> You find yourself in DEMON's office. It is characterized by enormous messiness. In the middle of all of this is a desk with a pile of papers and a Sinclair ZX Spectrum home computer with a Kempston Interface and a Quickshot joystick. The whole complex is connected to a National Panasonic cassette player.[81]

Malý proudly and diligently lists all the hard-to-get Western technology that is at his disposal. The description represents an ideal of a privatized, autonomous space for playing, programming, and tinkering, a space where messiness is allowed. The ambitious "Demon's office" label is, however, in marked contrast with the actual photograph of the room (see figure 2.8),

Figure 2.8
Martin Malý in front of his Spectrum; a TESLA portable black-and-white TV on the left. Photo courtesy of Martin Malý.

which shows a crammed, nonergonomic space whose walls are decorated with homely, rustic-patterned paint rollers typical for the period.

In spatial terms, a microcomputer was clearly a piece of Western influence encapsulated in socialist material culture. What we cannot see in photographs is the temporal dimension of computer use. Once a machine's place was established, both older and younger hobbyists spent long hours experimenting, working, or playing with it; for many of my interviewees, it occupied most—or nearly all—of their free time. Kids and teenagers were not the sole masters of their own leisure time and negotiated daily with parents; they had to take breaks for homework or chores. At the same time, the widespread belief among parents that computers were the future made computer use preferable to some other activities. So, surrounded by Czechoslovak material culture, the computer provided an ever-present instrument of imaginative travels. The everyday experience of microcomputer use was therefore neither socialist nor Western, but rather a deterritorialized mix of both.

This chapter has visited the intersections of the stories of computers as technological artifacts and the stories of their owners and users. These narratives played out on the terrain of the Czechoslovak shortage economy, where one had to *hunt* for a computer instead of simply purchasing it. This made the Czechoslovak computing experience not only delayed behind the West but also qualitatively different.

The British Sinclair ZX Spectrum, like other successful mass-market microcomputers, was essentially a populist machine—it was designed to be affordable by middle-class consumers. In Czechoslovakia, the need for hunting narrowed the domestic community to people who were exceptionally dedicated and persistent, or, alternatively, well-connected or otherwise privileged. Thus, the Czechoslovak micro user base was both a grassroots *and* an elite group. There were two prototypical categories of home computer users—members of the technical intelligentsia, and kids and teenagers (often sons of members of the technical intelligentsia). Micros were less likely to get into the hands of users who were socialized into having a more casual interest in technology, including girls and women or people without a technical background.

Due to the lack of nationwide advertising and retail networks, access to computers was structured by a variety of local factors. Youngsters who lived near a youth computer club had a greater chance to learn about micros and develop a fondness for them; people whose friends or family members had travel authorizations were much more likely to obtain a machine. All in all, one's social capital played a more critical role than one's income. Owning a computer was a privilege of sorts, and the widespread belief was that one should use it to its full potential—not only to play games but also to code or resolve real-life tasks. The price and scarcity of computers prevented users from moving on to more advanced machines, and as a result, many of them remained attached to their 8-bit computers until the late 1980s or early 1990s. The Spectrum and the Atari were therefore never fully defined as "toys" or disposable game machines as they were in the United Kingdom. Instead, they were simultaneously used for more serious work by the members of local hobby groups.

More so than acquisition, domestication followed many of the directions that have been observed in the West. Microcomputers—especially those popular in Czechoslovakia—were largely designed to be "personal" items that made the practice of family use inconvenient, even if there was social pressure to do so. At homes, just as in clubs, hobby computing was largely the domain of boys and men—a topic explored in the next chapter.

The relationship between the computer and interior design highlights the importance of materiality for the historiography of home computing.

Historical narratives of computer technology, especially those in the vein of platform studies, tend to focus on the computer or the console as their central artifact. This hardware tends to be well documented, understood, and historicized. However, a social historian should imagine the machine with all its connections and interactions with its immediate surroundings. Cables, aerial sockets, and tuning controls on TV sets, or the furniture used to house computers or display units—all of these could markedly alter the experience of coding or playing games.

Czechoslovak users worked with Western hardware, but their encounters with Western retail infrastructures were very rare. They could not have their machines serviced or buy peripherals in authorized stores. This contributed to the popularity of do-it-yourself repairs and hardware add-ons, explored in the next chapter. The nonexistence of domestic retail infrastructure also offers a part of the explanation—a material one—of why video game consoles were so marginal in the country. To take full advantage of 1980s consoles, one had to buy game cartridges, whose distribution was in turn largely dependent on a stable retail network capable of storing, displaying, and selling these objects—and no such network existed in Czechoslovakia. Microcomputers, on the other hand, allowed for a decentralized and emergent materiality of tape trading and informal distribution networks.

3 Our Amateur Can Work Miracles: Infrastructures of Hobby Computing

> I have claimed on many occasions that "our amateur can work miracles." Why am I writing this? Because it is the amateur who will play a very significant role here. I will not give away any secrets if I say that we have somehow slept through the development of electronics. … Nonetheless I believe that the amateur will overcome these obstacles and will be indispensable in the development of computer technology and its implementation into practice.[1]
>
> —From an editorial in the *Mikrobáze* newsletter, 1986

Popular digital game histories tend to follow a market-driven teleology of the game industry, in which the expertise and creative efforts of computer users are validated by commercial success, and in which game histories are written by authors, companies, or nations who turn their work into big profits. The British documentary *From Bedrooms to Billions* and the American popular history book *Smartbomb: The Quest for Art, Entertainment, and Big Bucks in the Videogame Revolution* embody this teleology in their very titles.[2] Business logic permeated Western computing and gaming cultures. Many of the British, American, and Scandinavian coders of the 1980s dreamed of writing a hit game.[3] Hardware and software industries, as well as computing and gaming magazine publishers, were motivated by their profit margins. And despite the important contributions of schools and other publicly funded institutions, much of the infrastructure that allowed gaming to flourish was established by commercial companies.

In Czechoslovakia, hobbyists were also eager to create, enjoy, and discuss hardware, software, and games. There was, however, no road leading to billions. Save for minor exceptions, the Czechoslovak state did not allow private enterprise, and therefore no privately owned companies could distribute hardware or games.[4] Nor did the state initiate any large-scale consumer-oriented IT projects of its own, as France did in the case of Minitel.[5] As Bohuslav Blažek concluded in 1990, "What powered the most massive commercial boom in America's history, was [in Czechoslovakia] a mere source of minor odd jobs."[6] The observation that gave this chapter its title was correct: it was up to the amateurs and enthusiasts to carry the flag of microcomputer culture, and to carry the burden of the scientific-technological revolution (STR). The state was not going to do it for them, and neither was the nonexistent commercial sector. The history of computer games in Czechoslovakia, therefore, cannot be a history of an industry—it has to be a history of the amateur.

This chapter will explore the achievements of Czechoslovak amateurs, and analyze the networks and infrastructures that they built or appropriated to work with computers. I use the term *amateur* (*amatér* in Czech) for two main reasons. First, it is sufficiently broad to explore the wide range of Czechoslovak microcomputing practices; according to a current dictionary definition, an amateur is "a person who engages in a pursuit, esp. a sport, on an unpaid basis."[7] Second, it is an emic term that participants themselves were using at the time. Despite its potential negative connotations, Czechoslovak technical hobbyists used it in a celebratory way. It was even in the title of their "bible," *Amateur Radio* magazine.[8]

The word can be understood in contrast to the *professional*, who undertakes comparable endeavors for financial remuneration. As discussed in chapter 1, being a computing professional was no longer an uncommon occupation in 1980s Czechoslovakia. However, professional programmers and hardware engineers were scattered in existing institutions and factories, and could not freelance or enterprise on their own.[9] The potential of the nation's technical intelligentsia was often mismanaged and not used to its full capacity. The amateur sphere was an environment where current and prospective members of the technical intelligentsia could channel their excess energy and realize their ambitions. Computer clubs were examples of *vnye* milieus as defined in this book's introduction: they were part of the state socialist system, while remaining independent of its norms and rules; they had the advantage of relative freedom, but due to their nonprofessional status, they were disconnected from material means of production, which limited their economic and social impact.

The rise of amateur hobby computing exemplified the boom of do-it-yourself activities typical of the normalization era's socialist consumerism. It also resonated with the national myth of *golden Czech hands*. In the influential 1996 study *The Little Czech Man and the Great Czech Nation*, anthropologist Ladislav Holý unpacks the ambivalent self-stereotype of "the little Czech." As the popular saying goes, Czechs have "golden Czech hands" that "manage to cope with everything they touch," making the little Czech "talented, skillful, and ingenious."[10] The notion of golden Czech hands celebrates self-reliance and inventiveness independent of the current political situation or ruling class, reinforcing the idea of Czechs as a nation of bricoleurs, prospering even when subjugated to hostile authorities. At the same time, the little Czech is viewed as "shunning high ideals and living his life within the small world of his home, devoting all his efforts to his own and his family's wellbeing."[11] The little Czech is a pragmatic tactician skeptical of ideologies and politics, traits we will explore in this chapter. These recurring self-definitions of the Czech nation shaped the norms that made certain practices and tactics acceptable or desirable. Although initially referring to Czechs, these norms likely filtered into the Slovak part of the federation during the decades in which the two shared a country. The belief in the amateur, voiced in the opening quote, can thus be read as an echo of the belief in golden Czech (or Czechoslovak) hands.

The amateur status in connection to microcomputers had a different meaning in Czechoslovakia than it did in the West. Amateur clubs of various shapes and sizes sprang up in different countries throughout the 1970s: the British Amateur Computer Club started operating in 1973; the California-based Homebrew Computer Club—attended by Stephen Wozniak, the cofounder of Apple—in 1975; and the Dutch Hobby Computer Club in 1977.[12] Wozniak claimed that "without computer clubs, there would probably be no Apple computers," suggesting a trajectory from homebrew culture to startup culture.[13] But the clubs' prominence was waning in the 1980s, when—at least in the United Kingdom—they seemed to be attended by a small minority of users.[14] This decrease in popularity can be attributed to various factors, including the growing availability of hardware and ready-made software, and the commercialization of the software industry.

Behind the Iron Curtain, however, amateur clubs played a critical role well into the late 1980s. In Communist-era Czechoslovakia, it was impossible for people to work and own equipment collectively without official backing, and amateurs therefore had to take advantage of existing state-sanctioned youth or paramilitary organizations. With formal backing from the state, clubs brought together amateurs of different ages, both hobbyists and gamers. Clubs were sites of collective, participatory activities—the building of hardware, creation and distribution of software, and publication and dissemination of information. Throughout the 1980s, they performed some of the functions that hardware and software markets and the hobby computing press were fulfilling in Western countries.

The following pages will follow microhistories of Czechoslovak microcomputer amateurs and examine their practices and tactics. I will start in education, looking at young people and their activist instructors, who went out of their way to bring cybernetics and computing into youth groups. The bulk of the chapter will focus on the apolitical *vnye* spaces of computer clubs affiliated with various socialist organizations, presenting their hardware, software, and organizational achievements, and explaining how they maintained their political autonomy. I will address the heavily gendered, masculine character of the clubs and hobby computing in general. Finally, I will compare clubs to the less impactful, but more radical efforts of individual do-it-yourself samizdat publishers, and the use of 8-bit computers by Czechoslovak dissidents, who positioned themselves outside of the club environment, but employed many of the same bricoleur tactics.

Cybernetics for Youth

The previous chapters have shown how inefficient central planning and economic isolation from the West hindered the development of home computing technology in Czechoslovakia. Nevertheless, the state gave material support for all kinds of hobby activities, including computing. As early as the 1950s and 1960s, the Czechoslovak leadership had allocated generous funding to technical and other hobbies, one of the proclaimed reasons being that hobbies were an "extension of extracurricular education."[15] Microcomputer hobby groups could therefore take advantage of existing state-funded hobby infrastructures.

In Czechoslovakia, as in Poland and the Soviet Union, education provided a convincing justification for hobby activities, and youth organizations were an ideal home to *vnye* milieus.[16] In addition, there was an even deeper affinity between kids and hobbyists. Children as well as hobbyists enjoyed hands-on experimentation, tinkering, and bricolage. They did not follow the teleology of the computer as a tool of industry automation. They wanted to be with computers, here and now. All this contributed to the fact that a large part of the local computer hobby movement converged around the institutions working with children and youth.

Especially important were the facilities run by the Pioneer Organization of the Socialist Union of Youth, or simply the Pioneer (Pionýr in Czech). Pioneer was a youth organization typically joined by children between ages eight and fifteen. Although membership was technically optional, there was strong social pressure on families to enroll their children, and the vast majority did.[17] Pioneer allowed the state to oversee and control children's free time, and to expose them to Marxist-Leninist ideology; however, much of its day-to-day routine was apolitical or inspired by "bourgeois" sources such as the Scouting movement.[18] Among other things, Pioneer ran a network of Houses of Pioneers and Youth (or simply "Pioneer houses"), which hosted workshops and hobby group meetings.[19]

Miroslav Háša (born 1944) was an instructor at Prague's Station of Young Technicians, a specialized branch of the local Pioneer house, located in the city's sixth district. A pedagogy graduate, Háša had always wanted to teach kids "who want to be taught," and therefore happily accepted a job at the Station. He also wanted to teach what he did *not* know, so that he could educate himself as well as his pupils.[20] His curiosity steered him toward cybernetics, and in October 1980, he launched the first extracurricular programming lessons for schoolchildren under the Station's newly founded Department of Cybernetics. Thanks to his enthusiasm and organizational skills, he soon convinced leading academics and computer experts to help him out with the lessons. This group included Rudolf Pecinovský, a mathematician and soon-to-be expert on the didactics of programming, and Eduard Smutný, then a design engineer at TESLA.[21] According to Háša's recollections, the latter first refused the invitation, citing a busy schedule of "doing scientific-technological revolution."[22] But he soon changed his mind and became a fervent proponent of children's computing groups. In 1982, he promoted the Station in an interview for *Amateur Radio*:

> I can confirm that as an expert, I have very much benefited from working with young people. Giving correct and precise answers to their thoughtful questions requires actual experience, knowledge of all causal relationships, and a true understanding of the matter. It is not enough to just quote catalogues and foreign articles.[23]

Around twelve kids, ages nine and ten, enrolled that year (see figure 3.1). At first, the group was short on computers, and most of the kids used paper

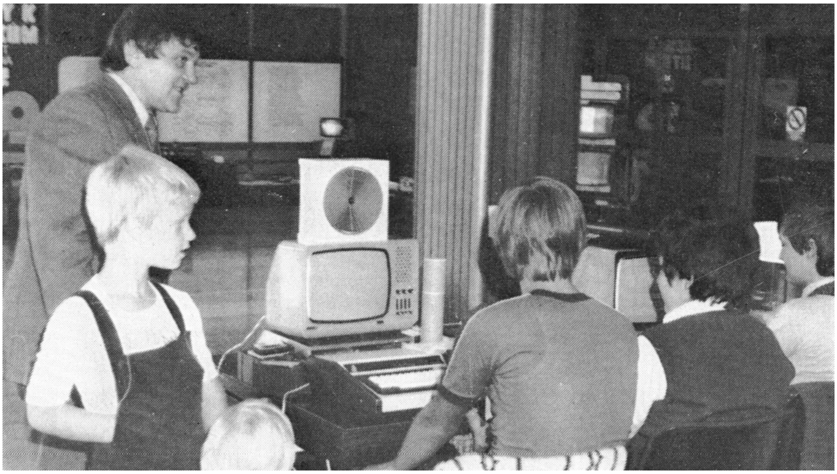


Figure 3.1
Station of Young Technicians' presence at the Roads to Tomorrow exhibition in October 1982. Miroslav Háša standing, František Fuka seated second from the right. Reprinted from *Amateur Radio*, year 1982, issue B2.

models.[24] Soon, the Station got hold of a Video Genie machine, a Sinclair ZX80, and Ohio Scientific's Challenger, a rare, US-manufactured micro. Guests and instructors were bringing components for soldering, and others donated or lent entire machines.[25] At that point, Háša had effectively turned the Pioneer house into a microcomputer think tank.

Young pupils learned at a breakneck pace, some of them soon surpassing their instructors. They became intimate with the machines and developed "their own language." When journalists came to report on the Station, they could not understand what the young whiz kids were saying.[26] The Station soon became a flagship of programming education and developed teaching methods and curricula for other institutions.[27]

Háša himself coauthored the popular paper model computer mentioned in the previous chapter. Even more importantly, the Station was the birthplace of the decade's most influential piece of Czechoslovak educational software—KAREL. Its story starts in the United States, where professor Richard E. Pattis created Karel the Robot—a tool to ease his students into writing code in PASCAL and grasping the principles of structured programming.[28] It featured a grid-based map and an anthropomorphic robot, whom the student controlled via simple commands that could be built up into more complex procedures.[29] The robot was named after Karel Čapek (coincidentally, a Czech writer), who had coined the word *robot*. Although Karel the Robot was originally a PASCAL add-on, Station affiliate Tomáš Bartovský re-created it as *KAREL* (capitalized), a stand-alone programming environment for the Sinclair ZX81. It was soon ported to other relevant platforms.[30] The team supplemented the software with a teaching curriculum, which spread across the nation through seminars and workshops. Leveraging young people's interest in games for the sake of programming education, KAREL was presented as "an entertaining game and a serious teaching tool at once."[31] It became a popular introductory-level instruction tool, competing with and likely beating the Logo language, which was standard in some other parts of the world.[32] Eventually, KAREL became more influential in the land of Karel Čapek than in the land of Richard Pattis.

Despite methodological sophistication, the atmosphere at the Station tended to be informal and relaxed. Schoolkids of various ages—mainly boys—mingled with more experienced teenagers and adults. Some played games, others programmed or built hardware; it was "all about the individual approach."[33] The Station, as well as other similar institutions and schools, also organized fondly remembered weekend and holiday retreats that combined outdoor activities with computing (see figure 3.2).[34]

Among the Station's alumni were some of the country's most prolific and influential game programmers, including Vít Libovický, as well as the trio of František Fuka, Miroslav Fídler, and Tomáš Rylek, who later formed the Golden Triangle homebrew collective. The three teenagers all lived in the sixth district and would often walk together along the nearby Vltava River, discussing software.[35]

As the 1980s progressed, youth hobby groups such as this one started to emerge in Pioneer houses in other towns and cities. Their numbers further multiplied thanks to Miroslav Háša's next endeavor. Knowing that the Socialist Union of Youth—the Pioneer's parent organization, and the Czech counterpart to the Soviet Komsomol—held considerable power and material resources, he convinced its Central Committee to form the Center for Youth and Electronics, with him as director. One of the Center's aims was to "promote the results of the work of the young generation against technical conservativism, and in the name of progressive methods and technologies."[36] Starting in 1983, the Center launched a network of dozens of its own Youth Clubs for Science and Technology. Throughout the decade, Háša

Figure 3.2
Unloading computers and TV sets on the first day of a youth computer camp, 1987. Photo by Vlastimil Veselý.

gained political support for youth computing from various institutions. As he told me in our interview, "We were always looking for pathways through which we could progress faster."[37] A skillful tactician, he capitalized on his connections to the media, and on his ability to navigate the structures of power. His efforts also made him an inadvertent godfather to the country's gaming communities, and he accepted games as an integral part of microcomputing.

Many other groups and workshops were launched by elementary schools and high schools, especially those with a mathematics or engineering specialization. One of them was led by another of my informants, Charles University mathematician Alena Šolcová (born 1950), at an elementary school in downtown Prague, and attended by about twenty kids every year. Simultaneously, Šolcová taught programming informally to a small group, including her own and her friends' children, at her home. Bottom-up initiatives such as the ones by Háša and Šolcová seemed more successful than the top-down Long-Term Complex Program of Electronization in Training and Education (see chapter 1), which failed to retrain a sufficient number of instructors to teach in programming.[38] But despite having plenty of practical experience, Šolcová was rarely consulted by those in charge of computing education.[39]

Repurposing the Paramilitary

The Station of Young Technicians in Prague's sixth district became a meeting place not only for young people but also for adult hobbyists, who soon outnumbered those they were supposed to instruct. Sometime around 1982, it was clear they had to move their activities somewhere else so they "did not get arrested for illegal gathering," as Háša put it.[40] To continue and develop their activities, they decided to formalize the group by joining another organization, which happened to use the same building as the Station and had a conveniently large lecture room. Around 1982, they became members of the 602nd Basic Organization of Svazarm, colloquially known as "the 602."[41]

Svazarm, or the Union for Cooperation with the Army (Svaz pro spolupráci s armádou), was the country's largest paramilitary organization. Its original goal was to train civilians, especially young people, for potential roles in the military. In reality, Svazarm had become an umbrella for a wide range of hobby activities, including motorsports, dog training, marksmanship, ham radio, aviation, hi-fi, and electronics, often with no direct military application.[42] In 1982, it had 978,326 members (90 percent of whom were men), meaning that about 1 out of 15 Czechoslovak citizens had joined Svazarm.[43] The organization was fraught with the "lasting contradiction between the requirement of the Army for the training of recruits and the defense of the rear, and the requirement of its members to secure conditions for their hobbies."[44] Although its military benefit was questionable, it captured much of the country's hobby activity, and therefore provided a degree of supervision and control.

In 1982, Svazarm officially declared its support for computing—games included—stressing both economic and military reasons:

> Svazarm will aim to track hobby activities in the field of computing, including construction of machines and equipment using digital integrated circuits, as well as building of various programs for calculations and games on programmable calculators and personal microcomputers, whose numbers have been rising both at workplaces and among individuals. It will aim to streamline this activity to

> increase the number of technical personnel, especially young people, who have good command of computers and will use it for the benefit of our national economy and for the defense of our homeland.[45]

In the same year, *Amateur Radio* magazine, published by Svazarm's Central Committee, introduced a regular eight-page supplement called *Microelectronics* (*Mikroelektronika*), which contained hardware schemata and descriptions, program listings, user tips, and reports. Despite the limited space, it became one of the central publication platforms of the domestic microcomputer hobby scene.

Svazarm was divided into thousands of local *basic organizations* (*základní organizace*), each of which was assigned an ordinal number when established (see table 3.1). Clubs sometimes referred to themselves using the numbers of their respective basic organizations, so expressions such as "the 602" were part of hobbyists' parlance. The emphasis on youth training was stipulated in the requirement that at least 90 percent of basic organizations that worked with electronics run a group for children under age fifteen.[46]

When a new club formed in a Svazarm organization, it was allotted premises by the local government or town hall. Svazarm usually covered the rent and some additional expenses, although the extent of the latter seems

Table 3.1
A list of influential Svazarm microcomputer clubs and their achievements relevant to the topic of this book.

Number	Headquarters	Relevant games-related achievements
415	Ostrava	Games (mainly conversions) for the PMD 85 computer, including *Flappy* (see chapter 6)
482	Prague	Games for the PMD 85 computer
487	Prague	A prominent Atari club; published the Turbo 2000 data storage system
602	Prague	The largest Svazarm computer club for all platforms; produced software and published the *Mikrobáze* newsletter/magazine; its alumni included prolific homebrewers
666	Prague	Splintered from the 602; published manuals, educational material, and the *Computer Games: Past and Present* book
Unknown[a]	Karolinka	Published numerous manuals for games and other software

[a] The club in the town of Karolinka did not use its number in its materials.

to have varied.[47] The space was not always fit for its purpose and sometimes had to be laboriously converted to a computer club workshop. The 415 in Ostrava, for instance, was regularly frequented by mice. After failed attempts at a mousetrap solution, the members gave up and ended up peacefully cohabiting the space with their rodent visitors—who, after all, did not damage their computers.[48]

Besides being a space to meet like-minded enthusiasts, clubs provided access to hardware, software, and information. Each member had to pay a monthly fee, usually in the tens of Czech crowns.[49] Although clubs usually owned some hardware, they invariably had more members than computers, and many users would bring their machines with them to the meetings.[50] Conversely, members could occasionally borrow club machines for a weekend. An Atari club based in North Moravia defined the rights and obligations of a club member as follows:

> Atari club membership is subject to the payment of a membership fee based on the designated budget and to active participation in the running of the club based on the member's capacities and the needs of the club management. It gives the member the right to participate in club events, the right to make free copies of programs, as well as access to the *Micro A* newsletter, other publications and technical information.[51]

The right to make free copies of programs extended to Western commercial software. Chapter 5 will return to this when discussing informal distribution.

The structure of clubs was shaped significantly by the hardware they used. The first clubs, such as the 602, welcomed everyone. However, as various platforms took root in the country, hobby groups began to splinter and specialize. The initial reason was the difference in architecture between platforms, and the need to focus on one platform in order to sufficiently master it. Most of the advanced microcomputer programming at the time took place in *machine code*—a set of instructions that were executed directly by the computer's CPU, without an intermediary programming language. What makes sense in terms of hardware design tends to be unintuitive for human coders, and writing in machine code meant producing a long series of hexadecimal numbers. It is rarely used today except by operating system and compiler programmers, but it was essential on 8-bit machines, where writing efficient code was much more critical. The more user-friendly *assembly languages* added a symbolic layer on top of machine code, and could be edited and compiled to machine code by a utility called an *assembler*. What is important for our story right now is that both machine code and assembly languages could vary significantly between platforms. Platform affilation thus had its practical reasons, but it soon became an identity marker. The latter half of the 1980s was a period of platform rivalry, especially between the two major Czechoslovak platforms—the Spectrum and the 8-bit Atari.[52]

The number of clubs rose steadily as the decade progressed. Based on directories published in club newsletters, we can estimate that there were at least a hundred computer clubs in the country in the late 1980s—at least one in every city and most larger towns.[53] Although Svazarm was the most prominent host of hobby computing groups, many clubs affiliated themselves with other organizations, such as the Socialist Union of Youth or the Czechoslovak Association for Science and Technology. In fact, while looking for institutional backing, the computer hobby expanded into almost all existing socialist organizations. The smallest clubs retained an informal group atmosphere, whereas the larger ones—such as the 602—went on to serve thousands of members, many of whom were correspondents rather than attendees. Together, these created the backbone of the local computing and gaming communities.

Activist Meshworks

Computer clubs were *vnye* milieus, dependent on the state socialist system but largely independent of its norms and rules. Day-to-day Svazarm activities were devoid of ideological content, and Czechoslovak authorities did not seem to know or care what exactly was happening in the clubs. We can observe this in a 1986 TV documentary commemorating the thirty-fifth anniversary of Svazarm. In one of the segments, a drily enunciated commentary reiterates stock propaganda statements about the goals of the organization: "In Svazarm collectives, youth learn to correctly understand the contemporary political situation, correctly assess the world's class divisions, and understand its contradictions. All activities of basic organizations develop proper relationships and attitudes to the construction and defense of socialism among citizens."[54]

Accompanied by this haughty monologue is footage of kids and teenagers playing *Daley Thompson's Decathlon*, a 1984 British sports game for the ZX Spectrum, and later enjoying some soldering work.[55] The disconnect between the voice-over and the images is striking. To some extent, we can treat this mismatch as an example of highly ritualized propagandistic rhetoric that has lost any reference to reality. At the same time, it suggests that authorities barely understood the activities that computer enthusiasts were partaking in. Conversely, members did not care about the goals of Svazarm, and their engagement with the organization was purely pragmatic. Many of my younger informants do not even remember which socialist organization they were members of, whereas the older ones routinely followed their hobby in several organizations at once.

It may seem paradoxical that entities within the paramilitary openly distributed pirate copies of British and American entertainment software, but this was due in part to the relative autonomy they had as hobby clubs. This autonomy owed in part to their peculiar position in the organizational structure of the state. In the case of Svazarm, they were nested in basic organizations, several steps removed from central control.[56] To understand the effect of this position, we can borrow some insights from Manuel DeLanda's philosophy of history. He distinguishes between two types of networks that emerged in various historical contexts: "self-organized *meshworks* of diverse elements, versus *hierarchies* of uniform elements."[57] In his view, Europe's growth in the medieval era was due in part to the meshwork nature of its towns, which allowed for numerous unobstructed flows of material, capital, and ideas. Chinese cities, on the other hand, were larger, more hierarchical, and more controlled, and they encouraged accumulation rather than multidirectional flows. To tie these network types to de Certeau's strategy/tactic dichotomy, we can associate hierarchies with strategies, and meshworks with tactics. Socialist Czechoslovakia had strong hierarchical governmental and party structures, but their inefficiency spurred the growth of the informal meshworks of the black market and hobby clubs. To gain legitimacy and access to material resources, clubs had to attach themselves to a hierarchy, usually Svazarm; but they continued operating as bottom-up meshworks, unrestricted by the organizational bounds of individual clubs. There were plenty of connections and collaborations—for example, 29.3 percent of the members of a prominent club were "guests" whose primary organizations were elsewhere.[58] Rather than being hierarchically organized, the resulting meshworks would cut through (predominantly) lower levels of multiple hierarchies. The amateurs' endeavors could then reach thousands of people across the nation while retaining a high degree of autonomy.

However, neither these tactics nor the rhetoric of STR gave computer clubs an automatic right to existence. Maintaining them required continuous effort on the part of group leaders and club organizers. Even in contemporary language, these were called "activists."[59] Hobby computing activists tended to interpret joining Svazarm as pragmatic infiltration of the regime. As Tomáš Smutný, a veteran of hobby clubs and a youth group instructor, recounts: "We didn't mind being under Svazarm and didn't care if it meant something for the army and the defense of the homeland and so on. We just unscrupulously took advantage of the regime to get to the things that we were attracted to and that we liked, and that we would otherwise never get."[60]

The relationship was often symbiotic rather than parasitic. Local officials of the respective socialist organizations benefited from the "miracles" achieved by computer clubs, especially if these were covered by the media, which were constantly hunting for any good news about Czechoslovak science and technology. A youth group's success in a high-profile international competition could, for example, be rewarded with a subscription to a foreign magazine.[61] But not all officials were open to the idea of computer clubs. Despite the party's and Svazarm's claims of support for hobby computing, some die-hard conservatives proved difficult to convince, as in the case of the country's first Atari club. The group had already advertised their club in *Amateur Radio*, only to learn that the local Svazarm chairman had suspended their membership, reportedly because they used unsavory American machines. The club later found its home in another city and another organization.[62]

Conservative bureaucrats became targets of criticism from Ladislav Zajíček, the outspoken editor of the 602's *Mikrobáze* newsletter. Zajíček was perhaps the most fearless activist of the Czechoslovak hobby scene. Born in 1947, he was originally a rock drummer, but gained prominence as the founder and chief manager of the Section of Young Music (Sekce mladé hudby) youth association. In the late 1970s and early 1980s, the Section explored and pushed the limits of what was legally possible in community life in normalization-era Czechoslovakia, organizing concerts and screenings, and publishing articles and books about artists otherwise excluded from official Czechoslovak media. All the while, Zajíček was extremely careful not to break the law, which set him apart from the "illegal" underground culture associated with dissent. In 1985, after persistent pressure from the Ministry of the Interior over the scope and nature of its activities, the Section—which then had about ten thousand members—dispersed.[63]

Soon after, Zajíček joined the 602 and used his activist chops in the services of his other interest—hobby computing.

A gaunt, long-haired, bearded rocker, he stood out among other computer enthusiasts. His aim was, among other things, to make computing accessible to people outside the technical intelligentsia—to people like himself. *Bits at Your Home* (*Bity do bytu*), his textbook of Zilog Z80 machine code, was a seminal resource to many 8-bit coders.[64] Between 1985 and 1988, he built the 602's newsletter, *Mikrobáze*, into a professional-looking magazine with original reporting, interviews, and editorials. Although it formally remained a members-only publication, its circulation reached six thousand copies in 1989. Zajíček never shied away from writing critically about the state of the Czechoslovak computer industry and bureaucracy. In 1989, this cost him the post of main editor of *Mikrobáze*, from which he was removed—according to his own later account—for political reasons.[65]

Zajíček's case highlights the connections between computer clubs and other hobbies and cultural activities. The struggle to build communities and follow one's interests required similar kinds of maneuvering within socialist organizations, regardless of the particular interest.[66] There were, however, crucial differences in political status among such activities. Nonconformist and alternative rock musicians suffered continual oppression, notably in the case of the Plastic People of the Universe, an avant-garde rock band whose criminal persecution inspired the Charter 77 dissident movement.[67] Computer clubs, on the other hand, were not systematically monitored by secret police, and their members only became targets of investigation in individual cases of suspected industrial espionage or smuggling, or for other unrelated reasons.[68] The same was true of the Soviet Union, where, according to Zbigniew Stachniak, the "Moscow regime did not consider computer hobbyism to be politically dangerous, an activity that could have facilitated ideologically undesirable activism."[69] Because the microcomputer hobby was not considered a part of "culture," it was not suspected of working with political messages. But even games could be explicitly political, as chapter 7 will discuss.

Tolerating the Man's World

We have seen that Czechoslovak hobby computing was an intergenerational endeavor. In terms of gender, however, it was strikingly homogeneous and overwhelmingly masculine. Partial statistical data suggests that the fraction of female hobbyists was substantially lower than that of female technical experts, which stood at about 20 percent, and staggeringly lower than that of professional programmers, which—as mentioned in chapter 1—reached over 50 percent. Men dominated the hobby across generations. Out of 138 readers who posted computer-related classified ads to the 1987–1989 issues of the *ABC of Young Technicians and Scientists*—a magazine that addressed both boys and girls—only one was female.[70] In 1987, one of the largest "all ages" clubs in the country reported 1,458 members, out of whom only 37 (less than 3 percent) were women.[71] While studying volumes of computer club newsletters, I never came across a single article signed by a woman; out of ove two hundred games that have survived from the period, only two list female coauthors, both in supporting roles.[72] Informants, too, concur that women made up a tiny minority of club members.

This imbalance has at least three complementary explanations. One has to do with upbringing and education, both of which reflected customary beliefs about children's talents, dispositions, and interests, common to much of the Western world. As Margolis and Fisher have demonstrated with US material, teachers and parents have traditionally looked for signs of interest in technology among boys rather than girls.[73] Even the otherwise progressive *ABC of Young Technicians and Scientists* reflected these assumptions by running a regular section for girls, called Girls' Fantasy Cookbook, later renamed to ABC of a Handy Girl, which focused on cooking, clothing, and handiwork.[74] Women in professional programming were—in absolute numbers—too few and not visible enough to become role models. The computer was thus presented and perceived as a boy's toy, as it was in the West. These stereotypes were further compounded by gender segregation in secondary education. Education policies emphasized applied, vocational education in demand by the industry, which led to a high degree of specialization already at the secondary level, perpetuating the existing division of boys' versus girls' fields. In the normalization era, nine out of ten pupils of vocational secondary schools studied in a gender-homogeneous environment.[75] And although general (unspecialized) grammar schools were more balanced, STEM-focused schools and classes—which were among the hotbeds of computer hobby—tended to be very masculine.

Second, the imbalance was a product of tradition within Svazarm and the electronics hobby movement. Svazarm was 90 percent male, and its *Amateur Radio* magazine cast hobby computing as an offshoot of existing hobbies that had already been codified as male, such as ham radio or electronics. This might have been discouraging to women who reached microcomputers through other avenues such as mathematics or education.

Finally, women had difficulties embedding the time-consuming hobby within their private lives. Despite several progressive gender policies, Czechoslovak society was a patriarchal one, and a relative gender equality in labor did not bring forth gender equality in leisure. Women behind the Iron Curtain tended to work two shifts, "toiling long hours in the factory and yet being expected, in a patriarchal society, to carry out the bulk of domestic familial duties."[76] According to a 1984 report, men had 16.1 percent of their time available for leisure activities, whereas women had only 7.7 percent.[77]

The prevailing masculinity of the computer hobby was stabilized through club practice and articulations in computer club newsletters. Several articles addressed the tensions between male hobbyists and their wives, resembling American texts about so-called "computer widows" and portraying clubs as fraternities separated from both work and family lives.[78] To celebrate the 1988 International Women's Day, a Prague Atari club published a piece called "Thanking Our Women." Its author wished to express his gratitude toward women for "tolerating the computer madness" of their husbands:

> Some marriages reportedly did not withstand the coming of the new "tenant"—at least that's what we hear from those who lost their wives to their computers. Elsewhere (as many desperately complain), we can hear words like "I'll throw this out of the window or I'll divorce you." Sometimes, the head of the family can save the situation by saying: "I'm doing this for the sake of our children. …" Some world-wise individuals argue: "Like I've taught her to drive a car, smoke and drink beer, I've also put a joystick in her hands … and now I can relax." But I consider the latter solution precarious, as it might lead to a total collapse of a household.[79]

Although decidedly hyperbolic, the text suggests that quiet tolerance is in fact the desirable mode of a woman's relationship to her partner's computer hobby. Otherwise, there is an imminent threat of either divorce or household collapse. The hypothetical option of the hobbyist taking over the household chores while his wife is using the computer is not even suggested; and the exclusion of women from hobby computing is considered justified.

When women were participating, they did not perform activities that would give them explicit or primary credit. In line with Haring's accounts of ham radio clubs, female club members tended to assume roles that were less visible.[80] Sylva Prokšová, who cofounded one of Prague's influential computer clubs, was a graduate of the Czech University of Life Sciences, with a degree in automated control systems. She was a professional first and an amateur second, initially encountering the hobby scene through her marketing job at TESLA Eltos. Prokšová played a critical role in keeping the club together: "I was never a superb programmer, but I understood what a programmer needed. My role was that of an intermediary."[81] While managing the club's budget and thoroughly enjoying its diverse activities, she did not intend to compete with (male) coding virtuosos who ended up being remembered for their games.[82] The mathematician Alena Šolcová taught programming to both elementary school pupils and university students, but did not join any organized hobby group. Instead, she preferred to pursue her hobby more privately, trading software and know-how at regular meetings with her friends (see figure 3.3).

Thus, beyond the actual statistical gender gap, the lack of women in my material may also underscore the fact that the historical record operates on

Figure 3.3
Mathematicians Alena Šolcová (left) and Pavla Polechová (with her back to the camera) teaching Logo to their kids in Šolcová's kitchen, around 1986 or 1987. At some of these sessions, they also played games. Šolcová is mending socks, showing that there was no clear line between hobby computing and household chores.

structures of appreciation and credit that favor those kinds of accomplishments that are more likely to be undertaken by men.[83] Other activities, such as programming education or club management, tend to stay under the radar. All in all, despite the different economic and social context and the prevalence of women in professional computing, the amateur hobby computing scene was just as masculine in Czechoslovakia as it was in the Western countries.[84] This was caused not by computing in general being defined as masculine, but by *technical hobbies* being defined as such. For homebrewers and gamers, computer clubs offered spaces where ambitions could be fulfilled. However, these spaces were mostly reserved for men and boys. This influenced the character of these groups, including the competitiveness and meritocratic dynamics (shown in the following chapters) shared with other male subcultures.

Build Your Own Peripherals

Hardware tinkering, a heritage of earlier ham radio and electronics hobby groups, was commonplace in Czechoslovak computer clubs. After all, many of its older members grew up soldering rather than coding. The Czech term for this sort of activity was *bastlení*, derived from the German *Basteln*, which can be translated as *tinkering*. Writing in 1987, a Czech linguist considered it a relatively new loanword and attributed its proliferation to the growing popularity of microelectronics and the influence of imported hobbyist literature in German.[85]

In the 1970s Western tradition, building one's own computer from the ground up was an essential part of the computer hobby.[86] The same could be said of some Eastern European countries. In Yugoslavia, eleven thousand hobbyists reportedly built their own Galaksija machines, whose blueprint was originally published in 1983; in the Soviet Union around the same time, at least hundreds of Mikro-80 machines were built.[87] Compared with these examples, assembling complete machines was a relatively marginal practice in Czechoslovakia. *Amateur Radio* magazine published two schematics based on a Z80-compatible CPU in 1985–1986 and 1988,[88] but both projects foundered due to continuous shortages of components.[89] Hunting down a mass-produced machine proved to be a safer bet than hunting down individual parts, and homemade computers were therefore very rare. On the other hand, the building of peripherals—also notoriously scarce—thrived. Instead of making another trip to the West or waiting for Tuzex to stock them, many people preferred making their own. Moreover, peripherals could be built using more common parts and assorted local resources.

Joysticks were the most popular ones. At least four joystick schematics were published in *Amateur Radio* in the 1980s, and several more in club newsletters.[90] In print, they were usually represented in the international visual language of abstract circuit board diagrams, but the resulting artifacts betray their origin in the local material culture (see figure 3.4). When making them, local bricoleurs would combine available TESLA-made electronic components with generic materials, household objects, and parts salvaged from other devices. The two 1980s joysticks found in the storage of the Station of Young Technicians in Prague feature calculator keys instead of buttons, and steel sticks fitted with common wooden furniture knobs.

Computer mice were also a favorite, often built around a table tennis ball, and sometimes even smaller parts—members of one computer club

Figure 3.4
A homemade joystick for the ZX Spectrum from the Station of Young Technicians in Prague. The switch on the left side of the chassis is a standard variety sold in electronics stores at the time. The joystick connected to the computer via a nonstandard interface, and therefore the Spectrum also had to be modified. The handwriting most likely describes the byte values that the joystick sent to the machine. Photo by Silvia Kolesárová.

challenged themselves to make one "out of a marble and a spray can lid."[91] In addition, *Amateur Radio* delivered two light pen schematics. Given that these peripherals were mostly designed to interface with the Spectrum—which had next to no software controlled by a mouse or light pen—their usefulness may be disputed. Nevertheless, the projects allowed hobbyists to demonstrate how good they were at *bastlení*, thus exhibiting their technical competence. Moreover, using a mouse could make a Spectrum user feel more in touch with unavailable state-of-the-art mouse-controlled operating systems on the Commodore Amiga or the Apple Macintosh. Besides peripherals, building hardware interfaces was essential to connect homemade devices or incompatible peripherals (especially printers), often used instead of unavailable compatible ones. Long before standardization of ports and drivers allowed for the proliferation of plug-and-play hardware, soldering was a common step in getting a device to work.[92]

Many if not most domestic users self-imported their computers from the West, and thus had no access to authorized service or replacement parts. Numerous hardware projects therefore aimed to extend the machines' lifespans. A common woe of local Spectrum enthusiasts was the machine's fragile keyboard. Both the rubber chiclet keyboard of the original model and the plastic one of the ZX Spectrum+ model tended to break under the strain of intensive use—an unwelcome outcome of the Spectrum's cost-saving design.

This mishap befell Tomáš Rylek, a participant at the Station of Young Technicians and a member of the Golden Triangle homebrew collective. Due to an agreement between their thrifty parents, Rylek shared his ZX Spectrum+ with his friend and neighbor Miroslav Fídler. Both wanted to occupy it for ten hours a day, and due to almost nonstop use, the keyboard membrane wore out after about a year. Of his bricoleur solutions (see figure 3.5), Rylek recalls:

> I made a new keyboard out of big round doorbell buttons, which was terrible, but provided a good finger muscle training—which I appreciated later when I started to play guitar. And then, there was another, "third generation." My dad, who worked at a chemistry research institute, brought keyswitches from a computer they discarded at work, and the Spectrum keys fit perfectly on top of them. I soldered them onto the circuit board, and … I had almost a proper PC keyboard. It was all a bit raised, so when typing, you looked like a dinosaur, but the tactile feedback was great for its time.[93]

Figure 3.5
Tomáš Rylek's ZX Spectrum+ (top) connected to a "third generation" external keyboard. The original keyboard space was covered with a plywood board, attached to the machine's plastic body with a steel screw. Photo by Silvia Kolesárová.

For Rylek, building a keyboard did not feel like an extraordinary achievement compared with some of his other friends' projects, highlighting how common such repairs were. Contributors to *Amateur Radio* introduced several means of protecting and replacing keyboards.[94] In fact, the magazine even presented joysticks as keyboard protection devices.[95]

In her analysis of early micro users' activities in Australia and New Zealand, Melanie Swalwell observes a swing from hardware tinkering toward the use and coding of software around 1985.[96] The continued dedication to hardware building in Czechoslovakia, which lasted well into the late 1980s, was in part born of necessity. The shortage economy gave hobbyists a reason to continue tinkering, but rather than assembling computers, they were building around and between them. The core of the machine—the CPU, custom chips, and main board—tended to remain intact, because those were the parts that Czechoslovakia could neither manufacture nor distribute in satisfactory numbers. However, the cores were often surrounded by improvised material artifacts with jagged edges and rough surfaces, interfacing with sleeker, mass-produced Western devices.

It is difficult to estimate what portion of users engaged in homemade hardware production at a given time. Given the ample coverage by club newsletters and its prominence in interviews, it is safe to assume that at least one-fifth to one-third of each club's membership was either interested or involved in hardware building. This suggests that histories of gaming hardware should stop focusing exclusively on mass-produced models and expand their scope to include unique, homemade objects that have proliferated especially in peripheral contexts. Unfortunately, many of these have been lost to history.

Amateur Entrepreneurs

Computer clubs provided their members with the items that socialist retail could not. Besides access to hardware, they maintained collections of software and documentation, the latter sometimes translated into Czech. Some clubs were publishing regular newsletters that bypassed the ban on self-publishing by virtue of being internal publications of socialist organizations. The newsletters chronicled club life and contained schematics, tips, tricks, and manuals for both games and serious software. Printed in runs of hundreds or even thousands, they maintained a certain do-it-yourself charm, combining computer printouts, line drawings, and typewritten pages.

Much of the club activity was governed by the ideals of collective work. Newsletter editors appreciated "selfless help" provided by club members, and encouraged them to "make the results of their own intellectual labor available to others."[97] But the bottom-up collectivism of *vnye* spaces was not necessarily the same brand of collectivism as what was promoted by the official socialist ideology. The former grew from small-group camaraderie and pragmatic concerns. Under conditions of shortage, sharing was a viable strategy for building necessary connections and infrastructure. That said, it was not an absolute imperative. Many were not content with the amateur status that was imposed on them. Several interviewees remembered frustration that their work could not have larger commercial or social impact, and spoke of the wasted potential: "Svazarm was full of smart people. And not only Svazarm. We could have beaten the West."[98] But private enterprise was not allowed in the country, and hobbyists could not spin off firms to enter the socialist economy. Nevertheless, several ambitious clubs started to outgrow the amateur model and sold their products and services for money. Their activities resembled the strategies of both Western commercial companies and the open-source movement.

Prague's 602 played a flagship role. Unlike other clubs that turned away new members once they reached a certain level of enrollment, the 602 reacted to the demand for its services with further expansion. As chronicled in editorials by Ladislav Zajíček, its members were frustrated by the nonexistence of a software market and the rigid policies preventing programmers from freelancing and enterprising.[99] The club's distribution service transformed into a de facto software publishing house in 1986, producing programming language implementations as well as educational and utility software.[100] Inspired by the club members' knowledge of the Western software industry, it was explicitly modeled after commercial companies:

> An integral feature of our new approach is the producer-user relationship, which is practiced by commercial firms, and which will be applied … in the organization-member relationship. All of our original productivity software and development tools will be discussed in this newsletter. Justified demands for changes or additions to the programs' essential functions will be addressed in subsequent versions.[101]

The original programs were no longer freely shared with the membership, and were instead sold for fixed prices. In exchange, customers received a guarantee of continued updates, provided by programmers who would in turn receive their share of royalties. The club even ran a phone help line manned by Ladislav Zajíček and other *Mikrobáze* contributors, and opened a retail location in downtown Prague, combined with a library of computer literature.[102] Besides software, the club sold literature and hardware, including five thousand units of a build-your-own computer mouse kit—with a table tennis ball included in the box.[103]

One of the greatest triumphs of local amateurs was the Turbo 2000 loader, a solution to the problem faced by local Atari users who did not have access to scarce and expensive disk drives, and had to use the notoriously slow proprietary Atari tape drive.[104] Invented around 1986 by Jiří Richter, then a student at the CTU, Turbo 2000 quadrupled loading speed while only requiring a piece of loader software and a bit of beginner-level soldering on the tape drive.[105] The standard's resounding success was due in part to the way it was disseminated. The specification, including hardware and software designs, was published in a special issue of the 487 club's newsletter and made available for further comments and improvements. Each club could adopt the standard and modify existing tape drives; some even provided the modification as a paid service. The invention was hailed by

Atari supporters as a blow to the hegemony of Spectrum users who had scoffed at Atari's slow loading times. It was soon adopted by the majority of the Czechoslovak Atari community, improving the efficiency of the tape-based informal distribution system that was already in place.[106] Similar loaders, also called "Turbo," also appeared in Poland, although it is not known whether they were directly related.[107]

In all the cases above, computer clubs moved beyond the realm of personal amateur endeavors, taking on complex projects that impacted thousands of users. They approached them in diverse ways, appropriating Western models or improvising their own. The 602's entrepreneurial drive suggests that many club members considered their amateur status involuntary and temporary and put forward the commercial model as a way of ensuring quality. On the other hand, the Turbo 2000 project resembles the open-source model, in which standards and source code are publicly available, but a range of "support sellers"—in our case, the clubs—help users implement them.[108]

We saw that some of the clubs, despite being nominally "amateur" communities, became quasi-startup companies enveloped within a paramilitary organization. Shielded by their presumed contributions toward education and STR, they became pockets of entrepreneurship within a socialist society, similar to chapter 1's farms that happened to manufacture computers. The concentration of creativity in amateur circles was not exclusive to the computer hobby, but rather a feature of late socialism. Automobile and aviation hobbyists were involved in similar or even larger-scale projects, and encountered similar tensions with the Svazarm management.[109] Amateurs' relationship to STR was not only rhetorical. In fact, the clubs' participatory nature and capacity for bottom-up innovation resonated with the prenormalization reformist visions of Radovan Richta, the original Czechoslovak theorist of STR, whom we met in chapter 1.[110]

Starting a Computer Fanzine

In the second half of the 1980s, clubs catered to a growing number of users. Thanks to their publishing output and the increasing amount of know-how circulating among users, attending a club in person was no longer essential. In that very period, authorities were becoming more lenient, and normalization's grip on the populace was loosening up in response to Soviet reforms. At that point, some users started creating vital new communication platforms independent of—but connected to—the club infrastructure.

Such was the story of David Hertl (born 1971), a teenage Spectrum enthusiast from Lenešice, a Northern Bohemian village relatively far from larger urban centers. At first, most of his interactions with other members of the community were carried out via mail. For many users, especially those who did not have access to or chose not to join a computer club, this was the primary distribution and communication channel throughout the decade. When Hertl started commuting to Prague to attend graphic design high school, he and his Prague-based friend Ondřej Kafka (born 1972) decided to visit the 602's Spectrum club. Hertl remembers his bafflement as he entered a messy room and saw a group of people copying programs amid a cacophony of voices and noises. Neither was he impressed by the club's newsletters. Disappointed by the 602's chaotic atmosphere, he decided to go back to communication via mail.[111]

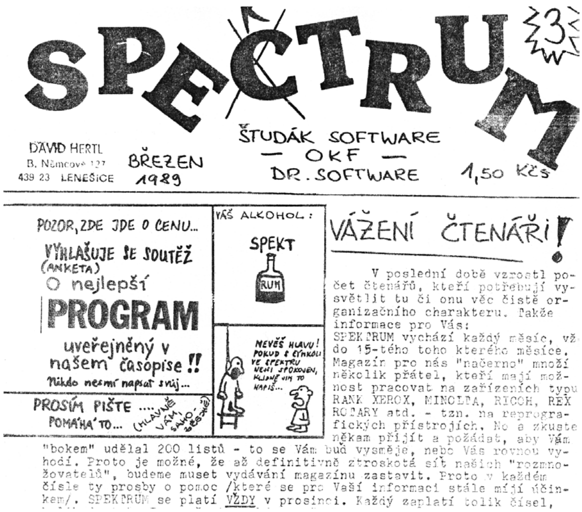
At that point, Hertl was determined to become a "mediator" himself—to facilitate the exchange of know-how and experience among Spectrum users, starting with his network of mail contacts. He found inspiration in the Polish magazine *Bajtek*, which he used to buy in the Polish Cultural Center in Prague. Reading Polish publications was not an uncommon practice, given the lack of domestic computing literature and the fact that Polish was partially intelligible to Czechs and Slovaks. Because it targeted youth, *Bajtek*'s tone was less technical than that of Czechoslovak newsletters.[112] "We found it fascinating that the Poles could have such a good magazine and we did not. And so we came up with the magazine idea."[113] Given the focus of their high school, Hertl and Kafka were familiar with the process of designing and printing a magazine, and eager to launch their own.

In June 1988, the duo released the first issue of a do-it-yourself magazine called *Spektrum* (based on the Czech spelling of the word *spectrum*), better known under its later title *ZX Magazine* (*ZX Magazín*). The magazine was unregistered with the authorities, and therefore technically illegal; it even boasted its independence in its tagline ("Independent magazine of ZX Spectrum users"). On the other hand, Hertl never openly advertised it, and only mailed it to his limited list of contacts because of his concerns over the magazine's legal status. "I was seventeen and afraid that some idiot would harass me and I wouldn't be allowed to graduate. So we kept a low profile."[114]

Until the Velvet Revolution in November 1989, the circulation grew from 15 to 105 copies. Despite the low circulation, individual copies were likely shared, copied, and redistributed. One of the benefits of Svazarm was its access to printing and photocopying machines, which were otherwise tightly controlled. The *ZX Magazine* team did not have one, but was helped by contributors from different regions of Czechoslovakia, who made copies at work and mailed or handed them back for Hertl to redistribute. As Kafka notes, these people were possibly risking even more than the publishers themselves.

The contents of *ZX Magazine* were assembled from contributions by a diverse range of readers, including students as well as veteran hobbyists, many of whom simultaneously participated in computer clubs. The magazine's collaborative, do-it-yourself writing, printing, and distribution process evokes two different points of reference: the *fanzine*, predominantly connected to Western fan cultures, and *samizdat*, the practice of underground publishing, mostly dedicated to the distribution of banned or antiregime texts.[115] The two tend to share similar technological and organizational solutions to material constraints. The layout of *ZX Magazine* reflected such constraints. In its initial stages, it was an assemblage of typewritten and handwritten material, embellished by cartoons and simple drawings (see figure 3.6). Its rather short articles were packed haphazardly onto a handful of A4 pages to make copying convenient; later the format changed to A5. Keeping with the spirit of bricolage, Kafka sometimes copied and pasted fonts and illustrations from West German magazines.[116]

The focus on short-form texts signaled a significant departure from Svazarm club newsletters, which could—and did—run long program listings and complex schematics due to their easier access to printing technology. *ZX Magazine*'s articles, on the other hand, included hints, tips, and short programming tricks rather than comprehensive software and hardware projects; the editorial style was casual rather than technical. *ZX Magazine* was the first Czechoslovak outlet that systematically documented the *pleasures* of using software made by others—including games. This shift in discourse, which stressed the enjoyment of ready-made software, reflected the changing status of games among microcomputer users, a topic discussed further in the next chapter. The style of writing resonated with users. As soon as the magazine was granted legal registration after 1989's Velvet Revolution, the circulation rapidly rose to thousands of copies (see figure 3.7). Kafka


SPEČTRUM 3

DAVID HERTL
B. Němcové 127
439 23 LENEŠICE

BŘEZEN 1989

ŠTUDÁK SOFTWARE
— OKF —
DR. SOFTWARE

1,50 Kčs

POZOR, ZDE JDE O CENU...
VYHLAŠUJE SE SOUTĚŽ (ANKETA)
O nejlepší
PROGRAM
uveřejněný v našem časopise !!
Nikdo nesmí napsat svůj...

PROSÍM PIŠTE .... POMÁHÁ TO... (HLAVNĚ VÁM, SAMO-ZŘEJMĚ)

VÁŠ ALKOHOL:
SPEKT
RUM

NEVĚŠ HLAVU! POKUD S ČÍMKOLI VE SPEKTRU NEJSI SPOKOJEN, KLIDNĚ VÍM TO NAPIŠ...

VÁŽENÍ ČTENÁŘI!

V poslední době vzrostl počet čtenářů, kteří potřebují vysvětlit tu či onu věc čistě organizačního charakteru. Takže informace pro Vás:
SPEKTRUM vychází každý měsíc, vždo 15-tého toho kterého měsíce. Magazín pro nás "načerno" množí několik přátel, kteří mají možnost pracovat na zařízeních typu RANK XEROX, MINOLTA, RICOH, REX ROTARY atd. - tzn. na reprografických přístrojích. No a zkuste někam přijít a požádat, aby Vám "bokem" udělal 200 listů - to se Vám buď vysměje, nebo Vás rovnou vyhodí. Proto je možné, že až definitivně ztroskotá síť našich "rozmnožovatelů", budeme muset vydávání magazínu zastavit. Proto v každém čísle ty prosby o pomoc /které se pro Vaší informaci stále míjí účinkem/. SPEKTRUM se platí VŽDY v prosinci. Každý zaplatí tolik čísel,

Figure 3.6
A part of the title page of the March 1989 issue of *ZX Magazine*, then called *Spektrum*. In these texts alone, the editors promote several participatory practices. They announce a programming contest, solicit user contributions, and ask readers who have access to copy machines to help them make copies of the magazine.

left in 1990 to build his own graphic studio, whereas Hertl continued for another year before quitting to focus on his university degree, transferring the brand to a company that continued to publish it until 1994.

The *ZX Magazine* story shows that independent amateur "miracles" were possible even without the patronage of computer clubs. But despite the magazine's differentiation from the Svazarm scene, it was still loosely connected to Svazarm's meshworks, collaborated with many of its members, and reprinted selected club materials.

Samizdat Research Institute

So far, this chapter has shown how microcomputers were embedded in the *vnye* spaces of computer clubs and in the fanzine-like practices of *ZX Magazine*. These spaces and practices did not directly engage with or challenge state power or ideology. But the use of micros by Czechoslovak dissenters is also worth exploring.

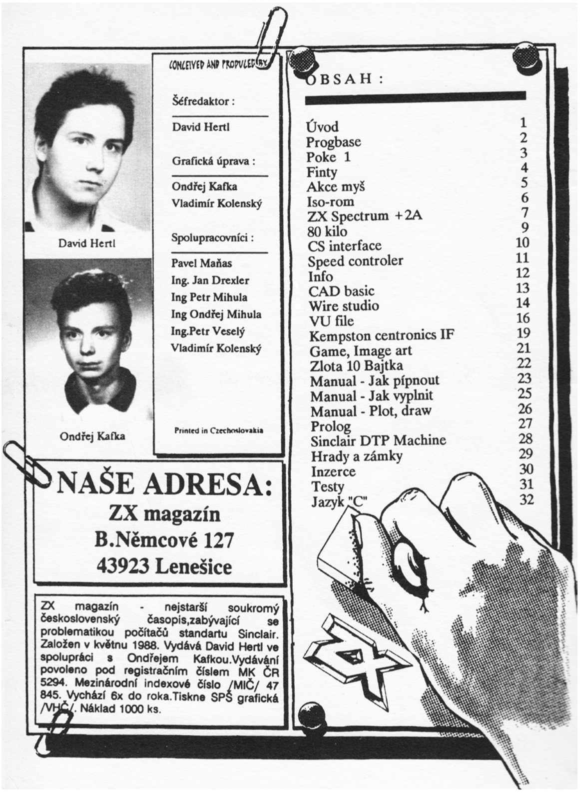


Figure 3.7
David Hertl and Ondřej Kafka pictured on the contents page of *ZX Magazine*, year 1990, issue 5, published soon after the magazine's "legalization." Note how many of their collaborators (*spolupracovníci*) have the "Ing." title given to graduates from technical schools.

The history of Central and Eastern European political dissent is a thorny topic. It has been by turns romanticized and deglamorized by both journalists and historians. Recent scholarly accounts of everyday life under communism tend to exclude dissidents because they were not representative of the whole population. This exclusion can give "the impression that dissidents had no everyday life of their own."[117] Understanding the samizdat's use of technology can therefore not only serve as a companion to the computer clubs' narrative but also contribute to a more nuanced portrayal of dissent. I will focus on two illustrative stories: that of Václav Havel's personal computer, and of the *Vokno* samizdat magazine.

Czechoslovak dissent in the 1980s comprised an incongruous group of people of different beliefs and methods united by their active stance against the regime. It included former reformist Communists expelled from the party after 1968, artists and writers, and members of the Christian intelligentsia—as well as anarchists, underground musicians, and assorted young rebels. Unlike in Poland, where dissent was connected to the Solidarity mass movement, Czechoslovak dissidents were a marginal group, and many citizens had not even heard of them.[118] They did not have access to radio, television, or printing presses, and were forced to improvise with a wide range of tools to spread their word both amid their own ranks and to the public.

Samizdat was a largely low-tech undertaking reliant on the tactical use of nineteenth-century analog technologies. Typewriters, carbon paper, hectographs, and mimeographs were all self-contained, portable, cheap technologies unlikely to be intercepted by authorities. Microcomputers—decades before their association with networked surveillance—promised to be another decentralized device of that sort. According to historian Jonathan Bolton, "Computers, dot matrix printers, and even memory typewriters had the potential to completely change the world of samizdat, but they were scarce until the last few years of Communist rule."[119] Besides scarcity, another obstacle to the use of computers was the humanistic rather than technical background of the dissident movement. The available machines were not particularly user-friendly, and writers, musicians, scholars, and artists—as opposed to technical intelligentsia—were not well versed in computer technology.[120]

Václav Havel was a typical example. Although he admired computers, he never came to terms with them and doggedly rejected their logic.[121] In the 1980s, he specifically asked foreign backers to send dissidents "fewer computers, more photocopiers."[122] Nevertheless, he was among the few Czechoslovak citizens to own an IBM PC machine, donated by his publisher in the West to use for word processing. Although the future president was a pariah in his own country, excluded from public life and constantly monitored by the police, he received substantial financial support from abroad, and his international fame protected him from excessive persecution. However, in October 1988, his computer was seized by the police on suspicion of criminal activity. Havel responded with a letter to Prime Minister Ladislav Adamec. One of his lesser known letters, it includes an acerbic and hilarious account of the reported incompetence of Communist police:

> I have to state that the computer was seized by persons who were unqualified, and no expert was present. The following proves their lack of qualification:
>
> a) As evident from the search protocol, Security mistook the keyboard for the personal computer, considered the computer itself an "amplifier" and left the monitor in my possession, because they considered it a television set. Only the printer was recognized as a printer. …
>
> I must also stress that I obtained my computer as a gift from my publisher who sent it to me so that I could, like Western authors, write and edit my works. It is in my rightful ownership, and an item of great financial value. I duly paid my gift tax as well as customs fees for it. If it is indeed seized, if its memory is tampered with or if it is damaged due to unqualified handling, it will be a preposterous breach of personal possession, which can bring negative publicity to the Czechoslovak government both domestically and abroad. If thousands of citizens who invested their long-term savings into a purchase of a computer are deprived of any guarantee that their machines cannot be at any time arbitrarily seized, this will lead to a further inhibition of technological progress in our country.[123]

Besides legal argumentation, Havel also defended his position by referring to the regime's own support for "technological progress" and the efforts of individuals to take part in it, highlighting the role of amateurs in technological culture. At the same time, the story suggests that the country's security forces were just as unprepared as many of the dissenters for the rise of microcomputers. Eventually, the police returned Havel's machine sometime before August 1989, but it is uncertain whether the letter had helped—or if the prime minister had even read it.[124]

Not all dissidents lacked computer skills. Pavel Lašák (born 1954) was an electronics enthusiast, renowned in samizdat circles for his technical expertise. He designed an improved version of the mimeograph that allowed him to print faster and more efficiently, and gained notoriety for building antennas that could block out jammers of foreign radio stations—hence his nickname Antenna (Anténa), which was also his codename in the secret police records. Lašák joined the group behind the *Vokno* (*Window*) samizdat magazine, one of the most politically radical and technologically progressive underground publications, which printed fiction, poetry, and reports from underground events as well as dissident manifestos. It was spearheaded by František Stárek (born 1952)—nicknamed "Hog" (Čuňas)—a remarkably stubborn activist hardened by serving jail time for publishing samizdat. According to Lašák, “Hog had this fierceness in him. He knew that possibilities were opening up and that we had to seize them.”[125] Noticing that microcomputers were proliferating among the population, Stárek, Lašák, and another colleague started the informal “Samizdat Research Institute” and investigated ways the new machines could prove useful to their cause (see figure 3.8). At the time, the use of computers for activism and liberation—notably formulated by Ted Nelson in the 1970s, but widely acknowledged only after the rise of social media—was still in its infancy.[126]

By that time, both Stárek and Lašák owned Sinclair ZX Spectrums.[127] In the mid-1980s, they started using them to improve their mimeographing process. The mimeograph duplicates pages by forcing ink through a stencil. Originally, the holes in the wax-covered stencil would be punched out on a typewriter. However, using an ink ribbon reduced the impact of keystrokes, so typists would usually impress the types directly into the stencil, unable to see what they were typing and therefore making plenty of typos. The “Institute” experimented with dot-matrix printers, often borrowed from friends or acquaintances.[128] Although dot-matrix printers found only limited use in reproduction itself because of their low speed and durability, they significantly optimized the production of mimeograph stencils. The first impressions from the Spectrum, seen in 1986’s issue 11 of *Vokno*, marked a clear improvement in typesetting quality.

Figure 3.8
Pavel Lašák at a ZX Spectrum+, probably working on one of *Vokno* collective’s publications. Photo courtesy of František Stárek.

The “Institute” found multiple uses for their Spectrums. They used them to create title sequences for their video magazine, circulated on VHS tapes starting in 1988. Realizing they could reach the estimated one hundred thousand Spectrum owners, the team was also developing an e-book reader of sorts by creating a text viewer program called *Book* (*Kniha*), driven by the idea that this would be “the future of samizdat.”[129] In addition, Stárek used the Spectrum to securely store the database of *Vokno* subscribers. Compared with a binder, a Spectrum cassette tape was much smaller and handier: “It was wrapped and hidden under a pile of coal. Whenever I needed it, I went to the coal chute, took the tape, sent away the kids who were playing some game, and loaded it up.”[130] To hide the fact that it contained data, the tape started with inconspicuous recordings of Beatles songs.

Despite the achievements of the “Samizdat Research Institute,” computers only played a very minor role in 1980s dissent. The publishers of *Vokno* were rare computer enthusiasts in a world that was still using analog technologies. Although their goals were different, the Institute’s uses of technology were much like those practiced by computer clubs and *ZX Magazine*. They engaged in *bastlení* and coding, and took advantage of tools and know-how that originated in the *vnye* milieus of Svazarm. Without ever joining Svazarm, whose power structures he found to be too close to those of the regime, Lašák frequented one of its computer clubs “to pick [the members’] brains.”[131] *Vokno* also used the *D-Text* word processor, an unauthorized Czech localization of the British commercial program *Spectral Writer,* administered and promoted by the 666th Svazarm club.[132] And dissidents played games, too. Stárek enjoyed the platform action title *Jumping Jack*, and Lašák was an avid player of *Bugaboo*, also a platformer.[133] So we see that computer-savvy dissidents did not inhabit an isolated world of their own. Instead, their everyday use of computer technologies largely overlapped with the practices of *vnye* milieus.

This chapter has shown how Czechoslovak hobby computing culture was kept alive by the continuous efforts of thousands of amateurs of all ages. Youth groups and computer clubs—which became hubs of the amateur scene—affiliated themselves with the existing infrastructures of educational institutions and socialist organizations. Clubs were *vnye* milieus, dependent on political patronage but apolitical in their everyday operation. Their members adapted these infrastructures to their needs, but also partially adapted *to* their requirements and policies. Amateurs worked outside these milieus, too. Amateur fanzine publishers and amateur dissidents engaged in activities that disregarded or confronted state power. In addition, computer enthusiasts among university students convened at dormitories, occasionally hosting all-night gaming marathons.[134] But although all these groups achieved miracles of their own, they, too, were directly or indirectly connected to the clubs.

Prominent members of the amateur scene were aware of the historical role they were playing and considered themselves a vanguard of microcomputing in an otherwise backward context. They rehearsed many of the features of the postindustrial information society. They moved flexibly between various clubs and activities; they merged work, entertainment, and activism; and they created value out of information and knowledge. In this context, we can see amateurs as being *postindustrial*, in contrast to *industrial* professionals—yet acknowledge that some individuals were in both camps.

Although intergenerational, hobby computing was mostly a male endeavor, a pattern that seems to have held in most of the Western world as well as behind the Iron Curtain. Technical hobbies, like hobbies in general, were a privilege of sorts—a privilege that many women could not reach. Besides the wider sociocultural context, this was due in part to Svazarm's paramilitary roots and its traditional focus on hobbies considered male, such as ham radio. However, future research on professional programming or computer education is likely to reveal more women who actively and creatively used both microcomputers and games.

The goals of computer clubs aligned with official policies, such as the government's "long-term complex programs of electronization" and Svazarm's microcomputing strategy. By supporting hobby groups, state officials basically outsourced a part of the programs to amateurs. Svazarm, in particular, successfully captured a significant part of hobbyist activities. The Svazarm clubroom became the equivalent of the American garage as a place where technological innovation took place. Clubs thus fulfilled some of the ideals that Radovan Richta put forward in his vision of STR, offering spaces for the bottom-up creative initiatives of individual experts and engineers within a socialist collective.[135] Despite the clubs' far-reaching impact, the direct economic benefit of the amateur scene was limited because of its disconnection from the means of production. As Ľudovít Barát, father of the Didaktik Gama computer, said of the 602—"They were ingenious guys, but did not have a manufacturing plant behind them."[136] Amateurs' hardware work was therefore mostly limited to individual bricolages. They were more successful in projects that were less dependent on material resources, such as education and publishing, as well as the creation, modification, and distribution of software.

Clubs played similar roles in other countries behind the Iron Curtain. As Stachniak wrote of the Soviet Union, "It was practically impossible to organize a hobby club and find a place for meetings without permission from the government and party authorities."[137] The club scene was perhaps not as strong in the region's more politically and economically liberal countries, such as Poland or Hungary. There, the gray market (the Polish microcomputer marketplaces, for example) might have played a larger role at the expense of formalized clubs. Regarding potential regional networks, there is little to no evidence of direct or continuous collaboration with computer clubs from other countries.[138] Czechoslovaks, for instance, read the Polish *Bajtek* magazine, but very rarely connected with Polish clubs, and the scene thus remained distinctly national.

Altogether, the amateur scene was responsible for most of the creative microcomputer work in the country. It created infrastructure and conditions for the practices of distributing and playing games. By reinforcing the idea of self-reliant *golden Czech hands* that could work computing miracles, it contributed to the boom of domestic homebrew computer game production. However, as the next chapter shows, the position of games within the wider hobby computing circles was far from uncomplicated.

4 Who's Afraid of Gameplay? Czechoslovak Discourses on Computer Games

> [Games] were until recently at the height of their fame. Many of them are interesting in many respects (graphics, tactics, puzzles, simulation, speed, etc.). Although computer users worldwide have got tired of games, and moved toward productivity software, we can be sure that quality games, based on clever ideas and programmed by professional teams, will never leave us.[1]
>
> —From the *Mikrobáze* computer club newsletter, 1985

In the previous three chapters, I rarely differentiated between gaming and other uses of microcomputers, because many early Czechoslovak hobbyists likewise did not. Initially, playing games was just one of the many things one could do with a computer. However, this changed in the mid-1980s, when an informal cassette-based software distribution network was established and large numbers of illegitimate copies of Western commercial games started to make their way into the country. More and more people played them—and some even seemed to prefer them to anything else a computer could offer.

Computer games became a matter of debate and controversy in hobbyist circles. How should the Czechoslovak amateur approach them? Should they be discussed in magazines and newsletters? Were they here to stay? What defined a good game? The answers were far from clear. As seen in the quote above—written from the perspective of an adult hobbyist—the discourse could oscillate between calling games a trivial fad, and celebrating them for their ingenuity, within one paragraph. This chapter will explore the position of games among other microcomputing practices. I will show how members of the amateur scene defined common and desirable kinds of engagement with computer games and how these definitions contributed to the emergence of a homebrew scene.

The development of discourse about games has notably been analyzed by Graeme Kirkpatrick. In his view, computer games had originally been considered just a nonserious subset of software. Around 1985, Kirkpatrick argues, specialist gaming press in the United Kingdom introduced the term *gameplay* and “the appraisal of games [took] on a limited independence from technical, educational and other normative criteria that get applied to other objects in the computer culture.”[2] Kirkpatrick attributes the lion’s share of responsibility to the gaming press, who profited from extensive game industry advertising, fully embraced games, and sold both their magazines and the games themselves to a narrower but more dedicated audience consisting mostly of teenage boys. Colorful and irreverent magazines such as *Computer & Video Games* (launched in 1981), *Commodore User* (1983), *Crash* (1984), and *Your Sinclair* (1984) presented games as consumer products that could be evaluated, purchased, and enjoyed without knowing how to program. As the magazines started to write more about commercial titles and less about programming, games gained autonomy from other cultural practices, including hobby computing. Along came “proprietorial emphasis on ownership of software and the demise of the *bricoleur* culture of copying and sharing,” eventually creating a divide between gamers on one side and producers on the other.[3]

Kirkpatrick was working with UK material, but his arguments have universalist ambitions, never limiting the projected scope of their applicability to the British context.[4] But as Gleb Albert points out in his review of Kirkpatrick’s work, the latter’s reliance on discourse analysis may generate blind spots in the narrative.[5] The focus on discourse, although illuminating, might divert our attention from the idiosyncratic economic, political, and social configurations of the 1980s United Kingdom—its powerful print industry; dense high-street retail infrastructure, which allowed for high penetration of games and magazines; and the extraordinary uptake of cheap microcomputers, which was among the highest—if not the highest—in the Western world.[6]

Evidence from Czechoslovakia will tell a different story. The previous chapter showed that the country’s amateur community largely relied on the political patronage of paramilitary and educational institutions. Users of all kinds and ages—educators, tinkerers, and gamers—gathered in a loose but complex tactical alliance that helped them cope with the limitations of the shortage economy and the restrictive regime. This alliance inhibited potential separatist tendencies. Debates over the definitions and values of games were never fully resolved, and a division between gaming and hobby discourses was never fully realized until the 1990s. Instead, gaming and hobby activities led an uneasy coexistence. Unlike in the United Kingdom, there were next to no media outlets that would mediate the discourse on computer games. Czechoslovakia did not have a mass-market specialist magazine dedicated to microcomputers, unlike Yugoslavia and Poland, where *Moj Mikro* and *Bajtek* launched in 1984 and 1985, respectively—both of which regularly covered games. With a few exceptions, the only places where one could read about games were computer club newsletters and several isolated books and brochures. Our case in this chapter can therefore serve as an example of a national discourse about games that developed with little involvement from commercial interests.

Despite the dearth of material, published discourse is the closest we can get to people's ideas about games in 1980s Czechoslovakia. Therefore, as Kirkpatrick has done, I will work with the discourse preserved on the pages of newsletters, magazines, and books. I will do so while remembering the limitations of this material. When Kirkpatrick writes of the "heroic intervention" by British magazines and their readers that was at the birth of gaming culture, he imbues discourse with an extraordinary impact. In the case of the United Kingdom, this impact could possibly be partially justified by a temporary alignment of interests between the press, its readers, and the game industry.[7] Czechoslovak discourse, on the other hand, was much more fragmented, with a handful of minor heroes putting forward their own visions of games. I will ground these visions in the historical context by connecting them to microhistories and personal narratives. This will shield my analysis from the potential danger of overstating the impact and representativeness of these discourses. In addition, I will look out for silences that may be more significant than words. The voices of players, for example, were almost nowhere to be heard in the Czechoslovak public discourse, and the experience of playing games was rarely publicly described.

This chapter will follow several individuals and groups and observe which forms of engagement with games they deemed desirable, or threatening. Party authorities did not comment on games beyond a couple of blanket statements about their use in education. We will therefore start with professionals, dwell more on hobbyists—whose relationship to games was notably spiky—and continue to educators and players. Finally, I will investigate how individual games and genres were evaluated by those who were writing about them.

I will refer to three approaches to computer technology, each tied to a certain type of subject and a teleology of computing. We have already seen the first two, borrowed from Lévi-Strauss—the *engineer* and the *bricoleur*.[8] Czechoslovak industrial managers tended to embrace the former category, setting up plans and projects based on rational reasoning. They promoted the official teleology of computers as catalysts of existing industrial and administrative processes. Hobbyists, on the other hand, tended to embrace the latter subject category, preferring hands-on tinkering and experimentation—although when they needed to legitimize their endeavors, they often used the language and argumentation of the engineer, better respected and more powerful within the socialist society. This chapter will introduce a third category—the *cyborg*. The *cyborg* is a subject position at which the man merges with the machine. In the words of Donna Haraway, it is a "taboo fusion"[9] between the machine and the human, in which it is no longer clear "who makes and who is made," and which blurs the line between the subject and the object.[10] According to Dovey and Kennedy, playing games is a cyborgean activity, because it relies on a continuous cybernetic feedback loop between man and machine. This kind of engagement was strange and new to engineers and hobbyists, who tended to view computers as *objects* of their projects and tinkering.

Playing with Computers

Even before computer games developed into an industry in the 1980s, play had long been a part of computing practice in the West as well as in the Soviet bloc.[11] However, playful or creative computing—which we could call playing *with* computers—was framed differently from playing ready-made computer games, or playing *on* computers.[12]

In December 1981, the popular science and technology monthly *Technology Magazine* published one of the earliest Czechoslovak articles about playful uses of computers. Called "Playful Circuits," it was written by Richard Bébr (born 1935), a communication networks engineer, veteran professional programmer, and dedicated promoter of what he called computer "diversions" or "entertainments" (*zábavy*).[13] Writing before the microcomputer boom, Bébr focused mostly on mainframes and minicomputers and gave a valuable overview of the avenues of playful computing known or practiced in Communist Czechoslovakia. His article covered all nonserious software, including the therapeutic chatbot precursor Eliza, chess programs, strategy and resource management games such as *Star Trek* and *Lunar Landing Game*, and computer-generated art and poetry, as well as biorhythms and horoscopes, which were particularly popular with the nonprogrammer public.[14] The lack of differentiation between these types of software was not unusual at the time; some software catalogs grouped games together with "demonstration" programs to indicate that they stood apart from the practical uses of the machines.[15]

Although he made it clear he did not wish to prioritize entertainment over critical tasks such as scientific calculations and data processing, Bébr listed benefits of computer "diversions." In his view, they contributed to advances in the fields of algorithmization and human-computer interaction, and demonstrated the possible uses of new displays and peripherals.[16] Games and diversions, wrote Bébr, could find their use in education. But above all, these pieces of software played an important role in developing the skills of programmers themselves:

> Game programs are almost always written by programmers in their free time, and therefore written out of love and with love. Every labor of love tends to be perfect in its own way.... Game creators subconsciously use rationalization procedures, which they admittedly avoid while working on their main jobs. It is astounding how elegantly they utilize all the possibilities of the machine and the programming language.[17]

In other words, "diversions" were associated with a liberated, playful, and creative approach to computing, which was often lacking in serious applications; they enabled an engineer to temporarily become a bricoleur and enrich his or her repertoire with new skills and approaches.

Microcomputers promised to make this creative play with digital technology accessible to the ordinary hobbyist. When the Czechoslovak hobby magazine *Amateur Radio* launched its microelectronics supplement in January 1982, it introduced a section called "Programs for Practical Application and Entertainment," which invited the readers to submit program listings:

> Together with the fast proliferation of personal computational technology into our industry and among individuals, we can observe production of a large quantity of diverse programs. In part, they help technicians and engineers fulfill their

> work tasks, in part they are products of the simple joy of programming, and enable, for example, various games. Even programming has therefore become a hobby of its own—a hobby very useful for the development of our economy.[18]

The quote justifies a focus on hobby computing using the official teleology of the "development of our economy." Also notable is the ontology it assumes. It does not speak of "games" but rather of "programs [that] enable games."[19] This confirms Kirkpatrick's observation that at this point, the idea of a "computer game" as a stand-alone category of objects had not yet stabilized.[20]

Before the section disappeared for undisclosed reasons later that year, three listings of BASIC games were published—*Artillery Combat*, *Grenade Target Practice*, and *Lunar Landing Game*. All of them were text-only, turn-based variations on ideas dating back to the mainframe and minicomputer era.[21] To start with, hobbyists' attitudes to gaming, as well as the game mechanics and templates, were thus inherited from the professional community. Their appreciation of games was born out of a wholesome vision of playful experimentation that cultivated the craft of programming—a vision compatible with the grander teleologies of computing. What they did not foresee was the impending industrialization of computer entertainment.

Forbidden Pleasures

From 1982 on, the Czechoslovak microcomputer user base was continuously growing. Although it lagged behind some Western countries, contemporary computer club estimates suggest that penetration by microcomputers tripled between 1985 and 1986, from around twenty thousand to sixty thousand machines.[22] Western commercial games started to enter the country via informal distribution channels (explored in chapter 5), and in 1985, clubs were offering their members unauthorized copies of British gaming hits for the Spectrum, such as *Manic Miner*, *Atic Atac*, or *Knight Lore*.[23] Many club meetings were turning into copying sessions where members exchanged the latest and hottest game titles. This new brand of games—substantially influenced by the arcade video game experience—was spellbinding. It offered audiovisual spectacle and fast-paced, compelling gameplay that BASIC programs such as *Lunar Landing Game* did not. *Manic Miner*, which became a bona fide classic to Czechoslovak players, was a 1983 platformer action game starring a hapless miner avoiding bizarre monsters within fantastic underground caverns.[24] Games such as these resisted framing in a rational, technical discourse; instead, they elicited emotional, passionate, and polarized reactions.

Most of the published reactions came from prominent hobbyists. Already in the very first issues published in 1985, the most influential club newsletter, *Mikrobáze* (see figure 4.1), ran articles that referred to playing as a "craze," "high," "hell," or "infatuation," and warned against the games' mind-numbing power over the user. These criticisms—soon to be joined by statements made in the most popular hobbyist magazine, *Amateur Radio*—-were not grounded in anti-capitalist rhetoric or a moral panic regarding new technologies. Neither were they linked to the issue of slot machine gambling, which was illegal in the country until 1990.[25]

To understand the controversy, we must distinguish between the pleasures of hobby computing and the pleasures of gaming. Traditional

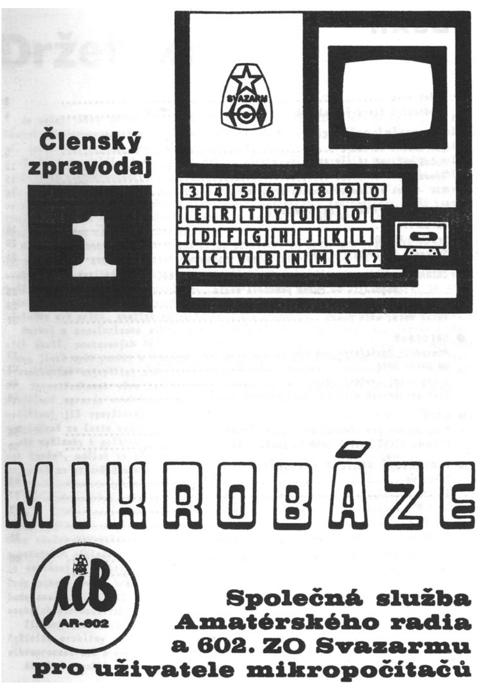


Figure 4.1
Cover of the first issue of the *Mikrobáze* newsletter, published by the 602nd Basic Organization of Svazarm in collaboration with *Amateur Radio* magazine. The Svazarm logo can be seen in the top right square.

hobbyists were bricoleurs who enjoyed construction and deconstruction of hardware and software artifacts, as well as the feeling of dominance over the machine.[26] Gaming pleasures have been described by game studies as quite different. Although part of their appeal is cognitive and rational, they have been linked to immersion and escapism, suggesting that they transport the player to—or engulf them in—a virtual environment.[27] Compared with programming, they are of a more distinctly bodily nature, with Lahti suggesting that "it is precisely the carnal pleasures of gaming that are being mobilized by producers and sought out by consumers."[28] Moreover, hobby electronics is associated with control, whereas games presuppose a certain degree of submission to predesigned rules. Writing in 1984, Turkle noted that "games combine a feeling of omnipotence and possession—they are a place for manipulation and surrender."[29] For Dovey and Kennedy, games offer "a pleasure that lies in an oscillation between activity and passivity. We actively participate in the creation of the game as we play it, while at the same time we passively submit to rules which limit our possible behaviours."[30] Gaming as an activity questions the delineation between the user (as a subject) and technology (as an object) due to the continuous cybernetic feedback loop between the game and the player.[31] The critics of computer games found this emerging kind of cyborg subjectivity deeply unsettling. The carnal aspect of video game play, along with the need to give up some control, were incompatible with the values of rationality and dominance, and allowed for framing of games in terms of addiction and health risks.[32]

The first and most outspoken among the critics of games and gaming was Ladislav Zajíček, former musician and activist editor of *Mikrobáze*. As a public-facing advocate of the amateur movement, he needed to legitimize his efforts by referring to the officially lauded values of purposefulness and rationality; it was in his and his colleagues' interest that clubs were not portrayed as gamer dens.[33] In his 1985 article for the first issue of the *Mikrobáze* newsletter, he admitted he had fallen for games in the past, describing this experience as a loss of control. He had once dominated the machine, but suddenly became a victim, subjected to the irrational urge to play more:

> How many of you have, just like me, succumbed to the almost fanatical collecting of various games? How many of you have cassettes full of these little (but sometimes rather long-lasting) diversions? No, don't say anything. I know this high and the lack of sleep quite well. And how long it had taken me before I awoke

> from this daze! Before I broke free from the situation where the computer became the user of the human; which is nonetheless the human's fault.[34]

According to Zajíček, this "gaming craze" had overtaken the overwhelming majority of ZX Spectrum users. In his opinion, *Mikrobáze* should serve as a cure by suggesting different, more productive ways of using computers. His colleague, writing in 1986 under the signature KŠ, took a similar stance. He welcomed *Mikrobáze*'s decision to stop distributing pirate copies of Western games:

> [This] will deal a substantial blow to the infectious snobbery of the pseudo-users of microcomputers, many of whom sink deeper and deeper into the imaginary world of gaming high, often with questionable content and goal. No offense meant to quality gaming software used for adequate relaxation for a reasonable amount of time.[35]

For KŠ, this "high" and snobbery eventually led to an infection of passivity:

> Of course, one can spend a long night until the break of dawn with a Spectrum while chasing after pearls as a diver, or while getting swallowed by a sea beast; skiing down snowy slopes, or driving a silvery Formula 1 sports car. Such programs can entertain, but one must realize that they originated under completely different conditions. One will get tired of them after a while. Unless I am just a passive user of sleepless nights and two sore fingers … I have never wrapped my head around the idea of computers used by passive users. The computer's nature is to stir up the desire for knowledge, embodied in one's creative work.[36]

These criticisms, followed by reiterations in later articles by both authors, contrasted the carnal gaming "high" with rational and purposeful hobbyist work. In their view, commercial games offered experiences that were dangerously immersive and addictive; they circumvented rational thinking and affected the player on the physical level. To make their point, the critics used medical vocabulary to illustrate both real and figurative game-related afflictions. The players had sore fingers and "red eyes in a phosphorescent face" from lack of sleep. They were "casualties wounded by software, glued to their keyboards and joysticks."[37]

The rampant references to drug use can be interpreted in two complementary ways. First, as Rob Cover has argued, games have a tradition of being likened to drugs because of their connections to unreal places and experiences.[38] Here, the critics were concerned that users might sink into imaginary worlds rather than work for the community. References to "sleepless nights" related to both actual all-night gaming binges and the strange new temporality of gameplay, which Cover describes as "unforeseeable, external to … clock time and beyond measurement according to our contemporary social criteria of time use."[39] Second, and more importantly, drug use implies loss of control—and control was one of the most important attractions of the electronics hobby. In a gaming "high," the hobbyist forgoes part of it to the game. Now, it is the game tasking the user, not the other way around—as Zajíček put it, "The computer becomes the user of the human."

For any hobbyist who took pride in his or her rationality and dominance over the machine, this might be difficult to accept without feeling either guilty or deceived. Hobbyists delighted in watching computers act, but only insofar as the machines acted upon the operators' commands. The anxiety over loss of control contrasted with passages in which the authors elaborated on more suitable or creative uses of computers in a language of calm rationality. Zajíček appreciated that "it takes one press of a button to print a document on a printer,"[40] and his colleague wrote about the need to educate "enough skilled and self-sufficient programmers who can work purposefully and effectively."[41] By stressing the importance of purposefulness and efficiency, the critics depicted a paragon of an active, stable, and controlling subject.

It is quite telling that hobbyists had not opposed games as long as these were wholesome pieces of programming folklore—open, modifiable, and occurring in manageable numbers. But commercial games were numerous and opaque, written in uninviting machine code by distant and unaccountable authors. Most of the games arrived in pirated copies through informal distribution networks, unaccompanied by instructional paratexts. In the critics' eyes, playing one was like taking candy from a stranger. It was exciting and addictive, but also illicit and suspicious.

In addition to enumerating their adverse effects on the individual, the critics also saw games as a threat to the community. As seen in the previous chapter, the amateur community—and especially the 602, *Mikrobáze*'s publisher—boasted of its self-reliance, achieved through sacrifices and activist work. Pure consumers of games were thus portrayed as frivolous snobs who did not contribute to the collective effort.

Note that neither critic dismissed games altogether; both were willing to spare some "quality" games, discussed later in this chapter. However, both of them hoped that, on both the individual and community levels, games were just a stage of engagement with computers—a stage one could mature from and leave behind. This belief resonated in later statements made by other authors in club newsletters and the hobbyist press:

- The interest in microcomputers and their use is on the rise. The wave of passionate playing of games of all kinds is ebbing away and people start thinking about really useful applications of microcomputers both at home and at work.[42]
- Most microcomputer owners are over the initial boundless excitement by various games and try to use the computer in other ways, too.[43]
- The times of the passionate craze over *Manic Miner*s, *Jetpac*s, *Underwurlde*s, etc., are irrevocably gone. Personal computers are being employed more and more frequently as helpers at work or at home.[44]

Looking back, the claims about the waning interest in games sound implausible in light of the rising accessibility of computers; some sources even suggest that in 1988, most micros in Czechoslovakia were being purchased primarily for gaming.[45] It is more likely that these statements hint at the continuing trend of differentiation of the user base. Whereas some (especially older) hobbyists truly abandoned games and moved to other pursuits, another group of users embraced them—young people who had no problem accepting a new cyborg subjectivity and who had plenty of time to collect, play, and dissect the avalanche of games that were coming from abroad.

Bringing Games under Control

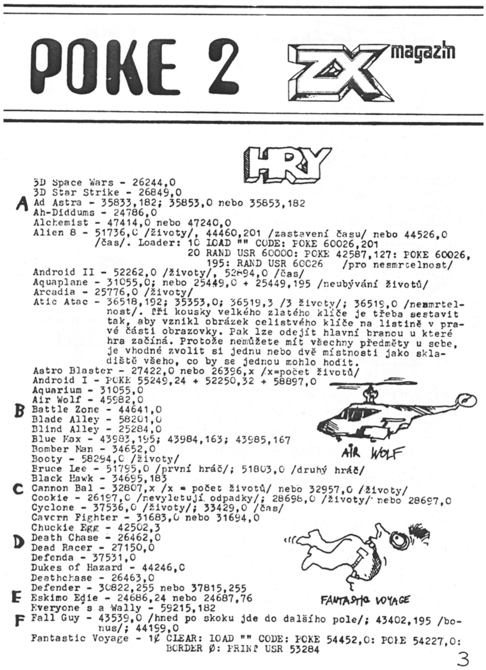
Despite the discursive tensions and the narratives of a temporary craze, games were here to stay. At the day-to-day level, games continued to play an important role in amateur groups and attracted young users to club activities. Hobbyist newsletters—including *Mikrobáze*—reluctantly accepted them and integrated them into hobby practice, occasionally publishing instructions or walkthroughs for selected titles. Although they did not show much appreciation for playing games, they promoted various other things one could do with games *besides* playing them. They were willing to accept games if they were treated as objects of bricolage.[46]

One of the most widely reported practices was that of building homemade controller hardware, already seen in chapter 3. In line with their attitude to games, the hobbyist press did not present joysticks as a means of enhancing the pleasure of play, but as lightning rods for the user's physical activity, protecting the computer from potential harm.[47] The schematics were accompanied by texts suggesting that the "joystick protects the keyboard from the impassioned user"[48] and calling for a "sturdy controller with unlimited durability, which would withstand continuous rough manipulation, for example by children."[49] Once again, games were associated with being "rough," physical, and "impassioned," as opposed to the neutral and measured technical hobby.

Instead of simply playing games, users were encouraged to disassemble or hack them. One newsletter author suggested that the best way to learn machine code was to dissect *Manic Miner*—and at least one of my informants did so.[50] As noted in the introduction, hacking can be understood as a type of bricolage, because of its *in situ* nature and its dependence on local resources. There are some differences between hobbyists and hackers—the latter tend to be more accepting of disruption and subversion, and concerned with tweaking rather than building.[51] Nevertheless, hobbyists did encourage hacking, and specifically two of its subcategories—*poking* and *cracking*.

A *poke* is a small alteration of the machine's RAM, usually applied to games and aimed at producing gameplay benefits, such as unlimited lives. On most 8-bit machines running BASIC, these small hacks were executed by the POKE command, which changes one byte of RAM to a given value.[52] Discovering poke addresses required at least basic knowledge of machine code; executing a poke was also a nontrivial matter. On the Spectrum and the Commodore 64, one had to intercept the game's BASIC loader (usually obfuscated to prevent hacking) and insert the POKE command; on the 8-bit Atari, the process was even more difficult. Nevertheless, pokes became a popular item in both Western and Soviet bloc magazines as well as Czechoslovak club newsletters (see figure 4.2). In one of its most extensive articles on poking, *Mikrobáze* justified its inclusion by "granting the blissful feeling of victory even to those who cannot or do not want to dedicate all their leisure time to entertainment."[53] Pokes could make gameplay more efficient, controlled, and reasonable: "If this article shortens your sleepless nights and returns you to your family and friends, then it has fulfilled its goal. Experience shows that even the greatest fanatics do not enjoy coming back to games that they already finished, no matter if unfairly."[54]

Although pokes could be considered cheating, *Mikrobáze* instead presented them as "enhancements of amusement," a rhetoric understandable



POKE 2 ZX magazín

HRY

3D Space Wars - 26244,0
3D Star Strike - 26849,0
A Ad Astra - 35833,182; 35853,0 nebo 35853,182
Ah-Diddums - 24786,0
Alchemist - 47414,0 nebo 47240,0
Alien 8 - 51736,0 /životy/, 44460,201 /zastavení času/ nebo 44526,0 /čas/. Loader: 10 LOAD "" CODE: POKE 60026,201
20 RAND USR 60000: POKE 42587,127: POKE 60026, 195: RAND USR 60026 /pro nesmrtelnost/
Android II - 52262,0 /životy/, 52894,0 /čas/
Aquaplane - 31055,0; nebo 25449,0 + 25449,195 /neubývání životů/
Arcadia - 25776,0 /životy/
Atic Atac - 36518,192; 35353,0; 36519,3 /3 životy/; 36519,0 /nesmrtelnost/. Tři kousky velkého zlatého klíče je třeba sestavit tak, aby vznikl obrázek celistvého klíče na listině v pravé části obrazovky. Pak lze odejít hlavní branou u které hra začíná. Protože nemůžete mít všechny předměty u sebe, je vhodné zvolit si jednu nebo dvě místnosti jako skladiště všeho, co by se jednou mohlo hodit.
Astro Blaster - 27422,0 nebo 26396,x /x=počet životů/
Android I - POKE 55249,24 + 52250,32 + 58897,0
Aquarium - 31055,0
Air Wolf - 45982,0
B Battle Zone - 44641,0
Blade Alley - 58201,0
Blind Alley - 25284,0
Blue Max - 43983,195; 43984,163; 43985,167
Bomber Man - 34652,0
Booty - 58294,0 /životy/
Bruce Lee - 51795,0 /první hráč/; 51803,0 /druhý hráč/
Black Hawk - 34695,183
C Cannon Bal - 32807,x /x = počet životů/ nebo 32957,0 /životy/
Cookie - 26197,0 /nevyletují odpadky/; 28698,0 /životy/ nebo 28697,0
Cyclone - 37536,0 /životy/; 33429,0 /čas/
Cavern Fighter - 31683,0 nebo 31694,0
Chuckie Egg - 42502,3
D Death Chase - 26462,0
Dead Racer - 27150,0
Defenda - 37531,0
Dukes of Hazard - 44246,0
Deathchase - 26463,0
Defender - 30822,255 nebo 37815,255
E Eskimo Edie - 24686,24 nebo 24687,76
Everyone's a Wally - 59215,182
F Fall Guy - 43539,0 /hned po skoku jde do dalšího pole/; 43402,195 /bonus/; 44199,0
Fantastic Voyage - 1Ø CLEAR: LOAD "" CODE: POKE 54452,0: POKE 54227,0: BORDER Ø: PRINT USR 53284

Figure 4.2
An illustrated alphabetical list of pokes for commercial Spectrum games from *ZX Magazine*, year 1989, issue 8. Image courtesy of Ondřej Kafka.

given the unforgiving difficulty of many arcade-inspired 1980s computer games.[55] But while "enhancing" the experience, they also erased some of the controversial pleasures of gaming by making games—or at least some of them—trivial. Pokes were interpreted as a triumph of hobbyist ingenuity and self-reliance over games and gaming highs, a moment of unshackling oneself from the rules imposed by the game designer. Poking distinguished "active" users from mere players—as another article put it: "Let us leave players to their playing, and let us instead invent tricks for their games."[56]

Mikrobáze similarly praised *crackers*—coders who removed copy protection from commercial programs. A 1986 report on new UK software releases, including games, reads: "Our decoders and dispellers of enchanted programs for the Spectrum are already flexing their neurons. Keep up the good work, boys!"[57] This gleeful endorsement of unauthorized break-ins into proprietary software signals the hobbyist approach to the Western game industries. Industry output was portrayed as material for hacker interventions rather than just gameplay, and cracking was considered participatory rather than illegitimate.

Computer Game Advocates

Unlike some Western countries, Czechoslovakia did not have a tradition of adult role-playing or strategic hobby gaming. *Dragon's Lair* (*Dračí doupě*), the local clone of *Dungeons & Dragons*, was only released in 1990.[58] People played chess, checkers, and various regional card games; the board game market was mostly limited to variations of Parcheesi.[59] Games were thus primarily associated with and relegated to childhood.[60] The influential and prolific writer Miloš Zapletal, author of several collections and encyclopedias of nondigital games, valued board games, card games, and sports games for their pedagogical value.[61] He saw them as a means of sharpening the minds of young people, strengthening their bodies, and building their character. Likewise, the rare examples of positive public evaluation of computer games tended to be justified by their benefits for children and youth.

Given the widespread belief in the pedagogical function of games and the state support for computing in education, one would expect that youth magazines would engage with computer games. Nevertheless, the two major science and technology magazines for young readers—*ABC of Young Technicians and Scientists* and *Science and Technology for Youth*—consistently avoided the topic. In line with policy emphases on the industrial application of computers, these magazines frequently reported on technological innovations and ran numerous articles on programming. But save for a few program listings and a couple of articles about educational games, they did not mention, let alone promote, entertainment uses of computers. In 1987, *Science and Technology for Youth* published a special issue called "Computer—Why and What For," an introduction to microcomputer hardware and programming. Speaking of the Sinclair ZX Spectrum, the authors admitted that the "most numerous programs are of course games of various types,"[62] but they never acknowledged any specific titles—instead presenting them as an amorphous mass that need not, or should not, be discussed.

This dismissive approach was staunchly criticized by Bohuslav Blažek, a Czechoslovak writer, academic, and vocal advocate of computer games (see

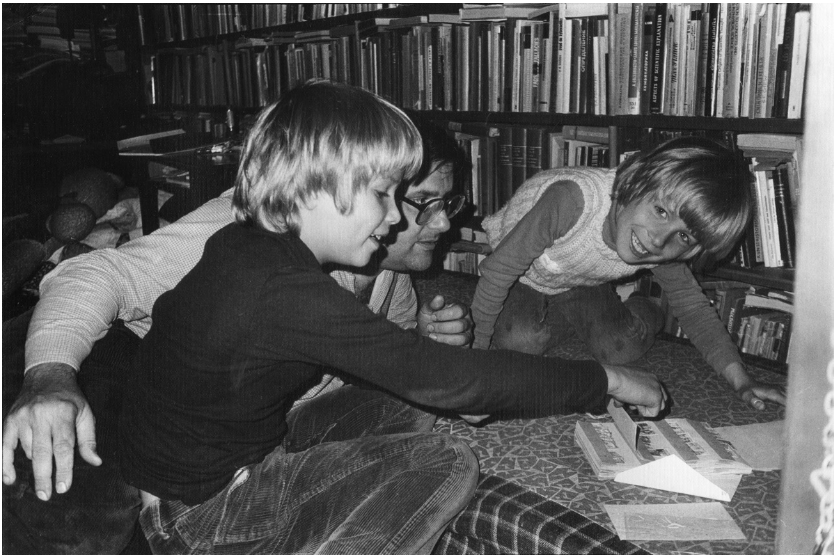


Figure 4.3
Bohuslav Blažek and his sons Kryštof (left) and Filip in the mid-1980s. Image courtesy of the Blažek family.

figure 4.3). Born in 1942, Blažek held degrees in film, philosophy, and psychology, and wrote on a wide range of topics, including computer games. Although he had long been a nondigital game enthusiast, he had no prior interest in computing, noting in his diary: "I only really feel inspired by electronics after I have found the way to make it participatory, dialogical, and playful not only in the consumerist sense."[63] After purchasing a Commodore 64 in Germany in early 1987, he started contributing to hobbyist publications and wrote a book called *Labyrinth of Computer Games*, published in 1990, but written before the Velvet Revolution. Blažek criticized the "Why and What For" brochure as a representative example of the country's hypocritical discourse about kids and microcomputers:

> [This brochure] from 1987, which could be labeled as the first but stillborn attempt to discuss microcomputers and children, reads like a badly written college textbook. Games are at least (I have counted three times) cautiously mentioned, but a playful relationship with the computer is taboo. … The absolute majority of children who have access to computers at home use them to play games in the same consumerist way as the less equipped kids who throw coins into arcade machines. … That is the unacknowledged reality, which is being pushed out of

> the public consciousness, the hidden backstage of our children's computer scene. That sheds a different light on the official extracurricular computer classes and the attempts at "mass implementation of computers in the educational process," ideally resulting in the mass production of little programmers.[64]

According to Blažek, teachers approached games as they did sex; they "were hiding their existence for as long as possible," leaving young people unprepared and uninformed.[65]

Inspired by the works of Seymour Papert—whose ideas were well known in select corners of the Soviet bloc intelligentsia—Blažek instead promoted children's playful interactions with computers, including games.[66] He approached computer games as parts of culture and argued that they "should not be understood as a scrapyard of ideas not serious enough for other spheres [of computing], but as the place where the density of ideas is at its highest."[67] Never a frequent player, he often chose to observe his two sons while they were playing. Nevertheless, his book included a short but sophisticated analysis of the "language of computer games," which foreshadowed many scholarly discussions in the discipline of game studies, including those of genre, historical poetics, ontology, and narrative. Using the example of *Defender of the Crown*, a strategy-action hybrid set in Robin Hood–era England, he even asked whether games were "stories, or machines," predating a similar debate that unfolded among game scholars in the early 2000s.[68]

Miroslav Háša, the tireless promoter of computing education and the man behind the country's first youth computer club, shared Blažek's views of computer games. He refused "futile" attempts to prevent kids from playing on computers, and instead proposed to "actively influence" such play.[69] He contributed to this effort by commissioning *Computer Games: Past and Present*, the only long-form publication dedicated to commercial games published during the Communist era—although it was technically a set of two low-circulation ring-bound booklets posing as teaching material. Háša contributed a foreword, but the text itself was written by František Fuka, who had already become the country's best-known collector and maker of computer games.[70]

Fuka was born in 1968 and started to frequent Háša's Station of Young Technicians in the early 1980s. Thanks to his articulate and enthusiastic personality, he was featured in several educational films and reports from the Station, assuming the role of a geeky whiz kid. He read English better than most of his peers, and had a ZX Spectrum and a subscription to the British *Your Sinclair* magazine thanks to his émigré uncle. His 1988 booklets chronicled the history of games for the ZX Spectrum from 1982 to 1988 and largely drew from the magazine, as well his own extensive gaming experience.

The publication outlined what we would now call a "canon" of Spectrum games, starting with *Manic Miner* and including the production of the British label Ultimate Play the Game (including games such as *Jetpac*, *Atic Atac*, and *Knight Lore*), which he described as "legendary" and "revolutionary."[71] Fuka styled himself as an insider with extensive knowledge of the UK game industry. He did not explicitly discuss the pedagogical or artistic value of games, but his writing bursts with amazement over the diversity of computer games and the ingenuity of their creators. A programmer himself, he mostly valued the work of other programmers in pushing the boundaries of the small 8-bit machine.

Blažek's and Fuka's books were too late or too obscure to reach a larger audience. But the lack of writing on games did not mean there was no demand—quite the opposite. Games were becoming a popular form of entertainment, but mainstream media did not acknowledge it. We can demonstrate this on the lone example of a mass-market periodical that regularly featured games. The *Chronicle* (*Zápisník*) was a biweekly news magazine published by the political bureau of the Czechoslovak People's Army. Embodying the logic of *vnye* in a media form, it offered a jarring combination of military propaganda with apolitical, sensational, and fashionable topics to make the magazine more palatable and appealing. In February 1988, the *Chronicle* started to publish a monthly two-page section about ZX Spectrum software and hardware, entitled Computer Enthusiast Club, sandwiched between series on other 1980s pastimes such as bodybuilding and science fiction. The response was overwhelming. So many requests and calls for help were pouring in that the editors had to refer them to local Svazarm computer clubs. Admitting the dire state of computing coverage, the editors recommended their readers buy the Polish magazine *Bajtek* at local Polish cultural centers and specialized newsagents.

Per requests by the readers, the section covered both games and other software. For the editors, balancing the two was not an easy task, "because the requests about the future focus of [the section] were diametrically divergent. On the one hand, there were calls for detailed descriptions of programming languages and system utilities, on the other hand calls for many more game walkthroughs."[72] The editors defended games coverage against detractors by claiming that "computers should not only be used for work, but also for entertainment," and argued that some games were "beautiful programs" that people should learn how to play.[73] In sum, the *Chronicle*'s coverage demonstrates that gaming was indeed competing with more serious practices, but that both had to live side by side.

The Appreciation of *Tomahawk*

Let me start this section with a personal recollection. In 1990, several months after the Velvet Revolution, my parents bought me a Sinclair ZX Spectrum–compatible computer, and soon thereafter my father—a programmer—went to acquire games for it. He paid a visit to his colleague, a hobbyist and computer technician, who gave him a cassette full of games. Among the first games my dad loaded and showed me was 1985's *Tomahawk*, the renowned and—for its time—impressively realistic helicopter simulator by the British company Digital Integration.[74] I was nine years old, and found the game neither attractive nor entertaining. My dad was disappointed, because *Tomahawk* was supposedly one of the best games out there. In fact, its popularity was so enduring that it appeared in Czechoslovak Top Ten games charts four years after its release.[75] But it was one of the games that were praised by *hobbyists*, and not necessarily by *kids*. The following analysis of discourse on games will focus on what—according to hobbyists and fans writing in the 1980s—made a good game and why *Tomahawk* was one of them.

Kirkpatrick points out that early UK game reviews positioned "games programs as computer software, lacking consistent terms of distinction and evaluation that would clearly separate them and hold them apart from other programs as games." The appraisal of games pivoted on the question of whether they were "well programmed." Around 1985, the word *gameplay* emerged as a new and specific measure of game evaluation.[76] In today's parlance it refers to "game dynamics," or "how it feels to play the game," and usually derives from a player's experience of interacting with the game's rules.[77] In 1980s Czechoslovakia, writing on games not only lacked any such term but often lacked any such evaluation at all. Most texts on computer games were decidedly utilitarian—catalogs, pokes, tips and tricks, instructions, or walkthroughs. Reviews—largely a market-driven genre—simply did not exist.

Perhaps the earliest mentions of individual commercial games date from the first issue of *Mikrobáze*. Published in 1985, it offered a catalog of the club library with short descriptions.[78] Given the editors' public struggle against the "gaming craze," it is unsurprising that they praised games that offered "more than just 'filler' of leisure time."[79] What they valued highest was the technical execution, realism, and complexity of the game systems. They marveled at the "amazing execution of graphics" in *Decathlon*, "the experience of actual driving" in *Deathchase*, or the "inventive, superbly animated three-dimensional graphics" of *Knight Lore*.[80] The authors seemed to revel in games that offered generous customization options, level editing, or complex controls. Speaking of the flight simulator *Tomahawk*, they wrote:

> One of the best programs for the ZX Spectrum. You are a pilot of a fighter helicopter, whose controls are a perfect simulation of actually flying a helicopter. … The game offers numerous options, such as the choice of night time or daytime combat, the height of clouds, turbulence, skill level and mission objectives.[81]

Tomahawk ticked many of the boxes that hobbyists considered important. It was adjustable, relatively open-ended, sophisticated, and—if we accept the idea that it could be used to train future helicopter pilots—even potentially useful for the military. As a guest on the *Computer Dilemmas* TV show said of another simulator: "We could even say that it's almost not a game—it's a flight simulator for light-sport aircraft."[82]

Simulators were never a dominant genre, but were often cited as examples of valuable games. Another such title, the skiing simulator *Ski Star 2000*, was admired for allowing the player to pick "a wide variety of factors," such as the "shape of skiing glasses" and "snowfall intensity."[83] These games offer not only challenges but also opportunities to tinker; and the player controls not only the helicopter or the skier but also the program. These are the games that most resemble serious software—games that are "almost not games." The same could be said about text adventure games, in which players input commands as if in an operating system, and which were praised for being "packed with logical problems and strategy."[84] Chapters 6 and 7 will cover these in much more detail.

Bohuslav Blažek's analysis was humanistic rather than technical, but his normative outlook was not far removed from that of hobbyist newsletters.

Stressing the importance of bricolage, he argued that games were at their best when they offered complex sets of options, editors, and construction kits. He summarily considered any game played on a computer an open and modifiable piece of software. This led to his uncompromising refusal of consoles and arcade machines. Admitting that console games could have identical rules and content, he criticized their closed nature:

> Game rules, graphics, narrative, characters, all of it is often indistinguishable, and connoisseurs keep on arguing whether the computer version is better than the original arcade one, or the other way around.… But the difference is precipitous and the similarity merely superficial.… A game in a box, that's always already a piece of hardware, inside of which software may be hidden, but one that you cannot modify.… Computer games, on the other hand, are gradually becoming full-fledged software, which, despite the initial attempts to prevent any intervention into it, beckons you to intervene and offers you tools to do so, which are so user-friendly and entertaining that they become another game, or a metagame.[85]

In this passage, Blažek sketched out an ontology of games that included components such as rules, graphics, narrative, and characters. But to him, in the end, a game was primarily a software product. In his view, *playing with the game* was more rewarding than *playing the game*. Considering the finding and application of pokes "another layer of the game," he ended his book with a thirteen-page list of pokes.

Fuka's book *Computer Games: Past and Present* stands out by being open to all genres of games and not shying away from evaluating them. When Fuka did so, he followed two basic criteria: quality of programming and originality. The latter meant a variety of things, including original settings, mechanics, controls, and innovative programming tricks. He wrote:

> So we first covered games that were not original, but were well done nonetheless. Then the games that were original, but not overly complex from the programmer's point of view. And now, we will look at those that are so original and complex that one cannot even believe that they were created by regular mortals for the regular Spectrum.[86]

Fuka saved the highest praise for the titles that combined complexity and originality, such as the 3-D space shooter *Academy*, the 3-D action-puzzle game *The Sentinel*, or the action-puzzle hybrid *Quazatron*, "an original, well-programmed game … what more could you desire?"[87] Without stating any clear preference, Fuka distinguished *action* games from *logical* games, with the latter including puzzles, text adventures, strategy games, and other titles that required logical thinking rather than reflexes; the two categories could blend, for example, in action adventure titles. He did analyze the games' content and challenges, but he simultaneously treated them as code, sprinkling his narrative with pokes and even including a machine code primer, so that readers could find pokes themselves.

Only seldom did authors attempt to write about the experience and pleasures of games, and when they did, they seemed to lack appropriate vocabulary. The *Mikrobáze* catalog made vague mentions of "game engagement"; Fuka mentioned "playability" in the sense of balanced difficulty, and noted that certain games "play well."[88] Blažek got closest to identifying the concept of gameplay when trying to describe the experience of playing *Krakout*, a *Breakout* clone.[89] He realized there was something missing in his ontology: "The concept of the game itself is not sufficient to make it successful. What is decisive is the execution of it, a certain charm of the game that cannot be explained by simply enumerating its components."[90] According to his sons, *Krakout* was a game he played for hours by himself rather than just watching others, experiencing what he described as "ecstatic concentration" and "zen-like emptiness of the mind."[91] We may assume that this pleasure was cyborgean, and that the indescribable *charm* was the design of the title's gameplay.

To sum up, whereas the British gaming press of the 1980s reportedly played an instrumental part in the division between games and other kinds of software, Czechoslovak advocates of games—despite playing the same titles—mostly continued to situate them within the category of computer software well until the early 1990s. They saw games as invitations for bricolage as much as (or more than) entertainment. For some of the writers, the best games were not the ones that were most *distinct* from serious software, but the ones that were most *like* it. Although their writing does not necessarily reflect their firsthand player experiences, it reflects the tastes and preferences of 1980s Czechoslovak gaming culture, and reveals the limitations of the underdeveloped terminology.

The previous chapters have shown that microcomputers were an ill fit for the Czechoslovak economy and the official teleology of computing, and largely became a guarded domain of enthusiasts and hobbyists. This chapter has shown that games had it even more difficult, because they did not seem to fit into the agenda of any of the important actors. There was neither space nor political support for a mass-market magazine that would cover games—with the late and unusual exception of the *Chronicle*. They did not fit into the agenda of youth magazines, who were preoccupied with educating young programmers. And hobbyists, at least initially, saw them as suspect objects that might erode the community and discredit computer clubs in the eyes of authorities.

Hobbyists, who had been the guardians of Czechoslovak microcomputing culture and portrayed themselves as the unsung heroes of the STR, were facing changes in user demographics and preferred uses of the microcomputer, and scrambled to promote and reinforce their own agenda. Despite their relative independence, they were also expected to conform—at least outwardly—to the official teleology of microcomputers, that is, their use to boost industrial processes. Commercial games did not fit into this teleology, and hobbyists hesitated to accept them. Their ascetic newsletters criticized games as carnal and irrational, and denounced playing on a computer as the least desirable, useful, and participatory computer activity. Having a computer was a privilege, and it was to be used to its full capacity. "Pure gaming" was seen by club elites as not befitting the golden Czech hands. Instead of promoting games, newsletters tended to promote practices that surrounded and subverted them, such as poking. Their ideal user was a self-reliant tinkerer and programmer, potentially also a cracker or homebrewer. Arising from a traditionally masculine milieu of ham radio and electronics hobby, this ideal was also implicitly—and, as seen in chapter 3, occasionally explicitly—articulated as male.

As work on the United Kingdom and the Netherlands shows, little about these tensions between gaming and hobby computing was exclusive to the Czechoslovak scene.[92] Reunanen notes that "pure gaming" was similarly "despised" even in the Western European demoscene subculture.[93] But what stood out about the Czechoslovak context was the remarkable influence of computer hobbyists, who, along with educators, presided over discourse on microcomputers. Hobbyists were pushing the idea that every user should be a tinkerer, an idea enthusiastically shared by many instructors of clubs and extracurricular classes. This belief likely contributed to the emergence of a prolific homebrew scene in the second half of the decade.

It is difficult to assess the impact of hobbyist discourse on actual user practices, values, and norms. In Czechoslovakia, gaming and computing practices developed in almost total media silence, and users instead relied on the information received informally from friends, acquaintances, and computer clubs. Despite the tensions between various groups of computer users, the debate about games was never fully resolved, and gaming and hobby cultures never fully separated before 1989.

This finding contrasts with Kirkpatrick's account of the UK history of games. Kirkpatrick traces a constitution of a gaming culture from within, based on the texts published in gaming and computing magazines.[94] But a comparison with 1980s Czechoslovakia suggests that the separation of gaming was not intrinsic and not applicable to all contexts. Without powerful players that drove the game industry and specialist press in the United Kingdom and some other Western countries, there was little incentive to portray games as separate from other software, and little agreement regarding *how* one should write about them.

In the end, foreign games were flickering on the screens in socialist homes, but no authority was giving instructions on how to use them—no one taught the users how to be good consumers. The impact of Western magazines on Czechoslovak discourse was mostly indirect because of the language barrier, and more importantly, their scarcity. As an interviewee said of *Your Sinclair*: "That was a treasure and the Holy Grail of magazines. If somebody let you borrow a dog-eared copy, they would be very careful that you return it."[95] Games and gaming experiences became the blinds spots of the local discourse on computing. With a bit of hyperbole, we could say that gaming was taboo—something that could not be mentioned, although everyone knew it was popular. Nevertheless, several individual authors were given the chance to write about games and portrayed them as unique achievements of creative work. It is perhaps not a coincidence that those who publicly defended them, such as Bohuslav Blažek and Miroslav Háša, had backgrounds in the humanities. Even Fuka, who studied cybernetics at a technical university in the late 1980s, soon quit the program and became a film translator and reviewer. These people were used to expressing their opinions and experiences in writing, but beyond that, their education and interests made them more likely to see games as more than just code.

The existing (albeit scant) research on other Soviet bloc countries shows both differences and similarities. The Soviets seemed to be similarly preoccupied with raising little programmers.[96] Polish computer experts also claimed that "computer toys are merely an unimportant episode" in the development of computing.[97] On the other hand, the Polish magazine *Bajtek* prominently covered computer games, although it did so to lure young readers, who were then expected to progress to more serious tasks. In Hungary, computer clubs seemed to be friendlier to games, with Beregi describing them as "pirate dens, where social life was sparkling around the sizzling floppy drives and tape recorders."[98] This is not to say these exchanges did not happen in Czechoslovakia, but there were efforts to balance them out with "serious" activities or at least keep them away from public sight.

After their voices had been ignored for such a long time, Czechoslovak players let themselves be heard during the explosive success of domestic gaming magazines in the early 1990s. Although some magazines continued in the hybrid gamer-hobbyist discourse, it was the purely gaming-oriented ones—such as *Excalibur* or *Score*—that were the most successful. Their discourse was often the exact opposite of the measured and rational critiques of the gaming craze. Their young authors reveled in flamboyant and hedonistic passages, celebrated excessive gaming, and wrote of gaming "orgies" and "ecstasies." In part a product of the unshackled everything-goes atmosphere of the 1990s, this was something that fans of games were eager to hear after years of media denying that games could be fun.

5 Lighting Up the Shadows: Informal Distribution of Game Software

> As far as we know, there are no programs for [Spectrum-compatible machines] available on the market. Before Christmas, copies of four games (probably sales items) for the ZX Spectrum were imported. They missed our Christmas season, however, because their prices had not been determined in time for Christmas.
>
> —From a report on the Spectrum-compatible Didaktik Gama computer, *Mikrobáze*, 1989[1]

> I have about 1,500 games and I have no doubt there are people who have twice or three times as many. Out of the games that I somehow acquire, I copy only one out of five—only the ones I like.[2]
>
> —From a 1989 interview with František Fuka, a creator and collector of games

In his monograph on the informal distribution of film, Ramon Lobato points out that "in many settings, informality is a norm, not an aberration."[3] As shown by the quotes above, this was certainly the case with computer game distribution in 1980s Czechoslovakia. The attempts to bring legal copies of software into retail at the tail end of the decade seem laughable given the copious amounts of Western games that some users had "somehow"—as the second quote vaguely puts it—acquired. The goal of this chapter is to illuminate this "somehow."

In the absence of official software distribution, informal distribution was an absolute necessity and a cornerstone of gaming cultures behind the Iron Curtain. Massive numbers of players were actively involved in it, collecting games and disseminating them to their friends and acquaintances. Distribution was at the heart of players' experiences. No social history of games in Czechoslovakia would be complete, therefore, without a thorough analysis of informal distribution. Rather than dwelling on its economic and legal implications, I will focus on its infrastructures and mobilities, drawing inspiration from scholars of film distribution such as Lobato and Brian Larkin, as well as John Urry's sociology of mobilities.

First, I will map out distribution *networks*. According to Lobato, informal distribution networks are decentralized and user-driven, largely underground, but extensive and efficient.[4] Compared with official networks, informal ones allow texts to "move through space and time with a lower level of interference from copyright law, taxes, tariffs and state censorship."[5] They can cross borders even when the official ones are confined to national spaces, although they are still shaped by the limitations imposed by regional geopolitics on the movement of people and goods. It was, for instance, difficult for a Czechoslovak computer user to reach out to a British one, but relatively easier for that same user to connect to a Polish person on the same side of the Iron Curtain. Significantly, software distribution networks were also platform-dependent. Major 8-bit platforms such as the Sinclair ZX Spectrum, Commodore 64, and 8-bit Atari were incompatible with each other, and their respective networks were therefore largely separate—an Atari owner had little incentive to connect to the Commodore 64 network. I will view these networks not as fixed infrastructures, but as sets of associations perpetually in the making, lacking the relative stability of regulated official infrastructures. Connections and acquaintances were emerging among users, and they disappeared once a person lost interest or switched to a different computer platform. However, some connections and distribution schemes were partially formalized and temporarily stable.

In addition to the structures of networks, I will examine the ways in which their nodes functioned as mediators. Far from being neutral transmission routes, distribution channels have significantly affected the cultural practices as well as the material culture around games. When looking for evidence of mediation, I will follow the distinction between *intermediaries* and *mediators* put forward by Bruno Latour. Whereas an *intermediary* "transports meaning or force without transformation," *mediators* "transform, translate, distort, and modify the meaning or the elements they are supposed to carry."[6] People engaged in informal distribution networks often act as mediators, adding subtitles to a film, or adding a poke and a new loading screen to a game. As Larkin puts it, "Pirate infrastructure is a powerful mediating force that produces new modes of organizing sensory perception, time, space and economic networks."[7] And, according to

Lobato, "The act of distribution also materially shapes the text itself."[8] Following these leads, I will examine the modifications that games undergo in the process of informal distribution, and the cultural practices and meanings associated with them.

These modifications—new filenames, cheat options, or appended messages—may seem subtle.[9] But despite being peripheral to what we may understand as the game "itself," they may significantly affect the game's reception. To understand the transformations, I will employ the concept of *paratextuality*, originally introduced by Gérard Genette.[10] In game studies, the term *paratext* has been applied rather loosely to various phenomena accompanying games, ranging from menus and patch notes to walkthroughs and strategy guides.[11] This chapter opts for a more rigorous use of the concept, informed by a framework introduced by Jan Švelch, who envisions paratextuality as "a tool that allows us to explore how various texts interact with socio-historical reality and other texts."[12] In his view, paratextual elements are situated on the periphery of a game and serve as indexical pointers to its sociohistorical context. Paratextuality is a relational feature, and it does not presuppose any clear boundary between the center and the periphery; filenames, loading screens, credits, and high score tables can all be considered paratextual. While being passed along the informal distribution network, games lose or accrue paratextual information. This can be read as a record of the modifications made on the path between the original producer and the player.

To explain the workings of informal distribution into and within Czechoslovakia, I will follow the story of one particular title: the 1987 British action game *Exolon* for the Sinclair ZX Spectrum.[13] Published at the peak of the Spectrum era, it was a typical hit game of the time, popular enough to spread behind the Iron Curtain in several different versions and along various distribution routes. Based on paratextual indices and archival research, much of the chapter will attempt to reconstruct its journey into the country, and more specifically into the collection of my informant Jan Lonský. Although it is impossible to pinpoint the exact paths of individual copies of games, I will present several possible scenarios. The last section will focus on arcades rather than computer games, outlining the shadow economy of arcade machine distribution in the country.

It is important to remind ourselves of the technical and material features of the informal networks we will be studying. In 1980s Czechoslovakia, the vast majority of software circulated on cassette tapes, the default storage medium of most 8-bit machines. The tape's major advantages were versatility, availability, and low price.[14] Software was stored on a tape as a sequence of impulses making up individual bits and bytes, usually distinguishing zeros from ones by lower or higher frequency. Listening to a data cassette by ear would therefore reveal a series of bleeps and grating noises. The piece of software that processed the data stored on the cassette tape was usually called the *loader*. Default loader routines were stored in the machines' ROMs, but could be replaced by nonstandard custom loaders that were potentially faster or more visually appealing, or that prevented further copying. Loading a game took several minutes—four to five on the Spectrum or the Commodore 64 and more on unmodified Atari 8-bit machines. The temporality of tape manipulation was a crucial aspect of the gaming experience. During loading, the user could be exposed to the loading screen image and other paratextual information. Loading and saving software was time-consuming, but the time could be used to engage in some other activity, and perhaps to talk with friends in the room.

For all cassette-based platforms, there were two basic ways of making copies—*analog* and *digital*. Analog copies—a possibility specific to audio tape storage—were made by dubbing the soundwave from one cassette to another using a double tape setup. Although the process was fast and convenient, analog copies had a high incidence of errors, and their quality deteriorated with each dubbing—a second-generation dub (a dub from a dubbed copy of the original) was already likely to be unusable. *Digital* copies were made by loading the software into the computer's memory and then writing it back to the tape, typically with the help of a dedicated copying utility. In this case, the quality of the copy was not inferior to the original, but copy protection mechanisms employed by game publishers could corrupt the copies.

From Yugoslavia with Cracks

In August 1987, the British company Hewson Consultants released *Exolon*, the new ZX Spectrum game by their rising star Raffaele Cecco, a twenty-year-old programmer based in London. A 2-D flip-screen action game, it starred a heavily armed astronaut fighting his way through the hostile environments of an alien planet. As usual for major titles of the time, the British release was accompanied by full-page ads in major gaming publications. Copies of *Exolon* were available in retail stores across the United Kingdom on factory-duplicated cassettes with a full-color inlay featuring instructions and a brief backstory. The game received critical acclaim for its "terrific graphics" as well as for being "instantly playable and highly addictive."[15] It became a sizable hit for Hewson, landing on the top spot of the Spectrum sales chart in November 1987. A couple of months later, a licensed and translated version was released in Spain by Erbe Software. The game's popularity, however, reached far beyond the national markets of the United Kingdom and Spain. In October 1988, it appeared in the charts of the Czechoslovak *ZX Magazine* fanzine, and stayed among the top five games (based on votes, rather than sales) until the summer of 1989. How did the game cross the Iron Curtain to players in Czechoslovakia, and what happened along the way?

For one thing, it was impossible to buy *Exolon* in retail. British companies made no effort to distribute their games legally in Czechoslovakia. Ordering a tape from Britain was also out of the question because there was no way to pay for it with Czechoslovak crowns. The first professionally made foreign games, mostly for the Sinclair ZX Spectrum, nonetheless started to appear in Czechoslovakia around 1982–1983, not long after the boom of the British game industry.[16] Some were brought in as original copies from the West by the few lucky travelers or "trade tourists." However, such direct import routes tended to be unreliable. When an "established" system of informal distribution fell into place around 1985, its routes tended to circumvent the more tightly guarded Western-Czechoslovak borders. Instead, a large percentage of imported games traveled to Czechoslovakia through the more open and liberal countries of the Eastern bloc—Yugoslavia, Hungary, and Poland. According to a 1990 *ZX Magazine* retrospective article, this did not necessarily impede distribution:

> Czechoslovakia has one huge advantage—it lies in the center of Europe, and two main routes of programs for ZX Spectrum intersected here. From the south to the north (Yugoslavia, Hungary, Slovakia, Poland) and the other way around (Poland, Czechia, Slovakia, and the same route southward). Most of our game and software dealers will attest to that. Whoever wanted to have the hottest new games had to go to Slovakia, because it was always supplied from the south.[17]

Stanislav Hrda, a player and homebrewer who lived close to the Hungarian border, does recall acquiring games from Hungary and Yugoslavia. At a certain point, he and his friends discovered the Yugoslavian hobbyist magazine *Moj Mikro*, which was publishing dozens of classified ads openly advertising the services of microcomputer software pirates. At the time, thanks to the lack of regulation and a relatively open border, Yugoslavia seemed to be a pirate's paradise. As Hrda remembers, "When somebody was lucky enough to go to Yugoslavia for vacation, they gave calls to these copyists and brought back fresh new games."[18]

Before Hrda or any other Czechoslovak player could copy *Exolon*, somebody in Poland, Hungary, or Yugoslavia had to bring a legitimate copy of the game from a Western country. Although this phenomenon has not been extensively studied, the little research that exists suggests that these people were often experienced "trade tourists" or smugglers instructed to bring in new games in exchange for commissions or access to games imported by others.[19] However, legitimate editions usually could not be digitally copied. Fighting piracy in their major markets, software companies introduced numerous copy protection mechanisms, many of them employing nonstandard loaders that rendered digital copying of the software impossible. *Exolon*, in particular, used the Hewson Slowload system, whose distinctive loader routine showed a countdown counter, tracking the four-minute loading time. Before this title could spread, someone in Eastern Europe had to *crack* the game—remove its copy protection mechanisms by altering its code.[20] The result of the cracker's effort was a new version of the game's code that could be copied losslessly and indefinitely. Eventually, even the most sophisticated copy protection schemes were cracked; introduced in 1986, third-party hardware add-ons such as Multiface enabled the user to store the RAM contents to tape, making the process significantly easier.[21]

As early as November 1987—the same month *Exolon* reached the top spot on the British charts—several Yugoslavian pirates were advertising cracked versions of the game through *Moj Mikro*.[22] It could take a few more weeks before some of these arrived in Czechoslovakia.[23] But who cracked them? This question takes us to the Russian-language fan website Full Tape Crack Pack, an invaluable source of material on informal distribution for the ZX Spectrum platform. Unlike the much more popular World of Spectrum fan archive, which contains digital versions of the *legitimate* releases of games, Full Tape Crack Pack collects the *unauthorized* ones. The latter contains seventeen different versions of *Exolon*.[24] The variety indicates that several crackers worked independently in different countries. Given the regional focus of the website, most of the versions come from Soviet crackers, but some appear to have traveled through Poland or Yugoslavia. All of them preserve the mechanics and in-game graphics of the game. Some versions have been translated into Russian, and one has even redrawn the logo to read "Planet of Death" (Планета смерти). Others included built-in optional *pokes* for unlimited lives and other gameplay benefits. Most versions added paratextual information such as nicknames and labels of their crackers. The appended credits demonstrate that games were being "cracked," "broken," or "destroyed" (a synonym for cracked); "fixed" or "restored" (after a game-breaking or otherwise damaging crack); and "trained" or "poked" (when adding cheat modes). As the games traveled across Europe, their code was modified and manipulated, in some cases repeatedly. In other words, they were subject to a wide variety of *coding acts*, as defined in the introduction and developed in the following chapters.

To identify which of these versions were likely to be available in Czechoslovakia, we can look for hints in preserved software collections, interviews, and other archival material. A particularly valuable contemporary source is the inconspicuous 1990 game *Piškworks*, a ZX Spectrum adaptation of the local variant of tic-tac-toe. In this title, the Czechoslovak coder Patrik Rak included a lengthy nondiegetic message, in which he comments on the quality of imported cracks by several Eastern European crackers (or cracking groups). He mentions the extremely prolific Polish cracker (or collective) working under the pseudonym Bill Gilbert (whose cracking style he considers "slimy"), as well as Yugoslavian labels Futuresoft ("pretty nice, but heavy-handed") and Jansoft (whom he pans for introducing game-breaking bugs and for "loaders that make one sick").[25] Flipping through the pages of *Moj Mikro* corroborates that both Futuresoft and Jansoft were indeed among the pirates who advertised in the Yugoslavian magazine. According to their classified ads, both were based in Ljubljana, which is now capital of Slovenia.[26] Furthermore, among the seventeen versions of *Exolon* available in the Full Tape Crack Pack archive, we find one by Ljubljana's Jansoft and one by the Polish Bill Gilbert.

The latter version deserves closer attention. On top of a credit reading "Distribution by Bill Gilbert © 1987," it introduced a brand-new loading screen. The image was credited to Maciej Stawicki, a Polish illustrator who collaborated with the Polish magazine *Bajtek* and moonlighted as a loading-screen artist for unauthorized copies of at least a dozen Spectrum games.[27] His work for *Exolon* portrays the game's protagonist, a hardened space hero.

Although it superficially resembles the official British cover, the pictures differ in the placement of weapons and armor components, suggesting that Stawicki did not have access to the original package and based his artwork on some other sci-fi illustration, possibly in combination with in-game graphics (see figure 5.1). The Polish screen is also more over-the-top, including the words “Kick Ass” scribbled on the astronaut’s chest armor, and “Let’s Rock!” on the helmet, ramping up the comics aesthetic, arguably to make it more decidedly Western.[28]

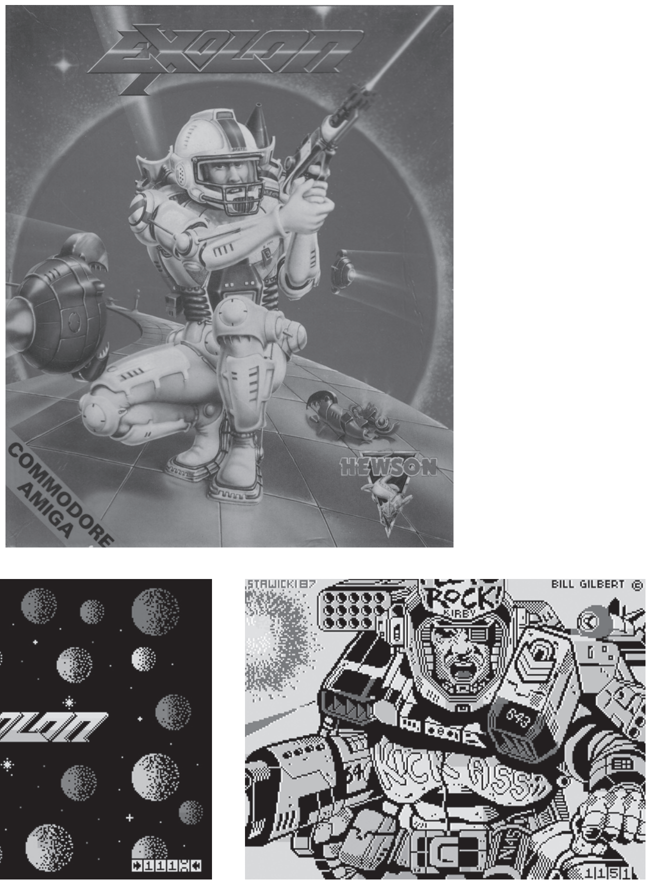


Figure 5.1
The official cover of *Exolon* by Steve Weston; original loading screen of the Hewson release; and Stawicki’s loading screen for the unofficial Polish release. Top image courtesy of D. Weston.

Some incarnations of *Exolon* may have been cracked in Czechoslovakia. Possibly the most prolific domestic cracker and modifier of games was František Fuka, who concurs that the "biggest pirates were somewhere in Yugoslavia."[29] He was in touch with a Yugoslavian contact who lived in Czechoslovakia and who would bring him raw analog copies of original games—presumably made from British originals—which Fuka cracked, trained, and further disseminated.[30] Writing in 1988, Fuka estimated there were up to five different versions of any given game circulating in the country. In theory, anyone could have created their own version of *Exolon*.[31] In the latter half of the decade, informal distribution networks seem to have been flooded with different versions of individual games, as a *Mikrobáze* newsletter writer urges readers not to "produce new modified versions of games," because "there is already big enough confusion in the world of microcomputing."[32]

Our reconstruction of a Spectrum distribution network paints a picture largely distinct from the existing accounts of the Euro-American "computer underground." In the West, especially continental Western Europe and Scandinavia, crackers started to form groups that competed for prestige and respect from peers. They honed their cracking skills, the timeliness of their releases, and the quality of "cracktros"—intro sequences packed with lavish visual effects and pumping music within as few bytes as possible.[33] They formed a subculture that defined itself in opposition to the commercial game industry while remaining wholly dependent on that same industry for its supply of games to crack.[34] Many of their activities were genuinely clandestine, and the shared thrill of illegality boosted their subcultural identity. In the late 1980s, the focus of this competitive creativity shifted from cracking to making demo sequences, and the "cracker scene" eventually evolved into a more creative "demoscene."[35] The present research on these communities highlights the interconnectedness and international nature of their networks.[36] This results in the frequent use of the term "the scene" in singular form, implying the existence of a united international community.[37] In reality, this scene's reach was limited in terms of both geography and platforms.

In March 1988, the *Illegal* zine—possibly the most influential scene periodical—published "The European Crackers Map, Anno 1988." Although the magazine listed contributors from Germany, Belgium, Denmark, Sweden, Holland, France, the United States, England, and Turkey, the map shows no crackers in Eastern Europe. As noted in the introduction, the whole region was blank, bearing only the tongue-in-cheek label “Communist in sight.”[38] In November 1988, the somewhat surprised editors discovered that demos were being made in Hungary, Poland, and the Soviet Union, but kept deriding Eastern Europeans for not having “new stuff” and playing games that were no longer “hot.”[39] Besides geopolitical reasons, this disconnect was partly due to platform differences. What is sometimes referred to as “the scene” is in fact a Commodore 64 (and later Commodore Amiga) scene that formed in Western Europe, and to some extent the United States.[40] In Czechoslovakia, Poland, and the Soviet Union, the Commodore 64 never achieved the status of a go-to 8-bit machine, and the ZX Spectrum and 8-bit Atari were more prominent.

The Euro-American Commodore scene and its Soviet bloc Spectrum equivalent had much in common. Eastern European crackers likewise exhibited their technicity. Cracking required knowledge of machine code and programming tricks, and was considered substantial enough to leave one’s stamp on the game, becoming a coauthor of sorts. But compared with the Euro-American “scene,” which became a full-fledged interconnected subculture with its own jargon, ethics, hierarchies, and niche media, the Eastern European one seemed to be a more pragmatic and casual affair. It could not discursively position itself against a nonexistent industry, and did not have to be clandestine and “illegal.” Instead, it was embedded in regular legal infrastructures. Unlike their Western European counterparts, who would often hide behind post office boxes, the Ljubljana crackers openly published their addresses and phone numbers in mass-circulation magazines, knowing that copyright was not enforced. Maciej Stawicki likewise did not hide the fact that he collaborated simultaneously with the shadowy Bill Gilbert and the legitimate, government-supported Polish magazine *Bajtek*. In Czechoslovakia, the little cracking that went on was mostly executed by members or alumni of state-sanctioned computer clubs.

One lingering question remains. Spectrum games circulating in Eastern Europe had usually been cracked in the region. But why had they not been cracked already in the United Kingdom, the country of their original release, and then exported? Piracy was certainly widespread enough in Great Britain to make Hewson Consultants and other companies employ copy protection. It is likely, though, that its proportion was significantly lower than in Eastern Europe. In the mid-1980s, companies such as Mastertronic and

Codemasters started releasing “budget” titles priced as low as £1.99, making originals affordable even to young players with limited pocket money. Widespread access to original retail copies (augmented by an infrastructure of game rental) contributed to the popularity of tape-to-tape analog dubbing, which obviated the necessity of cracking.[41] In other words, even if you did not own an original copy, it was likely that one of your schoolmates did. In addition, the Spectrum communities in Western Europe (mainly in the United Kingdom and Spain) were geographically separated from the Eastern European ones, making it more difficult to maintain stable connections for exporting unauthorized copies.

Following the possible paths of *Exolon* into Czechoslovakia, we have made several important observations. The Iron Curtain was “gamed” at multiple places and bypassed in multiple ways, depending on both political geography and the geographical distribution of platforms. Although the informal distribution was a cross-national undertaking, there was never just one European scene. Czechoslovak Spectrum users were dependent on both cracked and uncracked imports from Yugoslavia or Poland, whereas Atari and Commodore users were more likely supplied with games cracked by groups based in continental Western Europe, where these platforms were more prominent.

The Unregulated (Non)medium

To understand how *Exolon* could spread within Czechoslovakia, we must first examine the country’s regulatory policies. Lobato situates informal film distribution in the shadow economy—a space of “unmeasured, untaxed and unregulated” activity.[42] The effectiveness of the Czechoslovak informal distribution network was in part enabled by the almost complete lack of regulation of computer games.

Given the Communist government’s efforts to “normalize” and control culture and commerce, the lack of regulation of computer games may seem surprising. But in fact, the state had neither the reasons nor the means to regulate games. Generally, there are two basic motivations for regulation—*economic* control and *censorship*. Concerning the former, the state may exert economic and legal control over media distribution to extract taxes and curb copyright infringement. However, given that there was no legitimate software market, the state would not profit from curbing the informal one. Informal software distribution was but a small part of pervasive informal economic activities, some of which were arguably beneficial to the population because they alleviated the shortages of the command economy. Eggs and vegetables, artisanal work, VHS videocassettes, computers, computer games—all of these were being bought or exchanged in unmeasured, untaxed, and unregulated ways. According to a 1986 study, 60 percent of Czechoslovak citizens took advantage of shadow economies to procure otherwise unavailable goods and services.[43] In this context, going after computer hobbyists and teenagers would not bring much benefit.

The legal status of foreign-made copyrighted software was unclear at best. Czechoslovakia had signed several international copyright treaties that theoretically bound it to respect copyrights to works protected in other countries, but domestic legislation on software copyright was only introduced after the Velvet Revolution.[44] Prior to that, computer clubs mostly supposed that their software collections were in their collective ownership, and that sharing and copying of programs was permissible as long as it was free of charge and part of "organized club activities"[45]—an approach also documented in Hungary.[46] Individual users did not question the moral dimension of unauthorized copying. In their view, no harm could be done to the distant, mostly British and American companies that had produced these games, but had no official presence in the country. Local players believed that by passing games around, they were helping others, not harming the producers.[47] Even the Czechoslovak homebrewers played by the rules of informal distribution and—before the Velvet Revolution—never expected profits from their games. The situation was only slightly different in the rare cases of "serious" programs sold by Svazarm clubs.

As for *censorship*, I have found no documented instances of ideologically motivated restrictions on computer game use. As seen in previous chapters, the government viewed microcomputers as little more than toys or advanced calculators; games were rarely considered a part of "culture" and were dismissed as children's affairs. Moreover, censoring them would have required technological infrastructure and skilled staff—and as Václav Havel's story in chapter 3 showed, the country's security forces were not particularly well versed in computing.

Although the authorities did not censor computer games, they did on several occasions censor magazines and books *about* games, which was a practice that fit neatly into their routines. Writer and programmer František Fuka was lucky enough to have a subscription to the UK-published *Your Sinclair* magazine, but he occasionally discovered that some superficially subversive pages—for example, an article about a game based on the movie *Red Heat*, in which Arnold Schwarzenegger was pictured in a Red Army uniform—were torn out by censors.[48] Similarly, when writing his own 1988 book *Computer Games: Past and Present*, he was reportedly forbidden to include a screenshot featuring Winston Churchill, who arguably represented Western imperialism.[49]

There was, nevertheless, some concern among players over games with anti-Soviet content, such as *Raid over Moscow*, in which the player ends up bombing the Kremlin, and *Green Beret*, which features raids of presumably Soviet-aligned army bases.[50] The former game sparked a well-documented controversy in Finland, which was a neutral onlooker in the Cold War, yet was under constant political pressure from the neighboring Soviet Union. The game's Finnish release triggered a chain of events culminating in a diplomatic row with Moscow. The Soviets considered the game enemy propaganda and requested it to be banned. However, Finnish laws only allowed for restrictions on import in case a product was a health hazard. As a result, the game was never banned, and instead—thanks to the publicity—it became the country's top-selling game of 1985.[51] In Czechoslovakia, on other hand, *Raid over Moscow* and *Green Beret* only circulated in pirate copies. Czechoslovak users realized that playing these games in public settings might be problematic, and Fuka even remembers hacking Soviet iconography out of *Green Beret* so that it could be played in public.[52] Nevertheless, playing such games in private was safe. Stanislav Hrda recalls: "It was quite an experience to bomb the Kremlin. At first, we thought that this had definitely been banned. … Soon, I found out that I can make copies of *Raid over Moscow* for my friends and no one will report me."[53]

Overall, computer games themselves were probably the least censored medium in the country because those in power did not consider them a medium. The state's lenient approach to games contrasted with some other media, notably rock music. Illegal swap meets involving Western LPs were monitored and targeted by the police.[54] Rock music and samizdat publishing *were* deemed political, but games were under the radar, and their distribution faced minimal institutional and regulatory constraints.

Lightning-Fast Sneakernet

Once *Exolon* arrived in Czechoslovakia, it spread further along local routes. Given the restrictions on travel and the low mobility of most of the population, there were relatively few cross-national connections, but a dense network connected users within the country. With no official distribution channels to speak of, unauthorized distribution was the primary way, if not the only way, of getting software. We may reasonably expect that out of the thousands of people who played *Exolon* in Czechoslovakia, no more than a handful possessed an original copy. It was even more likely that there were *none*.

Informal distribution was based on physical transportation of cassette tapes rather than digital transmission through communication networks. We can thus classify it as a *sneakernet*—a term coined in the 1980s to differentiate the movement of physical media from the more recent computer networks.[55] The term is useful for my analysis because it evokes the image of people—not just inanimate objects—moving through space and time. Although Czechoslovaks did not wear Nike or Adidas sneakers, but rather a domestic brand such as Botas, the principle was the same. Every user could become a node in this peer-to-peer network. Thanks to lossless digital copying, code did not deteriorate as games were passed around. Any user could make a copy of equal quality and become a source for a potentially unlimited number of further duplicates. These affordances may sound trivial today, but they were very novel in the 1980s, as users' reflections attest. Although the common user interactions in the network seemed mundane and intimate, no one could see the network in its entirety and it seemed baffling when scaled up. As Blažek described it in his 1990 book: "[Software] becomes a circular letter of sorts, to which anyone can add their own bit. But instead of a circle, it moves along an indefinitely branching star shape. As if a message in a bottle was thrown into the sea and kept multiplying."[56] František Fuka felt compelled to explain in a 1989 interview: "As opposed to collecting stamps, collecting computer programs has a certain advantage—if somebody borrows one from me, they can copy it and return it back to me."[57] Despite its amateur nature and crude material conditions, tape-based distribution thus turned out to be surprisingly efficient, and has been described as "lightning fast."[58] Programs took as little as two weeks to traverse the country from one end to the other. Explaining how his games were distributed, Fuka continued: "I make … copies for my friends, they make copies for others and … this is how programs are copied and passed on here. The speed is unbelievable. One of the games that I released was offered back to me from Bratislava six days later."[59]

Besides the efforts of users themselves, operation of an informal distribution network was enabled by a variety of hardware and software tools produced by the domestic amateur community. Clubs often employed makeshift analog splitters—sometimes called “spiders”—that distributed the audio signal from one computer to multiple audio outputs at once, allowing several club members to plug in their own cassette players and record copies of games at the same time.[60] The Atari scene is associated with Turbo loaders, which were discussed in chapter 3. Combining hardware and software components, they significantly reduced loading times. Importantly, though, these loaders required all software to be converted into their specific format, eventually generating hundreds of new programs that were Turbo versions of existing titles.

Even more important was *copier* software. Writing an efficient copying utility was a challenging task, taken on by leading Czechoslovak coders. One of the most influential copiers was *TF Copy* (also known as *Tape-File Copy*), reportedly the first popular ZX Spectrum copier to use on-the-fly compression to fit more data into the machine’s memory.[61] Its first version was written in 1986 by Arnošt Večerka, a computer hobbyist and a professional programmer at the Palacký University in the city of Olomouc. As of 2018, Večerka continues to work as a lecturer at the university, maintaining his specialization in data compression. One of TF Copy’s prominent competitors was *BS Copy*, developed by Busysoft, a one-man label of Slavomír Labský, a proficient coder and one of the founders of the local Spectrum demoscene.

The informal distribution network comprised three major types of connections—*individual exchanges*, *club sharing*, and *for-profit piracy*. At the base level, Czechoslovak software distribution networks consisted of personal exchanges among friends and fellow hobbyists. This was the default method of obtaining software in the earliest Czechoslovak hobbyist community of Sinclair ZX81 users, as evidenced by the following claim from the *Mikrobáze* newsletter: “I propose … that [ZX81] programs be distributed in the usual way, i.e., from one of us to another.… I give the programs that I have away for free and expect the same from you. Each of us has written a program or bought it abroad and nobody has money to spare.”[62] Sharing one’s software with other members of the community was “usual” and desirable. In a 1985 survey of *Mikrobáze* readers, 98.5 percent of 298 respondents stated that they obtained software through "exchange."[63] These exchanges combined the principles of *gift economy*, as conceptualized by Lewis Hyde, with some elements of barter.[64] With copy protection removed, users did not "lose" a program upon giving it away, and there was nothing to gain by keeping it for themselves; people shared their software, expecting reciprocity from others in the community. Software was the lifeblood of the microcomputer communities and there seemed to be a strong imperative to keep it flowing.

We can safely assume that by the mid-1980s, most of the software traffic in the country consisted of Western computer games. As opposed to productivity software or programming tools, games are more akin to a consumer product—they can cease to entertain or be finished and discarded in favor of new ones; they can be "used up." As a result, the desire to acquire new games was one of the driving forces behind the informal distribution network. According to the 1985 survey, an average reader of *Mikrobáze* owned twenty-two games, which made up 54 percent of his or her software library.[65]

Personal exchange continued to be key throughout the 1980s, especially among young people. *Exolon* was likewise copied from one user to another. On its path into the collection of Jan Lonský, it may have changed hands dozens of times. According to his own recollections, the most probable source of this game was his cousin, a Didaktik Gama owner well-supplied with new software, whom Lonský's brother regularly visited. Here, we can see that the microcomputer hobby involved much more than sitting at a computer and playing or making games. Instead, computer fans walked or biked around their villages, towns, and cities. They would look for fellow users who owned a compatible computer and make new friends and acquaintances based on these platform affiliations. They visited each other at homes or met in computer clubs and spent hours copying games and discussing them. These meetings were more than software transactions—they were also social gatherings.[66] It was not even necessary to *own* a machine to be involved in the sneakernet. The Bratislava-based homebrewer Stanislav Hrda had taken on the role of an itinerant software disseminator, copying games to cassettes and bringing them over to other people's homes, even before he had his own machine. Besides enjoying these visits, he made connections that were indispensable for getting new software later. The more contacts one had, the "wider portfolio" of games one could choose from, because despite its efficiency, the sneakernet did not guarantee equal and equally fast distribution to every user.[67] Although it was based largely on gift exchange, informal distribution also created power relationships between users. Amassing large numbers of games gave collectors social capital and more opportunities to barter. This was one of the reasons why some users collected games they would never play.

Local computer clubs served as sneakernets' prominent hubs. Besides connecting individual users, some clubs ran free-of-charge distribution services for their members, some of them by mail. Even *Mikrobáze*—the country's most influential club newsletter—used to publish listings of available software until 1986. In fact, the very name *Mikrobáze* (short for *micro database*) was originally reserved for the club's software distribution system, and only later adopted as the newsletter title, highlighting the connection between the two principal missions of the clubs—the dissemination of information and programs.[68] Software libraries were among the clubs' most important assets, being a source of prestige as well as an attraction for potential members.[69]

As noted above, many users gave their games away for free or in exchange for other software. They were reluctant to distribute them for money, not over copyright issues but "out of basic decency, because these were not our products."[70] Others, however, traded unauthorized copies of software for money, and these people were already called "pirates." Jan Lonský's archive contains a letter from such a pirate, containing a listing of available games, utilities, and other software as well as his private address and phone numbers. Dated January 31, 1989, it includes *Exolon* (with a description reading "an astronaut on an alien planet"), making the pirate another potential source of the game besides Lonský's cousin. The prices of Spectrum software ranged from two to fifteen crowns per program, whereas a cinema ticket cost seven crowns and a loaf of bread was three.[71] Overall, however, the infrastructure that would allow for organized for-profit piracy was less evolved in Czechoslovakia when compared with Yugoslavia, where pirated software was massively advertised in hobby magazines, or with Poland, which was home to black-market-style computer street sales—physical sites of hardware and software trade.[72]

Homemade Tape Culture

Exolon traveled light. Before it reached Czechoslovak players, it was stripped of all the tangible paratextual material that had once accompanied a legitimate market copy—the price tag, the packaging with all the illustrations, the instructions sheet, and the cassette itself. At the same time, code could not circulate without "materializing" on a storage medium. Czechoslovak players recorded their own tapes and created an improvised material culture around tape trading. This section will continue my distribution-centered analysis by focusing on the practices and material artifacts that accompanied play.

Let us work from the image of the handmade cassette inlay of Jan Lonský's tape with *Exolon* (see figure 5.2). This historical artifact and the paratextual information within it can point us to the social and historical context into which the game entered, and uncover practices associated with gaming. These practices, including compiling, annotating, and embellishing cassettes, can be understood as *possession rituals*. According to Grant McCracken, such rituals provide a way in which "an anonymous possession … is turned into a personal possession that belongs to someone and speaks for them."[73] Through personalization, players asserted ownership over the cassette as well as the games; they made them their own.[74]

Tape

In the collection, *Exolon* is one of the fifteen items stored on an AudioStar HS-I 60 cassette of South Korean manufacture, with a total capacity of sixty minutes.[75] Lonský purchased it at the electronics store in his Western Bohemian hometown of Nýřany sometime in the late 1980s. Throughout his tenure with a Spectrum-compatible Didaktik computer, he used a series of tape decks, each of which had its own flaws and peculiarities. To read a tape from another tape recorder, one often had to experiment with tape head configurations or bass, treble, and volume settings. Rather than well-behaved intermediaries, cassette players acted as unpredictable mediators. Due to the low reliability and speed of tape storage, users of 8-bit micros were immersed in the materiality of computer games through various senses. Besides reading on-screen paratextual messages and touching various textures of plastic while manipulating the tapes, they also listened to the signal. Although the information was meant for the machine rather than the user, Lonský listened to the cassette's sound—as many others did—and remembers the distinctive tones and rhythms associated with particular types of data.

Figure 5.2
A selection of Jan Lonský's cassette tape inlays, with *Exolon* on the top right, and *Vexl* and *Werner* on the bottom right. Image courtesy of Jan Lonský.

Title

In this case, the title of the game as written on the inlay—EXOLON ’87—reflects the name of the file saved to the tape. However, the “87” (indicating the year of release) was not a part of the original title, but an addition appended somewhere along the distribution path. The addition was common enough that some Czechoslovak pirate listings and magazine articles treated it as part of the title. In general, renaming files while copying them was the most accessible form of game modification and personalization. One of Lonský’s sources liked making up his own titles, renaming *The Great Escape* to “Werner,” and *Elite* to “Vexl.”[76] As the scans show, Lonský’s inlays therefore list these made-up titles alongside the official ones.

The Compilation and the Counter

Whereas original tape releases usually comprised just one game, a standard sixty-minute audio cassette could fit many more. Each tape was therefore a compilation of sorts, sometimes a purposeful collection of selected games, but more often a record of the serendipitous order in which one acquired them. Lonský’s tapes included both games and utilities, of Western as well as domestic origin. They included early to mid-1980s “classics” such as the isometric action puzzler *Alien 8* or the text-based management game *Dictator* (in a Czech translation), as well as newer titles.[77] The digits on the margins of the inlay refer to the position of the respective games on the tape, measured by the tape player’s counter. The speed of the counter varied depending on the tape deck model. Writing down the positions therefore became both a necessity and a possession ritual. As a player remembers, “Whenever a new cassette arrived, I put in the tape deck and wrote down the exact position of each game.”[78] Tapes were then organized into collections. Successful collectors such as František Fuka had dozens of tapes and hundreds or thousands of games.[79]

Spine

The spines of Lonský’s tapes are embellished with colorful hand-drawn images. These include logos of actual game companies such as Digital Integration, Psion, Elite, or Ultimate Play the Game, but also crackers such as Ljubljana’s Futuresoft, Lonský’s own labels Gold Disc and Nagasoft (the latter of which he used when releasing his work in the early 1990s), and his favorite punk rock band, Plexis. Unlike many later pirate DVDs and CDs, Lonský's tape is far from being a counterfeit object attempting to pass for a legitimate copy. Hand-drawn logos of commercial companies could not bring a tape full of pirated software closer to the original, which remained distant and unattainable; but it did make the object more personal and unique.

Not all tape covers were as personal and colorful as Lonský's. As informal distribution was becoming more formalized and new technologies became available in the 1990s, computer-printed or photocopied inlays started to appear alongside the handmade ones. Nevertheless, both handmade and serially produced unofficial tapes reflected the diversity and complexity of the informal distribution networks. They were not just appendices to the gaming experience, but its integral parts. Whereas original cassettes serve as paratextual records of the game industry, the unofficial ones offer indispensable clues for the reconstruction of historical gaming cultures and user practices.

(Mis)understanding Games

Most games that were played in Czechoslovakia hailed from the West. Even after all the modifications, their mechanics, graphics, and sound effects remained mostly identical to the original versions. But they entered a very different context. Domestic players usually had limited (if any) English skills. Not knowing references to British, American, or Spanish popular culture was the least of their problems. In most instances, games arrived without manuals telling players how to play the game, and game software itself did not include instructions or tutorials due to memory constraints.[80] As a *Mikrobáze* article poetically observed: "Games are spreading throughout the exchange networks at the speed of light, but—alas—without manuals, it is not only difficult to find out the games' controls—sometimes even their very goal is obscured by impenetrable darkness."[81] Introducing a walkthrough of the 1984 British action adventure game *Pyjamarama* five years after its release, another writer noted that "most ZX Spectrum owners have surely come across this game, but only a few have grasped the point of the game and successfully finished it."[82] When buying software from pirate distributors, players had little information to go by, because catalog descriptions tended to be short and vague. In one of the listings, the text adventure *The Hobbit* is characterized as "a maze," and the description for the title *Defender* is simply the Czech translation of the word "defender."[83] Some clubs had access to foreign magazines or transcriptions of manuals and walkthroughs, but many individuals did not.

We can therefore reasonably expect that Czechoslovak players engaged with games differently from their Western counterparts, who were more likely to have access to original copies. How did they make sense of games, then? Game scholars have theorized several types of play that disregard the rules and authorial intentions—pragmatic ones such as *cheating*, aesthetically subversive ones such as *countergaming*, and advanced metagaming practices such as *speedrunning* or "*superplay*."[84] All of these, however, presuppose a purposeful decision on the part of a knowledgeable player. In our case, players simply did not know how to play games in the way their authors intended, making their practice more in line with Eco's concept of *aberrant reading*, a reading that arises not from deliberate subversion, but rather from a lack of shared code and knowledge.[85]

The misunderstandings could range from minor to fatal. *Exolon* was a standard action game and presented only a minor chance of confusion. Czechoslovak players might not have known that the main character's name is Vitorc, or they might have missed out on the instruction that one can gain 10,000 extra points for not using the "exoskeleton" power-up. A notable example of misinterpretation concerns the classic 8-bit platformer *Jet Set Willy*. The goal of the game was to collect all discarded glasses and bottles after a party in the titular character's huge new mansion. Until this task was completed—a feat literally impossible without pokes—a female character blocked the way into Willy's bed.[86] The original instructions identified her as Willy's Italian housekeeper Maria, but according to multiple Czechoslovak sources, Maria was actually Willy's wife.[87] This was a subtle change that, however, proposes an interpretation of the game that is based on sexual innuendo. More seriously, *Deus Ex Machina*, an experimental multimedia title that relied on synchronization with a bundled audio recording (resembling a radio play), was circulating without this recording, rendering the game's dystopian narrative incomprehensible to the Czechoslovak audience.[88]

Bohuslav Blažek, an academic and a prominent computer game enthusiast, criticized such aberrant uses as an unwelcome product of informal distribution. He likened the Czechoslovak scene, overflowing with various versions of pirated software, to a "computer swamp" dominated by "little software magnates" who did not actively seek out programs but simply awaited what the currents would bring, gluttonously collecting everything that floated by; they tried out a piece of software only to ditch it for a newer, more attractive one.[89] Blažek found this mode of engagement superficial and potentially dangerous: “Our willingness to use stolen, broken, virus-infested software, devoid of textual instructions eventually leads to a disconnection from the communication with the outside world.… As if we stopped being full-fledged participants in an erotic contact, and became mere voyeurs.”[90]

Despite his poignant observations—people did indeed collect many games they did not know how to play—his criticism of player practices is one-sided, conforming to hobbyist ideals about the correct uses of microcomputers, discussed in the previous chapter. I would argue that rather than voyeurs, players became amateur media archaeologists, collecting, examining, and sorting alien objects that arrived from distant places. Exploring games and learning how to play them was an active, demanding, and often exciting process.

When facing more complex games, players had to figure them out, often by themselves. As Fuka put it in 1988: “When I got *Shadowfire*, I knew absolutely nothing about it, and despite that I managed to win it. For me, the most fun part was to find out how the game works and which icon does what.”[91] Reminiscing about his player practices, informant Michal Hlaváč explicitly speaks of a “culture of experimenting”: “The moment of not knowing how it worked was the starting point, that’s where it began.… I remember going across the whole keyboard and trying out each key to find out what it does. That was not uncommon.”[92] Misunderstanding could result in metagaming practices such as the ones described by Stanislav Hrda. Hrda was fascinated by Western text adventures such as *The Hobbit*,[93] but his English skills were insufficient to solve its puzzles. So instead of trying to finish the game, he set himself another goal—to visit as many locations as possible: “Every time I played, I challenged myself to discover another screen.”[94] Others would go even further and disassemble games to examine the code and find out how the mechanics worked.[95]

To record and repeat in-game progress, players created textual artifacts such as notes, hint lists, walkthroughs, and maps. One of the most visually striking artifacts I have collected during my research is Patrik Rak’s map for *Saboteur II: Avenging Angel*.[96] Drawn with a red ballpoint pen on a square piece of folded yellowish paper, it is incredibly detailed and aesthetically

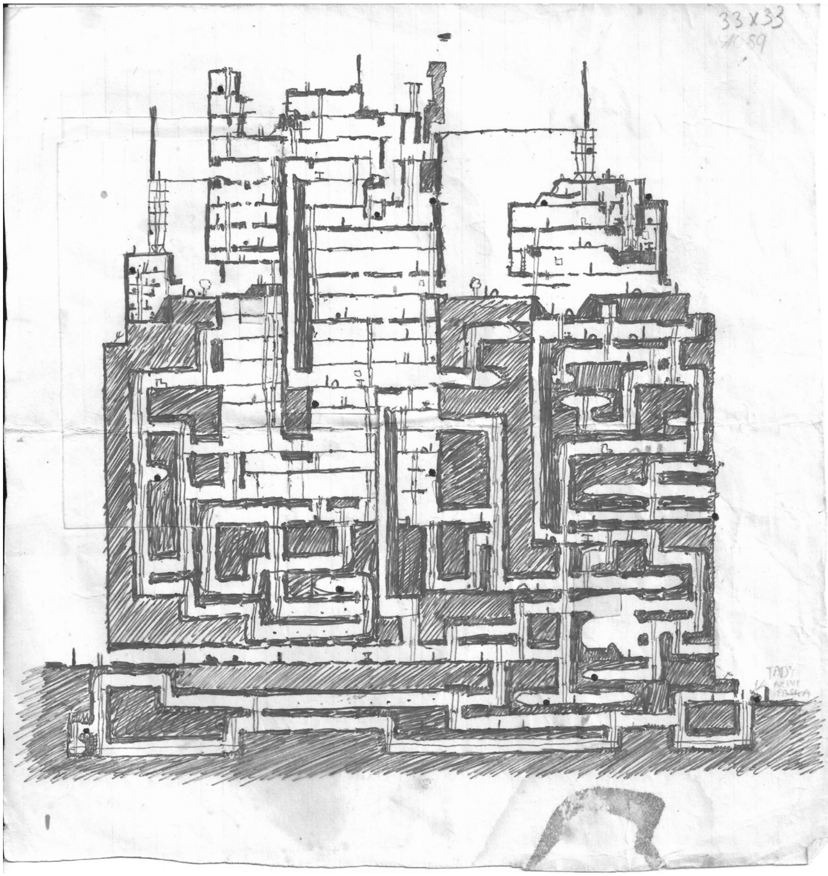


Figure 5.3
A map of *Saboteur II*. Image courtesy of Patrik Rak.

appealing (see figure 5.3). It was also useful, because *Saboteur II* took place in a large maze-like environment, as did many 1980s games.

When discussing walkthroughs, current game studies literature stresses their social functions, approaching them as a "formalization of [players'] knowledge and expertise into a guide for other gamers."[97] This presupposes their widespread availability to other players. However, Czechoslovakia's lack of media infrastructure limited the potential for sharing.[98] Rak does not remember sharing the map with anyone, apart from friends who visited and played the game with him. The map was a record of a personal journey rather than a beacon publicly displayed for other players.

In summary, 1980s Czechoslovakia's informal distribution system, along with the lack of paratexts and gaming press, brought about a specific kind of unknowledgeable, exploratory play. I am far from claiming that this practice was exclusive to the context I am studying. Rather, it is indicative of a larger number of peripheral contexts. The Czechoslovak example is valuable for reminding us to focus on the players who were—or are—in the dark.

A Cottage Arcade Industry

So far, this chapter has dealt with games for microcomputers, played and copied in the safety and relative coziness of bedrooms and computer clubs. However, Czechoslovak arcades were also notable parts of the country's informal economy, and therefore provide a point of comparison with the microcomputer scene. Arcades were even further removed from the official teleology of computing. Designed as nonprogrammable closed systems, arcade machines were not part of the mission to educate the youth, and did not get support from the paramilitary. Instead, they came to be embedded in a vastly different and grittier cultural tradition—that of traveling funfairs.[99]

The first documented arcade machines started to appear in Czechoslovakia in the early 1980s.[100] One could usually find them in trucks or buses converted into somewhat claustrophobic mobile arcades, crammed with machines of various ages in various states of (dis)repair (see figure 5.4). The vehicles traveled with other funfair attractions—carousels, rollercoasters, or shooting galleries—operated by an itinerant social group called *světští*, or "people of the world." From the little that has been written about them, we can infer that they share similarities with the German Yenish community and the Irish Travellers, organizing into close-knit family clans that travel around the country and offer amusements.[101]

By the 1980s, the funfair equipment formerly owned by the "people of the world" had been nationalized, and they themselves became employees of the so-called "Parks of Culture and Recreation," administered by regional governments, or, in some cases, by town halls. Theirs was a unique traveling *vnye* milieu; they avoided being "normalized" and retained their nomadic lifestyle while entertaining the socialist citizens at the behest of the Communist bureaucracy.[102] Like Tuzex stores, funfairs were windows into Western consumer culture. Thanks to their operators' connections to

Figure 5.4
Mobile video game arcades at a funfair in Prague, February or March 1989. The adults standing in front of the bus (bottom) are most likely waiting for their children or grandchildren (probably boys), who are playing inside. Photos by Karel Bucháček.

underground economies, they sold things and offered entertainments that were otherwise unavailable.[103] The "people of the world" quickly adopted new trends in the amusement industries, including arcade machines—compact and lucrative additions to their range of entertainment technology. Thanks to their novelty, the machines appealed especially to young people, including those who had micros at home but appreciated the better graphics and multiplayer capabilities of arcade games. In addition, traveling arcades made digital games temporarily available to demographics outside of the technical intelligentsia and computer club membership. However, although funfairs were technically open to all genders, interviews and photographs suggest that, like computer clubs, arcades too were predominantly male spaces.

Under heavy use, arcade machines tended to break. Most of them were secondhand (or black market) devices imported from the West, and there was no one around to repair them. The "people of the world" found an unlikely ally in Tomáš Smutný—a computer engineer, active member of Svazarm, and youth club instructor, as well as the twin brother of Eduard, the designer of the Ondra computer. Smutný recalls:

> In 1983, I happened to meet this funfair person in Benešov and what he had there were the first arcade machines. And I could see that it was a computer—at that time, I wasn't even aware that these were around. I told him: "Six of your machines are broken, let me fix them." And he said: "Mister, do not touch those, that's pointless—they're American, Japanese, you would break them."[104]

Driven by his curiosity and motivated by the challenge, Smutný persevered and employed his engineering skills, honed at a military school and Svazarm, to repair the strange new devices. Soon thereafter, he became the go-to repairman for a substantial part of the country's "worldly" community. He estimates there were thirty or forty mobile arcade trucks when he first got involved and that the number eventually increased to about a hundred. He quit his day job and started to drive around the country repairing arcade machines, sometimes even bringing his family with him; his son grew up playing the machines stationed in Smutný's workshop, beating all the competition in the mobile arcades. Smutný enjoyed this lifestyle because it allowed him hands-on work with advanced electronics and gave him considerable freedom. Moreover, it was lucrative. The machines swallowed two-crown coins at a breakneck pace, and the profits were "immense."

Smutný's involvement did not stop at repair. He and his "worldly" partners realized that instead of importing arcade machines for around fifty thousand Czechoslovak crowns (about a year's salary at the time), they could import just the main boards for thirty thousand, and build the cabinets in Czechoslovakia. Smutný started a cottage industry, in which he employed many tactics typical of the hobbyist practice of *bastlení*. He and his team built joysticks out of shower handles, and coins slots out of tin strips. Instead of the standard 56-pin JAMMA (Japan Amusement Machinery Manufacturers Association) connectors, they used telephone switchboard cables. They built their own power sources and coiled their own transformer windings. They obtained color TVs from Soviet soldiers stationed in a nearby base, who had been smuggling them into the country. All of this and more was housed in homemade cabinets built by a carpenter acquaintance. The resulting homemade arcade machines felt and played differently from legitimate ones. They lacked the marquees and the original controllers, and some gameplay modes were sometimes missing due to nonstandard wiring.[105]

Smutný says he built hundreds of such machines. At its peak, the operation employed ten people, fully within the realm of the shadow economy. Party authorities mostly tolerated these endeavors: "They never banned it, they were happy about it. Even though they knew that [the funfair people] were private entrepreneurs disguised as employees, they didn't mind, because the fairs gave kids and families the joy of play, which they always need."[106] However, the police thought otherwise. In late 1987, Smutný spent ten weeks in custody on suspicion of illegal enterprise because he was only licensed to repair, not to manufacture equipment.[107] The police brought in an expert witness to assess the extent of Smutný's economic activities. The expert, Vladimír Smejkal, ended up writing a favorable assessment in which he classified Smutný's endeavors as an "overhaul" rather than production, arguing that the board was the core of the machine and that neither Smutný nor anybody else in Czechoslovakia would have been able to produce such a board.[108] Soon thereafter, Smutný was released.

The assessment brings us back to the question of distribution and materiality. Deep inside Smutný's machines, there was an original board designed in Japan or the United States. During the "overhaul," however, the original package—as advertised and sold by the manufacturer—was reduced to the absolute minimum so that it could traverse the distance between the manufacturer and the funfair behind the Iron Curtain. The remainder was then encapsulated in locally improvised material housing. This reduction and replacement of the materiality of games mirrored the similar process in cassette-based informal distribution. Arcade machines were subject to hobbyist modifications, just as games were.

This chapter has put informal distribution at the center of my analysis and demonstrated the many ways in which it formed the experiences of Czechoslovak players. This is not to say there was a *complete* lack of official distribution of microcomputer software. The 602nd computer club of Svazarm and the Prague-based Program cooperative were selling several educational, productivity, and utility software titles, and Program also released several text adventures.[109] Czechoslovak Radio and even Czechoslovak Television were convinced by microcomputing pioneers to experimentally broadcast data as audio signal—similar to the United Kingdom, Poland, or Yugoslavia—but these experiments were rare and short-lived.[110] However, the overall impact of legitimate distribution was negligible, and *sneakernets* remained the number one source of programs for 8-bit machines. The plural form of the term emphasizes that different networks supplied software for different platforms. Although they coexisted with other networks of music and VHS tape trading, they were largely distinct, and employed different routes and data storage technologies. Demographically, the overlap between music fans, VHS owners, and micro enthusiasts was incidental rather than regular.

Czechoslovak sneakernets were structured around the geopolitical barriers and foreign trade regulations, but without any direct intervention from the authorities. The networks were improvisational and unruly, but also robust and reliable. Some contemporaries associated informal distribution with shadows and darkness. Perhaps a more fitting metaphor would be a cacophony of flickering, moving lights, representing people running around with cassettes and firing up their TVs to figure out new games. Sneakernets were experienced as *movement* rather than as a service or a set of formalized transactions.

Many Czechoslovak players—especially ZX Spectrum users at the end of the decade—lived in a state of paradoxical cornucopia, having access to numerous Western games despite being on the other side of the Iron Curtain. Activities associated with distribution—copying, collecting, and possession rituals—took up so much of players' time that some critics saw them as an impediment to meaningful interaction with the game.[111] Without informal distribution, however, the country would have had no microcomputer or gaming culture. Access to free software offset the exorbitant price of the machines themselves, and the widespread availability of games attracted young people to the world of microcomputers.

Informal networks were inhabited by unpredictable mediators rather than reliable intermediaries. In the course of their movement, games were being altered and modified, in line with Newman's claim that "the digital game is a potentially changing, unstable object."[112] Peripheral elements of the game were replaced by or appended with vernacular material—cracker credits, fan-made loading screens, or hand-drawn cassette inlays. Arcade machines were similarly disassembled before their journey and reassembled from domestic resources after coming into the country. In both cases, the original was unattainable and perpetually deferred. Yet despite the modifications, some things remained unchanged. It was a shared understanding that in-game graphics, game mechanics, and maps constituted the identity of a game. Adding poke options or loading screens resulted only in a new version, not a new game.

My focus on Czechoslovakia in the 1980s should not hide the fact that similar practices were also common elsewhere. Many of the features of the Czechoslovak distribution network are not necessarily typical of a Czechoslovak or Soviet bloc experience, but rather an unauthorized, peripheral experience. People who played games in South America or most of Asia may find my account familiar, and so might those who lived in Western Europe or the United States but could not afford (or chose not to purchase) legitimate copies of games. In Finland, for example, piracy was so rampant that the efforts to create a domestic games market foundered and Finnish programmers wrote for the British market instead.[113] In fact, in the 1980s, before international retail infrastructure and, later, digital distribution came into place, peripheries were arguably larger than centers, and much of the microcomputer world was running on pirated copies of games.

6 Bastard Children of the West: Establishing a Domestic Coding Culture

> Imagine that you're sitting at a computer, have BASIC at your disposal and you desperately want to program anything that moves. … Of course, there were inspirations. Somebody brought in the German *Chip* magazine. We couldn't read German, but there was a photo of an office and a game on the computer screen. You looked at it and said, hey, that's it. And made a game based on that tiny photo.
>
> —Programmer Vít Libovický explains the sources of inspiration for his early programming efforts[1]

Game history narratives have largely been dominated by talk of the founding fathers and their original contributions to canonical games. They tend to tell histories of "firsts."[2] But much of game culture, and of culture in general, is based on imitation of foreign templates. Roman poets imitated Greek classics, and European romantics imitated Dante.[3] Japanese role-playing games such as *Final Fantasy* started out by copying Western titles, and id Software—the creators of *Doom*—started out by imitating Nintendo's platformers.[4] Their makers may have "matured" to produce works we might call original, but imitation was a necessary part of their creative practice. However, by saying that cultures are built on imitation, I do not intend to diminish the value of such cultures. On the contrary, I wish to ascribe value to the processes of transformation and imitation, which tend to be underestimated in comparison to a romanticized ideal of an original work.

Discussion of imitation within the game industry has been framed by the issues of intellectual property and industry norms. The lines between copyright infringement, ethically unacceptable copying, and acceptable inspiration have been hotly debated, but remain fuzzy.[5] The overwhelming majority of 1980s Czechoslovak games were amateur productions created outside of an industrial context. Many of them, especially the early ones, were *ports*, *conversions*, or *clones* of existing works. Whereas the first two categories refer to reworkings of existing games for other platforms, a *clone* can be understood as a separate title that heavily draws from an existing game in terms of gameplay, art, or both.[6] Rather than setting ports, conversions, and clones aside as indices of the peripheral position of Czechoslovak game culture, I want to demonstrate that they all deserve the privilege of being analyzed as stand-alone, historically significant objects. This chapter will show how Czechoslovak programmers were imitating and transforming foreign influences and integrating them into their practice.

As seen in the previous chapter, the Czechoslovak microcomputer enthusiasts of the 1980s were immersed in foreign, mostly English-language software. This software arrived in unpredictable ways and did not originate in a single center. The center's location was shifting and platform-specific. Spectrum users mostly played British and Spanish games, Atari 8-bit owners played predominantly US titles, and the Commodore 64 was under a combined influx of North American and—especially later in the decade—numerous British and some German titles.[7] How did users relate to Western computing and gaming cultures? First and foremost, their knowledge of foreign game industries and cultures was limited by language and information barriers. Czechoslovak players' visions of Britain or the United States were fragmentary, assembled from scraps of popular culture, borrowed or photocopied magazines, hearsay, fantasy, and ideologically skewed reporting by the official media. For the last Soviet generation, writes Yurchak, the West was "an imaginary place that was simultaneously knowable and unattainable, tangible and abstract, mundane and exotic."[8] Western culture, including music, film, and consumer products, served as a marker of coolness, worldliness, cultural capital, and youthful revolt—or, as Urry puts it, a vehicle for "imaginative travel."[9] In everyday practice, wearing a Western T-shirt, watching American films on pirate videotapes, or playing British computer games did not express a straightforward affiliation with Western values. Rather, it was a link to an imaginary elsewhere that was deterritorialized and *vnye*—neither Czechoslovak nor Western. In computing and games, Western influence was even more pronounced, but also more depoliticized. Consumption of Western film or music was an act of deliberate choice because there was, after all, plenty of Czechoslovak and Soviet bloc music and film to choose from. However, in the case of computers, next to no domestic counterparts were available. For a microcomputer user, it was impossible to *not* engage with Western hardware and software.

I have previously likened the hobby programmer to the *bricoleur* who creates objects from "whatever is at hand." According to Lévi-Strauss, the bricoleur's selection of tools and resources is heterogeneous "because what it contains bears no relation to the current project, or indeed to any particular project, but is the contingent result of all the occasions there have been to renew or enrich the stock or to maintain it with the remains of previous constructions or destructions."[10] To paraphrase Henry Jenkins's observations about science fiction fans, Czechoslovak programmers stitched their works together from material washed ashore by the unpredictable currents of informal distribution, as well as from other elements of local and foreign popular culture.[11] They reused genre templates, source code, images, routines, and technical solutions from Western software, often arriving at unexpected combinations. The effective lack of copyright protection, combined with a tinkerer attitude, allowed for an uninhibited proliferation of modifications and appropriations. The resulting influence was less a product of cultural imperialism or a globalized media industry than a bottom-up work of individual bricoleurs, who filtered and twisted cultural flows. The modifications they performed did not necessarily result in legitimate and original releases, but in what I shall call *bastards*.

I adopt the term *bastard* for an illegitimate port, conversion, clone, or other appropriation unaccounted for by the authors and publishers of the source material. I build on a distinction developed by translation scholar Manuela Perteghella between the official translation (which is "the rightful inheritor" of the original) and unofficial adaptations. She calls the latter "bastard children"—"illegitimate" and "impudent," but "lively and devil-may-care."[12] Perteghella demonstrates how these bastard children are often better adjusted to their host culture. Along those lines, I aim to present how the transformation of foreign games was affected by the local cultural context. I am aware of the negative connotations of the term *bastard* and its connection to elitist discussions of "bastardization" of culture, but I will dispense with such normative biases. Instead of protecting the purity of the original, I want to celebrate the often wonderful "bastard children" and their role in the formation of peripheral gaming cultures. As Riccardo

Fassone notes in his study of an Italian version of the 1983 British game *Attack of the Mutant Camels*:

> [The version demonstrates] the existence of alternative local game histories that cannot be retraced only through exceptional cases (the occasional masterpiece, the notable example, etc.), but should be understood through the analysis of lower intensity processes and practices and of non-exceptional, often banal, cases such as adapted clones.[13]

Although the concept of *bricolage* fits the approach of Czechoslovak amateurs, it is too general to match the astounding breadth of transformative practices that resulted in these bastard children. I will therefore follow up on the previous chapter and adapt another concept from Gérard Genette's framework—*hypertextuality*. Ports, conversions, and clones manifest such hypertextuality, defined by Genette as a relationship between a preexistent text (a *hypotext*) and one that was derived from it (a *hypertext*).[14] The derivation can be either a "simple" *transformation* of an existing work, or an indirect *imitation*, essentially a new work that closely follows an existing template.[15] Although the boundary between the two is blurry, we can—at least for now—posit that ports and conversions are results of overt and holistic *transformation*, whereas clones or fan sequels (which Genette calls continuations) are works of *imitation*. Genette goes on to develop a broad inventory of categories of derived texts, which I will borrow from when discussing individual examples. Note that Genette's framework was primarily designed to deal with the medium of the codex book, whereas a computer game is arguably a structurally more complex, multilayered object. This is not necessarily an impediment to the use of his concepts, because we can extend the notion of hypertextual relationships to cover the levels of—for example—code, rule systems, written text, and audiovisual material.

This chapter will start with an overview of Czechoslovak homebrew scenes before presenting four case studies of local appropriation of Western or Japanese material. First, I will focus on unofficial ports and conversions and the ways these transformations built on and departed from previous versions. Then I will move to imitations of Western shooter games, made by local programmers to show off their coding chops. The last two sections will examine text adventure games and hacking games—examples of more indirect imitations of and homages to Western games and popular culture, which stood at the inception of specifically Czechoslovak trends.

Czechoslovak Homebrew Scene

To understand what kinds of games were being produced in Czechoslovakia and for what reasons, we need to shed light on the dynamics of its homebrew scene. The scene worked with foreign sources and imitated foreign practices, but it also had its regional and national specifics. The two best-documented points of comparison are the so-called bedroom coding typical of the early 1980s United Kingdom and the Western European cracker and demoscene.

Czechoslovak programmers were very much indebted to the United Kingdom's game development and admired British bedroom coders, who made their hit games after school or in their spare time. However, there was an important difference. British bedroom coding might have started out as a homebrew endeavor, but it almost immediately became interconnected with a bustling industry. The successful bedroom coders soon moved into offices.[16] This was not happening in Czechoslovakia. As programmer and writer František Fuka described it in his 1988 account:

> The few individuals that make games in our country can naturally hardly compete with teams of specialists, for whom making games is not only fun, but also a job (a paid one, of course). One person can hardly be a good author of a game idea, a programmer, a graphic artist, a musician and also have enough time to pull it all off.[17]

Instead, Czechoslovak production fits the five characteristics of *homebrew* game production put forward by Melanie Swalwell—"domestic location, amateur programmers, sole creators, local distribution, and experimental ethic"[18]—with one notable exception, which is that some games were made in clubs instead of homes.[19] Despite mostly consisting of "sole creators," the Czechoslovak homebrewer community formed what sociologists and cultural studies scholars would call a *scene*. Its members were connected to a "thematically focused cultural network," which overlapped with the pathways of informal distribution.[20] The scene revolved around a set of social "practices of consumption, of production, of interaction," a shared history and taste, much as independent music scenes did.[21]

Another useful point of comparison is the Western and Northern European cracker and demoscenes, encountered in the previous chapter. Although this scene revolved around different types of *coding acts*—cracks or demos as opposed to games—it had many features in common with the

Czechoslovak scene. Both were home to mostly young male enthusiasts who showcased their technical expertise. They positioned themselves outside the legitimate industry, participating in a meritocratic "honor economy" in which members sought recognition and social capital rather than money.[22] As Western crackers did in their crack intros, Czechoslovak amateurs incorporated greetings and shout-outs into the paratextual elements of their games. The Czechoslovak homebrew scene was nevertheless more porous and less fiercely competitive. As opposed to the shadowy existence of some Western groups, it developed in part within the *vnye* environment of legal, state-sponsored computer clubs. The rest of this section will lay out some of the basic features of this scene.

Formation

Homegrown software production was a part of hobbyist practice throughout the 1980s. The community underwent a significant evolution, however, as ownership of microcomputers rose from hundreds to tens of thousands. In the early 1980s, Czechoslovak programmers were not necessarily trying to "release" games. Besides private tinkering, they were adapting programs from BASIC listings and images in domestic and imported magazines.[23] When the informal distribution network became more dense and efficient in the mid-1980s, Czechoslovak authors started to clone, convert, and imitate Western commercial games, and in the process, they learned how to program and design their own. At the same time, the distribution network allowed them to release their work to an audience; they could perform coding acts addressed to an interconnected community that evaluated and appreciated their work. The resulting sense of a shared space was manifested in the first collaborations and paratextual greetings, which appeared around 1986. Releases could be motivated by several reasons, all of which were social—games could be a showcase of coding skills, an attempt to initiate communications with other members of the community, or an expression of their authors' experiences and opinions.[24] I will speak about all these motivations in more detail further on.

Demographics

Like bedroom coders, crackers, and demoscene members, Czechoslovak homebrewers were mostly young men, usually high school or university students. As Fuka noted in 1988, Tomáš Rylek—reportedly one of the best game programmers in the country—had to take a break from making games "because he was studying for his graduation exam."[25] Older hobbyists contributed fewer titles because they did not share their younger cohorts' enthusiasm for games and had less leisure time to spare. After all, trading, playing, and making games required continuous and serious time investment.

Most homebrewers had at least tentative connections to computer clubs, which served as proving grounds for collaboration and competition in programming. As observed in chapter 3, the clubs were overwhelmingly male, and the same could be said about the homebrew scene. Out of the preserved titles, I have found only two that were cowritten by women. Neither of the women were primary authors, and both were only marginally involved with computers and gaming. Hana Gregorová is credited as a coauthor of artificial intelligence in *Piškworks*, a computer adaptation of an analog tic-tac-toe–style game.[26] Her chance contribution materialized while her older brother was designing the game's pattern-matching algorithm, and Hana came by and suggested her own solution. The idea was core to the game's functionality, but she did not code it, nor was she involved in a collective that would have allowed her to capitalize on her contribution. The other woman coauthor, Alexandra Štorkánová, collaborated on the game *Six Hits into the Hat*, an adaptation of a TV quiz show, by recording and inventing trivia questions. The game was made by a group of teenagers who primarily engaged in amateur filmmaking, a less gender-biased pastime than computing.[27]

Although women were rarely involved, the scene's output does not indicate that it was misogynist. The 1980s homebrew games are mostly devoid of narratives and imagery that objectify or demean women, save for a few offhand jokes in scrolling messages, one strip poker game, and one erotic tile-matching game.[28]

Geography

The role of geography in any kind of scene is a matter of debate. Some music scenes—such as the Madchester scene—are tied to a single locale, but others are not.[29] The cracker and demoscenes are usually considered international.[30] What is striking about the Czechoslovak homebrew scene is its confinement to its national space. Although Czechoslovaks traded software with Polish, Hungarian, and Yugoslavian users, my material does not show evidence of persistent two-way communication or collaboration between Czechoslovakia and other Soviet bloc or Western countries. Due to a combination of travel restrictions and language barriers, the Czechoslovak scene was in an intriguing state of semi-isolation. Games flowed in but they rarely flowed out. This led to an emergence of game design elements and themes that were uniquely Czechoslovak—discussed later in this chapter and the next.

In terms of national geography, homebrewers worked at homes and clubs all over the country. Thanks to the concentration of technical intelligentsia and computer clubs, big cities dominated, with Prague in the lead. Some of the best-known domestic software for the Spectrum was made by authors connected to the Station of Young Technicians in Prague's sixth district, and to the 602 and 666 clubs. The Slovak capital of Bratislava was home to the Sybilasoft collective, makers of popular comedy, science fiction, and detective text adventures; and the Silesian city of Ostrava was home to the 415 club and its VBG Software team, who were authors of numerous conversions for the PMD 85 computer (see table 3.1 in chapter 3). Several programmers resided in small towns or villages, including Martin Malý (Demonsoft) from the Central Bohemian village of Chleby, and Viktor Lošťák from the Northern Moravian town of Nový Jičín. However, most of the country was densely inhabited, and even these two commuted to computer clubs in nearby towns.

Self-Stylization

One of the defining features of scenes is "collective self-stylization," built in this case on imitation of Western sources.[31] Although most homebrewers were members of official computer clubs, they rarely advertised this fact in their games' credits, a fact that highlights the pragmatic nature of their affiliation to Svazarm and other socialist organizations. Instead, they took inspiration from foreign developers, creating Western-sounding labels and logos such as Cybexlab Software, Demonsoft, or Hellsoft, and even referring to each other as "companies," "firms," or "labels" rather than individuals in their in-game greetings and nods.[32] However, having one's "firm" was a mimesis of Western templates rather than a realistic ambition. Even in cases where the game's form and content mimicked Western titles, its goal was not to address a Western audience, but to address a domestic audience familiar with Western games.

Platforms and Coding

Coding know-how and distribution networks for 8-bit machines tended to be platform-specific, and any regional or national scenes were therefore segmented along platform lines. In this book, I will refer to a general Czechoslovak scene as well as to platform-specific (sub)scenes. The latter were independent to an extent, but through clubs and friends, users would draw inspiration from the games available for other machines. This was an essential prerequisite for porting and converting.

The Spectrum scene was the most active and original. A continuous stream of games for the platform started appearing in 1985, totaling over 120 preserved titles made before 1990. Work for the PMD 85 also began in the mid-1980s, but access to this school- and club-only machine was limited, resulting in about thirty games. The Atari and Sharp MZ 800 platforms followed around 1986 with over forty each. Given the lack of official releases for the PMD 85 and the MZ 800, much of their catalogs consisted of conversions and ports of titles from other machines.[33]

As for programming languages used, the games ranged from simple BASIC programs to cutting-edge action games written in assembly language or machine code. The most sophisticated games were made for the Spectrum and the PMD 85. The Spectrum's design was especially open to software bricolage. Its lack of specialized chips resulted in a transparent and intuitive architecture, which allowed homebrewers to master it relatively easily without the help of detailed manuals. My interviewees who worked on the Spectrum have, for instance, praised the ability to "change the color of any pixel at will"[34] by simply writing a byte into RAM. An equivalent result required a significantly more complex operation on an 8-bit Atari or a Commodore 64.[35] Despite its outdated CPU and monochromatic graphics, a lot could be achieved on the PMD 85, too. Karel Šuhajda of Prague's 482 Svazarm club was the only Czechoslovak programmer to continuously use isometric 3-D graphics on any platform. He reached a pinnacle in the PMD 85 game *Hlípa*, strongly inspired by British titles such as *Knight Lore* and *Alien 8*, and usually considered the best nonconversion title for the platform (see figure 6.1).[36]

Output

Overall, fan archivists have gathered between two and three hundred games that can be identified—based on paratextual information and other evidence—as having been created in Czechoslovakia before 1990; the figure

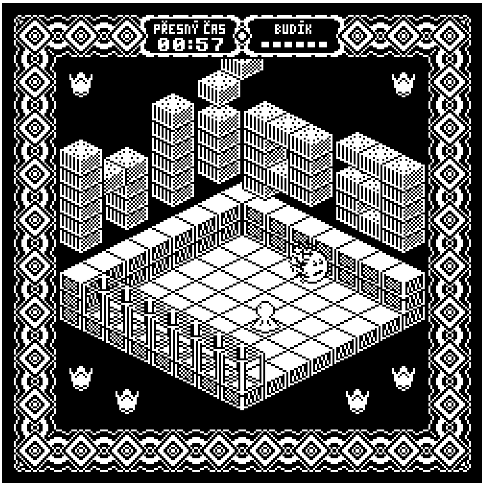


Figure 6.1
The opening room of *Hlípa*, a 1989 PMD 85 game by Karel Šuhajda. He later converted it to the Sharp MZ 800 and the Atari ST. Image courtesy of BBS Děčín.

includes conversions.[37] Naturally, any such collection is perpetually incomplete and unrepresentative of historical reality. Some games have not been preserved or discovered; every now and then, several new titles surface in online archives or discussion forums as veteran users digitize and share their collections. Many more games were never distributed because their creators produced them as private experiments for their own pleasure. Nevertheless, to get a sense of the scene's scale, consider how the figure compares with other 1980s scenes or industries. It is definitely smaller than the United Kingdom's, where over six thousand games were released for the Spectrum alone (though that figure includes both commercial and homebrew productions). Comparisons to other Eastern European countries are difficult because of the lack of reliable data. The only available, comparable figures are from international fan archives for 8-bit machines. Unfortunately, only two of these—World of Spectrum and Atari Mania—include country metadata. According to those, Czechoslovakia released by far the most games for the Spectrum and the second-most for the Atari within the region (see table 6.1). Comparing total number of games per country, regardless of platform, would bring little benefit because of incomplete data. Perhaps the only sensible platform-independent comparison is between Czechoslovakia

Table 6.1
Counts of games released 1980–1989 per country in the World of Spectrum and Atari Mania databases.

	Czechoslovakia	Poland	Yugoslavia	Hungary	USSR	UK	Spain
Spectrum	92	53	44	15	7	6,589	1,318
Atari	21	25	0	0	0	1,047*	1

Note: Other Eastern European countries (Bulgaria, Romania, Albania) have zero entries. Data for East Germany is missing. With the exception of the UK Atari figure (*), figures exclude educational titles. Data as of June 2017.

and Poland, where the Spectrum and Atari (for which we have data) were the top platforms. In this case, Czechoslovakia scored more games than Poland, despite lacking a software market and having a population two-fifths the size of its northern neighbor.

Overall, the existing data suggests that Czechoslovakia had one of the most productive homebrew scenes in the region. Of course, the alternative explanation would attribute the high number of preserved games to the dedication of its fan archivists, but such dedication would be unlikely without a history of an active scene.

Game Design

Although our focus is on games, homebrewers commonly engaged in a wider array of coding acts. They cracked, poked, and wrote other kinds of software. Writing one's own assembler, text editor, graphics editor, or music reproduction utility was also an achievement. Occasionally, there was even a chance of getting paid via commission from Svazarm to program "serious" software. However, *demos* did not play a significant role in the 1980s, possibly because the international demoscene revolved around the Commodore machines, then a marginal platform in Czechoslovakia.[38] Nevertheless, games were to the Czechoslovak scene what demos were to the Western Europeans—they provided the community's default genre of work.

To reiterate Lévi-Strauss's distinction, local homebrewers were software *bricoleurs* rather than software *engineers*, usually progressing without a detailed, premeditated plan.[39] Unlike many of their professional counterparts in the West, they did not have dedicated development tools or extra machines to use for debugging.[40] The moment when a game was finished was often determined not by fulfillment of the plan, but by the author running out of available memory or becoming exhausted or bored with the project.[41] Many games started out as proofs of concept—made to demonstrate a single technical or gameplay idea.[42] Given the homebrew nature of production and the young age of most creators, only a few games from the era are well-designed, user-friendly, or sophisticated enough to interest today's players for reasons beyond historiography or nostalgia. That said, the point of this book is not to unearth lost classics, but to write about *why* and *how* people made and played such games. After all, the same criticism could be raised about many Western games of the time, which were often repetitive, frustrating, and uninspired.[43]

Ports and Conversions

Lists of early Czechoslovak titles show that many of them were unauthorized "bastard children," namely *ports* or *conversions* of titles from other platforms. Despite deriving from existing games, their own influence was often immense, and the act of their making no less notable than that of the originals. As Gazzard writes of the related practice of "arcade cloning" in the United Kingdom, these projects allowed users to engage in the hobby community:

> Beyond the commercial aspect of licensed ports, the arcade game re-make or clone allowed for a sense of gaming community amongst home computer users for those wanting to learn how to program and develop games further. We can understand these coded practices as sites for shared expression amongst players and developers alike.[44]

Although it is often sidelined in histories of games, comparative analysis of ports and conversions has been a favorite pastime of retro gaming enthusiasts, ideal for demonstrating their knowledge of old machines.[45] Ports and conversions are just as valuable to scholars. As Fernández-Vara and Montfort suggest, "modifications, expansions, and omissions in ports can help us understand the affordances of different platforms" and "the defining features of a particular game."[46]

Unfortunately, the existing literature on the topic tends to collapse the categories of *ports* and *conversions*.[47] My material shows that the distinction is useful because it allows us to differentiate between two quite specific types of programming practice.[48] I will base my understanding of *port* on the concept of "portability," which was already in widespread use in software engineering literature of the 1970s. According to a 1978 definition, "Portability is a measure of the ease with which a program can be transferred from one environment to another; if the effort required to move the program is much less than that required to implement it initially, and the effort is small in an absolute sense, then that program is highly portable."[49] Portability is easier to achieve when using higher-level programming languages such as C; in such cases, the source code can be compiled for different targets. Games written in BASIC, too, could be relatively easily adapted to other machines' dialects of the language. However, most 1980s commercial titles for 8-bit computers were written in assembly language or machine code and directly accessed the machines' CPUs and hardware, making them difficult to port. Among those titles, a degree of portability was possible on platforms that shared the same CPU, such as the ZX Spectrum, Amstrad CPC 464, and Sharp MZ 800, all of which employed the Zilog Z80 processor.[50] In these cases, we can truly speak of *ports*.

In the Western European game industry, porting was especially frequent between the Spectrum and Amstrad platforms. Czechoslovakia, on the other hand, was home to prolific porting to the Sharp MZ 800, a Japanese computer released in 1985 that remained obscure in most of the Western world. Thousands of these machines were imported into Czechoslovakia in the second half of the decade (for details, see chapter 2). In a shortage economy, even an unknown and unsupported machine such as the MZ 800 sold quickly, resulting in a sizable exclave of Sharp users who soon discovered that almost no one was making original games for their machine. Luckily, the MZ 800 had the same CPU and roughly the same display resolution as the ZX Spectrum, allowing for convenient porting of Spectrum games. In the mid-to-late 1980s and early 1990s, Czechoslovakia became a Sharp gaming superpower, as various groups and individuals from different parts of the country collectively created a library of over a hundred ports for the machine. Due to differences in display handling, the "lazier" ports were black-and-white, but many others were even faster and more colorful than the Spectrum originals. The library of unofficial releases included Spectrum classics such as *Knight Lore* as well as later hits such as *Exolon*, the game whose journey we traced in the previous chapter and which surfaced on the MZ 800 in two separate versions by two different programmers.

We may postulate that, in contrast to *ports*, the code of *conversions* is written from scratch. In the 1980s game industry, programmers would often be tasked with creating a conversion of an arcade game without having access to documentation, source code, graphics files, or even the arcade machine itself. They had to infer game mechanics from their own gameplay and laboriously redraw images from video recordings.[51] If a faithful re-creation of the original was impossible due to hardware limitations or insufficient information, the converters had to make their own creative decisions. A well-known example is the acclaimed Spectrum conversion of the arcade game *Midnight Resistance*, which replaced realistic character graphics with more cartoonish, chunkier ones in order to accommodate the target's display limitations.[52]

In Czechoslovakia in the 1980s, ports and conversions were motivated not by commercial interests, but by the desire to demonstrate one's technical expertise, or by the shortage of games for a platform. One of the earliest preserved Czechoslovak titles is a remarkable conversion of the 1983 British hit *Manic Miner*, from the Spectrum to the Sinclair ZX81, a machine that was—to simplify—technologically one generation behind the Spectrum. In the summer of 1984, Aleš Martiník (born 1964), a student at the Brno University of Technology, took up the challenge. His motivation for writing the game was "twenty percent the desire to play *Manic Miner*, even if I only had a ZX81. … And the remaining eighty percent was the desire to prove to myself, but mainly others, that I am good enough to pull it off, at both of which I succeeded."[53] He succeeded to the degree that his effort continues to impress fans of 8-bit computers more than twenty years later.[54] This is mainly because the game replicates the relatively high-resolution 256 × 192 pixel graphics of the Spectrum on the inferior ZX81, which, by default, only allows the use of monochromatic character-based graphics.

Martiník's work-around requires at least a short and simplified explanation.[55] The ZX81 refreshed the display, line by line, at a frequency adjusted to the refresh rate of a regular TV set. Each ZX81 character had 8 × 8 pixels, so for each single line, eight pixels of each character were drawn—for instance, the horizontal line at the top of the letter *T*. All the character graphics data was stored in the machine's ROM. The display was monochromatic, so each pixel could be represented by either a one or a zero, which meant that the eight pixels corresponding to one line of a character could be represented as one of the integers from 0 to 255, equivalent therefore to one

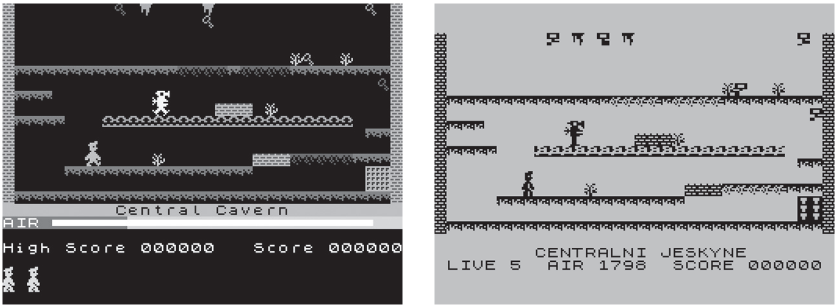


Figure 6.2
First level (the “Central Cavern”) of the original *Manic Miner* (left) and the ZX81 conversion (right). Note the slightly jagged, irregular edges of sprites, an effect of the game’s nonstandard pseudo-high-resolution graphics. The irregular graphics also made collision detection less precise, rendering the conversion more difficult—hence the need for five instead of two lives and a more convenient layout of in-game objects. (*Manic Miner* TM & © 1983–2017 B. B. Siddiqui. Use authorized by Elite Systems Group Ltd. Developed by Matthew Smith.)

byte of memory. However, Martiník wanted to draw graphics, not letters. To achieve this, he replaced the display routine with a custom one and executed it before each horizontal line was drawn on the screen. Instead of drawing the top of the letter *T*, the routine would fetch a byte from elsewhere in the ROM to draw the eight pixels it needed to assemble *Manic Miner*’s cave and sprite graphics. This was repeated for each character and line. Unfortunately, the ROM did not contain the whole range of values from 0 to 255, and if the correct value was missing, the program had to resort to using the closest possible value instead. This resulted in slightly jagged graphics with misplaced pixels, as shown in the screenshot (see figure 6.2).

What Became of Flappy

The previous section discussed the process of porting to the Sharp MZ 800. Games were converted in the opposite direction, too. That was the case with *Flappy* for the PMD 85, which is—thanks to its author’s extensive notes and diaries—possibly the most thoroughly documented Czechoslovak conversion (see figure 6.3).[56] Its example will show how converting was far from a straightforward technical operation, and how deeply the process was embedded in the local context.

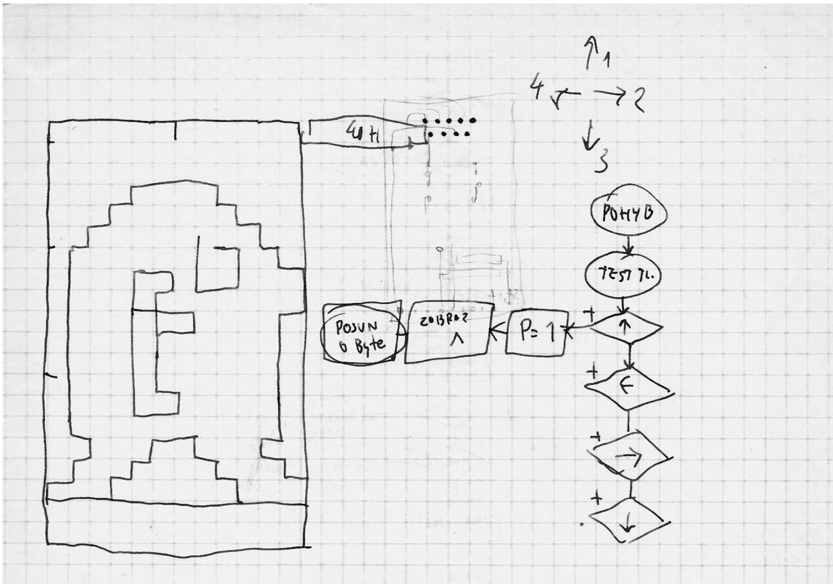


Figure 6.3
A page from a design document of the PMD 85 version of *Flappy*, showing the main character sprite and the structure of the character control routine. Note that the position of hands/wings was not clear from the outset. Personal archive of Vlastimil Veselý.

Just before Christmas 1986, the 415th club of Svazarm in the city of Ostrava purchased a Sharp MZ 800 computer at the local Tuzex store. The machine came bundled with a copy of *Flappy*, an action-puzzle game by the Japanese company dB-soft.[57] Club members were impressed by its elegant and intricate gameplay, consisting of the titular character pushing a block to its destination in two hundred maze-like screens. In late January of 1987, Vlastimil Veselý (born 1963), one of the club's more experienced programmers, set out to write a conversion of *Flappy* for the Czechoslovak "school computer" PMD 85, another platform available in the club. Because the PMD 85 was based on a different CPU, straightforward porting of machine code was out of the question. Veselý remembers:

> It was a bet of sorts. "Bet you wouldn't pull that off!" And I said I would. So I copied all the critters from the screen with a pen. I translated the drawings into code and just started writing the game. Two months later, I gave up, because initially, I hadn't given it much thought and started out wrong. … Then I rethought it and started over, until it was done.[58]

As were many of his homebrewer peers around the world, Veselý was motivated to showcase his skills. A conversion was, in a sense, a more accessible coding act than creating one's own title because the game's design was already in place. This was convenient for Veselý, who admits he has always been a better programmer than designer or screenwriter.[59]

The game was released on June 30, 1987, under the VBG Software label, a group that included three of the club's members: Veselý, Libor Bedrlík, and Ladislav Gavar.[60] Although written from scratch, Veselý's *Flappy* is a remarkably faithful conversion, unlike Gavar's more liberal reworkings of *Boulder Dash* and *Manic Miner*.[61] The only minor gameplay difference awaits on level 160, whose layout—originally arranged to read "SHARP"—was replaced with one that read "TESLA" as a nod to the PMD 85's Czechoslovak manufacturer.

Although in black-and-white due to the PMD 85's display limitations, the game's visuals are also very faithful (see figure 6.4). But focusing on individual character sprites, we can see another change, which—although seemingly subtle—begets questions about the identity of the main character. In both versions, the character's movement animation is achieved by cycling through two basic frames, each of which is distinguished by different leg and hand positions. In the original, the hands move downward, but in the conversion, they move upward, making them look like wings rather than tiny hands. In addition, Veselý replaced the original's blue stone with a white one to compensate for the lack of color. The conversion's paratexts

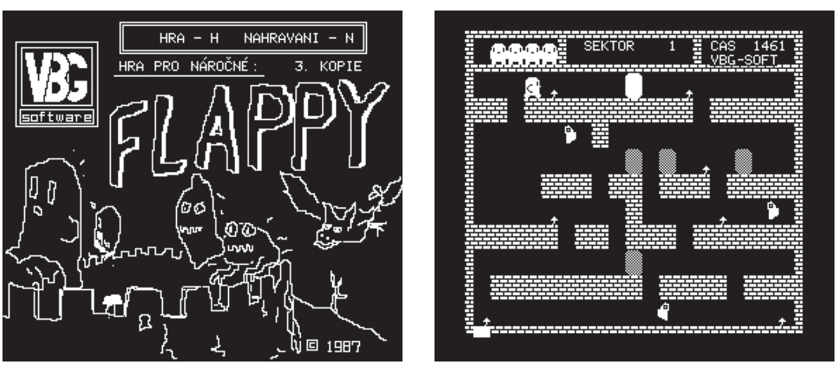


Figure 6.4
Loading screen (left) and in-game (right) screenshots from Vlastimil Veselý's *Flappy* conversion.

also diverge from the original. Whereas the Japanese packaging illustrations (which were probably not included in the Czechoslovak bundled version) depict Flappy as a mole-like creature pushing a blue rock in a mine, the PMD 85 version's loading screen shows a more bird-like Flappy chased by monsters on castle battlements (see figure 6.4). The latter's instructions even read: "Help the chicken called Flappy get the white egg onto the white platform." This claim about the character's species had little bearing on the gameplay itself, but likely affected subsequent conversions.

Thanks to its availability at schools equipped with PMD 85s, *Flappy* started gaining recognition as a classic even outside the Czechoslovak Sharp community, and ranked among the most converted titles. Three more Czechoslovak conversions followed—for the Atari 8-bit, ZX Spectrum, and Commodore 64—each taking more liberties than Veselý's version. The Atari conversion was the only one of the three that replicated the level layouts, but it replaced the Flappy sprite with a completely different frog-like creature.[62] The Spectrum and Commodore versions attempted to preserve the sprite, but featured different levels.[63] Both were probably based on the PMD 85 version, and they reiterated the "chicken and egg" narrative. The Spectrum version went even further, narrating that "Flappy the chicken" is on a mission to deliver its unhatched brother to the mother hen—a story far removed from alien mines of the Japanese original.

The Flappy case illustrates the transformations that games underwent as they were unofficially converted from one platform to another. Thanks to conversions, an obscure Japanese game was given a second lease on life in a completely different context. In an environment without copyright control and with original documentation unavailable, features were lost and added as in a game of telephone. Some parts of the game were localized; the unknown and unspecified aspects of the original were replaced by familiar, localized interpretations. A mole became a chicken, and a mine was replaced by a castle, a typical Czechoslovak landmark.

These transformations show what the authors "imagined to be the most important aspects of the game and what [they] thought was expendable."[64] In other words, they offer a range of emic understandings of what makes up the identity of the game. The basic mechanics were preserved; there was less consensus about the main character and the level layouts; and the backstory was considered tangential. Before we read this as a confirmation of the ludologist thesis about the primacy of rules, note that some conversions may preserve the story and replace mechanics.[65] For example, the graphic adventure *Leisure Suit Larry in the Land of the Lounge Lizards* underwent a Czechoslovak "conversion" into *Lazzy Larry*, a text adventure with completely different puzzles in more or less the same setting.[66] We might therefore conclude that in story-based titles, narrative—and not mechanics—could be considered a defining element of the game. All abovementioned transformations highlight the fact that conversions, and to a lesser extent ports, are acts of creation, not mechanical transfer. To reiterate our point from the previous chapter, the makers of conversions were *mediators*, not *intermediaries*.[67]

Forging the Shooter

When the first (mostly British) commercial games for the ZX Spectrum started to arrive in Czechoslovakia in 1982–1983, they were greatly admired. In the competitive environment of the booming UK game industry, coders had the incentive to outperform each other and take full advantage of the limited hardware. Anyone who wanted to implement arcade-paced animated action on 8-bit computers had to give up on higher programming languages and work directly in machine code. The task of learning machine code was daunting to many, but some Czechoslovak coders started to imitate Western games and, in the process, learn and hone their craft.

In the mid-1980s, a trio of talented Czechoslovak teenagers, later known as the Golden Triangle, rose to the challenge. Tomáš Rylek, Miroslav Fídler, and František Fuka had been meeting at the Station of Young Technicians in Prague's sixth district (see figure 6.5). As seen in chapter 3, the Station was *the* place to learn about microcomputers, and its extracurricular computer courses for young people were overseen by some of the country's leading and most forward-thinking experts. In the Station's open, experimental, and competitive atmosphere, the trio engaged in various coding acts, mainly on the ZX Spectrum, often staying up late into the night. They started working in BASIC but soon realized it was painfully slow.[68] While looking for "pokes" and "cracks," they delved into the depths of machine code.

Rylek (born 1971) was the youngest and the most analytical-minded of the three, and, as he puts it himself, "always a reverse engineer of sorts."[69] He enjoyed peering at listings of disassembled machine code and learning other people's tricks. One title that particularly impressed him was *Terra Cresta* (a

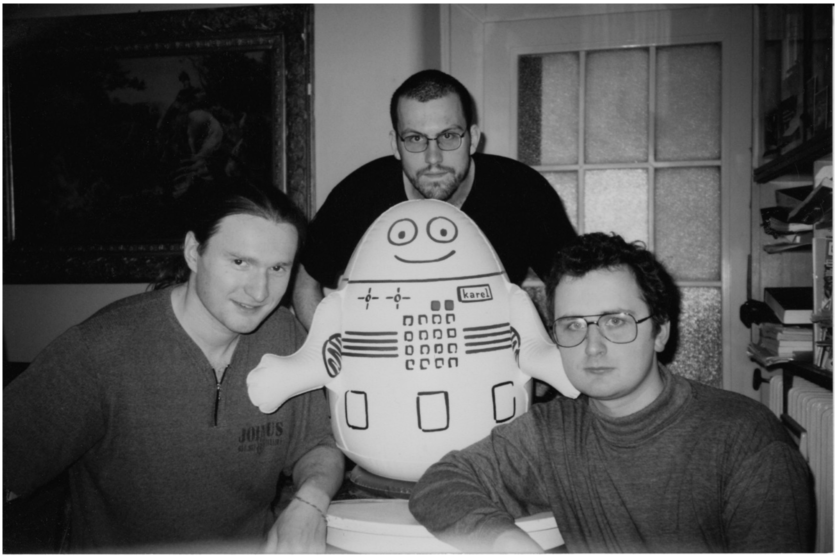


Figure 6.5
The Golden Triangle in the 1990s, with an inflatable Karel the Robot toy (see chapter 3) in the middle. From left to right: Tomáš Rylek (TRC), František Fuka (Fuxoft), and Miroslav Fídler (Cybexlab). The photograph was taken in Fuka's apartment. Image courtesy of František Fuka.

rather loose conversion of the Japanese shoot-'em-up arcade game), written in 1986 by the British programmer Jonathan F. Smith, who was revered by the trio as a top Spectrum coder.[70] Given the Spectrum's limitations—it was designed to handle neither sprites nor scrolling graphics—*Terra Cresta* was quite an achievement. Rylek, with help from his colleagues, discovered that such animation was only possible when the sprites were placed into the video memory in a specific order, so that they were always displayed before the light beam of the TV set refreshed the screen. In other words, he had to "race the beam," much as the programmers on the Atari 2600 console did.[71]

Rylek's own 1987 shooter games *Star Fly* and *Star Swallow*, released on his TRC label, sought to replicate the achievements of *Terra Cresta* while trying to pass for Western games (see figure 6.6).[72] Neither bears any surface signs of being produced in Czechoslovakia, like bastard children hushing the Czechoslovak part of their lineage. Today's players might identify them as *clones*, but taking a hint from Genette's framework, we might also classify them as *forgeries*, or imitations "whose challenge would be to pass for an

Figure 6.6
Tomáš Rylek's *Star Swallow*. The static screenshot cannot capture the fast animated graphics.

authentic text in the eyes of absolute and infallible competence."[73] Once again, I am not using this term to disparage or belittle these works. Instead, I want to focus on the art of masterful imitation. The games succeed admirably in replicating the fast-paced animation of *Terra Cresta*. However, there are at least two giveaways. First is their imperfect English, twisted into strangely poetic turns of phrase, similar to the so-called Engrish of hastily translated Japanese arcade and console games.[74] *Star Swallow* instructs the player to "Fight bravely but carefully," because "they [the enemies] are angry," and its hall of fame is entitled "The Best at Terror." Instead of a simple Game Over screen, *Star Fly* features a poem that is worth citing in its entirety:

> There where the resting
> Carcasses are
> There are you sleeping
> Fly of the star!
> Break, hero, the deathly gain
> Press a key and try again!

Besides language, a more significant mark of forgery is the lack of attention to gameplay. The games offered neither original nor balanced design. As Rylek admits: "I more or less exhausted myself by the graphics engine, and the rest was quickly patched-up window dressing."[75] As a result, *Star Fly* and *Star Swallow* were primarily technical demos, written to prove that their coding would hold up in the context of Western commercial production. Having achieved this, Rylek stopped making his own games and became an author of crucial development tools such as assemblers and disassemblers, and a frequent contributor to his colleagues' projects.

Miroslav Fídler (born 1970), publishing under the moniker Cybexlab Software (a portmanteau of Cybernetic Experimental Laboratory), was a "synthetic" counterpart to Rylek's analytical approach. He was a prolific author, writing eleven games between 1985 and 1989. In contrast to Rylek, Fídler was more dedicated to design, never considering a game ready for release unless he could play it from start to end himself.[76] Employing Rylek's machine code routines, he wrote *Maglaxians* and *Itemiada* in 1985, clones of the British platformer hits *Manic Miner* and *Jet Set Willy*, respectively.[77] He continued with a sophisticated scrolling shoot-'em-up game called *Galactic Gunners* in 1987, but it is his 1988 flip-screen shooter *Jet-Story* that best demonstrates the growing self-confidence of domestic coders.[78] Besides the names of all members of the Golden Triangle, the credit screen of *Jet-Story* also includes a Czechoslovak flag. Considered one of the top Czechoslovak action games for the Spectrum, it was rereleased as a commercial title in 1992, and its pirate copies made their way into the former Soviet Union. Rather than being a straightforward clone, it capitalizes on the experience gained by years of learning by imitation. It is an original title within an international genre, proudly carrying the banner of the domestic community.

Second Lives of Indiana Jones

The third member of the Golden Triangle, František Fuka (Fuxoft), was possibly the single most influential homebrewer in 1980s Czechoslovakia. As seen in chapter 4, this was due in part to his enthusiastic personality. His cohort Rylek remembers: "I always loved visiting him, because he was always so energetic, as if he was constantly high on cocaine. Whenever I left his house, I was so full of energy, of a thousand ideas about what to do next."[79] Another reason was structural, and related to Fuka's cultural capital. The son of a classical musician, he had learned to play piano at an early age, and had always been interested in and influenced by music and film. He could read English better than most of his peers, and thanks to his uncle in the United States, he had subscriptions to Western magazines, thereby gaining easier access to foreign inspiration. Combined with his membership in

Prague's top computer club, he was in a unique position to become a trendsetter for the community.

Like his two younger friends, Fuka was proficient in machine code, and well-known as a cracker and hacker of ZX Spectrum software. His trademark modification was the addition of new music tracks to existing games, which he then redistributed. Some of the tracks were his own compositions, and others were his "covers" of Commodore 64 tunes, which he recorded to an audio tape and rewrote for the Spectrum.[80] Fuka created several action games, but these did not match the impact of his work within the genre of text adventure. In 1985, he made *Treasure 2*, a partial conversion of a lost Czechoslovak ZX81 title that was, in turn, most likely a translation of a Western game. Later that year, he released a more distinctive and original adventure, inspired by a film he had just seen in one of Prague's movie theaters—*Raiders of the Lost Ark*.

Even in Czechoslovakia, select Hollywood films were regularly screened at theaters, although "capitalist" productions could not make up more than 30 percent of screenings, and films tended to be shown years after their US premieres.[81] *Raiders of the Lost Ark* was a box office hit even behind the Iron Curtain, selling almost two million tickets in a country of fifteen million, and becoming the third-highest grossing Hollywood title shown in normalization-era Czechoslovakia.[82] It left a lasting impression on Fuka, too, as he recounted in a 1991 interview:

> As soon as I left the theater, I realized that this film was so great that I had to make a game based on it. The character of Dr. Jones and the film's setting are a perfect fit for a text adventure game. I conceived the game during a vacation in Romania (it was raining all the time) and called it *Indiana Jones and the Temple of Doom*, although I had not seen that movie by then.[83]

His game was a relatively simple text adventure written in BASIC.[84] In adventure games, according to Montfort, "a player explores a challenging simulated and fictional world to understand this world and to exhibit this understanding through in-game actions."[85] The genre's name and basic features descend from the text-based game *Adventure*, written in FORTRAN on a university computer in 1976.[86] In their canonical form, text adventures (sometimes labeled with the more recent term "interactive fiction") feature a protagonist controlled by commands typed in by the player. The commands usually take "verb + object" form, for example "EXAMINE LETTER," and are decoded by a subprogram called the *parser*.

Boosted by the popularity of Indiana Jones, an iconic hero of Western popular culture, Fuka's *Indiana Jones and the Temple of Doom* was one of the foundational titles of the genre in Czechoslovakia (see figure 6.7).[87] However, as is typical of the homebrew scene, it treated the source material of the film with considerable liberty. Fuka had not even seen the movie before he made the game, which made his *Temple of Doom* a piece of *fan fiction* rather than an adaptation. In Genette's classification, it might fall into a category of unauthorized *continuation*, albeit in a different medium; transmedia theory would see it instead as a user-generated transmedia extension.[88]

In the game, Indy finds himself in Egypt on a mission to infiltrate the Temple of Doom, find the golden mask of the Sun God, and escape. Gameplay consists of a series of relatively sensible puzzles along with maze navigation. Interestingly, it also features overt references to the *Ghostbusters* franchise.[89] One of the few hostile enemies in the game—a ghost—can be eliminated with a Ghostbusters proton pack. *Ghostbusters* was not screened in Czechoslovakia until January 1989, and Fuka therefore must have seen it on a pirated videotape—a fact that further highlights the importance of informal networks to the formation of Czechoslovak gaming culture.

Indiana Jones quickly became a household name even among players who had not seen any of the films. In total, seven unlicensed Indiana Jones games were released for the Spectrum, only three of which were made by Fuka himself, who revisited the character in 1987 (see figure 6.7) and 1990 sequels. For the 8-bit Atari, at least four fan-made conversions of the Spectrum originals appeared, as well as two unrelated stand-alone titles.

Figure 6.7
The loading screens of *Indiana Jones and the Temple of Doom*, with the ghost in the center (left), and *Indiana Jones 2*.

In addition, Jones was featured as a "guest character" in at least two other games, one of which—*Indiana Jones on Wenceslas Square*—will be discussed in more detail in the next chapter. In contrast, only one licensed Indiana Jones text adventure was produced in English, the obscure 1987 title *Indiana Jones and the Revenge of the Ancients* for the Apple II and IBM PC.[90]

The text adventure (or *textovka* in Czech and Slovak) went on to become the country's dominant homebrew genre. Text adventures made up over 50 percent of all preserved 1980s Czechoslovak games for both the Spectrum and Atari 8-bit platforms, as opposed to 17 percent and 5 percent in the largest international archives for the respective platforms.[91] This was due to a combination of reasons. First, they were relatively easy to produce. Second, as Fuka himself acknowledged at the time, Czech- and Slovak-language text adventure games became a niche in which local programmers did not have to compete with Western professionals.[92] Text adventures were also, along with simulators, among the genres considered valuable by hobbyists. In fact, the leading hobbyist newsletter *Mikrobáze* encouraged translations of foreign titles, announcing: "We welcome (and are ready to remunerate) any fruitful effort that would help include these specific, strategically and logically challenging games into the Mikrobáze collection."[93] Only a handful of translations appeared—possibly due to the generally poor English skills of the Czechoslovak population—but dozens of original games followed. As the next chapter will show, they often drew from local popular culture and everyday life.

The influence of Western text adventures on their Czechoslovak equivalents was in no way straightforward. The range of titles available in the country's informal distribution system was limited by platform. In the United States, the genre is often associated with Infocom, the company behind the *Zork* series.[94] However, *Zork* was never converted to the ZX Spectrum. Instead, Czechoslovak players were more likely to play *The Hobbit* (an Australian game published in the United Kingdom) or early adventures by the British company Artic.[95] Most domestic programmers knew little English and could neither understand nor replicate the intricate puzzles of Western titles. More often, local creators were following up on previous domestic efforts, creating a chain of influence that was uniquely local. Understandably for homebrew efforts, domestic games tended to be shorter and less complex than commercial Western ones, and usually did not include graphics.

The prototypical Western text adventure tends to be associated with a parser-based user interface. However, numerous Czechoslovak text adventures used a simplified menu-driven interface instead, in which the player constructed the command from a finite list of verbs and noun phrases. This control setup was similar to that of early point-'n'-click adventure games such as *Maniac Mansion*.[96] Introduced simultaneously in 1987 by two separate titles—one of which was Fuka's *Indiana Jones 2*—this new interface made text adventures both easier to make, because it eliminated the need to write sophisticated parsers, and easier to play, because the users did not get stuck looking for the right phrase.[97] Others (but not all) followed suit, creating a distinct variation of the genre. Miroslav Fídler's popular Tolkien-inspired 1989 adventure *Belegost* even featured an icon-based interface, making it essentially a point-'n'-click adventure without graphics.[98] In the end, we may argue that the Czechoslovak *textovka* was to the Western text adventure what the Japanese role-playing game is to the Western computer role-playing game—a relatively distinct, semi-indigenous subgenre inspired by Western sources but developed in a specific, partially isolated environment.[99]

Hacking Games

The Czech label *textovka* sometimes covered not only text adventures but also other types of games presented in text form. These included text-based management and strategy titles derived from the minicomputer game *Hamurabi*, such as the British *Dictator*, popularized in the country through translations and conversions.[100] It also covered a subgenre specific to the 1980s Czechoslovak scene—the *hacking game*.

Its origins lay in Western popular culture. In the 1980s, hacking was in vogue thanks to films such as *Tron* and *WarGames*, and Steven Levy's book *Hackers: Heroes of the Computer Revolution*.[101] At least two Western commercial 8-bit titles attempted to capitalize on this trend—1984's British *System 15000* and 1985's American title *Hacker*.[102] These games simulated the experience of networked computing in a fictional world by replacing the user interface of the player's micro with the simplistically simulated operating system of a more powerful machine. In other words, the Spectrum suddenly behaved as if it were connected to a modem. Gameplay usually consisted of dialing numbers, receiving messages, connecting to distant computers, and breaking into proprietary databases by means of information gathering and puzzle solving. In the West, very few such titles were produced and the fad appeared over by 1986.

However, the genre survived in an unlikely place. In 1986, František Fuka released *The Sting 3* (*Podraz 3*), a game that combined at least two seemingly incongruent inspirations (see figure 6.8).[103] The first was the abovementioned *System 15000*. The other reference was to a film rather than another game. As the title suggests, Fuka presented the game as a futuristic sequel to the American caper movie *The Sting II*, which opened in Czechoslovak theaters in October 1985, soon after *Raiders of the Lost Ark*.[104] Although *The Sting 3* takes place in 1989 New York and focuses on computer crime rather than bank robberies, it follows some tropes of the caper movie genre, and even opens with a ragtime tune very much like the ones in the film. The game invites the player to assume "the role of the unemployed programmer Tim Coleman,[105] equipped with a Timex 2097 computer and an RS-2368 modem, and try to rob other robbers with a little help from your friends."[106]

Although the narrative is clearly set in the United States, the game contains several references to the Czechoslovak computing experience. The police record of the game's villain, for example, says he has had "extensive contacts with Yugoslavian software thieves." Like *System 15000*, the game is

Figure 6.8
Loading screen of *The Sting 3*. The subtitle reads "The First Great Computer Robbery," and the text in parentheses announces that it is a "logical" (puzzle) game.

best played with a pen and notepad at hand to note phone numbers and passwords. The narrative, told through electronic mail messages, manages to capture the thrill of Western hacking games and movies. Upon player completion, *The Sting 3* displays a "secret code" to unlock the next game, but Fuka never made a sequel himself.

Nevertheless, the game proved to be remarkably influential. Between 1986 and 1990, *The Sting 3* spawned at least twenty-five unofficial sequels and clones, mostly for the ZX Spectrum and Atari 8-bit computers. Anton Tokár from the Slovak town of Revúca directly continued Fuka's story in sequels *The Sting 4–6*; other authors independently contributed another *The Sting 4,* a separate *The Sting 5* and, finally, *The Sting 7*.[107] In 2002, *The Sting 3* became one of the few text-based Czechoslovak games to receive an English translation, made by the Total Computer Gang retro gaming group.[108] The multiplicity of sequels shows that many homebrewers saw games as material that was open to adaptation and continuation within a culture of uninhibited hypertextuality. Programmers from different towns and cities worked concurrently on their continuations, only aware of each other's works when they arrived through the informal distribution networks.

The popularity of hacking games among homebrewers has several complementary explanations. First, like text adventures, they did not require extraordinary coding skills. Hacking games sidestepped the technological limitations of the 1980s computers by narrating mainly via email messages and text files downloaded from fictional databases. This allowed them to be audiovisually realistic while remaining reasonably easy to make. Second, they appealed to the *technicity* of computer users by letting them experience their hacker power fantasies, as evidenced by the very titles of games such as *Expert* (see figure 6.9) or *Cracker*.[109] The games invariably allowed the player to control machines much more powerful than the Spectrum or 8-bit Atari—real or fictional 16-bit machines with networking capabilities, a far cry from what most hobbyists owned. In addition, hacking had the flavor of an activity that was both Western and subversive.[110] Allowing one to perform ingenious and transgressive acts abroad using a computer, hacking games offered an ultimate case of what we have described as *imaginative travel*.

Whereas the original *The Sting 3* took place in a near-future New York, in some other hacking games, the settings were "localized" and the narratives were moved to Czechoslovakia, featuring casts of friends and

Figure 6.9
A screenshot from the 1988 hacking game *Expert* by Ondřej Kafka, the copublisher and graphic designer of *ZX Magazine*. The game includes a simulation of an operating system inspired by MS-DOS.

acquaintances. Like Indiana Jones, Tim Coleman became the main character of many more Czechoslovak games. Both text adventures and hacking games became templates that domestic amateurs used to participate in the creation of homebrew software.

This chapter has shown how Czechoslovak homebrewers worked with Western and Japanese material, integrating it into local programming practices and creating unique lineages of bastard children. Their treatment of source material, both Western and domestic, was more akin to folklore than the modernist tradition of authorship. As Jenkins noted of fan culture, "It was the application of folk culture practices to mass culture content."[111] Speaking of folk culture, folklorist Kenneth Pimple has argued that "those cultural phenomena that require a higher number of layers of artificiality for their creation and continued existence are less folk."[112] He adds that the notion of "individual control" over creation and distribution of content allows folklore to proliferate. Thanks to its bottom-up infrastructures, the Czechoslovak homebrew scene left much control to the creators at their decentralized microcomputers. There was no copyright holder around to protest local versions of *Manic Miner*, and no George Lucas filing lawsuits against fan-made Indiana Jones games. As a result, homebrewers could take the products of commercial industries and turn them into a kind of national digital folklore. But unlike folklore, the scene valued personal contribution and authorship thanks to its meritocratic principles—the possibility of attaching one's name to a piece of software was considered empowering and desirable.[113]

Czechoslovak homebrewers may not have respected copyright and the integrity of computer game code, but through imitation, they paid respect to the skills of Western (and Japanese) programmers, artists, and designers. On some occasions, they tried to be more British than the British, as Tomáš Rylek of the Golden Triangle put it.[114] This continuing relationship of dependence between centers and peripheries in software development echoes the findings by Yuri Takhteyev, who writes about Brazilian coders in the 2000s. In his view, "Local participants orient themselves toward such meccas in an attempt to draw on their symbolic power and to bring the local practice closer to the remote standards."[115]

Although much of domestic production was based on foreign templates, local influences also played an important role. As mentioned, the use of local language—either Czech or Slovak—was part of the appeal of domestic homebrew production. Some authors wrote text adventures inspired by Czech history or popular culture; others worked in the attractive niche of adaptations of local nondigital games.[116] Examples of the latter include *pexeso*, the Czechoslovak variant of the memory game, and *piškvorky*, the local variant of tic-tac-toe, played on a significantly larger and potentially unlimited grid.[117]

Each of the case studies described in this chapter demonstrates that the appropriation of Western sources was a very localized practice, addressing the specific gaps, deficiencies, or interests of the Czechoslovak microcomputer scene. Ports and conversions remedied the lack of software for some domestic platforms or brought games from obscure platforms to more popular ones. Clones of action games provided an outlet for young programmers to show off their coding skills. Hacking games and Indiana Jones text adventures filled the niche of narrative-driven games in Czech and Slovak. The next chapter will focus on games that, beyond being made for local players, also addressed local topics and served as a means of self-expression for Czechoslovak programmers.

Before moving on, we can make a few comparisons. As noted above, the flow of software into Czechoslovakia was mostly one-way. No Czechoslovak game from this period was commercially released outside the country, and only a handful of titles seem to have made it into other Soviet bloc countries in an informal way. This was not the rule in all of Eastern Europe. In Hungary, Yugoslavia, and Poland, all of which had comparatively more liberal state socialist regimes, local programmers continuously produced software for the international market.

In Hungary, the Budapest-based import-export company Novotrade took advantage of cheap and talented domestic labor, struck a deal with British publishers, and recruited young Hungarian mathematicians and programmers to produce games for the Western European markets. These included popular and innovative titles such as *Caesar the Cat* (1983), *Eureka!* (1984), and *Scarabaeus* (1985).[118] British magazines and newspapers acknowledged the quality of Hungarian programs, likening their success to that of the Rubik's cube.[119] However, the production routines resembled outsourcing or contract work rather than export. Some games were designed in the United Kingdom and merely coded by Hungarian geniuses; in several cases, Hungarian programmers were not even credited for their contributions. All this points to the fact that Hungary, despite its programmers' achievements, was still treated as a periphery.

The runaway success of *Tetris* owed to Novotrade's UK connections. Originally designed by computer scientist Alexey Pajitnov at the Soviet Academy of Sciences in Moscow, the game's IBM PC version was discovered by Hungarian-born British entrepreneur Robert Stein during his visit to Budapest, and later licensed by Mirrorsoft, a large UK publisher. Given how rare and expensive IBM PCs were at the time, very few Eastern European players could ever run the Soviet version. Instead, they were much more likely to play the ZX Spectrum conversion produced in Britain, or the British-commissioned, Hungarian-made Commodore 64 one.[120] Thus the game's popularity in the region would have been impossible without the contribution of the British game industry. All things considered, *Tetris* was a very exceptional case, and no other Soviet titles enjoyed such success.[121]

Poland had a small but active homegrown software market in the 1980s. According to Budziszewski, the first commercial release was 1986's *Pandora's Box*—a text adventure that even advertised on its cover the fact that it was in Polish: "Having trouble with *The Hobbit*? Don't know English well enough? Buy *Pandora's Box*!"[122] Parallel to Novotrade's efforts in Hungary, the aptly titled Polish company California Dreams based their business on programming games for export into the United States. Besides commission work—including conversions for the famous publisher Strategic

Simulations Inc.—it published a series of original titles for the PC, including the Tetris-inspired 3-D puzzler *Blockout* and the racing game *Street Rod*, both released in 1989.[123]

Starting in 1984, Yugoslavian programmer Duško Dimitrijević and his colleagues published three Spectrum games—*Kung Fu*, *Movie*, and *Phantom Club*—with British publishers Bug-Byte, Imagine, and Ocean.[124] According to contemporary reports, and unlike in the Novotrade case, Dimitrijević dealt with the publishers independently.[125] At the same time, Yugoslavia had a self-sufficient commercial game industry of its own, led by the Croatian publisher Suzy Soft, a subsidiary of the record company Suzy Records. As was the case in Czechoslovakia, many of their games were text adventures in one of the local languages. Unfortunately, next to no academic work has been published about the fascinating Yugoslavian scene.[126]

The Golden Triangle's example shows that Czechoslovakia had comparable talent, but such collaboration and coproduction with Western publishers of computer games was unheard of. It is difficult to assess the extent to which the lack of professional opportunities encouraged amateur production, but it certainly did not prevent the formation of one of the region's most vibrant homebrew scenes. Although no money could be made by programming games, the prestige of being an author was a coveted source of social capital.

7 Empowered by Games: Games as a Means of Self-Expression and Activism

> If you have something to comment on, if you have an idea or a unique perspective, the microcomputer culture will gladly and earnestly accept them. … Any one of you can become a hero of microcomputer culture—you only need to have something to say.
>
> —Bohuslav Blažek, in a 1989 essay[1]

> Pure thought creates a new reality.
>
> —Miroslav Fídler, from a 1997 retrospective interview[2]

In 1997, former members of the Golden Triangle homebrew collective took part in a retrospective interview for the alternative culture magazine *Živel*. They recounted their memories of writing games for 8-bit computers in the 1980s and mused about microcomputers. Explaining why he enjoyed coding more than hardware tinkering, Miroslav Fídler (also known as Cybexlab Software) said that in programming, "pure thought creates a new reality."[3] The trio collectively laughed at the lofty statement, and half-jokingly proposed it as the interview's takeaway message. But hyperbole aside, this statement is true—coding is indeed a unique way of creating new or alternative worlds.[4] It can empower users not only to solve mathematical problems but to communicate and express themselves.

Previous chapters have shown the ways users gained advantages through their machines. The computer offered a seemingly infinite space for exploration and self-realization in a society that tended to close off opportunities rather than open them. Through a range of coding acts, programmers could put their names out there, attached to pieces of distributable code. Moreover, homebrewers could fashion games, and specifically text adventures, into their *zines* or *samizdat*. This chapter will show how Czechoslovak amateur programmers used computer games to say something about themselves, their friends, their favorite movies, their frustrations, or their opinions on current affairs. Their tactics ranged from sharing autobiographical details in a game's nondiegetic messages to outright political activism, including text adventures about antiregime demonstrations.

Ted Nelson conceptualized the idea of empowerment through computers in the United States in his 1974 book *Computer Lib/Dream Machines*. By "computer lib" he meant both liberation *of* computers and liberation *by* computers through personal expression. He encouraged people to "see computers for what they really are: versatile gizmos which may be turned to any purpose, in any style. And so a wealth of new styles and human purposes are being proposed and tried, each proponent propounding his own dream in his own very personal way."[5] His writing influenced many microcomputing pioneers and became an important part of the Silicon Valley cultural milieu, although the dream of personal empowerment was soon incorporated into neoliberal ideology and reinterpreted as entrepreneurial ambition.[6]

In Czechoslovakia, the academic Bohuslav Blažek—who could have been inspired by Western sources—held views similar to Nelson's. Writing in 1989, he argued that "microcomputers are a grand conspiracy against arrogant professionals who abuse their monopoly." Published on the eve of the Velvet Revolution, this statement could easily be read as an indictment of government bureaucrats who attempted to impose *scientific-technological progress* from above. For Blažek, a new culture was emerging from below thanks to microcomputers, which allowed one to create text and graphics and share them with others. This was "not an improvement of existing culture, but a profound change of its criteria." In his view, "any one of you can become a hero of microcomputer culture."[7] What he was describing was very close to Henry Jenkins's idea of a *participatory* culture, an array of grassroots activities that obviate the power of institutional communicators and mass media, and result in personal and communal—often playful—transformations of popular culture phenomena.[8] In the Czechoslovak context, homebrewers were the avant-garde of this participatory microcomputer culture.

The use of computers for self-expression was, in de Certeau's terms, a *tactic* that developed in the terrain dominated by the socialist state's strategies of censorship and propaganda. In normalization-era Czechoslovakia, unauthorized media tactics such as samizdat and record trading were routinely intercepted by state power. But the terrain of microcomputers generated less friction—mainly because it had never been claimed by state power in the first place, flourishing instead in *vnye* milieus. In computer clubs and at homes, thousands of people were left alone with their computers—presumably to train for their industry and military jobs—and in the process co-created a grassroots communication network that allowed them to address thousands of others.

Besides investigating user tactics, I will also view games as *tactical media*. Besides its apparent terminological connection to de Certeau's theory, the term has a history of its own. It was defined in 1997 by David Garcia and Geert Lovink as both a descriptive category and a blueprint for artistic and critical interventions:

> *Tactical media* are what happens when the cheap 'do it yourself' media, made possible by the revolution in consumer electronics and expanded forms of distribution (from public access cable to the internet) are exploited by groups and individuals who feel aggrieved by or excluded from the wider culture. Tactical media do not just report events, as they are never impartial they always participate and it is this that more than anything separates them from mainstream media.…Tactical media are media of crisis, criticism and opposition.…Their typical heroes are[:] the activist, Nomadic media warriors, the pranxter, the hacker, the street rapper, the camcorder kamikaze.[9]

The term came to be used for localized, bottom-up, noncommercial uses of digital technologies for activist purposes, and included hacktivism and other related practices.[10] As this chapter will show, some Czechoslovak coders used their micros (which were not "cheap" by Czechoslovak standards, but accessible to thousands of individuals) in similar ways—to comment on a crisis, and to voice criticism and opposition.

Tactical media, as well as the related practice of *culture jamming*, are associated with humor and playfulness.[11] We can connect their Czechoslovak predecessors to the tradition of resistance through humor, which had long been considered a part of the Czech (and by association Czechoslovak) national identity. This aspect of the national self-image is embodied in the character of Good Soldier Schweik from the eponymous novel by Jaroslav Hašek—one of the best-known works of Czech literature. On the eve of World War I, Schweik (Czech spelling Švejk) expresses unusual eagerness to fight for the Austro-Hungarian Empire, but "through kind-mannered idiocy and over-the-top obsequiousness, he proceeds to subvert army precision and discipline." As Robertson notes, "Schweikian passive resistance has been a common tactic in dealing with repressive rulers."[12] Hašek himself was a lifelong anarchist prankster, who even founded his own mock political party—the Party of Moderate Progress within the Bounds of the Law.

In Czechoslovakia, programming of microcomputer games for self-expression started to blossom around 1987–1989. This was a period when two important concurrent developments took place, one in society at large, and the other in the microcomputer community. Regarding the former, the mid to late 1980s—described by McDermott as the "pre-terminal period" of Communist Czechoslovakia—were an era of piecemeal liberalization of cultural and economic life in response to Gorbachev's reforms in the Soviet Union.[13] Censorship was not as strictly enforced, and some previously banned writers and musicians were allowed to publish and perform. The authorities kept control over the media but accepted and tolerated careful criticism and satire targeted at current social ills. Western artists started to visit the country. In 1988 the band Depeche Mode, who had a strong fan base in the country thanks to bootleg tape copies, played a show in Prague—the first major Western pop group concert since the 1969 Beach Boys performance. As a magazine review commented, thanks to this landmark show, "our country has stepped out of the periphery of the rock music scene," even if it did so later than Hungary or Poland.[14]

At the same time, everyday life was becoming increasingly politicized. Both party officials and ordinary citizens discussed "reconstruction" (*přestavba*)—the Czechoslovak attempt at perestroika. But few Czechoslovaks still believed in the dream of communism. The party was privately mocked because it failed to keep up with the West and respond to Soviet reforms. The growing distrust in the party was documented by a series of opinion polls conducted by the Institute for Public Opinion Research. In the Czech part of the federation, the percentage of respondents who believed that the Communist Party's leading role was "properly performed" dropped from 57 percent in 1986 to 40 percent in 1988, and stood at only 31 percent in June 1989.[15] The most commonly cited reasons were privileges of party members, the party's nondemocratic leadership, and mismanagement.[16] Slow liberalization did not mitigate the citizens' disenchantment, instead making them even more determined to protest and criticize the regime. State socialism was losing the backing of the technical intelligentsia, too, whose expertise made it possible to run the socialist economy.[17]

The other development was the growing accessibility of microcomputing, and the resulting upsurge in homebrew activities. In 1987, significant numbers of users could join in thanks to the launch of the Didaktik Gama and shipments of Atari and Sharp machines; at the same time, the informal distribution networks became dense and efficient. In 1988, *ZX Magazine* started to chronicle the homebrew scene, publishing interviews with game programmers and positioning them as "authors." At the same time, the Czechoslovak *textovka* was firmly established as a genre that provided homebrewers with a storytelling template.

A testament to the status of *textovka* is the 1989 project *City of Robots*, likely the only Communist-era domestic title that received an official nationwide release (see figure 7.1). At its core was a challenging science fiction text adventure for the ZX Spectrum, Sharp MZ 800, PMD 85, IQ 151, and Ondra platforms, programmed by "the 602" club alumnus Vít

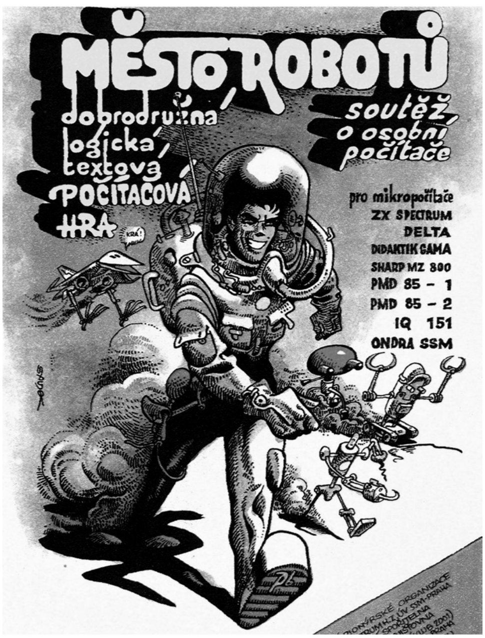


Figure 7.1
A promotional image for *City of Robots*, drawn by the renowned Czechoslovak comics artist Kája Saudek. In the 1980s, Saudek, whose work was heavily indebted to Western comics, was effectively banned from publishing through mainstream channels. This led to his collaboration with *vnye* communities, such as science fiction clubs and microcomputer hobby groups. Image courtesy of the Kája Saudek estate.

Libovický.[18] Although it originated in the computer club milieu, the game was connected to a nationwide contest, organized with the help of the Socialist Union of Youth, and announced by nationwide media, including Czechoslovak Television and Radio. According to the game's manual, five thousand prizes (including six Didaktik computers) were available for the players who finished the game. Only the owners of an original copy—which sold for a relatively high price of ninety-nine crowns—could take part.[19] Although the contest itself was compromised by a technical issue, the whole event was covered by the media and closely watched by many domestic players, appearing prominently in their recollections.[20] Its massive scale hints at the organizers' belief that a text adventure could become a major cultural event. Similar contests had already taken place in the United Kingdom in the early 1980s and in Yugoslavia in 1984,[21] but *City of Robots* was the first Czechoslovak attempt.

Hello World!

In the context of the commercial game industry, the personality of the programmer is often suppressed in favor of corporate identity and the product's entertainment function. A classic example from video game history is Atari's treatment of programmers working for their Video Computer System console. In order to have his signature included in his game *Adventure*, Atari programmer Warren Robinett had to hide it in a secret room, and famously created the first of the software "Easter eggs"—defined by Nooney as "digital objects, messages, or interactions built into computer programs by their designers" that are "intended as a surprise to be found by the user, but they are not required in order to use the program."[22]

Czechoslovak coders, not limited by industry standards, were free to use their programs as vehicles for personal or community messages. Nevertheless, inserting hidden messages into the code, both in BASIC and machine code, was "very much in vogue" because it played to the competitive bricoleur practice of disassembling and analyzing other people's programs.[23] Homebrewers would come up with new ways of obfuscating and encrypting their code and hide messages for those who could crack the puzzle and reach the end. The ludic nature of this endeavor suggests that, like cracking, this was a metagame to the games themselves. Breaking into the BASIC code of the 1988 crime text adventure *Katanga*, for example, one learns that it was "written as an original form of revenge" against some of the author's acquaintances.[24] Knowing this, we may understand the real illocutionary intent behind the game as a coding act.

Perhaps more often, domestic homebrew games would communicate through nondiegetic *scrolling messages* (or *scrollers*). Already in the early 1980s, some commercial 8-bit games—including hits such as *Manic Miner*—used these to present credits and instructions on the main menu screen. Later, scrollers became best known as a hallmark of "crack intros" or cracktros produced by the Western European Commodore 64 cracker and demoscenes.[25] In Czechoslovakia, scrolling messages started appearing regularly around 1987 in most games released by the Golden Triangle collective. We will now examine one of them, from the 1987 shooter game *Galactic Gunners*, written by Miroslav Fídler.

Upon loading, a main menu screen pops up and music starts playing. It is a "cover" of one of the most popular chiptunes of all time—the music for the Commodore 64 title *Monty on the Run*, by Rob Hubbard—admirably squeezed into the original ZX Spectrum's one chirping channel.[26] Simultaneously, the message starts scrolling from right to left on the bottom of the screen. It starts with a perfunctory copyright notice, followed by Fídler's address and telephone number, all leading to a villa on the outskirts of Prague's sixth district. Although the menu and all in-game text are in English, the language now switches to Czech and we can read that Fídler is looking for collaborators on his next projects, and proposes to divide all his profits fifty-fifty, admitting that so far his profit has been nil.[27]

Over the years, some homebrewers started to add elaborate greetings and in-jokes, usually unrelated to the content of the game.[28] A case in point is the scrolling message—over a thousand words long—in Patrik Rak's (Raxoft) title *Piškworks*, an adaptation of the regional variant of tic-tac-toe. In the message, Rak managed to do the following:

- He told the origin story of the game and praised Hana Gregorová, his friend's sister, who invented the game's basic AI algorithm even though "she is only in fourth grade."
- Advertised his older titles (two of which were unsolicited sequels to other authors' homebrew games), as well as his upcoming releases.
- Greeted sixteen fellow ZX Spectrum homebrewers, most of whom he had never met in person. Said hello to "all the friends from our beloved 602"—the computer club he frequented.

- Condemned the incompetence of four Yugoslavian and Polish crackers (see chapter 5).
- Asked for the whereabouts of his lost cassette tape ("Which scoundrel has my cassette no. 12?!").
- Asked his friends for a microswitch component so he could repair his joystick.
- Confessed that it was three weeks before his graduation exam: "I might be going crazy."

This is, of course, an extreme example, made possible by the fact that the game itself was light on data, leaving ample space for additional text. However, it provides notable historical evidence of what concerned young coders of the period.

The messages appended to games as paratextual and nondiegetic elements have been compared by my informants to "shouts in the dark" and to exclaiming "Hello world!" on today's social media.[29] Their inclusion relates to the observation that games are coding acts, which may convey many things other than entertainment. Scrolling messages enabled the members of the community to greet or insult concrete people, but also allowed them to reach out to other, unknown users. This idea of a potentially unknown audience—phrased by an informant as "the idea that somebody is using it, playing it, that somebody knows me and notices me"—was thrilling to homebrewers, and made them feel like "real" media producers.[30] Some coders fondly remember getting telephone calls or letters from the other side of the country.[31] A few of them—such as František Fuka—were even swamped by them. In the scrolling message of his 1990 title *Indiana Jones 3*, Fuka wrote: "Please, if you get stuck in this game, don't send me letters—I'm already drowning in letters. If you can't help it, call this number …, but not at eight in the morning (I like to sleep in!)"[32] All this allows us to conclude that there was an attentive audience of Czechoslovak microcomputer users, awaiting more messages, more games, and more stories.

Adventure in Your Home

Most of the games in this chapter are text adventures. The genre's popularity was due to the tradition started by František Fuka, as well as its formal and technological properties. As opposed to animated graphics, the written word proved to be a remarkably versatile mode of communication that required relatively little labor.[33] By way of text adventures, homebrewers could rapidly craft “new realities” of various kinds, while using established forms of narration, often with the help of verbal humor. We can demonstrate both this diversity and the importance of humor in the example of the Sybilasoft collective.

Besides his Golden Triangle cohorts, Sybilasoft were the only other ZX Spectrum homebrewers that František Fuka found worth mentioning in his 1988 book *Computer Games: Past and Present*, opining that “their games are neither complex nor sophisticated, but one can have a good laugh while playing them.”[34] Sybilasoft were five high schoolers living in the Slovak capital of Bratislava. They visited each other at home or met in one of the city’s computer clubs; from time to time, the five of them would spend a few days at a weekend cottage belonging to the parents of one of the members. Having extensively explored and discussed computers and computer games, they soon started making their own. In 1987 alone, they released six text adventures.

Although members tended to contribute to each other’s games, the core of each title was usually written by one or two authors. Martin Sústrik was the most proficient coder, and the author of the first Sybilasoft game, *Plutonia*, which was a variation on František Fuka’s *Treasure 2*; Sústrik then went on to create science fiction adventures such as *Tria* and *Stensontron*.[35] Stanislav Hrda’s games included *Fuksoft*, with a partially autobiographical narrative, and *Shatokhin* (a parody of the *Rambo* movies), which we will discuss later.[36] The Hlaváč brothers, Michal and Juraj, worked with literary sources, adapting a Sherlock Holmes short story into a text adventure, and making a game featuring Pouch the Beetle, a popular anthropomorphic bug from a Czechoslovak children’s book.[37] The fifth member, Pavol Čejka, did not own a computer, but contributed ideas and writing to other members’ games.[38] The thematic content of Sybilasoft production was eclectic, yet firmly rooted in the 1980s Czechoslovak teen culture landscape. As one of the members recalled: “We wrote things that made sense to us and that made sense to our audience—our friends.”[39] They drew from their own cultural experiences, mixing Western-infused novelty with domestic familiarity. For a 1980s teenager, *Pouch the Beetle* could have been just as formative as the pirated *Rambo* movies.[40]

Besides drawing from shared cultural experience, many Czechoslovak games referred to shared experiences of everyday life. Hrda’s 1987 title *Fuksoft* was likely the first Czechoslovak homebrew title to feature autobiographical elements within the game's narrative. Its title was a play on Fuxoft, the software "company" of František Fuka. Hrda attests that Fuka was a major influence on Sybilasoft:

> I'd played many Czech and a few Slovak games and I was always looking forward to, for example, Fuka releasing a new Indiana Jones game. It was exactly like the moment when your favorite writer has published a new book and you are leaving the bookstore with a copy. … I had not met Fuka in person by then, but I looked up to him, because he was always a few steps ahead of us and released the first [Czechoslovak] text adventures. So [*Fuksoft*] was, among other things, an homage to him.[41]

The homage did not stop at the title. The main character, Tim Coleman, was borrowed from Fuka's *The Sting 3*; and the goal of the game itself was to rescue a fictionalized František Fuka. Unlike Fuka, who set his games in exotic and foreign locales true to the original fictions, Hrda opted for a more mundane setting—a Czechoslovak-style apartment building, which he populated with not only Fuka and Coleman but also his friends and acquaintances. These included his "platonic love" Lenka, his schoolmates, and one of his high school teachers, as well as his Sybilasoft partner Pavol Čejka and other fellow ZX Spectrum fans. The building even housed a computer club and the fortified den of the notorious Yugoslavian smuggler and software pirate "Satansoft." The game's fictional world is a prototypical bricolage, assembled from "bits of jokes" and "spontaneous ideas" that Hrda collected. It can also be read as a synecdoche representing the microcosm of Hrda's everyday life, spiced up with a bit of fantasy and condensed into one apartment building. The gameplay is based on map navigation and puzzle-solving against a time limit, and is controlled by keyboard shortcuts rather than typed-in commands to "make it easier for people to enjoy the story."[42] When making the game, Hrda was primarily looking forward to the reactions of the people featured in the narrative. But the title became a hit of its kind, spawning an Atari 8-bit fan conversion and a fan sequel entitled *Fuksoft II*, and inspiring other programmers to write about themselves and their friends.[43]

Martin Malý, the fifteen-year-old "mastermind" behind the Demonsoft label, had already released a few text adventures for the Spectrum when he wrote *Demon in Danger* in September 1988. The introductory text reveals a plot indebted to *Fuksoft*:

> On November 13, 1988, a strange person appeared in the village of Chleby. He was wearing an army uniform and a military hat. Those who know games for the ZX Spectrum would recognize him as Dan Dare, a comics hero.[44] He was an agent of the studio Fuxoft, constructed by František Fuka. His mission is to plant a bomb into the apartment of the competing studio Demon, that is, into the apartment of Martin Malý.[45]

The goal is to deactivate the bomb before Malý gets home from school. This time, Fuka is not a victim, but a villain, threatening to exterminate a fictionalized version of the author himself. Whereas *Fuksoft*'s setting was a synecdoche of Hrda's everyday life, *Demon in Danger* takes place in a faithful recreation of the apartment where Malý used to live with his parents. Its gameplay is simple, but offers the voyeuristic pleasure of wandering around a 1980s *bytovka* apartment when nobody is home. In place of illustrations, it provides detailed, almost superfluous descriptions of mundane items. You can learn about the contents of the family's medicine cabinet, or find out that a doormat cost fifty crowns. As mentioned in chapter 2, the game gives us a unique peek into Malý's bedroom, described by the game as "Demon's office." Here, one can even turn on a computer and play around in a simulated operating system—an overt reference to the hacking games mentioned in the previous chapter. Thus, *Demon in Danger* is not just a game that was made locally and uses local influences—it is *about the local*, about a particular building in the Central Bohemian village of Chleby, street number 17.

Taking inspiration from the concept of hyperlocal media, I call such games *hyperlocal* and define them as games created by people from a particular place, about that place and about the people who inhabit it, written mainly (but not exclusively) for the local community.[46] As Laine Nooney has demonstrated in the case of games by Roberta Williams, domesticity has been largely written out of the main narratives of computer game history.[47] This has as much to do with the prevalent masculinity of gaming cultures, as with the interest in commercial titles over amateur ones. Domesticity, however, remains preserved on the margins of game histories, including Czechoslovak hyperlocal games. The earliest hyperlocal game I have come across in my research is the 1984 Yugoslavian text adventure *XIV*, which takes place at the 14th Grammar School in Belgrade (hence the title).[48] It is likely that similar hyperlocal games were produced by other homebrew communities outside of Czechoslovakia, but so far these have been neglected by game historians.[49]

Hyperlocal games became particularly popular in the country in the early 1990s, a period from which over twenty have been preserved. They offered a convenient pathway into game making, without the obligation to contrive elaborate fictions; they served as coding acts through which their authors could say something about themselves and their surroundings, and impose their rules on their everyday spaces. The games appealed to many other microcomputer users, many of whom were part of the relatively monolithic Czechoslovak boy culture. After all, fans of 8-bit computer games often lived in a *bytovka* like Malý's or navigated social microcosms like Hrda's.

Spreading Unofficial Culture

Let us now return to Sybilasoft's *Fuksoft* and its apartment building packed with friends and cultural references. When the game loads into one's Spectrum, it plays a melody from a song that has a complicated history of its own—the number "When They Took Me in the Army" by the Czech folk singer Jaromír Nohavica. The opening lines go:

> When they took me in the army
> They snipped my hair bare
> I looked as if I were a moron
> Like everyone around there[50]

Another Nohavica reference can be found later in the game, when the main character has to bring a videotape with Nohavica's performance to one of Hrda's schoolmates, a metalhead who "besides metal, also enjoys listening to artists such as J. Nohavica or V. Vysotsky."[51] Nohavica's songwriting was in fact heavily influenced by the Russian folk singers Vladimir Vysotsky and Bulat Okudzhava; Nohavica wrote poetic, often sarcastic and self-deprecating lyrics that likewise laid bare the "alienating and dehumanizing aspects" of the late socialist reality.[52] Although he was not a dissident, his commentary was frowned upon by the state establishment. Like Vysotsky, he had difficulty finding venues to perform and putting out records, but gained widespread popularity through privately traded bootleg tapes.

Thanks to their unregulated nature, homebrew games could adapt content that was not officially available through mainstream channels. As seen in the previous chapter, František Fuka alluded to Western films never officially released before 1989. Nohavica, on the other hand, was part of the grassroots domestic singer-songwriter culture. Listening to Nohavica, like listening to heavy metal, signaled one's refusal of regime-supported mainstream culture, and concurrently, engagement in an unofficial and "cool" culture—serving both as a sign of cultural capital and a mild subversion of authority. The song "When They Took Me in the Army"—first performed in 1982, but only released after the Velvet Revolution—was especially likely to resonate with young men who were facing one or two years of military service after high school or college.[53]

The inclusion of the episode gains new meaning when we look at the photograph of Hrda from around the time he made the game. His head was cut clean just as in the song, a result of a school ban on long hair, not unusual and widely despised by the youth in 1980s Czechoslovakia (see figure 7.2).[54] Because his long-haired metalhead friend was forced to cut his hair, so did Hrda, out of solidarity. Including the song can therefore be interpreted as a small but conscious subversion, rooted within the author's lived experience.

One of the most fondly remembered text adventure games of the decade is 1988's *The Mystery of the Conundrum*. Incidentally, its author—Petr Mihula, a fresh cybernetics graduate from the city of Brno—wrote it while in military service. He was granted official permission to take his ZX Spectrum with him, and made the game at nights to keep himself occupied with creative work. The game takes place within the fictional universe of *Fast Arrows* (*Rychlé šípy*), a popular comic for youth, whose publication history mirrors the development of freedom of press in the country. Created by writer and Boy Scouts instructor Jaroslav Foglar, the comics chronicled the adventures of a group of five boys, known as the Fast Arrows (see figure 7.3). Foglar's goal was to entertain and educate the youth (primarily boys) by promoting the values of truth and honor as well as physical and intellectual fitness. The comic's frequent and explicit moralizing notwithstanding, it became massively popular among the youth thanks to its use of humor, adventure, and mystery. Its far-reaching influence over young people and its association with the Western-originating Scouting movement did not sit well with the oppressive regimes that ruled the country. First published in the late 1930s, it was banned by the Nazis in 1941, resumed after World War II, was shut down by the Communist establishment in the 1950s, resumed during the Prague Spring, and was banned again in 1972 by the "normalizers."[55] In

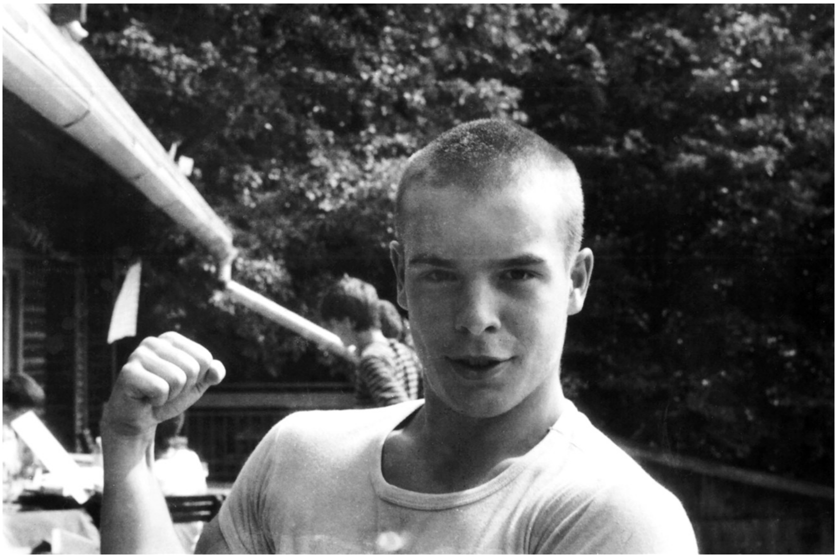


Figure 7.2
Stanislav Hrda around 1987 at a model airplane club getaway with Sybilasoft's Hlaváč brothers. Image courtesy of Stanislav Hrda.

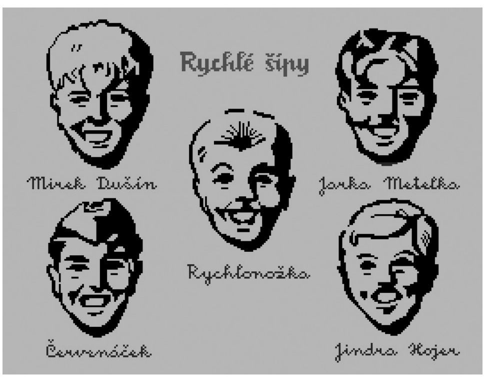


Figure 7.3
Loading screen of Mihula's *The Mystery of the Conundrum*, featuring the five members of the *Fast Arrows* club. Long before image scanning was an accessible technology, Mihula transferred the graphics from the original comic to graph paper while holding both over a lamp, and stored the resulting binary pixel values into the computer.

the 1980s, *Fast Arrows* was a cult favorite, still read by young people from dog-eared copies handed down by their fathers or grandfathers.

Mihula's game was not an adaptation of the comics' stories, but of the Foglar-penned tie-in novel *The Mystery of the Conundrum* (1941), well suited to be remade into a text adventure because of its focus on exploration and puzzle-solving. In 1988, the game was a rare piece of new *Fast Arrows* content released into informal distribution—obviating the official ban. With an extensive map, tiny but evocative illustrations, and numerous puzzles, it ranked among the most substantial domestic adventure games and was commercially rereleased, along with a sequel made by Mihula's brother, in 1990.[56]

In the late 1980s, inclusion of unofficially circulating or "banned" material was not a dissident act; it was nothing that would be outright penalized or criminalized. After all, Mihula wrote his game in army barracks. But it was nonetheless a political act, signaling an affiliation with unofficial culture, and a communication act, highlighting the possibility of spreading this unofficial culture through the medium of the computer game.

Small Subversions

In the 1980s, Communist propaganda was still omnipresent in Czechoslovak public spaces. Red stars, hammers and sickles, banners with slogans, and portraits of Lenin and Czechoslovak Communist heroes all adorned the buildings and hallways of institutions and factories.[57] Propaganda was heavily standardized, reduced to a series of token images and phrases. Each Labor Day (celebrated on May 1), each Victorious February and Red October, one could see the same symbols and read the same slogans, ritualistically repeating year after year (see figure 7.4). According to Yurchak, this rigidness enabled these signs to be reinterpreted in various ways:

> Performative replication of the precise forms of authoritative representation rendered the constative meanings associated with this representation unanchored, increasingly unpredictable, and open to new interpretations, enabling the emergence of new and unanticipated meanings, relations, and lifestyles in various contexts of everyday life.[58]

Figure 7.4
A snapshot from an early 1980s May 1 parade on the Letná plain in Prague. Photo by Karel Bucháček.

The last Soviet generation was exposed to a barrage of Communist symbols, but did not necessarily connect them to their intended ideological meanings. The stale symbols of an outmoded regime turned into material for pranks and subversions, which became common in the 1980s as the legitimacy of the Communist regime was eroding. Not even the parades organized to manifest the support of the regime were spared. As Yurchak points out about the Soviet Union, the iconography and ideological content of these parades were rarely taken at face value, and many young people experienced them as fun events where they could hang out with friends.[59] Stanislav Hrda remembers:

> Participation in the May 1 parade was mandatory, and we were supposed to carry the flags of friendly [socialist] countries. Our high school was quite old and we found out that there was an Albanian flag. At that point, Albania was no longer

> a friendly country and its flag was no longer used in the parades. So I grabbed it, and enthusiastically waved the Albanian flag at the May 1 parade, in front of the platform with all the party officials.[60]

In a similar fashion, he and fellow coders included pranks that decontextualized and subverted Communist iconography. An early example is the 1985 text adventure *Space Saving Mission,* written by the Golden Triangle's Tomáš Rylek. The game was distributed under the acronym *SSM*—also the acronym of the Socialist Union of Youth.[61] A counterpart to the Soviet Komsomol, the Union was the only major Czechoslovak youth organization, and it was also a mother organization to the Pioneer Movement, which ran the Station of Young Technicians that Rylek attended. The title's loading screen featured a massive and lifelike image of the Pioneer emblem (see figure 7.5). However, the game's science fiction narrative was in no way related to the Socialist Union of Youth or the Pioneer. According to Rylek, the inclusion of the logo might have been provocative, but he notes that he and his computing buddies "certainly weren't dissidents."[62]

It is tempting to liken this symbolic work to the practices of the punk subcultures, analyzed by scholars such as Dick Hebdige. In the late 1970s punk style, "union jacks were emblazoned on the backs of grubby parka anoraks or cut up and converted into smartly tailored jackets." The insignia of power thus became "'empty' fetishes."[63] However, computer homebrewers were not as confrontational as the punk subculture. Perhaps a more fitting

Figure 7.5
The loading screen of *Space Saving Mission*. The game was credited to "Ultravideo Software," unlike Rylek's later games, which used the "TRC" label.

concept is that of *excorporation*, introduced by John Fiske, describing the process in which "the subordinate, in a reversal of incorporation, steal the discourse of the dominant and use its signifiers for their own pleasures, their own identities."[64] Whereas the mainstream fashion industry had eventually *incorporated* the aesthetics and signifiers of the punk subculture and disconnected them from their subcultural meaning, Fiske gives the counterexample of the pop artist Madonna's "Into the Groove" video, which *excorporated* Christian imagery (such as the cross) by combining it with the aesthetics of pornography. According to Fiske, this "new combination does not 'mean' anything specific, all it signifies is [the artist's] power over discourse, her ability to use the already written signifiers of patriarchal Christianity, and to tear them away from their signifieds so that she is not subjected by the discourse as she uses it."[65] Although Czechoslovak homebrewers seem to be light-years away from Madonna, the way they worked with signs was similar. Through gratuitous use of the Pioneer logo, as well as other symbols, the authors claimed the power to digitize and reuse those symbols in a setting that they controlled—in the world of computer games.

The 1988 text adventure *Shatokhin* by Sybilasoft's Stanislav Hrda (with graphics by Michal Hlaváč) similarly—but more extensively—subverted Soviet culture and iconography.[66] At that time, promotion of Soviet culture was part of Czechoslovak cultural policies. In the 1980s, for example, Soviet films accounted for twenty percent of theatrical releases, only topped by domestic Czechoslovak production. In addition, theaters had to ensure that Soviet movies accounted for at least twelve percent of total attendance, which they achieved with the help of special screenings for schools or factory collectives.[67] The fact that consumption of Soviet culture was so often compulsory made it vulnerable to mockery, especially because the Soviet iconography was inextricably connected to the Communist ideology, which young people found unconvincing and outdated.

Shatokhin's titular character, Major Shatokhin, originally appeared in the 1985 Soviet film *Solo Journey*. The film itself was an example of late Soviet military propaganda, featuring the heroic major infiltrating a US military base and disrupting a CIA-orchestrated provocation aimed at derailing the upcoming disarmament talks between the two countries. Using stylistic and thematic elements of American action movies of the era, and casting several American actors, it has been described as a Soviet attempt to match the *Rambo* series. Although today's viewers mostly watch it as a parody, original promotional materials suggest that it was serious propaganda.[68] The members of Sybilasoft were familiar with both *Solo Journey* and *Rambo*, although the latter was only available on pirated videotapes. In the game, Hrda pits Shatokhin against Rambo during the Vietnam War in a series of dramatic duels. Its narrative contains plenty of self-aware humor and parodic elements. As Hrda explains, it "posed as an exemplary proregime game, but all these jokes inside of it twisted it around."[69]

The game opens with two introductory screens. The first one features a portrait of the main character and his last name written in Cyrillic type (see figure 7.6, left), and the second is a gratuitous full-screen rendition of the hammer and sickle symbol, a staple of Soviet iconography (see figure 7.6, right).[70] Hrda had previously smuggled a hammer and sickle into *Fuksoft* as a red herring. As an item that could be picked up but did not serve any function, it was—in his words—"just a useless piece of junk," highlighting its status as an empty signifier. In *Shatokhin*, Sybilasoft "came up with the idea to put in as large a hammer and sickle as possible, right in your face."[71] They knew that among their audience, no one would interpret this as a sign of overzealous propaganda. Unlike TV and radio, computer games contained no pro-Soviet discourse; the only time one would encounter Soviet symbols in games would be in Western titles such as *Raid over Moscow* or *Green Beret*, in which Soviets were the enemy. To a Spectrum user at the time, *Shatokhin*'s excess of Soviet symbolism would appear the way it was intended—as an absurd joke.

Figure 7.6
The two intro screens of *Shatokhin*. The game's program and story were the work of Stanislav Hrda, and the graphics were drawn by Michal Hlaváč.

Within the game's narrative, the player may lead Major Shatokhin to a heroic victory over Rambo. At the same time, the game exploits the potential for failure—specific to the medium of computer games—to punish the player for wrong moves, and subjects Shatokhin to humiliating and spectacular deaths at the hands of John Rambo, for instance, being "charred to bits" in a burning helicopter. The theme of humiliation is even more pronounced in another, hidden part of the game, effectively an Easter egg. If the player goes to the controls menu and redefines the keys to *K*, *G*, and *B*, he or she enters a minigame called "Mini Shatokhin."[72] Its story takes place twenty-four years after the events of the main game; Major Shatokhin is interned in a sanatorium for disabled veterans, and Rambo comes to get his revenge. A confrontation between the two elderly ex-heroes can have two major outcomes—Rambo killing Shatokhin with a fork, or Shatokhin suffocating Rambo with his "historic" socks. Like the hammer and sickle, the brooding, serious, and heroic Shatokhin becomes fodder for the author's playful subversion. The aesthetics and plot elements of Communist propaganda and both American and Soviet war films have been *excorporated* to create an entertaining, slightly transgressive violent comedy.

A Protest of Sorts

Shatokhin and *Fuksoft* contained nods and in-jokes for an audience that enjoyed the subversion of Communist iconography, but these games never explicitly called for action. However, a handful of other titles employed similar tactics to openly oppose the regime.

The earliest preserved example is the 1988 text adventure *RECONSTRUCTION*.[73] Attributed to a fictional "Central Committee Software" (probably referring to the Central Committee of the Communist Party), it claims to have been released "to commemorate the twentieth anniversary of the liberation of Czechoslovakia by allied [Warsaw Pact] forces." The game itself starts out in a decrepit, abandoned building, without instructions or a clear goal. Eventually, the player discovers that to make progress, the main character must destroy and deface symbols of the regime. Although straight-faced, the game uses satire to great effect. To light the way through a dark tunnel, the character sets ablaze a copy of Marx's *Capital*—a book whose one-sided and dogmatic misreading was at the heart of the Communist Party's Marxist-Leninist ideology. When burning, the tome "emits the light of progress." But setting it on fire is not an easy task, because the game equips the character with a Soviet-made lighter that only works with thirty percent probability—a jab at the quality of Soviet products. The game's ultimate goal is to tear down a statue of Lenin by detonating a bomb in an anonymous, nondescript town. This contrasts with the title, which refers to Gorbachev's liberal reforms in the Soviet Union. The game thus implies that a "reconstruction" of state socialism is impossible, and a metaphorical destruction of its calcified ideology is the only way out. Throughout the game, several notorious slogans are displayed on the bottom line of the screen, ironically framing the game's destructive narrative:

> Workers of the world unite
> Forever with the Soviet Union
> Five-year plan today—Communism tomorrow
> Socialism—the way of tomorrow
> Glory to the Leninist Politics of the Party

RECONSTRUCTION employs the unique traits of the computer game medium by giving the player agency, or complicity, in all these subversions. But the game does not stop there. Upon its successful completion, the player is explicitly invited to take part in a real-life demonstration commemorating the twentieth anniversary of the invasion (and occupation) of Czechoslovakia by the Soviet army: "Congratulations once more. We will all meet on August 21 in Old Town Square … (Or anywhere else.)." The gathering on August 21, 1988, ended up being the first of the series of demonstrations leading up to the Velvet Revolution.

In the discipline of game studies, many authors have theorized about games made for persuasive, political, or otherwise serious reasons. One of the more popular approaches, Bogost's *procedural rhetoric*, presupposes that the procedures or mechanics of these games express opinions by modeling selected processes in a way that reveals the author's argument.[74] That might be the case in games such as Molleindustria's *McDonald's Videogame*, which simulates the controversial business practices of the fast food giant.[75] But the concept can hardly be applied to Czechoslovak "protest" games, which rarely model any real-life procedures. It is more fruitful to frame our cases in terms of their function rather than formal properties. Rather than a simulation, *RECONSTRUCTION* can be likened to an activist satirical pamphlet and a piece of tactical media. It was most likely to be played by people who shared similar opinions about Czechoslovak state socialism. Its most powerful message was not a commentary on how this system works, but an assurance that protest is possible.

For obvious reasons, the game was released anonymously. But soon after the Velvet Revolution, the identity of Central Committee Software was revealed. In the scrolling message of František Fuka's widely popular *Indiana Jones 3* (1990), everyone could read that *RECONSTRUCTION* was the work of his Golden Triangle cohort Miroslav Fídler. Looking back, Fídler acknowledges the game as a "naïve form of protest." As a young programmer, he was frustrated by Czechoslovak state socialism: "I had always wanted to make a living programming and making games, and the regime, in a way, made that impossible." Unlike his other games from that time, which used sophisticated machine code routines, *RECONSTRUCTION* was a game he wrote purely in BASIC, primarily to hide his identity: "I was so ingenious that in order for the State Security not to catch me, I made the game awfully primitive." The game was released inconspicuously, only to select people. "If the State Security had come, we would have just told them we found it somewhere, because there is no [information about authorship] in the code."[76]

It is quite telling that Fídler chose to hide his technical abilities and his label—the two things that Czechoslovak homebrewers tended to boast about—to express his political frustrations.[77] At a time when scenesters competed with each other in encrypting their code using "trademark" routines, he made *RECONSTRUCTION*'s code completely transparent, and thus paradoxically more anonymous. Besides hiding his identity, the choice made the game slightly easier to port and convert—as evidenced by the existence of an 8-bit Atari conversion.

Taking to the Streets

RECONSTRUCTION's open call to join the August demonstration resonated with the growing participation in public gatherings in the last years of state socialism. Small subversions, such as waving an Albanian flag at a parade, were common, and as the regime's tight grip on power was loosening up, people became less afraid to voice their opinions. Ordinary citizens, and especially students—who did not have firsthand experience of post-1968 repressions—started to join the dissent in demonstrations, usually organized to mark an important anniversary. Some of them believed in the cause, some were curious, and others were in it for the thrill—such as František Fuka: "I went to [demonstrations], because it was exciting. Just like young kids play 'cops and robbers,' we played 'cops and protesters.'"[78] No matter the individual motivations, in late 1988, demonstrations had already become a part of lived experience for significant groups of young men and women in larger cities.[79] Almost none of these gatherings were approved by authorities, prompting riot police to come and disperse the crowds. During these police interventions, persons identified as dissidents or organizers tended to be harassed and arrested, but "ordinary" people were largely left alone. This changed with the so-called "Palach Week."

In January 1989, several dissident and opposition groups organized demonstrations to commemorate the twentieth anniversary of the self-immolation of Jan Palach, a student who had set himself on fire to protest the post-1968 political developments in the country.[80] On January 15, thousands of people gathered in Wenceslas Square, downtown Prague's widest and most spacious boulevard and a traditional site of public events. Citizens' efforts to pay respects to Palach and protest the oppressive regime were met with "disproportionate police response, including the use of tear gas and water cannon against peaceful protesters and even curious bystanders," arousing "widespread public indignation" and discrediting the authorities (see figure 7.7).[81] Undeterred by the police, people continued to gather for the following four days.

After two decades of less visible oppression, the blatant violence of the police and the People's Militia (the Communist Party's paramilitary force) during Palach Week started an open and public confrontation between the state and its citizens. Official news media tried to hush the events' significance but could not hide them altogether. As one of the main organizers reminisced twenty years later: "It was a breaking point because of the determination of the participants. And because the whole society noticed it."[82]

Among the responses was a text adventure called *The Adventures of Indiana Jones in Wenceslas Square in Prague on January 16, 1989* (henceforth *Indiana Jones in Wenceslas Square*).[83] The game puts the iconic Western hero of Czechoslovak text adventures in the shoes of a protester. The goal is to escape the square and return to "your homeland, America." The game is

Figure 7.7
Riot police in action in Wenceslas Square during Palach Week. Photo by Karel Bucháček.

aware of its absurd premise and builds on it with irony and hyperbole. In its instructions, the game signals that it is aimed at "experienced players," and it is, indeed, quite challenging. Although geographically faithful, it portrays Wenceslas Square as a battlefield where every careless move is punished with death, where tear gas, water cannons, and policemen await on every corner. Like *Shatokhin*, the game revels in spectacular failures, but this time, they serve to forefront and exaggerate police violence:

> You are standing at an unobstructed entrance into the subway. As soon as you showed up, an officer came to you and searched you. Having found nothing, he called on his "comrades" and they beat you senseless. As they were running away to deal with some woman with a baby carriage, one of them lost a machete. You crawled for it and committed hara-kiri. INDIANA JONES IS DEAD.

While drawing attention to the violence as such, the game also allows the player to relive the demonstration as a power fantasy. To finish the short but difficult game, Indiana Jones must brutally execute members of the Communist police as well as the People's Militia, as in the following scene:

> You are standing in front of the Grocery House department store. The entrance into the subway is fortunately clear. An annoying man (probably a communist) is looking out of a balcony and happily watching the good work of the members of the Public Security. You can go down, to the right and inside. You see a cop.
>
> > USE AXE
>
> You drove your axe so deep inside his skull that it cannot be pulled out. You see a dead cop.

Indiana Jones prevails by improvisation and unconventional methods, such as swinging the axe or hurling stones and iron bars; the only gun in the game lacks ammunition and will cause trouble more than it helps. Rather than equipping weapons of those who represent power, Indy uses weapons of the streets. This not only makes for more interesting puzzles but also clearly positions the character as one of the protesters. Unlike in the case of *RECONSTRUCTION*, the author of the game remains unknown. It is cryptically credited to a "Susan Reborn," although the affixes in the pseudonym are male, making the author's gender unclear. The game's highly confrontational, angry tone suggests its creator could have had his or her stakes in the actual demonstration, which would make the game partially hyperlocal.[84]

Ten months after Palach Week, on November 17, several of Prague's student associations organized a march through the city to commemorate the fiftieth anniversary of the closure of Czechoslovak universities and the execution of student leaders by the Nazis in 1939. (Another demonstration—similar, but smaller and less eventful—had already taken place in Bratislava on November 16.) The gathering attracted an unprecedented number of people—estimated at more than ten thousand—bearing banners such as "Democracy and Law," "Genuine Perestroika," and "Free Elections." A group of around five thousand protesters deviated from the approved route and marched toward Wenceslas Square, but they were flanked by police on the nearby Národní třída boulevard. In the ensuing turmoil, around six hundred students were reported as beaten up by heavily armed riot police—despite crying "We have empty hands" and "We don't want violence."[85] This large-scale unprovoked violence was to be the last blow to the legitimacy of the regime, marking the start of the peaceful Velvet Revolution. The state-controlled media downplayed the scope of the November 17 events as well as the following days' protests, but the news spread quickly among the population thanks to improvised tactical media such as pamphlets, leaflets, samizdat periodicals, and simple personal communication. Student leaders and opposition activists traveled to all corners of the country to deliver the message. The games based on these events likewise shifted their focus to finding out and delivering their true account.

According to its paratextual information, the game entitled simply *11/17/1989* was released two days after the demonstration, on November 19.[86] Its loading screen reiterates the most powerful slogan of the Velvet Revolution, "We Don't Want Violence," and the instructions make it clear that the game is intended as a "protest against the brutal attack of the riot police." Sticking out of the intro screen is the face of a grotesquely aggressive-looking policeman, wearing a helmet and holding a shield and a baton, the equipment of the riot police squads responsible for the November 17 violence. A closer inspection reveals that the face is—pixel for pixel—identical with the face of a karate fighter on the introductory screen of *The Way of the Exploding Fist*, a 1985 Australian-made fighting game. The creators of *11/17/1989* most likely redrew the image and replaced the kimono with a hastily drawn police uniform and helmet (see figure 7.8). There is some irony in repurposing a screen from a fighting game to oppose violence, but reusing graphics and sound from commercial games was common. In this case, the authors likely rushed the game to release a timely, almost immediate response to the events.

Like *Indiana Jones in Wenceslas Square*, the game re-creates the events of the demonstration—but unlike the Indiana Jones title, it does not dwell too

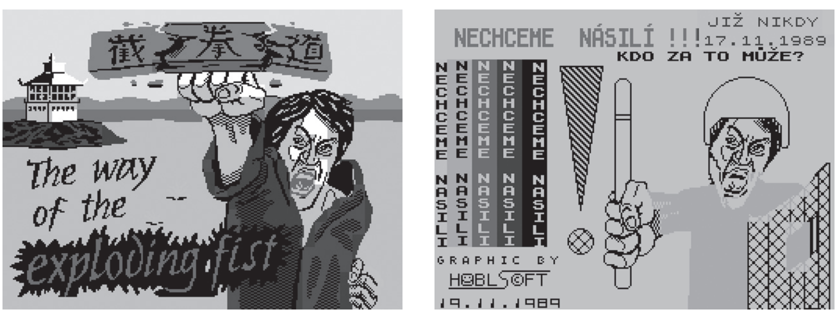


Figure 7.8
Loading screens of *The Way of the Exploding Fist* by Beam Software (left) and *11/17/1989* (right). The face is in the same position, suggesting that the original screen was loaded into a graphics editor and redrawn.

much on the violence itself. At the beginning, the anonymous main character is being shoved forward with the crowd, and has to escape the flank into a nearby building, find video recording equipment, and record evidence of police violence. A similar plot features in another Velvet Revolution–themed game, *Strike*, which reacts to the general strike of November 27 that demanded the end of the Communist Party's monopoly on power. The goal of this very short adventure is to get access to an agency news item reporting that the Federal Assembly removed the clause about the "leading role of the Communist Party" from the Czechoslovak Constitution.

Both games focus on journalistic work. Unbiased reporting was rare in the early days of the Velvet Revolution, and people were hungry for news unfiltered by official propaganda. Besides analyzing these titles as tactical media, we can also understand them as *newsgames*, described by Bogost, Ferrari, and Schweizer as "a broad body of work produced at the intersection of videogames and journalism."[87] *Strike* and *11/17/1989* are *newsgames* in two respects—they not only cover current events and thus supplement journalistic practice, but they also focalize the very act of reporting news.

By spreading the word of discontent through microcomputer technology, activist games fulfilled some of the goals of the "Samizdat Research Institute" from chapter 3—although there are no traces of their authors' connections to dissident circles. These titles were not "quality" products when measured by industry standards. They were often short, simple, and buggy text adventures, limited by unsophisticated parsers or menu-driven systems. But once again, they should not be evaluated as separate and bounded artifacts. They were unlikely to convince or entertain people by themselves, but instead added to the chorus of the Velvet Revolution's tactical media.

In total, at least four "protest" games by four different authors were created in late 1980s. We can understand them as a culmination of two trends—political activism becoming an everyday concern, and games becoming an outlet for personal expression. Such early and continuous use of computer games as tactical media has so far not been documented anywhere else in the world. In the literature on serious and persuasive games, Bogost provides the earliest example, the 1982 US title *Tax Avoiders*, which mounts "an interesting and relatively complex procedural rhetoric about tax avoidance strategies."[88] However, its goal—to become a millionaire—is very much in line with the neoliberal ideology of the Reagan era, and is therefore not as subversive as it might seem. In the United Kingdom, several games from the first half of the decade engaged with current social and political issues, likewise through parody and satire. According to Wade, "Games were [at that point] still able to challenge the status quo and engage in political discourse."[89] He gives the example of *Hampstead*, a 1984 text adventure that satirized "people who treat living in the fashionable north London suburb as the ultimate badge of social success."[90] In the popular platformer *Monty on the Run*, the titular anthropomorphic mole Monty runs from the authorities after his involvement in the 1984–1985 miners' strike.[91] These British games did comment on politics, but rather than being tactical media in the activist sense, they were essentially commercial releases whose topical nature was used to spark audience interest.

Another early example of a political game is the controversial 1989 Israeli action title *Intifada*, which reacted to the Arab uprising in the territories occupied by Israel.[92] In the game, the player "is an Israeli soldier on patrol in an Arab city. Palestinians throw rocks and Molotov cocktails at him. His job is to stop the Arab rioters while staying within army orders against opening fire, which become more lenient as the game progresses."[93] The blatantly anti-Palestinian game was criticized for trivializing Arab deaths and encouraging Jews to shoot Arabs. Despite its deeply problematic politics, it nonetheless shares some characteristics with Czechoslovak examples—it was by made by a single author as a political statement, and re-creates real-life events. Outside of Czechoslovakia, no similar socialist-era examples have been described in the former Soviet bloc. However, this may be due to the general lack of comprehensive national histories of games in the region.

Examples of "classic" political games in recent literature include *September 12th*, an antiwar game made by game scholar and designer Gonzalo Frasca in 2003, and *McDonald's Videogame*, made by the Molleindustria collective in 2006.[94] The striking difference between these games and my examples is that the newer titles are relatively high-brow, self-reflective efforts by accomplished activists and intellectuals, whereas the Czechoslovak examples were likely to be (and *RECONSTRUCTION* certainly was) products of regular homebrewers.

To repeat the point Blažek made in 1989, "Any one of you can become a hero of microcomputer culture—you only need to have something to say." And users did have things to say—about themselves, their communities, and politics. This chapter's narrative has shown how games were embedded in the life and communication practices of ordinary homebrewers, from harmless in-group nods to small subversions and explicit political statements. We can sum up this mentality in two complementary thoughts: If something is on your mind, why not make a game about it? And: If you want to make a game, why not make it about something that is on your mind? In the given context, making political games on 8-bit computers was not a rarefied revolutionary practice. Quite the opposite, it shows how naturally homebrewers treated games as a means of communication and self-expression.

In the introduction, I set out to view games as a subset of a wider category of coding acts. This brings us to a pressing question. Given the existence of the informal distribution network, why would users not choose another way of expressing themselves than specifically *games*? Why, for instance, did they not use text files? On 8-bit computers, distribution of text files was certainly possible. Some manuals and walkthroughs were distributed this way, and the "Samizdat Research Institute" even planned to digitally circulate samizdat. But trading text files on a cassette tape was impractical because of the need for rewinding to load a specific file.[95] In addition, 8-bit computers were conventionally connected to a specific set of practices. Fans were used to copying games, so anything that resembled a game was a suitable host for messages. Each game could then be a complex coding act that delivered multiple messages in addition to the gameplay. Text adventures could fulfill this function with extreme efficiency, minimalizing the extent of programming work and highlighting statements and narratives. As the more contemporary genre label "interactive fiction" suggests, text adventures were very close to literature, and some Czechoslovak text adventure authors wrote or considered writing short stories as an alternative.[96] An additional advantage of text adventures was that they addressed users in second person—potentially making it easier for users to identify with the character than when mediated through an on-screen avatar. That could be especially powerful in hyperlocal and protest games.

Games such as *Fuksoft* might have reached thousands of players, and their popularity can be demonstrated by their conversions and sequels, whereas titles such as *Indiana Jones in Wenceslas Square* were likely to make a smaller impact due to their sensitive subject matter, low production values, and ephemerality. Protest games were relatively rare (four out of over two hundred preserved titles), and microcomputer users still accounted for less than 2 percent of the population,[97] meaning that folk and punk rock songs that expressed the same sentiments were much more influential than these games. Overall, it would be foolish to think that computer games brought down the Communist regime—but they were certainly considered a suitable instrument for subversion and protest, and utilized as a versatile, full-fledged medium of expression.

Conclusion

This was a story of how a country where computer stores did not sell computers gave birth to a lively amateur movement. Its members “gamed” the Iron Curtain and created a national gaming scene, whose homebrewers used computer games in ways that would be considered inventive and extraordinary even by today’s standards. Each chapter documented a crucial point in that story. Below is a summary of these seven key points.

1. *Separation of state and microcomputers*. Despite the talk of scientific-technological progress, Communist authorities never claimed the territory of 8-bit home computers, which they found of little significance. They were not interested in the social or cultural impact of digital technologies, but only in their benefits for the command economy. The state’s indifference then pushed microcomputer activity into *vnye* milieus.
2. *Emergent mobilities of computer hardware*. Despite the state’s indifference and numerous other obstacles, hardware made its way into the country, at first thanks to dedicated individuals who imported their machines individually. As foreign computers were brought into Czechoslovak homes, programming and computer games became a part of everyday life for tens of thousands of people, mainly young men and members of the technical intelligentsia.
3. *Computer club network*. Although the state itself did not claim the territory of home computers, its socialist organizations granted patronage to clubs, which became hubs of relatively autonomous user networks. Clubs in turn offered services that were otherwise (in capitalist contexts) mostly performed by commercial companies. By granting them patronage and facilities, the authorities essentially outsourced the development of home computing to amateurs.

4. *The bricoleur approach*. In their newsletters and publications, Czechoslovak hobbyist communities promoted self-reliance and creativity over a consumerist approach to technology. This, and the fact that 8-bit machines offered easy gateways to programming and hardware tinkering, contributed to the cult of the computer bricoleur. The same approach was applied to games. Although an increasing number of people bought computers solely to play games, disassembling them or making one's own was considered a more desirable practice.
5. *Informal distribution networks*. Local micro users developed efficient tactics for moving software around even without a legitimate software market. Games arrived in the country with relatively short delays, but usually did so without any documentation and in variously modified versions. Therefore, players tended to treat them as material for playful experimentation and subsequent coding acts, rather than playing them according to the designers' intentions.
6. *Creative transformations and imitations*. Unrestrained by copyright, amateurs felt free to convert, clone, and imitate foreign games. In the process, they created a body of work and an inventory of templates that drew from foreign inspirations, but adapted them to the skills and needs of the domestic scene. The Czechoslovak *textovka* and *hacking game* emerged as distinctly local variations of Western genres.
7. *Games as tactical media*. As computers and games became more widespread among the general populace in the late 1980s, people began to use them not only for entertainment but also as a medium of communication. From personal messages and small subversions to political statements, games became a go-to medium of expression among an interconnected group of young technophiles.

The order of the chapters and the summary does not imply that each preceding point was a necessary prerequisite for the next. We could, for example, ask: Would games be treated as a medium of expression without the contribution of computer clubs? It is certainly possible, although less likely. Nor is this list a normative judgment about some games or practices (e.g., games as tactical media) being more valuable than others. Nevertheless, I see subversive political games as a point of culmination; a point at which an originally Western form, performed on mostly Western machines, became truly embedded in the Czechoslovak sociopolitical context; a point at which Czechoslovak kids and amateurs discovered the expressive potential of the medium of computer games. The book's epilogue will sketch a continuation of this narrative.

Each of the chapters attempted to cover the decade from the early 1980s to the Velvet Revolution. My aim was to leave them free from arbitrary periodizations that could have brought in more confusion than insight. However, now is a good moment to reflect on the overall narrative and identify its crucial phases. (Note that these phases often overlapped and proceeded at different speeds in different places—big cities tended to be ahead of the rest of the country thanks to a higher concentration of resources and expert know-how.)

1. *Mainframe and mini phase* (until about 1980). During this time, computer games were reserved for programmers and operators of industrial and institutional machines.
2. *Early adopter phase* (1980–1982). A period of when microcomputers first appeared among the technical intelligentsia, first cybernetics youth groups started to gather, and type-in games started to circulate.
3. *Consolidation phase* (1982–1985). In this phase, users started to organize and connect to each other. In addition, ZX Spectrum emerged as the primary machine of the Czechoslovak hobby scene. The year 1985 marked a significant turning point. This was when the *Mikrobáze* newsletter was launched, along with the first critiques of the computer game craze, and when the PMD 85 computer was released to schools and computer clubs. It was also the year when the members of the Golden Triangle collective released their first games, laying a foundation for the homebrew scene.
4. *Imitation phase* (1985–1987). In this period, Western games became widely available, the informal distribution network was well established, and the Czechoslovak homebrewer elite started to imitate Western titles.
5. *Scene phase* (1987–1989). During this phase, access to computers improved thanks to the launch of Didaktik Gama and imports of Atari and Sharp machines, and an interconnected network of homebrewers created original titles.

Bricoleurs and Tacticians

In the introduction, I connected my project to several subfields and bodies of literature. I now return and evaluate how this book speaks to them, starting with regional history and the history of technology, and then moving through game studies to game histories.

Seen from the perspective of regional history, this book has documented an intriguing, if niche element of the period's everyday life and leisure. Computer games in 1980s Czechoslovakia represented a point where computer technology met everyday life and the East met the West. The historical narrative has extended our knowledge about do-it-yourself practices, and the scale and nature of mobilities between the two sides of the Iron Curtain. More importantly, it has revealed the need to study regional and national histories more holistically. So far, histories of technology, as well as economic histories, have often been separated from cultural histories and media histories. This is a result of disciplinary politics, and more generally, of the modernist split between the realm of things and the realm of man, well described by Latour.[1] This book has highlighted the need to treat technology *as* culture.

The culture that arose around microcomputers in Czechoslovakia was inextricably connected to the features of microcomputer hardware. In the words of Ted Nelson, a micro is a "versatile gizmo," a portable and decentralized piece of technology that could easily be taken out of its country of sale and brought behind the Iron Curtain, that functioned without any additional infrastructure and support, and that was difficult to control from the center. User bases in the Soviet bloc were largely dependent on Western imports, but national foreign trade and monetary policies created idiosyncratic computer cultures through a combination of import barriers and individual deals, such as in the case of the Sharp MZ 800.

The history of computing and technology has often, quite understandably, focused on high-profile projects and artifacts. My material has allowed me to sketch out a people's history of computers, exploring the tiny interventions, poaching, imitations, and transformations that users engaged in. Chapter 1 documented a clash between two teleologies of computing: the official teleology of industry automation, which saw computers as mere tools in achieving grander goals, and the hobbyist teleology, for which the computer was not a means, but an end in itself—an object to be explored and tinkered with. These two teleologies correspond to the technocratic and technophile approaches to technology, to engineer and bricoleur subjectivities, and to strategies and tactics. In *The Savage Mind*, Lévi-Strauss describes bricolage as a type of activity that predates abstract modern science; in his view, bricolage is a "science of the concrete."[2] Czechoslovak hobbyists oriented themselves toward the concrete. Although they used technology that seemed to embody the triumph of modernity, their tactics diverged from the modernist paradigm. They shunned both the socialist and the consumerist ideology, focusing on artifacts and short-term goals rather than on grand ideas. Their work with hardware was mostly do-it-yourself, and their approach to designing games was reminiscent of folklore.

Tactical meshworks of computer clubs were indispensable to the advancement of microcomputer hardware and software in Czechoslovakia. They allowed user communities to overcome the limitations of state socialism and explore the possibilities of computers and computer games through tinkering and improvisation. The clubs became sites of activist participatory work, somewhat resembling the eighteenth-century coffee houses that Jürgen Habermas put forward as examples of a public sphere.[3] But like those coffee houses, computer clubs were not equally accessible to everyone, and their lack of gender diversity remains a more disputable part of their heritage. As shown in chapter 3, women accounted for as little as 3 percent of computer club membership, due to a combination of structural factors—such as traditional gender stereotyping, gender segregation in higher education, the Czechoslovak women's double burden of work and household duties—as well as cultural factors, including the clubs' fraternity culture and their association with masculine pursuits such as ham radio.

We Have Always Been Indie

Many of the games discussed in this book would be considered "bad" by contemporary players, because of their technical deficiencies and uneven game design. Perhaps Czechoslovak homebrewers did not *know* how to make a quality game, or perhaps they did not try. However, the lack of knowledge was also liberating. Czechoslovak homebrew production in the 1980s is amazing in its thematic diversity. People wrote games about Indiana Jones and Flappy the mole/chicken—but they also wrote about their friends, their favorite songs, and antiregime protests. They programmed in a carefree atmosphere, without pressure to conform to industry norms—although with a bit of peer pressure. Their titles may not have excelled in gameplay, but they abounded in immediacy and homegrown charm.

Today, we might label them "indie" games, a term that denotes a counterpoint to mainstream corporate production.[4] In her 2012 book *Rise of the Video Game Zinesters*, the prominent independent designer Anna Anthropy encourages readers to "take back" the art form of video games and create games unshackled by industry standards and expectations—to use the medium to express themselves. In a section entitled "What to Make a Game About?" Anthropy answers: "Your dog, your cat, your child, your boyfriend, … your friends, your imaginary friends, your summer vacation, your winter in the mountains, your childhood home, your current home, your future home."[5] This is very close to the approach of the Czechoslovak homebrewers. Who were they if not "zinesters"?

Anthropy's rhetoric of "taking back" an art form implies longing for a bygone time when games were unspoiled by the industry, and anybody could make a game about anything. She suggests that this might have been the early 1990s era of semi-amateur shareware games such as *ZZT*, about which she also wrote a book.[6] In his research on UK gaming history, Kirkpatrick discovers yet another period of creative explosion—early 1980s Britain, where hundreds of bedroom coders experimented with their computers, releasing games about the Bible, lawnmowers, and the royal baby. As the industry consolidated, this carefree anything-goes atmosphere diminished in favor of following successful templates—a process that Kirkpatrick calls "containment of ludic imagination."[7] This echoes Horkheimer's and Adorno's 1930s critique of the cultural industry, in which they lament the commercial takeover of radio, writing that "private transmissions [are confined] to the apocryphal sphere of 'amateurs,'" and criticizing commercial radio for sanitizing "any trace of spontaneity."[8]

Today, the dominance of industrial production is often interpreted as a given—as a default configuration of creative work in the medium. However, that was not always the case. Early computer games such as *Adventure* were not made for profit.[9] Before there was an industry, people programmed games for themselves or their friends. Czechoslovakia in the 1980s was, arguably, a reservation where this preindustrial logic was maintained longer than in market-driven contexts. Rather than making games solely as pieces of entertainment, local homebrewers treated them as coding acts—envelopes with messages about themselves, their experiences, and their worldviews. The fact that their output resembles today's indie games is not coincidental. Instead, it hints at an ever-present undercurrent of amateur, homebrew, and bedroom production. Our focus on the Czechoslovak context has made this kind of production more visible because there was no commercial production to overshadow homebrew efforts.

I believe there is a lot to learn from homebrewers' tactics and their open-minded approach to computing and games. Of course, I would never wish to replicate the backwardness and oppression of normalization-era Czechoslovakia. There is, however, an argument to be made for *vnye*-style environments that are controlled neither by state power nor by commercial companies. That is one of the reasons I believe that game jams and school-based collaborative game design projects, perhaps today's closest equivalents of computer club activities, deserve public attention and support.

Toward Comparative Histories

The next big question to ask in a conclusion concerns regional and temporal specificity. Which of these findings are exclusive to Czechoslovakia, and how much can be generalized to other game histories? Over the years, I have presented my research at various international venues, and many audience members seemed to identify shards of their own memories in the Czechoslovak narrative, despite coming from places as diverse as Poland, Spain, Finland, Austria, India, and Singapore. The Czechoslovak story does not speak only to a Soviet bloc experience—it also speaks, more generally, to the *peripheral* and *marginal* experience. Many of the basic contours of the history presented in this book—such as the role of the amateur, the operation of informal distribution, and the imitation of influences arriving from the center—are likely to be common to a variety of peripheral contexts. In many of the countries mentioned above, games were similarly "poached" from more technologically advanced regions and adapted to local preferences. Different sets of economic and political conditions have often led to similar outcomes. Wherever access to hardware, software, or know-how is limited or obstructed, we can expect informal economies and amateur movements to arise. Moreover, some of the practices I have described did also occur in more spatially "central" contexts, but were marginal or marginalized. Illegal copying was common in the United Kingdom, too, but frowned upon by the press and the industry.[10] Work by the American designer Roberta Williams reflected on domestic spaces, but neither she nor her games have been fully accepted into the male-dominated canon.[11]

How typical is the Czechoslovak narrative for the Soviet bloc in general? Once again, the basic contours are likely to hold. *Vnye* spaces played important roles in Poland, Hungary, and the Soviet Union, although their infrastructures varied.[12] Polish and Hungarian engineers experienced similar excitement, but ultimately Sisyphean struggles to build domestic microcomputers for schools or homes.[13] But there were differences, too. Countries such as Poland and Hungary—and especially Yugoslavia, a state socialist country that had, however, defected from the Soviet bloc—were more open and liberal, offered more publication options, and hosted rudimentary software markets, enabling commercial game production—see the overview in chapter 6.[14] Czechoslovakia, on the other hand, represented an intriguing case of an industrialized socialist country with a satisfactory standard of living, decent education, and a strong technical intelligentsia, but without any capitalist-style market dynamic.

From the little comparable material that is available, the Czechoslovak hobbyist and homebrew groups appear more communal and bricolage-oriented than the ones in Poland or Hungary. A major difference is that the use of games for political activism and ideological subversion has not yet been described in other Soviet bloc countries in this period. This could be due to the experimental nature of Czechoslovak hobbyist and homebrew groups. Alternatively, it could be connected to the fact that the Czechoslovak protests of 1988–1989 were heavily attended by young people, and therefore resonated with the demographic of microcomputer enthusiasts. More likely, though, these games did exist in other countries but have not been discovered by historians.

Another useful point of comparison is the United Kingdom, the industry center that many Czechoslovak users looked up to. Unsurprisingly, Czechoslovakia was at least temporally lagging behind the United Kingdom. Whereas Haddon and Gray have situated the British boom of interest in 8-bit machines in the early to mid-1980s, a sizable community only formed in Czechoslovakia in the second half of the decade.[15] The discursive split of games from other forms of software, whose beginning Kirkpatrick dates to 1985, did not occur in Czechoslovakia until the 1990s; the first gaming magazine entered the market with a ten-year delay. In Czechoslovakia, the Spectrum remained a viable platform until the early to mid-1990s—at that point, British gamers had largely moved on to 16-bit hardware such as the Commodore Amiga or Sega Mega Drive.

However, this “lag” cannot be uniformly applied to all areas of computing and gaming. Czechoslovakia—and arguably the rest of the Soviet bloc—was not simply *delayed*, but created its own, independent timeline. There was a significant difference between hardware and software. Due to low purchasing power and constraints on foreign trade, Soviet bloc users adopted hardware more slowly than software. We can visualize this cross-national flow as a bottleneck in which a physical computer could get stuck, but which allowed a piece of software to pass through, because the latter could move from one cassette to another with less friction. The first Czechoslovak conversion of *Manic Miner* appeared in the summer of 1984, a year after its British release.[16] Toward the end of the decade, it would only take a couple of weeks before a new game arrived in the country. In the early 1990s, Czechoslovak programmers for 8-bit machines drew inspiration from games for more advanced platforms, for example, creating 16-bit-style point-’n’-click adventure games for the ZX Spectrum. Concurrently, local users extended the lifetimes of their computers by repairing obsolescent machines and equipping them with new homemade peripherals.

Despite these attempts at comparisons and generalizations, the truth remains that our knowledge of regional and local game histories is extremely limited. One of my goals with this book has been to sketch out a possible blueprint for future national or regional studies. That was one reason I opted for a thematic, rather than chronological, structure—dedicating separate chapters to technological policies, acquisition and domestication of hardware, communities, discourse, game distribution (together with gameplay), and production (which spanned two chapters). As for methodology, oral histories and analysis of community material have proven to be a good fit for studying a peripheral, grassroots computer game culture. Each local or thematic history will, however, require its own set of methods. Most of the territory is still uncharted, and I am looking forward to the histories that will follow.

Preserving the Peripheral

Finally, the findings in this book give us an opportunity to reflect on the practices of game preservation. So far, any research on 8-bit games in Czechoslovakia has relied on the remarkable work of fan archivists.

At present, their websites and repositories are absolutely essential to the work of historians. Besides thanking and crediting them (which I do in the acknowledgments and bibliography), we should also consider what happens if—for technical, financial, or personal reasons—these invaluable archives shut down. As Melanie Swalwell has suggested in a piece inspired by her work in New Zealand, it is the national or local cultural institutions who are best equipped to preserve indigenous game cultures.[17] In Czechoslovakia, and most of Eastern Europe, institutional preservation efforts are only just beginning.

In Slovakia, the Slovak Museum of Design is poised to take up the challenge, and it has already kicked off with a 2017 exhibition of Slovak 8-bit games (see epilogue). In the Czech Republic, no concentrated efforts have been made—not for lack of interest, but more often due to lack of funding, or because of other, more pressing priorities. The hybrid nature of games, which are pieces of both technology and culture, often leads to games falling "between the cracks of what different cultural institutions [feel] they should be collecting."[18] Nonetheless, the National Technical Museum and the National Film Archive have both expressed interest. Whoever proceeds with their plans should, in my opinion, not only consult academic historians and preservationists, but also reach out to the most dedicated fan archivists and use (and reward) their expertise.

Throughout this book, I have made several observations about the types of objects that are worth preserving. Their selection must arise from a careful study of historical user practice. Given what we learned about gaming in Czechoslovakia, we must gather more than just original and legitimate pieces of hardware and software. Regarding software—unofficial, cracked, and broken versions were often closer to the players' everyday experience than originals, but tend to receive less attention from international fan and institutional archivists. Fan archives of Commodore 64 cracks and demos, such as Demozoo and the C-64 Scene Database, are massive and invaluable, but they mostly focus on the Western European scene.[19] In the former Soviet bloc context, besides the Russian-language Spectrum-oriented Full Tape Crack Pack website, a much smaller fan archive of Yugoslavian cracked versions is hosted by the RetroSpec retro gaming website.[20] Collecting these versions should be on the agenda of both academic and institutional archivists—doing so may help us paint more precise and evocative social histories of computer games.

In his monograph on game preservation, James Newman rightly argues against the primacy of the original video game “text.” He suggests that archivists should strive to preserve not only games as playable objects but also records of “live experience of gameplay,” which would show how games were played at the time of their release.[21] I concur about the immense value of such records, but this strategy poses another tricky question. Would we not replace the primacy of the original text with the primacy of an original experience? How do we pick the site or sites of such gameplay? Historical evidence shows that tens of thousands of people in Eastern Europe and beyond played cracked and poked versions of games, often on unofficially cloned computers, with homemade joysticks, several years after the original release dates. Their non-original experiences are in no way less historically significant. We should therefore be aware of the fascinating multitude of contextualized gaming experiences.

To describe these experiences, we must conduct interviews, collect both official and subaltern publications, and perhaps—per Newman’s recommendation—look for shards of gameplay recordings. Even more importantly, we must preserve the everyday material culture of gaming. Homemade joysticks and handwritten cassette inlays are not just artifacts but also witnesses. From their structures and aesthetics, from materials, textures, and components, we will be able to read the stories of amateur bricolage—the stories of kids and hobbyists taking soldering irons, ballpoint markers, or simple everyday objects, and making games and computers into parts of their own lives. I hope this book will play its part in convincing the historian and archivist communities that stories of Soviet bloc hobby computing and gaming are worth preserving.

Epilogue: After the Curtain Fell

In the late 1980s, Soviet-backed state socialist regimes in most Eastern European countries started to collapse. Czechoslovakia's Velvet Revolution of 1989 turned out to be a surprisingly fast process. During the "pre-terminal period" of Communist rule, the rigid and conservative leadership had already lost the support of large portions of the population, including the technical intelligentsia.[1] The Czechoslovak regime, which was among the most conservative in the Soviet bloc, was also arguably the most quickly and radically dismantled. Prominent reformers from within the party—both the disgraced veterans of the Prague Spring, and the current proponents of perestroika—were sidelined. Václav Havel, who not too long before had been a relatively powerless dissident and a banned author of absurdist plays with little mass appeal, became—somewhat reluctantly—a figurehead of the opposition. He found an unlikely supporter in Marián Čalfa, a junior Communist minister, who led the negotiations with the opposition and eventually defected from the party line. In part thanks to Čalfa's lobbying, the Czechoslovak Federal Assembly—still dominated by normalization-era Communists—elected Havel president on December 29, 1989.[2] Censorship was abolished and the country became a representative democracy, with the first free elections taking place in 1990.

The transition to a market-driven economy was just as rapid. Despite 1989 polls that showed the population's preference of a middle way between socialism and capitalism,[3] the minister of finance (also future prime minister and Havel's successor as president), Václav Klaus, secured political support for his neoliberalism-inspired "shock therapy" approach to transformation, and most of the nationalized economy was privatized. As critics have argued, the speed and scope of the process were preferred at the expense of a robust legal framework and ethical considerations, leading to many cases of asset stripping and embezzlement.[4] After a series of conflicts between the Czech and Slovak governments, the latter led by a nationalist coalition, the federation was peacefully dissolved on January 1, 1993.[5] Despite the demise of the federation, Czech and Slovak game industries and gamer communities remained remarkably interconnected, in part thanks to student and labor mobility between the two countries enabled by bilateral agreements and the mutual intelligibility of the Czech and Slovak languages.

All in all, the atmosphere of the early 1990s was chaotic, but exhilarating. The newfound liberty brought a surge in publication activity; books and records by previously banned authors were coming out in large numbers, and many upstart newspapers and magazines—including ones about computers and computer games—competed in a crowded market. Following long years of state socialism and inspired by the proliferation of neoliberal policies and ideas, numerous Czechs and Slovaks searched for their inner entrepreneurs and started business of all shapes and sizes, many of them failing as the economy started consolidating.

But looking specifically at computing and computer games, we see that some infrastructures and laws may have changed almost overnight, yet player and maker practices were transforming more slowly, following the momentum that started in the mid-1980s. The Velvet Revolution and the start of the new decade did not bring a sudden switch from 8-bit to 16-bit hardware. Instead, 8-bit machines coexisted with more powerful computers, and homebrew tactics intermingled with early entrepreneurial efforts. Western hardware and software companies took their time before entering the Czechoslovak market, creating a window of opportunity for local entrepreneurs, many of whom were former amateurs.

This short epilogue will conclude some of the narratives from the previous chapters. First, I will follow the themes of chapters 1–5 and briefly sketch out their development in the period of Czechoslovakia's transition to a free market economy. Those readers interested in more in-depth analysis of gaming communities in this era can consult some of the accounts I have published elsewhere.[6] Later on, I will focus specifically on the production side of games, investigating the 1990s 8-bit commercial game development, and tracing the genealogy of current game industry trends to the 1980s homebrew scene. Finally, I will return to the personal stories of computing pioneers and show how they fared in the decades after the Velvet Revolution.

Computers and Games in Transition

Chapter 1 chronicled the attempts to manufacture home computers in the 1980s. Despite the influx of legal bulk imports that followed the Velvet Revolution, Czechoslovak micros did not simply disappear. Schools kept their PMD 85s and IQ 151s for several years after the Velvet Revolution, until they became mementos of the socialist computer industry. In the 2000s, the PMD 85 experienced a renaissance in retro computing circles—to the extent that thirteen new titles were released for it in the 2010s, compared with the roughly thirty that survived from the 1980s. Eduard Smutný, the man behind the Ondra, died prematurely in 1993 at the age of forty-nine. Retro fans, too, keep making games for his machine, and his twin brother Tomáš continues to promote his legacy. The Didaktik factory, renowned for its popular Gama computer, transformed into a full-fledged commercial enterprise in 1990. Taking advantage of its know-how and the popularity of the Spectrum platform, it continued to produce several more models of their Didaktik machine—1990's Didaktik M and 1992's disk drive–equipped Didaktik Kompakt. In total, the company produced around one hundred thousand machines (including the pre-1989 Gamas), considerably extending the viability of the Spectrum platform in the country by domestic clones.[7] According to a 1994 *ZX Magazine* survey, two-thirds of the magazine's readers owned a Didaktik rather than a British Spectrum model.[8] But overall, domestic manufacture ceased in the 1990s because of the availability of relatively affordable PC hardware imported from countries such as Taiwan.

In the 1990s, people did not have to stand in queues to get a computer in the way I described in chapter 2. As soon as the borders opened, anyone could legally travel to the West and buy both old and new machines and peripherals. Despite the new possibilities, the buying power of both the population and the institutions was rising only slowly, and cheaper 8-bit machines continued to be an affordable entry option. Many young players from the former Soviet bloc, myself included, thus started discovering the world of 8-bit games retrospectively, at a point when their platforms were already becoming obsolete. An influx of secondhand machines contributed to the rise of an active Commodore 64 scene, which created dozens of games in the early 1990s, while also taking inspiration from the Western European demoscene. Some of their production was published on cover disks in German magazines.[9]

At that point, the Commodore Amiga (and to a lesser extent the Atari ST) had become the dream platform of most young players, soon to be replaced with the IBM PC. Following its modest deployment under state socialism, the PC became a widespread standard for professional use, and building and reselling PCs from imported parts became a lucrative business. Just like their predecessors, many PCs were shared by the whole family or worked double shifts as work and entertainment machines. Around the mid-1990s, they became the dominant gaming platform, owing in part to the popularity of first-person shooters such as *Doom*.[10] Thanks to their embeddedness in informal software distribution, and the late and halfhearted entry of console importers into the market, PCs remained—according to the latest nationwide statistics from 2011—more widespread as gaming machines than consoles.[11]

Chapter 3 followed the narratives of computer clubs affiliated with socialist organizations. Most of these organizations were transformed or shut down soon after 1989. Svazarm officially dissolved in March 1990, followed by a complicated redistribution of assets. Each club could choose whether to affiliate with the successor organization, the Association of Technical Sports and Activities.[12] The Socialist Union of Youth, tainted with its close ties to the Communist Party, was forced to transfer its humongous assets to the state.[13] These organizations' demise, however, did not bring a significant blow to the practices of microcomputer users. The clubs' critical role had stemmed from the impossibility of running either nonprofit or for-profit entities under state socialism. When these options opened up, some clubs continued to meet informally, others became nonprofit civic associations, and yet others transformed into commercial businesses.

Svazarm's prominent 602 club, which had adopted entrepreneurial practices even before 1989, splintered into several commercial companies, the most notable being Software602, a successful producer of word processing and spreadsheet software that proudly kept the club's number in its name. Its Text602 word processor, first introduced in June 1989, became the Czechoslovak standard for the MS-DOS platform until the mid-1990s.[14] Its dominance was cemented by Western software's initial lack of support for

Czech and Slovak diacritics. Another major effort of the 602 club, the *Mikrobáze* newsletter, found a continuation in *Bajt*, the licensed Czechoslovak version of the US magazine *Byte*, led by *Mikrobáze*'s most influential editor, Ladislav Zajíček. Given Zajíček's mid-1980s critiques of games, it is not surprising that *Bajt* only rarely covered them.

The Czech Republic in general has grown to become an important hub of software development. For example, three renowned antivirus software suites, ESET, Avast, and AVG, whose combined worldwide market share was about thirty-seven percent in 2017, are produced in former Czechoslovakia.[15] Their histories date back to the late 1980s and early 1990s, when Czechoslovak PCs were particularly vulnerable to computer viruses due to widespread reliance on informal software distribution.[16] Czech IT successes are owed in part both to the tradition of good technical education and to hobby clubs. During the 1980s, the clubs provided informal training in microcomputer hardware and software that proved invaluable in the 1990s, when personal and home computing became a profitable business opportunity. The shift toward personal and home computing can also help explain why the participation of women in IT was declining throughout the 1990s.[17] The feminized data-processing sector dwindled after the privatization of state-owned industries, and new PC-based industries were frequently headed by alumni of hobby clubs, which were overwhelmingly male.

Chapter 4 discussed the published discourse about computer games in 1980s Czechoslovakia. I concluded there was very little of it, and it was driven by people who tended to be critical of games rather than game enthusiasts. This changed rapidly after 1989. Among the first magazines dedicated primarily to games were the Slovak *Bit* and the Czech *Excalibur*, both of which launched in 1991. Whereas *Bit* was more hobbyist-oriented, *Excalibur* catered primarily to players and promoted a playful, humorous, and transgressive way of writing and appreciating games, similar to the British *Amiga Power* magazine, described by Maher as "garish and excitable."[18] *Excalibur* and its successor *Score* ironically appropriated mainstream critique of the computer game medium, wrote enthusiastically (and jokingly) of gaming "orgies" and "ecstasies," relished video game violence, and embraced the fantasy of becoming cyborgs, hooked up to their machines and exploring new virtual worlds. In contrast to the engineer and bricoleur discourses dominant in the 1980s, mastery of technology was sidelined in favor of hedonistic enjoyment of games.[19]

Informal distribution networks, discussed in chapter 5, continued to play a crucial role—even though explicit copyright protection of computer programs was introduced into copyright law in March 1990.[20] Legal retail copies of games tended to be priced for Western markets and therefore unaffordable. Piracy thus kept the community in touch with trends in other countries. As the IBM PC started to dominate, innocuous exchanges between friends were supplemented by the services of numerous for-profit pirates, who sold PC games cracked by Western groups.

The prominence of informal distribution under socialism, however, did not necessarily lead to high piracy rates in later periods. In 1994, "pirated" software accounted for 66 percent of installed software in the Czech Republic, compared with 77 percent in Spain.[21] In 2015, unlicensed software installations made up 33 percent of installations in the Czech Republic and 36 percent in Slovakia—the lowest figures in the former Soviet bloc, and lower than Italy, Spain, and Portugal.[22]

Tomáš Smutný, who ran an illicit arcade machine manufacturing operation in the 1980s, continued servicing these machines (and slot machines) in the 1990s. However, a substantial part of his hardware was lost in the 2002 flood that destroyed Prague's largest fairground.[23] At that point, arcades were already in worldwide decline.

A Belated Cottage Industry

One of the recurrent misconceptions appearing in fan and journalistic histories of Czechoslovak games is the claim that the first domestic commercial game was 1994's *The Secret of Donkey Island* for the PC. The error may stem from the tendency to see the 1990s as a 16-bit decade, or from the false equation between "commercial" production and the PC platform.[24] However, it does not take too much digging to find many earlier 8-bit commercial titles. For the Spectrum alone, dozens of domestic games were released in the early 1990s. This period of the homegrown Czechoslovak game industry resembled the origins of the British "bedroom coding" scene—albeit a decade later and on a much more modest scale. It also welcomes comparison to the similar developments in 1990s Poland.

Without much experience and without a stable distribution infrastructure, early game industry pioneers experimented with various business models. Following the success of 1989's *City of Robots*, a group of entrepreneurial enthusiasts based in the city of Pilsen attempted a similar project—a text adventure connected to a contest with prizes for the fastest players. Called …*What the Heck?!*, it boasted a unique hypertext interface and a quirky science fiction narrative about a journalist uncovering an international conspiracy. Around 1,750 players bought the game through mail order, and the project ended as a commercial success.[25]

Out of the more traditional publishers, two became important players in the 1990s—Proxima, based in the Northern Bohemian city of Ústí nad Labem, and Ultrasoft, based in the Slovak capital of Bratislava. Both published games for the Spectrum. Although they had to compete with the vast library of widely available pirated games, they could take advantage of the lack of new titles produced in the West. Initially, they secured licenses to older but still popular homebrew titles. František Fuka, for example, granted nonexclusive licenses to his games to both Ultrasoft and Proxima for a fixed price, reserving the right to freely distribute those titles himself.[26]

Besides the Indiana Jones adventures and a couple of action games, Fuka's oeuvre also included *Tetris 2*, the 1990 unofficial sequel to the Soviet game.[27] The game—whose scrolling message includes a "respectful dedication" to the authors of the original *Tetris*—cleverly develops the original's mechanics by including competitive split-screen action and ninety-nine levels of diverse predefined challenges. Its appealing gameplay helped it spread into other postsocialist countries, as evidenced by the existence of a Russian translation.[28]

At that time, however, the country's top coders (such as Fuka) were already moving to more powerful platforms, and publishers therefore started soliciting software from fresh talent. Ultrasoft and Proxima published a combined output of around forty commercial titles, mostly puzzle games and platformer-adventure hybrids inspired by the British *Dizzy* series.[29] Despite the continuing prevalence of informal distribution, individual titles sold hundreds of copies. And although one could not make a living solely by programming games for 8-bit computers, the royalties provided a welcome source of extra income for authors, most of whom were still students.[30]

The vision of financial profit motivated creators to make their games more substantial and polished, but nevertheless they followed in the footsteps of homebrew tradition. Unofficial ports, conversions, and clones were commonplace. On the Sharp MZ 800, a couple of small companies sold unofficial Spectrum ports.[31] On the Spectrum, 1990's *Atomix*, for example, was an unlicensed, modified conversion of the Amiga puzzle game of the same title.[32] *Sherwood*, released in 1992, was a partial clone of the 1986 Amiga title *Defender of the Crown*, with some in-game images faithfully redrawn from the original game, and others from screencaps of the (then recent) Hollywood blockbuster *Robin Hood: Prince of Thieves* from 1991.[33] The last domestic title to be published for the Spectrum during its initial lifespan was the 1995 *Twilight: Land of Shadows*, an ambitious science fiction point-'n'-click adventure and the final release by Ultrasoft.[34]

Czechoslovak games of the 1990s were generally not distributed in other countries, with one important exception—the work for the short-lived Sam Coupé platform. The relatively powerful, vaguely Spectrum-compatible machine (manufactured in the United Kingdom by the small company Miles Gordon Technology, and later by SAMCo) was advertised as the ultimate 8-bit computer, with a fast 3.5″ disk drive and graphics capabilities close to those of the Commodore Amiga. Although it arrived too late to succeed on the UK market, it managed to attract a sizable following in Czechoslovakia. Alan Miles of SAMCo traveled to Prague to solicit software from local programmers,[35] resulting in the puzzle game *Hexagonia* being published in the United Kingdom and reviewed by *Your Sinclair* as the "first Sam game to come out of Eastern Europe."[36] At the same time, František Fuka contributed music to several Sam titles, including the conversion of *Prince of Persia*. However, SAMCo went bankrupt in 1992, ending the brief and late honeymoon of Czechoslovak authors with the Western 8-bit industry.

Homebrew Lives On

The fact that games could make money did not mean that all amateurs took that opportunity. Homebrew continued to be a major force, especially on the 8-bit Atari, where over a hundred noncommercial homebrew games were released in 1990–1992. Because animated graphics started to be required of any game that would call itself "professional," text adventures largely remained within the homebrew domain. Hyperlocal titles continued to be popular, and around twenty have been preserved from after 1989.

In terms of both popularity and influence, none of them surpassed *Multi Pascal 2.7 or The Revenge of the Insane Atari User*. Written by Viktor Lošťák and Zdeněk Polách from the Atari club in the backwater town of Odry, it took place in that very club. The protagonist of the game, an anonymous

Atari user, takes his revenge against the club members after his machine has been destroyed in a fight over new games. Released four months before the Velvet Revolution, this title directly inspired at least ten 1990s games, mostly for the 8-bit Atari, but also for the Spectrum and IBM PC platforms. Many of these text adventures continued in the tradition of "small subversions" mentioned in chapter 7. They invited the main character to partake in transgression against authority, be it parents, teachers, more experienced community members, or social norms in general.

In the later periods, homebrew production might have been overshadowed by commercial production, but it never disappeared. It offered an outlet not only for amateur and unfinished work but also for content that would have been too transgressive or experimental for commercial release. A good example is the *Life Is Not Beautiful* series of point-'n'-click adventures, started by Martin Pohl in 2002. Between 2002 and 2007, Pohl released seven installments featuring badly drawn sex, gore, violence, and morbid humor.[37] Originally targeted at a niche online audience, the series was rediscovered when Pohl gained public notoriety as a cartoonist and rapper around 2010.

Games continue to be used to comment on political issues, most famously in the case of 2014's *PussyWalk*. Inspired by ragdoll physics games such as *QWOP*, it features the Czech Republic's then president Miloš Zeman as the player character. Zeman has been portrayed by the media as a heavy drinker, and the game emulates his reportedly wobbly walk by giving the player purposefully counterintuitive control of his limbs. This results in many instances of emergent slapstick humor. In addition, the game satirizes Zeman's ties to Russia and his incorrect pronunciation of the English word "pussy" when criticizing the Russian activist group Pussy Riot.[38] A sequel, *PussyWalk II*, was released in late 2017, before the presidential elections, in which Zeman ended up narrowly securing a second term.

The Game Industry Today: Adventures, Army, and Automation

As of 2018, former Czechoslovakia has a relatively stable and self-sufficient game industry. But this is a result of a series of contingent events rather than a heritage of the 1980s homebrew scene. As the example of Hungary shows, a booming 8-bit and 16-bit scene does not automatically create a solid foundation for later periods.[39] As mentioned in chapter 6, Hungarian programmers created numerous titles for Western publishers in the 1980s. This trend continued in the 1990s—for example, unbeknownst to most mainstream audiences, the 1992 Sega Mega Drive hit *Ecco the Dolphin* was developed by a Hungarian team.[40] Despite the cutting-edge know-how, Hungary's game industry largely disintegrated in the late 1990s.[41]

There is little direct personal continuity between the 1980s homebrew scene and the current Czech and Slovak game industries. But although the market and the people changed, some features of domestic production have persisted, allowing us to follow two threads that connect 1980s homebrew games and current titles. One of the threads concerns the language barrier and the resulting focus on domestic audiences. The ability to make and play games in one's mother tongue contributed to the boom of homebrew text adventures in the 1980s. Similarly, in the 1990s, many commercial developers set out to create point-'n'-click adventure titles for the domestic market. The 1994 game *The Secret of Donkey Island*, a parody of Lucasfilm's *The Secret of Monkey Island*, was initially an enthusiast effort by a group of high school students, but received nationwide distribution and a degree of acclaim.[42] Its playful humor, in part inspired by its source material, set the tone for much of the local adventure game production.[43] Released in 1998, the cop comedy adventure game *Polda* started the longest-running Czech game series, with a seventh installment announced in 2015. It continues to be aimed primarily at a domestic audience, and one of its attractions has been voice acting by well-known local comedians.

Jakub Dvorský, possibly the country's most influential indie game designer, started his game development career while in high school, drawing graphics for early to mid-1990s adventure games. He went on to study at the Academy of Arts, Architecture, and Design, graduating with a project that was later released as the 2003 game *Samorost*.[44] Soon thereafter, he founded the Amanita Design studio, which became one of the stalwarts of both the domestic and international independent scenes. Their releases include *Machinarium*, *Botanicula*, and 2016's *Samorost 3*, the latest installment in the series.[45] All of these are playful and poetic point-'n'-click adventures. Although the domestic popularity of early Czech point-'n'-click games was partly due to the use of the Czech language, Amanita Design's games use no verbal language whatsoever, instead communicating through visuals, music, and gestures. Nevertheless, their idiosyncratic audiovisual style and frequent use of collage points to the rich tradition of Czech illustration and animation, represented by Jiří Trnka and Jan Švankmajer.[46]

The other thread connecting 1980s games with today's is the tinkerer attitude. Around 2000, it became clear that the domestic market was too small to sustain larger development projects, and local companies started developing new franchises for the international PC market, and later, the console market. Their design approach reflects some of the observations made in chapter 4 when explaining the Czechoslovak fame of the 1985 British helicopter simulator *Tomahawk*.[47] The local audience, largely consisting of technical-minded hobbyists, appreciated the range of options it gave the player, its realism, and its complexity. This mode of enjoyment and the resulting design approach has become typical of much of Czech and Slovak production. *Hidden & Dangerous* (1999) by Illusion Softworks and *Operation Flashpoint* (2001) by Bohemia Interactive (whose roots lie in the Atari 8-bit and ST scene) were both realistic war-themed games that boasted complex simulations and a variety of weapons and vehicles at the player's disposal.[48] They were, in many ways, games by tinkerers for tinkerers. After losing the rights to the *Operation Flashpoint* franchise to its publisher, Bohemia Interactive turned to self-publishing and launched its own successful series, *ArmA*, which remains one of the most realistic mass-market military simulations.[49] Yet another example of this tinkerer attitude is *Euro Truck Simulator 2* by Prague-based SCS Software, a notable surprise hit of 2013, followed in 2016 by *American Truck Simulator*.[50] Once again, these are complex, adjustable games that take pride in realism.

If we were to pick an ultimate Czech or Slovak tinkerer game, it would be Wube Software's *Factorio*—a real-time strategy game about factory automation.[51] Although still in beta version as of the spring of 2018, it has already attracted worldwide popularity and critical acclaim. One of the reviews reads:

> It pushes the concept of resource-management to such an extreme that anything that does anything less risks seeming underwhelming, while anything which tries to go further could well be overwhelming. We're talking dozens of conveyor belts, hundreds of robotic arms, thousands of chunks of ore, and constructions quite possibly in the millions. It's a remarkable thing. A ridiculous thing.[52]

Given that socialist industry planners saw hobby computing in the light of its potential contribution to industry automation, *Factorio* can be interpreted as a virtual fulfillment of their fantasies. Although excessive and incomprehensible at first sight, the game's systems are ultimately understandable and manageable. It models an industry that—unlike the Czechoslovak command economy—*can* be centrally controlled by an omniscient and omnipresent player.

Where Are They Now?

Throughout this book, I have traced a web of personal and collective narratives. Following them after 1989 not only completes the personal stories but also reveals wider contextual and structural reasons that made people get involved in games. In the 1980s, games stood at a convergence of several interests—programming and computers in general, but also art, communication, entertainment, and activism. Computer club alumni were generally well equipped for the transition to a free market economy. They understood computers, which was still a rare skill in 1990. They tended to have good English skills, which became useful in an increasingly global economy. Participation in clubs broadened their social network and connected them to the country's technological elite. And last but not least, many of them had learned to self-manage and work on complex projects.

Many of the people in my story were attracted to games because of their more general inclination to programming, rather than to games in particular. None of my interviewees is currently making games professionally, and only a few continue to do so on an amateur basis. Miroslav Fídler of Golden Triangle fame, as well as Patrik Rak, worked on several game projects at different periods, but later switched to other, more lucrative, avenues of programming. The skills that homebrewers rehearsed in the 1980s allowed them to get prestigious jobs in IT, often abroad. Tomáš Rylek of the Golden Triangle is a principal software engineer at Microsoft. Michal Hlaváč, one of the members of Sybilasoft, received a master's degree from the MIT Media Lab, and as of 2018 works as a senior designer, also at Microsoft.

For other users, games were mainly a means of or facilitators of expression. František Fuka, author of *The Sting 3* and the *Indiana Jones* text adventure series, never considered himself a superb programmer. His games often built on his interest in popular culture in general, and he enjoyed film and music just as much as he enjoyed games. In the late 1980s and early 1990s, he was using his English skills to dub pirated VHS copies of Western films.

Today, he translates subtitles for official editions of English-language films and runs a crowdfunded film review blog, occasionally composing music and making experimental games.[53] David Hertl, the publisher of the *ZX Magazine* fanzine, earned a degree in history and returned to journalism. After a stint in a regional newspaper, he has worked for Czech Radio, the country's public service broadcaster.[54] His collaborator Ondřej Kafka started his own graphic design studio in 1990, is still successfully managing it, and publishes *Font* magazine, dedicated to graphic design. Martin Malý, author of several text adventures, including the bedroom "thriller" *Demon in Danger*, worked in IT after 1989, but kept being drawn to writing. In 2003, he founded the bloguje.cz blogging platform. As of 2018, he works as a deputy editor-in-chief for a major Czech news media company, keeps blogging, and occasionally writes political commentary.[55]

The educators behind the computer clubs and school groups that we discussed remained dedicated to education. Miroslav Háša of the Station of Young Technicians now runs a small publishing house focusing on financial literacy; Alena Šolcová teaches several subjects at the Czech Technical University, including history of mathematics and history of technology. Bohuslav Blažek, one of the most original Czechoslovak thinkers writing about computers in the 1980s, dedicated most of his post-1989 work to environmental science, but kept writing essays about the media and the Internet until his untimely death in 2004.

One of the attractions of microcomputers in the 1980s was their relative simplicity and the control one could have over them. Some of my interviewees lost interest in games and programming when machines and operating systems became too opaque. Suddenly, mastery of the machine was difficult to demonstrate, and the hobbyist urge to explore every corner of the computer was gone. It is no surprise, then, that several 1980s hobbyists are enthusiastic about small open-source computers such as the Raspberry Pi, which they see as a throwback to the times of the Spectrum and Atari 8-bit machines.[56]

A couple of members of the community remain genuinely committed to the retro gaming scene. The 602 affiliate Patrik Rak has developed ZXDS, an accurate and user-friendly ZX Spectrum emulator for the Nintendo DS, and maintains its English website, including a long list of game recommendations for newcomers to the Spectrum. Vlastimil Veselý is one of the rare examples of people who have held the same job since the 1980s. Still a technician at the city of Ostrava's public transportation company, he has returned to programming for the PMD 85. He made a pattern-matching puzzle game called *Pontris* in 2010 and collaborated on a conversion of the 1990 British Amiga hit *Lemmings* in 2014.[57]

I have mentioned that there were few direct personal connections between the 1980s homebrewers and the current game development community. But the 1980s amateur production has not been forgotten, and at least some of the games and authors are acknowledged as part of an ongoing narrative of Czech (and Slovak) games. In 2017, the Slovak Museum of Design held a compact exhibition of selected Slovak titles from 1987 to 1993, entitled *Pirates and Pioneers*. Besides four other titles, it included Sybilasoft's *Shatokhin*, running on original Didaktik hardware. Accompanied by posters that explained each game's unique contribution to game design, the exhibition was enthusiastically received by both the press and the public.

In the same year, the Czech Games (České hry) association, which organizes the annual Czech Game of the Year award ceremony, introduced its Hall of Fame. The intention behind the Hall of Fame is to induct one person each year. But in its very first year, it welcomed three inductees at once—the Golden Triangle's Miroslav Fídler, František Fuka, and Tomáš Rylek. Let us hope that such events, as well as this book, will garner institutional support for game and gaming culture preservation, which is still sorely lacking in both countries, despite the amazingly rich computer game history that they share.

Read More on the Book's Website

Visit the book's website at ironcurtain.svelch.com to find updates, extra content, additional images, and links to playable versions of games discussed in this book.

Appendix: Important Dates

February 1948: “Victorious February” and the start of the rule of the Communist Party of Czechoslovakia

1953: The death of Stalin brings “thaw”; repressions by the regime are slightly relaxed and some power starts moving from the party apparat to experts

1956: SAPO, the first Czechoslovak-made computer, launches

1966: Radovan Richta’s *Civilization at the Crossroads* published, popularizing the term *scientific-technological revolution*

1968: Alexander Dubček becomes General Secretary of the Communist Party and initiates the Prague Spring reforms

August 1968: Invasion of Warsaw Pact troops and start of normalization

1979: Federal Ministry of Electrotechnical Industry founded to coordinate research and production of electronics

1982: Communist Party Congress and Svazarm leadership declare support for microcomputers

1982: *Amateur Radio* magazine launches the *Microelectronics* supplement and publishes the first type-in games

1984: Long-Term Complex Program of Electronization of Czechoslovak National Economy adopted

1984: *Manic Miner* for ZX81 released, the earliest reliably dated Czechoslovak game whose code has been preserved

1985: Long-Term Complex Program of Electronization in Training and Education adopted

1985: *Mikrobáze* newsletter launched by Svazarm’s 602 computer club

1985: PMD 85 computer launched

1985: František Fuka's *Poklad 2* and *Indiana Jones and the Temple of Doom* released—probably the first original Czechoslovak text adventures

1986: Ondra computer launched

1987: Didaktik Gama computer launched

1987: Sybilasoft and VBG Software start releasing games for the Spectrum and PMD 85, respectively

June 1988: *Spektrum*, later renamed *ZX Magazine*, launched—the first computing fanzine published outside the computer club infrastructure

Summer 1988: *RECONSTRUCTION*, the oldest preserved game that openly criticized the Communist regime, released

January 1989: Palach Week demonstrations in Prague

Summer 1989: *City of Robots* released, possibly the first Czechoslovak game with an official nationwide release

November 1989: Velvet Revolution, starting with a student demonstration on November 17, leads to the dismantling of the Communist regime

Glossary

COMECON: *see* Soviet bloc.

Communism: *see* state socialism.

Communist Party of Czechoslovakia: The ruling party in Czechoslovakia between 1948 and 1989. Founded in 1921, it had close ties to the Communist Party of the Soviet Union and was part of the Communist International. Led by Klement Gottwald, it won thirty-eight percent of the vote in the 1946 free elections, and seized power in the country in a 1948 coup called Victorious February (Vítězný únor). Soon thereafter it removed political opponents both outside and within the party in purges and show trials. After the economic decline of the early 1960s, a reformist wing came into power, leading to the Prague Spring reforms of 1968, which were quashed by the Soviet invasion in August of that year. Another wave of purges and screenings followed, consolidating the power of the "normalizers" within the party. In the 1980s, the party was divided between a conservative wing and reformists, the latter supporting implementation of Gorbachev-style reforms. The power in the party was concentrated in its Central Committee.[1]

Czech and Slovak languages: Both Czech and Slovak are Southern West Slavic languages with a high degree (over ninety percent) of mutual intelligibility.[2] In the 1980s, Czechoslovak television was broadcasting programs in both languages. The next closest major language to both Czech and Slovak is Polish.

Czechoslovak geography: Czechoslovakia was a landlocked country in Central Europe, with mountain ranges (and, to a lesser extent, rivers) providing natural borders with West Germany to the west, East Germany and Poland to the north, the Soviet Union to the east, and Hungary and Austria to the south. Its shape was elongated in the east-west direction. After the federalization of 1969, it consisted of two republics—the Czech Socialist Republic and the Slovak Socialist Republic. The former comprised three historical regions: Bohemia (the largest) in the west, Moravia in the east, and the relatively small Silesia in the northeast, bordering Poland.

Czechoslovak governments: In Communist Czechoslovakia, the government was an executive body formally running the country, although it mostly implemented decisions made within the Central Committee of the Communist Party of Czechoslovakia. After the federalization in 1969, there were three governments—federal, Czech, and Slovak. Some of the ministries, such as the Federal Ministry of Electrotechnical Industry, existed only on the federal level.

Iron Curtain: *see* Soviet bloc.

Normalization: Normalization refers to the political processes following the Prague Spring reforms and the 1968 invasion of Czechoslovakia by Warsaw Pact forces (see chapter 1). After the invasion, Soviet-backed party members took the helm of the country and initiated massive purges and screenings. Protagonists of the Prague Spring were expelled from the party, and thousands of reform supporters, especially culture workers, lost their jobs. The goal of the normalizers was to pacify the population and prevent future crises. This was achieved by a combination of censorship, monitoring by secret police, and support for apolitical consumerist activities. The regime started loosening up after the mid-1980s in response to Gorbachev's reforms in the Soviet Union, and the normalization era ended with the Velvet Revolution in 1989.

Pioneer: Pioneer was a youth organization typically joined by children between ages eight and fifteen. It was part of the Socialist Union of Youth. Although membership was officially optional, there was strong social pressure on families to enroll their children, and the vast majority did.[3] Although Pioneer allowed the state to oversee and control children's free time, and to expose them to Marxist-Leninist ideology,[4] much of its day-to-day routine was apolitical or inspired by "bourgeois" sources such as the Scouting movement. Among other things, Pioneer ran a network of Houses of Pioneers and Youth (or simply "Pioneer houses"), which hosted workshops and hobby group meetings.[5]

Scientific-technological revolution: An approach to science and technology policies adopted in some Soviet bloc countries in the 1950s and 1960s as part of de-Stalinization. Inspired by the work of the Marxist historian of science John Desmond Bernal, it was conceived as a "consultative process in which political leadership was to be joined by technocratic expertise."[6] Its tenets were most famously formulated in Czechoslovakia by philosopher Radovan Richta, who envisioned a reformed socialist society thriving on democratic implementation of cutting-edge technologies in economy and culture.[7] Scientific-technological revolution (STR) was part of 1968's Prague Spring reform project. During the normalization era, many of its important tenets, such as its participatory nature, were abandoned.

Socialism: *see* state socialism.

Soviet bloc: The term Soviet bloc generally refers to countries in the Soviet sphere of influence during the Cold War era, all of which were governed by some form of state socialist regime. In this book, the term refers mostly to Central and Eastern European countries such as Czechoslovakia, Poland, Hungary, East Germany (or the German Democratic Republic), and the Soviet Union.[8] The Soviet bloc was divided from Western Europe by the so-called Iron Curtain, a heavily guarded line of fences and border defenses. In a metaphorical sense, the term Iron Curtain also refers to all efforts to limit the movement of people, goods, and ideas between the West and the East.[9] The Soviet bloc was formalized through two international agreements—the military Warsaw Pact and the economic Council for Mutual Economic Assistance, or COMECON. One of the aims of COMECON was to increase self-reliance and competitiveness with the West through collaboration within the Soviet bloc, and to overcome the adverse effects of partial isolation from international trade. It is crucial to note that not all Central and Eastern European state socialist countries were members of the Soviet bloc. Tito's Yugoslavia split from the Soviet line in the 1940s and started the Non-Aligned Movement in 1961. Albania defected in 1961.[10]

State socialism: Czechoslovakia of the 1970s and 1980s has been called both a *socialist* and a *communist* country. Both in academic and lay discourse, these terms tend to be used interchangeably, but their specific origins and meanings should not be forgotten. Strictly speaking, the declared goal of communist parties around the world, at least initially, was to achieve a *communist* society, that is, one in which all resources are shared and all inequalities have been abolished. However, none of the countries governed by communist parties ever reached this goal. Czechoslovakia therefore tends to be called a *communist* (or *Communist*) country strictly because it was ruled by the Communist Party of Czechoslovakia, not because of its actual economic or governance system. In the discourse of communist parties, *socialism* was an intermediate stage in the evolution of the society that was heading toward communism. Czechoslovakia officially declared that it had finished transition to socialism by adopting the adjective *socialist* in its official title in 1960. A more precise academic term to describe the Czechoslovak system is *state socialism*, which means the economy was state-owned and centrally planned. For the sake of stylistic flexibility, I am using the terms *Communist* and *socialist* to denote the historical period, but I will use the term *state socialism* when speaking about the system of governance and the country's economy.

Svazarm: Svazarm, or the Union for Cooperation with the Army (Svaz pro spolupráci s armádou), was Czechoslovakia's largest paramilitary organization. Its original goal was to train civilians, especially young people, for potential roles in the military. In reality, Svazarm became an umbrella for a wide range of hobby activities, including motorsports, dog training, marksmanship, ham radio, aviation, hi-fi, and electronics, often with no direct military application.[11] Svazarm's benefit to the military was questionable, but it concentrated much of the country's hobby activities, and therefore provided a degree of supervision and control. It was home to many of the influential Czechoslovak computer clubs.

TESLA: The major electronics manufacturer in socialist-era Czechoslovakia, with output ranging from single components to consumer electronics and industrial machines. Most electronics on the Czechoslovak market were produced by TESLA in one of its over thirty branches. The one based in the Slovak spa town of Piešt'any is the most important for this book because it produced microcomputer CPUs and the PMD 85 computer.[12]

Tuzex: A hard currency shopping chain, a counterpart to East Germany's Intershop and the Polish Pewex. The original intention behind these stores was to offer Western tourists luxury goods, trinkets, and souvenirs in order to collect their precious hard currency.[13] In the 1970s and 1980s, Tuzex transformed into a chain of luxury stores for Czechoslovak citizens. Once a citizen earned Western money—by working abroad, for example—he or she had to exchange it for "consumption coupons" (colloquially called *bon*, plural *bony*), which could be spent in a Tuzex store. One could also buy *bony* on the black market. Tuzex intermittently sold 8-bit microcomputers, and was one of the important sources of hardware for Czechoslovak users.

Notes

Introduction

1. Jeff Smart, "The European Crackers Map, Anno 1988," *Illegal* 3, no. 26 (1988): not numbered.

2. Andromeda Software, *Tetris*, ZX Spectrum (Mirrorsoft, 1987). The original game was made in Moscow. For more about *Tetris*, see chapter 6. Dan Ackerman, *The Tetris Effect: The Game That Hypnotized the World* (New York: PublicAffairs, 2016).

3. Aphra Kerr, *The Business and Culture of Digital Games: Gamework/Gameplay* (London: SAGE, 2006), 1.

4. Ibid., 20.

5. Henry Lowood and Raiford Guins, eds., *Debugging Game History: A Critical Lexicon*, Game Histories (Cambridge, MA: MIT Press, 2016), xv.

6. Steve L. Kent, *The Ultimate History of Video Games: From Pong to Pokémon and Beyond: The Story behind the Craze that Touched Our Lives and Changed the World* (Roseville, CA: Prima Publishing, 2001); Van Burnham, *Supercade: A Visual History of the Videogame Age 1971–1984* (Cambridge, MA: MIT Press, 2003); David Kushner, *Masters of Doom: How Two Guys Created an Empire and Transformed Pop Culture* (New York: Random House, 2004); Heather Chaplin and Aaron Ruby, *Smartbomb: The Quest for Art, Entertainment, and Big Bucks in the Videogame Revolution* (Chapel Hill, NC: Algonquin Books, 2006); Jeff Ryan, *Super Mario: How Nintendo Conquered America* (London: Portfolio, 2013).

7. See for example: Laine Nooney, "A Pedestal, a Table, a Love Letter: Archaeologies of Gender in Videogame History," *Game Studies* 13, no. 2 (December 2013), http://gamestudies.org/1302/articles/nooney; Carl Therrien, "Inspecting Video Game Historiography through Critical Lens: Etymology of the First-Person Shooter Genre," *Game Studies* 15, no. 2 (December 2015), http://gamestudies.org/1502/articles/therrien; Lowood and Guins, *Debugging Game History*.

8. Erkki Huhtamo, "Slots of Fun, Slots of Trouble: An Archaeology of Arcade Gaming," in *Handbook of Computer Game Studies*, ed. Joost Raessens and Jeffrey H. Goldstein (Cambridge, MA: MIT Press, 2005), 3–22.

9. Patryk Wasiak, "Computing behind the Iron Curtain: Social Impact of Home Computers in the Polish People's Republic," Tensions of Europe Working Paper 2010, no. 8 (2010): 1–17; Patryk Wasiak, "Playing and Copying: Social Practices of Home Computer Users in Poland during the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. Gerd Alberts and Ruth Oldenziel (London: Springer, 2014), 129–50; Zbigniew Stachniak, "Red Clones: The Soviet Computer Hobby Movements of the 1980s," *IEEE Annals of the History of Computing* 37, no. 1 (January 2015): 12–23, https://doi.org/10.1109/MAHC.2015.11; P. Konrad Budziszewski, "Poland," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 399–424; Tamás Beregi, "Hungary," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 219–234; Andreas Lange and Michael Liebe, "Germany," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 193–206; Alexander Fedorov, "Russia," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 439–449.

10. Vacek makes unsubstantiated claims about state authorities' critical stances toward arcade and computer games. In my research, I have not found any evidence of such stances. The Czechoslovak government's 1985 Long-Term Complex Program of Electronization in Training and Education even supports the use of computer games at the elementary school level (see chapter 1). The 1986 quote critical of computer games that Vacek presents on page 149 is misattributed to *Elektronika* magazine (published by the Ministry of Electrotechnical Industry starting in 1987), and was in fact published in *Mikrobáze*, a hobby computing club newsletter. Vacek's entry also claims that the "first professional Czech video game" was 1994's *The Secret of Donkey Island*. Besides lacking a definition of "professional," this statement disregards dozens of commercially published games for 8-bit computers in the early 1990s (see epilogue). Patrik Vacek, "Czech Republic," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 145–58.

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13. Nooney, "A Pedestal, a Table, a Love Letter"; Carly A. Kocurek, *Coin-Operated Americans: Rebooting Boyhood at the Video Game Arcade* (Minneapolis: University of Minnesota Press, 2015).

14. Patryk Wasiak, "'Illegal Guys': A History of Digital Subcultures in Europe during the 1980s," *Zeithistorische Forschungen/Studies in Contemporary History, Online-Ausgabe* 9, no. H. 2 (2012), http://www.zeithistorische-forschungen.de/site/40209282/default.aspx#zitieren; Markku Reunanen, Patryk Wasiak, and Daniel Botz, "Crack Intros: Piracy, Creativity and Communication," *International Journal of Communication* 9 (March 26, 2015): 20.

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18. Ibid., ix.

19. Ksenia Tatarchenko, "The Anatomy of an Encounter: Transnational Mediation and Discipline Building in Cold War Computer Science," in *Communities of Computing: Computer Science and Society in the ACM*, ed. Thomas J. Misa (New York: ACM, 2016), 199–227, https://doi.org/10.1145/2973856; Ksenia Tatarchenko, "'The Computer Does Not Believe in Tears': Soviet Programming, Professionalization, and the Gendering of Authority," *Kritika: Explorations in Russian and Eurasian History* 18, no. 4 (2017): 709–739, https://doi.org/10.1353/kri.2017.0048.

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21. Vítězslav Sommer, "Scientists of the World, Unite! Radovan Richta's Theory of Scientific and Technological Revolution," in *Science Studies during the Cold War and Beyond: Paradigms Defected*, ed. Elena Aronova and Simone Turchetti (New York: Palgrave Macmillan US, 2016), 177–204, http://dx.doi.org/10.1057/978-1-137-55943-2_8; Jiří Hoppe et al., *"O nový československý model socialismu": čtyři interdisciplinární vědecké týmy při ČSAV a UK v 60. letech*, Sešity Ústavu pro soudobé dějiny AV ČR 49 (Prague: Ústav pro soudobé dějiny AV ČR, 2015); Michal Pullmann, *Konec experimentu: přestavba a pád komunismu v Československu* (Prague: Scriptorium, 2011); Miroslav Vaněk, *Byl to jenom rock'n'roll?: hudební alternativa v komunistickém Československu 1956-1989* (Prague: Academia, 2010); Miroslav Vaněk and Pavel Mücke, *Velvet Revolutions: An Oral History of Czech Society*, Oxford Oral History Series (New York: Oxford University Press, 2016).

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Getaways: The Chata, the Tramp, and the Politics of Private Life in Post-1968 Czechoslovakia," in *Socialist Spaces: Sites of Everyday Life in the Eastern Bloc*, ed. David Crowley and Susan E. Reid (Oxford: Berg, 2002), 123–140; Paulina Bren, "Tuzex and the Hustler: Living It Up in Czechoslovakia," in *Communism Unwrapped: Consumption in Cold War Eastern Europe*, ed. Paulina Bren and Mary Neuburger (Oxford: Oxford University Press, 2012), 27–48.

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27. Clara Fernández-Vara and Nick Montfort, *Videogame Editions for Play and Study*, TROPE 13-02 (Cambridge, MA: MIT Trope Tank, June 5, 2014), 5, http://dspace.mit.edu/handle/1721.1/87668.

28. I used ZXSpin and ZXDS to emulate the Sinclair ZX Spectrum, Altirra for the Atari 8-bit, WinVice for the Commodore 64, MZWin for the Sharp MZ 800, and PMD 85 Emulator for the PMD 85.

29. Peter N. Stearns, ed., *Encyclopedia of Social History* (London: Routledge, 1993), ix, http://www.tandfebooks.com/isbn/9780203306352.

30. My use of the term and its lack of capitalization follow the authoritative English-language anthology of *alltagsgeschichte*: Alf Lüdtke, ed., *The History of Everyday Life: Reconstructing Historical Experiences and Ways of Life*, Princeton Studies in Culture/Power/History (Princeton, NJ: Princeton University Press, 1995).

31. Ibid., 15.

32. Ibid., 14.

33. See Vaněk and Mücke, *Velvet Revolutions*.

34. Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network-Theory*, Clarendon Lectures in Management Studies (Oxford: Oxford University Press, 2005), 43.

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36. McDermott, *Communist Czechoslovakia, 1945–89*, 169.

37. Michel de Certeau, *The Practice of Everyday Life* (Berkeley: University of California Press, 1984), 37. The term *nowhere* has been italicized for clarity.

38. Ibid., 40.

39. It has also been a focus of *alltagsgeschichte* researchers, who have been interested in "reappropriation" of the world and society by historical subjects; and in the "styles of behavior which establish and screen off spheres of their own—unobtrusively perhaps, yet with great effectiveness." Lüdtke, *The History of Everyday Life*, 17.

40. Pullmann, *Konec experimentu*; McDermott, *Communist Czechoslovakia, 1945–89*; Vaněk and Mücke, *Velvet Revolutions*; Martin Husák, "Rock Music Censorship in Czechoslovakia between 1969 and 1989," *Popular Music and Society* 40, no. 3 (March 30, 2017): 1–20, https://doi.org/10.1080/03007766.2017.1295503.

41. Alexei Yurchak, *Everything Was Forever, until It Was No More: The Last Soviet Generation* (Princeton, NJ: Princeton University Press, 2006), 141.

42. Ibid., 146.

43. Ibid., 148.

44. Ibid., 154.

45. Ibid., 168.

46. Swalwell, "The Early Micro User."

47. Mark Deuze, "Participation, Remediation, Bricolage: Considering Principal Components of a Digital Culture," *The Information Society* 22, no. 2 (April 1, 2006): 63–75, https://doi.org/10.1080/01972240600567170.

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52. Lévi-Strauss, *The Savage Mind*, 33.

53. Ibid., 19.

54. Ciborra, *The Labyrinths of Information*, 48.

55. For a classical overview, see: Raymond Williams, *Television: Technology and Cultural Form*, Routledge Classics (London: Routledge, 2003).

56. Latour, *Reassembling the Social*, 72.

57. Turkle, *The Second Self*, 52.

58. Petri Saarikoski, Jaakko Suominen, and Markku Reunanen, "Pac-Man for the VIC-20: Game Clones and Program Listings in the Emerging Finnish Home Computer Market," *Well Played* 6, no. 2 (2017): 7–31.

59. I have developed this argument in my previous article: Jaroslav Švelch, "Keeping the Spectrum Alive: Platform Fandom in a Time of Transition," in *Fans and Videogames: Histories, Fandom, Archives*, ed. Melanie Swalwell, Helen Stuckey, and Angela Ndalianis, Routledge Advances in Game Studies 9 (New York: Routledge, 2017), 57–74.

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61. Melanie Swalwell, *Homebrew Gaming and the Beginnings of Vernacular Digitality* (Cambridge, MA: MIT Press, forthcoming).

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67. Reunanen, Wasiak, and Botz, "Crack Intros."

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1 Micros in the Margins

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31. As one of my interviewees, Viktor Lošťák, put it: "Back then, you were born and there was a path laid out for you. You had to be a Spark, a Pioneer, a Socialist youth, a Communist, then take part in the socialist economy, so that you have a secure life and eventually retire." Viktor Lošťák and Zdeněk Polách, interview by Jaroslav Švelch, January 26, 2016.

32. McDermott, *Communist Czechoslovakia, 1945–89*, 169, 202.

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37. Alexej Bálek, "Československá ekonomika v osmdesátých letech 20. století," *Acta Oeconomica Pragensia* 2007, no. 7 (2007): 45–54, https://doi.org/10.18267/j.aop.176.

38. Ibid.

39. Anders Åslund, "The Functioning of Private Enterprise in Poland," *Soviet Studies* 36, no. 3 (July 1, 1984): 427–444, https://doi.org/10.1080/09668138408411542.

40. Pullmann, *Konec experimentu*.

41. McDermott, *Communist Czechoslovakia, 1945–89*; Bálek, "Československá ekonomika v osmdesátých letech 20. století."

42. Slava Gerovitch, *From Newspeak to Cyberspeak: A History of Soviet Cybernetics* (Cambridge, MA: MIT Press, 2002); Benjamin Peters, *How Not to Network a Nation: The Uneasy History of the Soviet Internet* (Cambridge, MA: MIT Press, 2016).

43. Petr Kovář, "Historie výpočetní techniky v Československu" (master's thesis, Charles University in Prague, 2005), http://www.historiepocitacu.cz/o-projektu-historie-pocitacu.html.

44. Stefan Guth, "One Future Only: The Soviet Union in the Age of the Scientific-Technical Revolution," *Journal of Modern European History* 13, no. 3 (2015): 355, https://doi.org/10.17104/1611-8944-2015-3-355.

45. Guth, "One Future Only."

46. Radovan Richta and collective of authors, *Civilizace na rozcestí: Společenské a lidské souvislosti vědeckotechnické revoluce* (Prague: Svoboda, 1967), 54.

47. Ibid., 60.

48. Ibid., 142.

49. Ibid., 148.

50. Ibid., 145.

51. Jindřich Filipec and Radovan Richta, *Vědeckotechnická revoluce a socialismus* (Prague: Svoboda, 1972), 92.

52. Karel Löbl and Ladislav Říha, *Vědecko-technický pokrok a socialistický rozvoj v ČSSR* (Prague: Melantrich, 1976); Karel Löbl and Ladislav Říha, *Vědecko-technický pokrok a intenzifikace* (Prague: Melantrich, 1988).

53. Gerovitch, *From Newspeak to Cyberspeak*.

54. Sommer, "Scientists of the World, Unite!," 193–194.

55. For more period publications manifesting the official technological policies, see: Löbl and Říha, *Vědecko-technický pokrok a socialistický rozvoj v ČSSR*; Miroslav Frk, Vladimír Hrbek, and collective of authors, *Československý elektrotechnický a elektronický průmysl 1948–1988* (Prague: SNTL, 1988).

56. Pullmann, *Konec experimentu*; Löbl and Říha, *Vědecko-technický pokrok a socialistický rozvoj v ČSSR*.

57. Pavel Urbášek, *Vysokoškolský vzdělávací systém v letech tzv. normalizace* (Olomouc, Czech Republic: Univerzita Palackého v Olomouci, 2008).

58. Ibid., 173.

59. The pre–World War II Soviet Union, for instance, was a notable user of American and German tabulation machines. James W. Cortada, *Before the Computer: IBM, NCR, Burroughs, and Remington Rand and the Industry they Created, 1865–1956*, Princeton Studies in Business and Technology (Princeton, NJ: Princeton University Press, 1993).

60. Ted Friedman, *Electric Dreams: Computers in American Culture* (New York: New York University Press, 2005).

61. Kovář, "Historie výpočetní techniky v Československu."

62. British Government, "Consolidated List of Goods Subject to Security Export Control," *British Business* 10, no. 1 (1980): 1–28.

63. Kovář, "Historie výpočetní techniky v Československu."

64. Helena Durnová, "Sovietization of Czechoslovakian Computing: The Rise and Fall of the SAPO Project," *IEEE Annals of the History of Computing* 32, no. 2 (April 2010): 21–31, https://doi.org/10.1109/MAHC.2010.7.

65. Peters, *How Not to Network a Nation*.

66. Jan Kaše, "Výpočetní technika v bývalé ČSLA," *Computerworld* 6, no. 14 (1995): 1, 33; Václav Budil, "Z historie automatizace velení a řízení v armádě," *Computerworld* 6, no. 6 (1995): 17.

67. Jiří Hofman and Jan Bauer, *Tajemství radiotechnického pátrače TAMARA* (Prague: Sdělovací technika, 2003).

68. In 1966, the TESLA Pardubice factory started manufacturing its TESLA 200 based on technology licensed from the French Bull-General Electric company. Frk,

Hrbek, and collective of authors, *Československý elektrotechnický a elektronický průmysl 1948–1988*.

69. In Czechoslovakia, the Uniform System was known under the acronym JSEP. For a discussion of industrial espionage, see: Amy Wilson, "Computer Gap: The Soviet Union's Missed Revolution and Its Implications for Russian Technology Policy," *Problems of Post-Communism* 56, no. 4 (August 2009): 41–51.

70. The Czechoslovak acronym for the System of Small Electronic Computers was SMEP. Helena Durnová, "JSEP—Jednotný systém elektronických počítačů," in *Věda a technika v Československu od normalizace k transformaci*, ed. Ivana Lorencová (Prague: Národní technické muzeum, 2012).

71. Another quarter (554 units) was produced in other COMECON countries. The statistics cover "registered" computers, referring to those that were run in state-owned factories and institutions. These did not include personal machines. Český statistický úřad, *Stav a využití výpočetní techniky v roce 1986 v ČSR* (Prague: Český statistický úřad, 1987).

72. Regarding the numbers of machines produced, the available official figures are 116 for TESLA 200, 240 for the JSEP-based EC 1025, and 300 for its successor EC 1027. Frk, Hrbek, and collective of authors, *Československý elektrotechnický a elektronický průmysl 1948–1988*; Kovář, "Historie výpočetní techniky v Československu."

73. Wilson, "Computer Gap."

74. Vladimír Jelínek, "Rozvoj aplikací výpočetní techniky v 8. pětiletce," *Elektronika* 1, no. 1 (1987): 2; Kristina Šedivá, "Elektronika a mladá generace," *Elektronika* 1, no. 3 (1987): 2. Special focus was placed on "automated systems of management" within factories, similar to the Soviet Union.

75. Amatérské radio, "Mikroelektronika," *Amatérské radio* 31, no. A1 (1982): 17.

76. Malec inherited his interest in industrial management from his father, general secretary of the interwar republic's Metalworking Industry Association and, as Malec ironically puts it, a "servant of the capitalists." Due to his capitalist background, Malec had difficulties applying for university education, but eventually graduated from the University of Political and Economic Sciences. Ivan Malec, interview by Jaroslav Švelch, July 8, 2015, and November 5, 2015.

77. Ibid.

78. Frk, Hrbek, and collective of authors, *Československý elektrotechnický a elektronický průmysl 1948–1988*.

79. Michal Petrov, *Retro ČS 2: Jak jsme si to (u)žili za reálného socialismu* (Brno, Czech Republic: Jota, 2015).

80. As Pullmann notes, this was not the only issue on which the government and the Central Committee could not find common ground. Especially in the late 1980s,

the federal government generally played the role of the relatively more progressive inciter of changes, whereas the Central Committee tended to oppose them. Pullmann, *Konec experimentu*.

81. Malec, interview.

82. The former program has its counterpart in the *Complex Program of Scientific-Technical Progress and its Socioeconomic Consequences for the Years 1976–1990*, discussed by Guth. The latter was possibly inspired by *Measures in Providing for the Computer Literacy of Secondary School Pupils and Wide-spread Application of Computers to Educational Process*, ratified in the Soviet Union in May 1985, as discussed by Tatarchenko. Guth, "One Future Only"; Ksenia Tatarchenko, "'A House with the Window to the West': The Akademgorodok Computer Center (1958–1993)" (PhD diss., Princeton University, 2013).

83. In the context of the whole program, the part about "the application of electronics in products and devices for personal consumption" seems almost like an afterthought, and focused primarily on household appliances, although it also mentioned electronic musical instruments and toys. Jelínek, "Rozvoj aplikací výpočetní techniky v 8. pětiletce."

84. MŠ ČSR, "Program elektronizace," *Učitelské noviny* 89, no. 11 (March 13, 1986): 11.

85. Ibid., 11–12.

86. Gazzard, *Now the Chips Are Down*; Graeme Kirkpatrick, "Meritums, Spectrums and Narrative Memories of 'Pre-virtual' Computing in Cold War Europe," *The Sociological Review* 55, no. 2 (May 2007): 227–250, https://doi.org/10.1111/j.1467-954X.2007.00703.x.

87. Ivan Malec and (r) [editorial staff], "Mikropočítače z 'druhé strany,'" *Elektronika* 1, no. 1 (1987).

88. Fr. Penczek, "Jsem skeptikem," *Amatérské radio* 37, no. A5 (1988): 183; Malec and (r), "Mikropočítače z 'druhé strany.'"

89. Miroslav Šrol, "Plnenie realizačného programu elektronizácie vo výchově a vzdelávaní na základných školách," *Pedagogika* 38, no. 4 (1988): 397–400.

90. Eden Medina, *Cybernetic Revolutionaries: Technology and Politics in Allende's Chile* (Cambridge, MA: MIT Press, 2011); Peters, *How Not to Network a Nation*.

91. George Kolankiewicz, "The Technical Intelligentsia," in *Social Groups in Polish Society*, ed. David Lane and George Kolankiewicz (London: Macmillan Education UK, 1973), 180–232.

92. Urbášek, *Vysokoškolský vzdělávací systém v letech tzv. normalizace*.

93. Federální statistický úřad, *Historická statistická ročenka ČSSR* (Prague: SNTL, 1985).

94. Petr Vysoký, "Professor Svoboda, Professor Trnka and First Courses on Computers at the Czech Technical University," in *Computing Technology: Past & Future*, ed. Jaroslav Folta, vol. 5, Prague Studies in the History of Science and Technology (Prague: National Technical Museum in Prague, 2001).

95. Jan Blatný, "Stručná historie katedry," Fakulta informačích technologií VUT v Brně, 1985, http://www.fit.vutbr.cz/FIT/history/history86.html.cs.

96. The sense of identity was compounded by the technical intelligentsia's reproductive nature. Socialist Czechoslovakia was less egalitarian than its official ideology might suggest. Apart from the 1950s shake-up following nationalization, general social mobility was not significantly higher in Czechoslovakia than it was in the West. The party's drive for equality in higher education clashed with its wage policies, which favored manual workers, and therefore discouraged working-class youth from going to university. Students entered higher education not because of its economic value—which was negligible—but because of its cultural value. The total number of students was rising, but secondary and tertiary education tended to run in a family. Natalie Simonová, "Vzdělanostní nerovnosti a vzdělanostní mobilita v období socialismu," in *Nerovné šance na vzdělání: vzdělanostní nerovnosti v České republice*, ed. Petr Matějů and Jana Straková (Prague: Academia, 2006), 62–92; Marek Boguszak, "Socioprofesní mobilita v Československu: Srovnání s vyspělými zeměmi a vývoj," *Sociologický časopis/Czech Sociological Review* 27, no. 3 (1991): 255–275.

97. In 1989, for example, an average state-employed computer programmer earned 3,070 crowns a month, slightly below the average salary. Český statistický úřad, *Stav a využití výpočetní techniky v roce 1989 v ČSR* (Prague: Český statistický úřad, 1990).

98. Kolankiewicz, "The Technical Intelligentsia."

99. Mark Lipovetsky, "The Poetics of ITR Discourse: In the 1960s and Today," *Ab Imperio* 2013, no. 1 (2013): 109–139, doi:10.1353/imp.2013.0010.

100. Kolankiewicz, "The Technical Intelligentsia."

101. Women were rarely represented in political leadership. Women's organizations became mere tools for promoting the party's agenda among the female part of the population. Denisa Nečasová, "Women's Organizations in the Czech Lands, 1948–89: An Historical Perspective," in *The Politics of Gender Culture under State Socialism: An Expropriated Voice*, ed. Hana Havelková and Libora Oates-Indruchová, Routledge Research in Gender and Politics 2 (London: Routledge, 2014), 57–82.

102. Simonová, "Vzdělanostní nerovnosti a vzdělanostní mobilita v období socialismu."

103. Květa Jechová, "Postavení žen v Československu v období normalizace," in *Česká společnost v 70. a 80. letech: sociální a ekonomické aspekty*, ed. Oldřich Tůma and Tomáš Vilímek, *Česká společnost po roce 1945* (Prague: Ústav pro soudobé dějiny AV ČR, 2012), 176–246.

104. Federální statistický úřad, *Historická statistická ročenka ČSSR*; Miroslava Šolcová, *Postavení ženy v socialistické společnosti* (Prague: Horizont, 1984).

105. Český statistický úřad, *Stav a využití výpočetní techniky v roce 1986 v ČSR* (Prague: Český statistický úřad, 1987). The figure would drop slightly to 54.3 percent when including additional 1,497 operating system programmers, who were considered a separate category. The figure corresponds to interview material and anecdotal photographic evidence, both of which suggest that women accounted for up to half of the programming staff at the Czech postal service's IT branch in the 1970s, and about half of the students who worked on university mainframes in the early 1980s. Richard Bébr, interview by Jaroslav Švelch, March 7, 2014; Sylva Prokšová, interview by Jaroslav Švelch, October 16, 2015; FIT VUT, "Historické fotografie," *Fakulta informačích technologií VUT v Brně*, 2010, http://www.fit.vutbr.cz/FIT/history/foto.html.cs.

106. Janet Abbate, *Recoding Gender: Women's Changing Participation in Computing*, History of Computing (Cambridge, MA: MIT Press, 2012).

107. For scholarship on the history of gender and computing in Western countries, see: Thomas Haigh, "Masculinity and the Machine Man," in *Gender Codes: Why Women Are Leaving Computing*, ed. Thomas J. Misa (Hoboken, NJ: Wiley, IEEE Computer Society, 2010), 51–71; Nathan Ensmenger, "Making Programming Masculine," in *Gender Codes: Why Women Are Leaving Computing*, ed. Thomas J. Misa (Hoboken, NJ: Wiley, IEEE Computer Society, 2010), 115–141; Abbate, *Recoding Gender*; Marie Hicks, *Programmed Inequality: How Britain Discarded Women Technologists and Lost Its Edge in Computing*, History of Computing (Cambridge, MA: MIT Press, 2017).

108. As Tatarchenko has pointed out about the Soviet Union, numerous female programmers were recruited from the field of mathematics: "Due to subordinate status of applied mathematics in mathematical departments, programming was often the second choice for male students and the first choice of their female peers." Ksenia Tatarchenko, "'The Computer Does Not Believe in Tears': Soviet Programming, Professionalization, and the Gendering of Authority," *Kritika: Explorations in Russian and Eurasian History* 18, no. 4 (2017): 733, https://doi.org/10.1353/kri.2017.0048.

109. Ladislav Zajíček, "Hovory o programování: Femina Quae Programmat," *Mikrobáze* 5, no. 1 (1989): 2.

110. Český statistický úřad, *Stav a využití výpočetní techniky v roce 1986 v ČSR*.

111. The PP 01–PP 06 series of microcomputers was developed under the SMEP framework, but only in very small runs. Their influence on the Czechoslovak microcomputer scene was negligible. Karol Horváth, "Osobný mikropočítač PP-01," *Amatérské Radio,* 35, no. A12 (1986): 457–461.

112. Raymond G. Stokes, *Constructing Socialism: Technology and Change in East Germany, 1945–1990*, Johns Hopkins Studies in the History of Technology (Baltimore: Johns Hopkins University Press, 2000), 204.

113. Trojan, "Jsme schopni vyrábět mikropočítače?"

114. For administrative tasks, some factories and institutions purchased GDR-produced Robotron 1715 microcomputers, equipped with text-only displays and the CP/M operating system. Ivan Malec, "Jestem sceptykiem aneb Jak jsem se stal skeptikem," *Elektronika* 2, no. 5 (1988): 37–38.

115. Malec and (r), "Mikropočítače z 'druhé strany.' "

116. When attributing years to computer models, I refer to the start of serial production. However, many of these machines were prototyped in small quantities months or years before production.

117. Although research on other Soviet bloc countries is scant, some period Czechoslovak sources suggest that Hungarian computer production was similarly unorganized. Marcel Derian, "Mikropočítače v MLR," *Amatérské radio* 33, no. A7 (1984): 260.

118. Miro Kern, "V Tesle tajne postavil prvý osobný počítač—legendu PMD 85, Piešťanský mikropočítač displejový," *Denník N*, June 2015, https://dennikn.sk/150664/v-tesle-tajne-postavil-pocitac-legendu-pmd-85-teda-piestansky-pocitac-mikrodisplejovy/.

119. 30 let osobních počítačů v Československu (1/4) - Roman Kišš: PMD-85: Jak to začalo, proběhlo a nakonec skončilo, 2013, http://www.youtube.com/watch?v=LitYDyvJwjM.

120. Ibid.

121. (VV) [pseud.], "PMD-85: Verzia 85-1 & 85-2," *Elektronika* 3, no. 1 (1989): 32.

122. Out of the three school computers, 1984's IQ 151 stood apart as the only machine that was directly commissioned by the government—a team at the CTU designed it at the request of the Ministry of Education. Despite having only 32 kB of RAM, it was remarkably modular and capable of networking, but also unreliable, unwieldy, and notorious for heat leaks. According to a widespread but possibly apocryphal story, some teachers even used it as a stove. Its massive metal chassis betrayed its origins at Industry Automation Works' Nový Bor factory, which produced industry-grade machines rather than consumer electronics. See Jiří Ježek, "Školní počítač IQ151," *Amatérské radio* 33, no. A12 (1984): 459–460. Production volume estimates are based on period ministerial sources and the interview with Bořivoj Brdička, who worked in the machine's distribution in the 1980s. Malec and (r), "Mikropočítače z 'druhé strany' "; Bořivoj Brdička, interview by Jaroslav Švelch, January 4, 2012.

123. Jan Klabal, "Náš interview s ing. Františkem Hamanem," *Amatérské radio* 36, no. A5 (1987): 161–162.

124. Vít Libovický, interview by Jaroslav Švelch, April 13, 2011.

125. Eduard Smutný, “Ondra,” *Amatérské radio* 35, no. A3 (1986): 92–93.

126. Přemysl Engel, “Náš interview s ing. Eduardem Smutným,” *Amatérské radio* 35, no. A3 (1986): 81.

127. Československá televize, “Report from the Electronization and Automation Exhibition” [in Czech], *Televizní noviny* (Prague: Československá televize, November 15, 1985).

128. Antonín Vomáčka, “Televizní klub mladých 1/1987,” *Televizní klub mladých* (Prague: Československá televize, January 25, 1987).

129. Antonín Vrba, “Informatika a výpočetní technika—první zkušenosti s novým předmětem na gymnáziu,” *Pedagogika* 38, no. 4 (1988): 401–404.

130. The Komenium national enterprise, the country’s dominant manufacturer of school equipment, was in charge of distribution of the IQ 151 computer and offered a relatively narrow selection of educational software and games. Miloslav Feil, *Hry pro IQ 151* (Prague: Komenium, 1986); Bořivoj Brdička, interview by Jaroslav Švelch, January 4, 2012.

131. He also complained that “various interest groups, which have formed behind these products,” were forcing their continued production—the Ministry of Education, for example, kept asking for more IQ 151s to fulfill the Long-term Program of Electronization in Education.

132. Malec and (r), “Mikropočítače z ‘druhé strany,’ ” 5.

133. The difficulties of the Federal Ministry of Electrotechnical Industry were even discussed in the *Rudé právo* (*Red Justice*) newspaper, the mouthpiece of the Communist Party’s Central Committee. Vladimír Čechlovský et al., “O rozvoji československého elektrotechnického průmyslu,” *Rudé právo* 68, no. 60 (March 12, 1988): 3.

134. Československá televize, Report from the Electronization and Automation Exhibition.

135. Ibid.

136. Jiří Franěk, “My chceme počítače!,” *Mladý svět* 29, no. 13 (1987): 13.

137. Ladislav Zajíček and -kš- [pseud.], “Dalibor výpočetní techniky,” *Mikrobáze* 3, no. 6 (1987): 2–4.

138. Bohuslav Blažek, *Bludiště počítačových her* (Prague: Mladá fronta, 1990), 38.

139. Ivan Malec, “Vážení čtenáři …,” *Elektronika* 1, no. 2 (1987): 1.

140. Ivan Malec, “K další perspektivě výroby malých počítačů,” *Elektronika* 3, no. 12 (1989): 20.

141. The interview was conducted in Prague after a seminar for international journalists. The article was published in *Bajtek* and later translated into Czech and published in the Czechoslovak *Amateur Radio* magazine, likely without Malec's knowledge. Malec later disputed the interview as inaccurate. However, he did confirm he made the statement about "immature society," but only as a personal opinion, not an official statement. Penczek, "Jsem skeptikem"; Malec, "Jestem sceptykiem aneb Jak jsem se stal skeptikem."

142. Malec, interview.

143. Ibid.

144. Kristina Šedivá, "Didaktik Gama: Československý Sinclair?," *Elektronika* 2, no. 3 (1988): 5.

145. Jindřich Nevrtal, "Rentabilita přidružené výrobní činnosti v jednotných zemědělských družstvech na okresu Blansko," *Sborník prací Filozofické fakulty brněnské univerzity*, G, Řada sociálněvědná 36, no. G31 (1987): 31–44.

146. Robert Sedláček, *František Čuba: Slušovický zázrak*, TV documentary (Česká televize, 1999).

147. Even Eduard Smutný, the man behind the Ondra, was eventually picked up by an ambitious agricultural enterprise, State Farm Klíčany. While working for the farm, he designed his last pre-1989 computer that went into production, the 8-bit FK-1 model, unofficially codenamed "Farm Computer 1."

148. Gábor Képes, "Hungary: Computers behind the Iron Curtain," *Hungarian Museum of Science, Technology and Transport*, accessed October 30, 2016, http://www.mmkm.hu/index.php/computers#OTODIK.

149. Lojzo, *Lojzo: Pridružená výroba*, 1987, https://www.youtube.com/watch?v=aPcvK3njYD0.

150. Ľudovít Barát, interview by Jaroslav Švelch, October 25, 2016.

151. Ibid.

152. Šedivá, "Didaktik Gama: Československý Sinclair?"

153. Michal Bechyně, "Ještě jednou Didaktik Gama," *Mikrobáze* 5, no. 2 (1989): 25–27.

154. Vladislav Čacký, "Televizní hry s tranzistory," *Amatérské radio* 26, no. A10 (1977): 369–373; Jaroslav Budínský, "Televizní hry," *Amatérské Radio* 28, no. A1 (1979): 19–22; Kompjutry, "Sbírka historické výpočetní techniky," 2016, http://www.kompjutry.cz/pongyvse.html.

155. Peter Smith and Joseph Fanfarelli, "An Exploration of the Historical Contexts of *Nu, Pogodi!*, a Soviet Era LCD Game," *Well Played* 6, no. 2 (2017): 72–89.

156. These machines can be seen in Moscow's Museum of Soviet Arcade Machines.

2 Hunting Down the Machine

1. pb, “Na návštěvě ve Spálené,” *Mikrobáze* 4, no. 5 (1988): 6.

2. These embargoes were coordinated by CoCom—the Coordinating Committee for Multilateral Export Controls. An example list of controlled goods can be found in: British Government, “Consolidated list of goods subject to security export control,” *British Business* 10, no. 1 (1980): 1–28.

3. János Kornai, *The Socialist System: The Political Economy of Communism* (Oxford: Clarendon Press, 2000), 233.

4. The “Delta” computer sold in Czechoslovakia in this period bears all the signs of being repackaged and relabeled unsold stock of the Sinclair’s ZX Spectrum+ model, manufactured in the United Kingdom. It was bundled with a Czech translation of the original British Spectrum manual. See Peter Turányi, “Český originál príručky užívateľa pre ZX Spectrum+,” Softhouse.Speccy.cz, 2007, http://softhouse.speccy.cz/documents/ZX_Spectrum_plus_prirucka_uzivatela.htm.

5. pb, “Na návštěvě ve Spálené,” 6. Contrary to the deputy manager’s answer, the Didaktik Gama was not a Czech product, but a Slovak (or Czechoslovak) product.

6. Daniel Meca, “Z domova,” *Mikrobáze* 5, no. 1 (1989): 30.

7. Alex Wade, *Playback: A Genealogy of 1980s British Videogames* (New York: Bloomsbury, 2016), 153.

8. Roger Silverstone, *Television and Everyday Life* (London: Routledge, 1994).

9. John Urry, *Sociology beyond Societies: Mobilities for the Twenty-First Century*, International Library of Sociology (London: Routledge, 2000), 64.

10. Melanie Swalwell, “Questions about the Usefulness of Microcomputers in 1980s Australia,” *Media International Australia,* no. 143 (May 2012): 64.

11. Mia Consalvo similarly stresses the impact of cosmopolitan citizens on the exchanges between US and Japanese game design and player cultures. Mia Consalvo, *Atari to Zelda: Japan’s Videogames in Global Contexts* (Cambridge, MA: MIT Press, 2015).

12. Tomáš Smutný, interview by Jaroslav Švelch, January 23, 2013.

13. ÚV Svazarmu, *Svazarm od 6. do 7. celostátního sjezdu: fakta o činnosti od r. 1978 do r. 1983* (Prague: Politicko-organizační oddělení ÚV Svazarmu, 1983); Jan Klabal, “Do Nového roku,” *Amatérské radio* 38, no. A1 (1989): 1–2. The magazine was published in two series: the primary A series, which contained shorter schematics and listings as well as news and editorial articles, and the B series, which ran longer, more technical material. To make this clear in the references, I am including the series designation in the issue numbers.

14. Kristen Haring, *Ham Radio's Technical Culture* (Cambridge, MA: MIT Press, 2007), 160.

15. Sherry Turkle, *The Second Self: Computers and the Human Spirit*, 20th Anniversary Edition (Cambridge, MA: MIT Press, 2005), 164.

16. Alena Šolcová and Jakub Šolc, interview by Jaroslav Švelch, November 17, 2017.

17. Martin Campbell-Kelly et al., *Computer: A History of the Information Machine*, 3rd ed., the Sloan Technology Series (Boulder, CO: Westview Press, 2014); Alison Gazzard, *Now the Chips Are Down: The BBC Micro*, Platform Studies (Cambridge, MA: MIT Press, 2016).

18. David Hertl, "První z prvních," *ZX Magazín* 4, no. 4 (1991): 19–26.

19. Vít Libovický, interview by Jaroslav Švelch, April 13, 2011.

20. Patrik Rak, interview by Jaroslav Švelch, December 8, 2014.

21. Programming on paper was also common in Soviet programming education. Ksenia Tatarchenko, "'A House with the Window to the West': The Akademgorodok Computer Center (1958–1993)" (PhD diss., Princeton University, 2013).

22. Jiří Rada, Miroslav Háša, and Martin Pilný, "Programovaný počítač a protihráč CGS," *ABC mladých techniků a přírodovědců* 25, no. 7 (1981): D6–7. For more about the history of *Lunar Landing Game*, see: Benj Edwards, "Forty Years of Lunar Lander," Technologizer, published July 19, 2009, http://www.technologizer.com/2009/07/19/lunar-lander/.

23. Brian L. Stuart, "CARDIAC," Computing Museum, Drexel University website, accessed December 13, 2015, https://www.cs.drexel.edu/~bls96/museum/cardiac.html.

24. František Fuka, interview by Jaroslav Švelch, August 28, 2008, and February 2, 2017; František Fait, interview by Jaroslav Švelch, April 26, 2011.

25. Sylva Prokšová, interview by Jaroslav Švelch, October 16, 2015.

26. Vlastimil Veselý, interview by Jaroslav Švelch, February 11, 2013.

27. As chapter 5 will show, many wandering computer fans had collected software even before they had their own machines. Stanislav Hrda, interview by Jaroslav Švelch, October 4, 2016.

28. In the second half of the 1980s, Prokšová attended the Eltos stand at three large national events per year, and at numerous regional or local ones.

29. Prokšová, interview.

30. Sean Fenty, "Why Old School Is 'Cool': A Brief Analysis of Classic Video Game Nostalgia," in *Playing the Past: History and Nostalgia in Video Games*, ed. Zach Whalen and Laurie N. Taylor (Nashville: Vanderbilt University Press, 2008), 19–31.

31. Michal Hlaváč, interview by Jaroslav Švelch, November 20, 2012.

32. Ultimate Play the Game, *Jetpac*, ZX Spectrum (ACG, 1983).

33. Petr Bednařík and Irena Reifová, "Normalizační televizní seriál: socialistická konstrukce reality," *Sborník Národního muzea v Praze, řada C—Literární historie* 53, nos. 1–4 (2008): 71–74.

34. Hlaváč, interview.

35. Ladislav Zajíček, "ZX nostalgie," *Mikrobáze* 4, no. 6 (1988): 1.

36. As Kirkpatrick observed in Poland, what mattered was often "ownership for its own sake." Graeme Kirkpatrick, "Meritums, Spectrums and narrative memories of 'pre-virtual' computing in Cold War Europe," *The Sociological Review* 55, no. 2 (May 2007): 244, https://doi.org/10.1111/j.1467-954X.2007.00703.x.

37. The Atari ST and Commodore Amiga computers started to appear among dedicated hobbyists in the late 1980s. Martin Ludvík, "Když se řekne Amiga," *Informace pro uživatele mikropočítačů* 1, no. 1 (Počítač přítel člověka) (1989): 5.

38. These figures are based on classified ads in *Amateur Radio* from 1985–1986 and 1989, and on interviews. Average salary figures were discussed in chapter 1.

39. Alek Myslík, "mikro PF 86," *Amatérské radio* 35, no. A1 (1986): 17–18.

40. A studio recording was released on Ivan Mládek's 1985 album "Banjo, Out of the Bag!" ("Banjo, z pytle ven!")

41. David Hertl, interview by Jaroslav Švelch, February 10, 2013; Tomáš Rylek, interview by Jaroslav Švelch, January 15, 2015.

42. Hlaváč, interview.

43. Urry, *Sociology beyond Societies*; Patryk Wasiak, "Playing and Copying: Social Practices of Home Computer Users in Poland during the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. Gerd Alberts and Ruth Oldenziel (London: Springer, 2014), 129–50.

44. zav, "Celní úleva a počítače," *Haló sobota* 68, no. 11 (March 19, 1988): 4.

45. Ladislav Zajíček, "Než nám ujede šestnáctka," *Mikrobáze* 3, no. 9 (1987): 2–4.

46. Hertl, "První z prvních."

47. Bořivoj Brdička, interview by Jaroslav Švelch, January 4, 2012; Libovický, interview; Jan Opl, interview by Jaroslav Švelch, February 5, 2013; Šolcová and Šolc, interview.

48. Myslík, "mikro PF 86"; zav, "Celní úleva a počítače."

49. Oldřich Burger and Pavel Poláček, interview by Jaroslav Švelch, February 12, 2013.

50. Jiří Knapík and Martin Franc, "Tuzex," in *Průvodce kulturním děním a životním stylem v českých zemích 1948-1967*, ed. Jiří Knapík and Martin Franc (Prague: Academia, 2011), 963–965.

51. Paulina Bren, "Tuzex and the Hustler: Living It Up in Czechoslovakia," in *Communism Unwrapped: Consumption in Cold War Eastern Europe*, ed. Paulina Bren and Mary Neuburger (Oxford: Oxford University Press, 2012), 33.

52. Česká televize, "Tuzex," *Retro* (Česká televize, 2009), http://www.ceskatelevize.cz/porady/10176269182-retro-tuzex/208411000360521/video/.

53. Patryk Wasiak, "Computing behind the Iron Curtain: Social Impact of Home Computers in the Polish People's Republic," Tensions of Europe Working Paper 2010, no. 8 (2010): 1–17.

54. Michael Mastanduno, *Economic Containment: CoCom and the Politics of East-West Trade*, Cornell Studies in Political Economy (Ithaca, NY: Cornell University Press, 1992).

55. Kornai, *The Socialist System*. For a period discussion of the problem in Czechoslovakia, see: Miroslav Brdek and Vojtěch Krebs, "Dovoz spotřebního zboží pro československý vnitřní trh," *Zahraniční obchod* 39, no. 11 (1986): 6–10.

56. Tony Smith, "Sord Drawn: The Story of the M5 Micro," The Register, Winter 2013, http://www.theregister.co.uk/2013/04/23/feature_the_sord_m5_home_micro_is_30/; Daniel Dočekal, interview by Jaroslav Švelch, December 2, 2016. The hobbyist press, who diligently reported on the state of Tuzex supplies, criticized the Sord machine for not having any third-party software. Libor Štolc and Daniel Dočekal, "Osobní počítač Sord M5," *Amatérské radio* 33, no. A10 (1984): 377–378.

57. Petr Trojan, "Jsme schopni vyrábět mikropočítače?," *Informace pro uživatele mikropočítačů* 1, no. 1 (Počítač přítel člověka) (1989): 1–4.

58. Vít Libovický and Daniel Dočekal, "Domácí počítače, s nimiž se (možná) setkáte," in *Proč a nač je počítač: Kousněte si do jablka poznání (Magazín VTM pro příznivce informatiky a výpočetní techniky)* (Prague: Mladá fronta, 1987), 22–26.

59. Adrian Graham, "Enterprise Computers," Binary Dinosaurs, 2017, http://www.binarydinosaurs.co.uk/Museum/Enterprise/index.php.

60. Burger and Poláček, interview.

61. Kirkpatrick, "Meritums, Spectrums and narrative memories of 'pre-virtual' computing in Cold War Europe."

62. Ibid.

63. Ladislav Zajíček, "Hovory o programování: Trojúhelník žena-muž-počítač—Bohuslav Blažek," *Mikrobáze* 4, no. 4 (1988): 2–4.

64. Kimberly Elman Zarecor, *Manufacturing a Socialist Modernity: Housing in Czechoslovakia, 1945–1960*, Pitt Series in Russian and East European Studies (Pittsburgh: University of Pittsburgh Press, 2011).

65. Czech Statistical Office, "Tab. 806: Obydlené byty podle právního důvodu užívání bytu, počtu obytných místností, obytné a celkové plochy v m^2 a počtu bydlících osob, podle složení bytové domácnosti a počtu osob v bytě a podle materiálu nosných zdí," *Czech Statistical Office*, 2011, https://vdb.czso.cz/vdbvo2/faces/cs/shortUrl?su=33a00b8a.

66. Cyril Říha, "V čem je panelák kamarád," in *Husákovo 3+1: bytová kultura 70. let*, ed. Lada Hubatová-Vacková and Cyril Říha (Prague: Vysoká škola umělecko-průmyslová, 2007), 17–38.

67. Note that many of the normalization-era research institutions produced unbiased quality research critical of the state of the economy or society, but much of it remained unpublished. This particular research was conducted by the Commerce Research Institute. Vlasta Štěpová, Růžena Komárková, and Ivana Čapková, *Předměty dlouhodobé spotřeby a bydlení—Souhrnná zpráva o výsledcích výzkumu bydlení* (Prague: Výzkumný ústav obchodu, 1979).

68. Lada Hubatová-Vacková, "Interiér panelového bytu: lidový sloh 70. let, jeho zrna i plevy," in *Husákovo 3+1: bytová kultura 70. let*, ed. Lada Hubatová-Vacková and Cyril Říha (Prague: Vysoká škola umělecko-průmyslová, 2007), 93–118.

69. Štěpová, Komárková, and Čapková, *Předměty dlouhodobé spotřeby a bydlení—Souhrnná zpráva o výsledcích výzkumu bydlení*; Hubatová-Vacková, "Interiér panelového bytu: lidový sloh 70. let, jeho zrna i plevy."

70. Czech Statistical Office, "Československé sčítání lidu 1980," *Czech Statistical Office*, 2010, https://www.czso.cz/csu/sldb/ceskoslovenske_scitani_lidu_1980.

71. Hlaváč, interview.

72. Hertl, interview.

73. Zajíček, "Hovory o programování: Trojúhelník žena-muž-počítač—Bohuslav Blažek."

74. Hlaváč, interview.

75. Šolcová and Šolc, interview.

76. Bohuslav Blažek, *Bludiště počítačových her* (Prague: Mladá fronta, 1990).

77. Emil Svárovský, "Ještě jednou o stolcích pro počítače," *Atari zpravodaj (Olomouc)*, nos. 4–5 (1986): 9–10.

78. Rylek, interview.

79. Wade, *Playback*.

80. This formulation is inspired by Nick Montfort's unpublished talk "Bringing the Home into the Computer," presented at the Home Computer Subcultures and Society before the Internet Age workshop on March 24, 2017.

81. Martin Malý, *Demon in Danger*, ZX Spectrum (Chleby, Czech Republic: Demonsoft, 1988).

3 Our Amateur Can Work Miracles

1. kš [pseud.], "Na prahu páté generace," *Mikrobáze* 2, no. 3 (1986): 4.

2. Anthony Caulfield and Nicola Caulfield, *From Bedrooms to Billions*, film documentary (Gracious Files, UK, 2014); Heather Chaplin and Aaron Ruby, *Smartbomb: The Quest for Art, Entertainment, and Big Bucks in the Videogame Revolution* (Chapel Hill, NC: Algonquin Books, 2006).

3. Kristine Jørgensen, Ulf Sandqvist, and Olli Sotamaa, "From Hobbyists to Entrepreneurs: On the Formation of the Nordic Game Industry," *Convergence: The International Journal of Research into New Media Technologies* 23, no. 5 (December 2, 2015): 457–476, https://doi.org/10.1177/1354856515617853.

4. Until 1988, when small enterprises were allowed, as long as they conducted business in their place of registration.

5. Julien Mailland and Kevin Driscoll, *Minitel: Welcome to the Internet*, Platform Studies (Cambridge, MA: MIT Press, 2017).

6. Bohuslav Blažek, *Bludiště počítačových her* (Prague: Mladá fronta, 1990), 15.

7. Angus Stevenson and Christine A. Lindberg, eds., *New Oxford American Dictionary*, 3rd ed. (Oxford: Oxford University Press, 2010), http://www.oxfordreference.com/view/10.1093/acref/9780195392883.001.0001/acref-9780195392883.

8. The positive meaning is also documented in a 1986 Czechoslovak dictionary, which supplies an almost identical definition, adding that "unlike the dilettante," the *amateur* often has "good training in the subject." Lumír Klimeš, *Slovník cizích slov* (Prague: Státní pedagogické nakladatelství, 1986), 17.

9. Ladislav Zajíček, "Než nám ujede šestnáctka," *Mikrobáze* 3, no. 9 (1987): 2–4.

10. Ladislav Holý, *The Little Czech and the Great Czech Nation: National Identity and the Post-Communist Social Transformation* (Cambridge: Cambridge University Press, 1996), 73.

11. Ibid., 65.

12. Frank Veraart, “Losing Meanings: Computer Games in Dutch Domestic Use, 1975–2000,” *IEEE Annals of the History of Computing* 33, no. 1 (January 2011): 52–65, https://doi.org/10.1109/MAHC.2009.66.

13. Stephen Wozniak, “Homebrew and How the Apple Came to Be,” in *Digital Deli: The Comprehensive, User-Lovable Menu of Computer Lore, Culture, Lifestyles, and Fancy*, ed. Steve Ditlea (New York: Workman Pub, 1984), 74.

14. Leslie Haddon and Peggy Gray, “Explaining ICT Consumption: The Case of the Home Computer,” in *Consuming Technologies: Media and Information in Domestic Spaces*, ed. Roger Silverstone and Eric Hirsch (London: Routledge, 1994).

15. Jiří Knapík, “Zájmová činnost,” in *Průvodce kulturním děním a životním stylem v českých zemích 1948-1967*, ed. Jiří Knapík and Martin Franc (Prague: Academia, 2011), 1050–1052.

16. Graeme Kirkpatrick, “Meritums, Spectrums and Narrative Memories of ‘Pre-virtual’ Computing in Cold War Europe,” *The Sociological Review* 55, no. 2 (May 2007): 227–250, https://doi.org/10.1111/j.1467-954X.2007.00703.x.

17. Andrew Lawrence Roberts, *From Good King Wenceslas to the Good Soldier Švejk: A Dictionary of Czech Popular Culture* (Budapest: Central European University Press, 2005).

18. Knapík, “Zájmová činnost.”

19. Zdeněk Svatoš, ed., *Pionýrská encyklopedie* (Prague: Mladá fronta, 1978).

20. Miroslav Háša, interview by Jaroslav Švelch, July 10, 2012.

21. Rudolf Pecinovský, “Historie mimoškolní výuky programování u nás,” in *Tvorba softwaru 2009* (Ostrava, Czech Republic: VŠB-TUO Ostrava, 2009).

22. Háša, interview.

23. L. Kalousek, “Náš interview s ing. Eduardem Smutným (dokončení),” *Amatérské radio* 31, no. A7 (1982): 242.

24. Soviet programming education efforts faced similar problems. Ksenia Tatarchenko, “‘A House with the Window to the West’: The Akademgorodok Computer Center (1958–1993)” (PhD diss., Princeton University, 2013).

25. Pecinovský, “Historie mimoškolní výuky programování u nás”; Háša, interview; Miroslav Fídler, interview by Jaroslav Švelch, November 18, 2014; František Fuka, interview by Jaroslav Švelch, August 28, 2008, and February 2, 2017.

26. Háša, interview.

27. Jan Opl, interview by Jaroslav Švelch, February 5, 2013.

28. Contemporary Soviet programming education methods included writing algorithms for a fictional robot named Robik, although it is unclear whether these were an inspiration for the Czechoslovak version of KAREL. Tatarchenko, "A House with the Window to the West."

29. Richard E. Pattis, *Karel the Robot: A Gentle Introduction to the Art of Programming* (New York: Wiley, 1981).

30. Pecinovský, "Historie mimoškolní výuky programování u nás"; Mikrobáze, "Programová nabídka Mikrobáze (1/1985)," *Mikrobáze* 1, no. 1 (1985): 61–82.

31. Mikrobáze, "Programová nabídka Mikrobáze (1/1985)," 74.

32. Logo was also used by some Czechoslovak educators. Logo Foundation, "Logo History," Logo Foundation website, 2015, http://el.media.mit.edu/logo-foundation/what_is_logo/history.html; Alison Gazzard, *Now the Chips Are Down: The BBC Micro*, Platform Studies (Cambridge, MA: MIT Press, 2016).

33. Tomáš Rylek, interview by Jaroslav Švelch, January 15, 2015.

34. Ivo Slávik, "Rozhovor s autorom: František Fuka," *Bit* 1, no. 10 (1991): 38; Vít Libovický, interview by Jaroslav Švelch, April 13, 2011.

35. Fuka, Personal interview, Prague, 28. 8. 2008.

36. L. Kalousek, "Náš interview—s Miroslavem Hášou a Štefanem Kratochvílem," *Amatérské radio* 32, no. A7 (1983): 241.

37. Háša, interview.

38. Tatarchenko, "A House with the Window to the West."

39. Alena Šolcová and Jakub Šolc, interview by Jaroslav Švelch, November 17, 2017.

40. Háša, interview.

41. Pecinovský, "Historie mimoškolní výuky programování u nás."

42. Petr Roubal, "Svazarm," in *Průvodce kulturním děním a životním stylem v českých zemích 1948-1967*, ed. Jiří Knapík and Martin Franc (Prague: Academia, 2011), 903–904.

43. ÚV Svazarmu, *Svazarm od 6. do 7. celostátního sjezdu: fakta o činnosti od r. 1978 do r. 1983* (Prague: Politicko-organizační oddělení ÚV Svazarmu, 1983).

44. Roubal, "Svazarm."

45. Amatérské radio, "Svazarm a výpočetní technika," *Amatérské radio* 31, no. A4 (1982): 140.

46. Amatérské radio, "Svazarm v roce 1983," *Amatérské radio,* 31, schematics supplement (1982): 1–2.

47. Výbor, “Základní údaje o naší ZO po jednom roce činnosti,” *Zpravodaj Atari klubu (487. ZO Svazarmu)* 1, no. 6 (1987): 3–7; Sylva Prokšová, interview by Jaroslav Švelch, October 16, 2015.

48. Vlastimil Veselý, interview by Jaroslav Švelch, February 11, 2013.

49. Výbor, “Základní údaje o naší ZO po jednom roce činnosti”; Prokšová, interview.

50. Jan Lonský, interview by Jaroslav Švelch, January 17, 2016; Fuka, interview.

51. Mikro-A, “Pro nové majitele Atari,” *Mikro-A* 1, nos. 3–4 (1988): 2.

52. To compare to other platform rivalries, see: Petri Saarikoski and Markku Reunanen, “Great Northern Machine Wars: Rivalry between User Groups in Finland,” *IEEE Annals of the History of Computing* 36, no. 2 (April 2014): 16–26, https://doi.org/10.1109/MAHC.2014.20.

53. Mikro-A, “Adresář,” *Mikro-A* 1, nos. 3–4 (1988): 10.

54. Ján Fila, *Jubilant: Tridsiate piate výročie Zväzarmu*, TV documentary (Bratislava: Československá televize, November 4, 1986).

55. Paul Owens, Christian F. Urquhart, and F. David Thorpe, *Daley Thompson's Decathlon*, ZX Spectrum (Ocean Software, 1984).

56. ÚV Svazarmu, *Svazarm od 6. do 7. celostátního sjezdu: fakta o činnosti od r. 1978 do r. 1983*.

57. Manuel DeLanda, *A Thousand Years of Nonlinear History* (New York: Zone Books, 1997), 32.

58. Výbor, “Základní údaje o naší ZO po jednom roce činnosti.”

59. Amatérské radio, “Jak v roce 1989?,” *Amatérské radio* 38, no. A1 (1989): 17; Oldřich Burger, “Vážení ATARI fans, …,” in *Seznam programů ATARI XL/XE*, ed. Josef Kubelka (Rožnov pod Radhoštěm: FLOP - Vičar, 1990).

60. Tomáš Smutný, interview by Jaroslav Švelch, January 23, 2013.

61. Ibid.

62. Oldřich Burger, “Počítač Atari 600 XL/800 XL,” *Amatérské Radio* 35, no. A1 (1986): 23–24; Oldřich Burger and Pavel Poláček, interview by Jaroslav Švelch, February 12, 2013.

63. Petr Hrabalík, “Sekce mladé hudby,” *Česká televize - Bigbít*, accessed March 9, 2013, http://www.ceskatelevize.cz/specialy/bigbit/clanky/192-sekce-mlade-hudby/.

64. Ladislav Zajíček, *Bity do bytu* (Prague: Mladá fronta, 1988); Martin Malý, interview by Jaroslav Švelch, August 12, 2014; Patrik Rak, interview by Jaroslav Švelch, December 8, 2014; Pavel Lašák, interview by Jaroslav Švelch, January 22, 2016; Lonský, interview.

65. Ladislav Zajíček, "BAJT náš vezdejší," *Bajt* 1, no. 1 (1990): 1.

66. Science fiction fandom, also burgeoning in the 1980s, faced identical organizational challenges. Ivan Adamovič, "In the Grey Zone of Sci-Fi," in *I Shout "That's Me!": Stories of the Czech Fanzine from the '80s till Now*, ed. Miloš Hroch (Prague: Page Five, 2017), 110–43.

67. Miroslav Vaněk, *Byl to jenom rock'n'roll?: hudební alternativa v komunistickém Československu 1956-1989* (Prague: Academia, 2010).

68. According to an unpublished report produced at my request by the Security Services Archive, document no. ABS-S 193/2017.

69. Zbigniew Stachniak, "Red Clones: The Soviet Computer Hobby Movements of the 1980s," *IEEE Annals of the History of Computing* 37, no. 1 (January 2015): 16, https://doi.org/10.1109/MAHC.2015.11.

70. Data from unpublished research by Robert Jameson.

71. Výbor, "Základní údaje o naší ZO po jednom roce činnosti."

72. For more details, see chapter 6.

73. Jane Margolis and Allan Fisher, *Unlocking the Clubhouse: Women in Computing* (Cambridge, MA: MIT Press, 2002).

74. The Czech titles were "Kuchyně dívčích přání" and "Abeceda šikovné dívky," respectively.

75. Květa Jechová, "Postavení žen v Československu v období normalizace," in *Česká společnost v 70. a 80. letech: sociální a ekonomické aspekty*, ed. Oldřich Tůma and Tomáš Vilímek, Česká společnost po roce 1945 (Prague: Ústav pro soudobé dějiny AV ČR, 2012), 176–246.

76. Kevin McDermott, *Communist Czechoslovakia, 1945–89: A Political and Social History*, European History in Perspective (New York: Palgrave Macmillan, 2015), 83.

77. Miroslava Šolcová, *Postavení ženy v socialistické společnosti* (Prague: Horizont, 1984).

78. Gerald A. Bird, Rosemary C. Goss, and Gloria W. Bird, "Effects of Home Computer Use on Fathers' Lives," *Family Relations* 39, no. 4 (1990): 438–442, https://doi.org/10.2307/585225; Kristen Haring, *Ham Radio's Technical Culture* (Cambridge, MA: MIT Press, 2007).

79. Jan Hlaváček, "Poděkování ženám," *Zpravodaj Atari klubu (487. ZO Svazarmu)* 2, no. 2 (1988): 73.

80. Haring, *Ham Radio's Technical Culture*.

81. Prokšová, interview.

82. Ibid.

83. This vein of game history critique has been developed by Nooney: Laine Nooney, "A Pedestal, a Table, a Love Letter: Archaeologies of Gender in Videogame History," *Game Studies* 13, no. 2 (December 2013), http://gamestudies.org/1302/articles/nooney.

84. For a discussion of masculinity in Western gaming cultures and home computing subcultures, see, for example: Jonathan Dovey and Helen Kennedy, "From Margin to Center: Biographies of Technicity and the Construction of Hegemonic Games Culture," in *The Players' Realm: Studies on the Culture of Video Games and Gaming*, ed. J. Patrick Williams and Jonas Heide Smith (Jefferson, NC: McFarland & Co, 2007), 131–154; Patryk Wasiak, "'Illegal Guys.' A History of Digital Subcultures in Europe during the 1980s," *Zeithistorische Forschungen/Studies in Contemporary History, Online-Ausgabe* 9, no. H. 2 (2012), http://www.zeithistorische-forschungen.de/site/40209282/default.aspx#zitieren.

85. Although this is a credible hypothesis, the author does not back it up with empirical evidence. Jarmila Bachmanová, "Bastlovat, bastlit," *Naše řeč* 70, no. 4 (1987): 222–223.

86. Melanie Swalwell, "Movement and Kinaesthetic Responsiveness: A Neglected Pleasure," in *The Pleasures of Computer Gaming: Essays on Cultural History, Theory and Aesthetics*, ed. Melanie Swalwell and Jason Wilson (Jefferson, NC: McFarland & Co, 2008), 72–93.

87. Bruno Jakić, "Galaxy and the New Wave: Yugoslav Computer Culture in the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. G. Alberts and Ruth Oldenziel (London: Springer, 2014), 129–150; Stachniak, "Red Clones."

88. Amatérské radio, "Mikro-AR," *Amatérské radio* 34, no. A11 (1985): 417–420; Aleš Juřík and Radim Hlucháň, "Ještě jednou mikropočítač programově kompatibilní se ZX Spectrum," *Amatérské radio* 37, no. A8 (1988): 297–299.

89. Amatérské radio, "Jak v roce 1989?"; Juřík and Hlucháň, "Ještě jednou mikropočítač programově kompatibilní se ZX Spectrum."

90. Tomáš Mastík, "Programovatelný ovladač pro ZX Spectrum," *Amatérské radio* 35, no. A2 (1986): 57–59; Antonín Hofmann and Zdeněk Stuchlík, "Kempston joystick," *Amatérské radio* 36, no. A9 (1987): 337–340; Karel Novotný, "Joystick," *Amatérské radio* 37, no. A5 (1988): 177–178; Miroslav Kozák, "Kempston joystick s MHB 8255A," *Amatérské radio* 37, no. A12 (1988): 462; Pavel Dočekal and Jiří Suchánek, "Výroba joysticku (řídící páky) pro počítače ATARI 600, 800 XL," *Atari zpravodaj (Olomouc)*, nos. 4–5 (1986): 12–15.

91. Prokšová, interview.

92. Building teletext receivers was also popular in the 1980s when most domestically produced TV sets could not display Czechoslovak Television's teletext signal.

93. Rylek, interview.

94. Ivan Horsák, “Jednoduchá tlačítková klávesnice k počítačům Sinclair ZX-81 a ZX Spectrum,” *Amatérské radio* 37, no. A3 (1988): 104.

95. Hofmann and Stuchlík, “Kempston joystick.”

96. Melanie Swalwell, “The Early Micro User: Games Writing, Hardware Hacking, and the Will to Mod,” in *Proceedings of DiGRA Nordic 2012 Conference: Local and Global—Games in Culture and Society* (Tampere, Finland: DiGRA, 2012), http://www.digra.org/dl/db/12168.37411.pdf.

97. -er [pseud.], “Poznámka redakce AZ,” *Atari zpravodaj (Olomouc)*, nos. 4–5 (1986): 34.

98. Prokšová, interview.

99. Zajíček, “Než nám ujede šestnáctka.”

100. The club also ported the CP/M operating system to the Spectrum.

101. Mikrobáze, “Programová nabídka Mikrobáze (3/1986),” *Mikrobáze* 2, no. 3 (1986): 55.

102. Based on a leaflet for the *Datafon* help line, which I acquired from the collection of my informant Petr Mihula. It was probably inserted in one of the issues of *Mikrobáze*.

103. The instruction manual for the mouse kit was co-designed by Miroslav Háša. Mikrobáze, “Akce myš,” *Mikrobáze* 4, no. 5 (1988): 27.

104. The 602 even manufactured cartridges for the 8-bit Atari in a project coordinated by leading Atari enthusiast Oldřich Burger. The Atari machines had a cartridge slot, which allowed instantaneous execution of software from a read-only medium. The production of cartridges, however, required memory chips and factory assembly. According to Burger, he had the chips manufactured by TESLA’s Piešt’any plant outside of the approved production plan. To mask their origin, he screen-printed the fake label “IECO Korea” on them. Among the first programs to be distributed on the locally produced Atari cartridges was—unsurprisingly—the Turbo 2000 loader. Burger and Poláček, interview; JK [pseud.], “Tipy přílohy Mikroelektronika pro rok 1989,” *Amatérské radio* 37, no. A3 (1988): 422–424.

105. Jiří Richter and Jan Hlaváček, “TURBO 2000: Systém zrychleného přenosu dat z magenetofonů ATARI,” *Zpravodaj Atari klubu (487. ZO Svazarmu)* 1, no. Příloha 2 (1987): 1–26.

106. Jiří Richter, interview by Jaroslav Švelch, January 8, 2016. It can also be deduced from the frequency of ads offering games in this format in the post-1989 advertising papers. See: Annonce, “15-2 Programy (23. 5. 1990),” *Annonce* 1, no. 21

(May 23, 1990): 15; Annonce, “15-2 Programy (4. 1. 1991),” *Annonce* 2, no. 118 (January 4, 1991): 26–27; Annonce, “15-2 Programy (3. 1. 1992),” *Annonce* 3, no. 265 (January 3, 1992): 3–4.

107. Tomasz Mazur, “Atari Super Turbo,” *Bajtek* 5, no. 4 (1989): 10–11.

108. Steve Weber, *The Success of Open Source* (Cambridge, MA: Harvard University Press, 2004).

109. Česká televize, “Svazarm,” *Retro* (Česká televize, December 2012), http://www.ceskatelevize.cz/porady/10176269182-retro/211411000360003/video/.

110. Radovan Richta and collective of authors, *Civilizace na rozcestí: Společenské a lidské souvislosti vědeckotechnické revoluce* (Prague: Svoboda, 1967).

111. The story was published in the form of comics in a 1989 issue of *ZX Magazine*. ZX Magazín, “Comix zvaný ‘spektrácký,’” *ZX Magazín* 2, no. 10 (1989): 2; David Hertl, interview by Jaroslav Švelch, February 10, 2013.

112. Ladislav Zajíček, “Bajtek,” *Mikrobáze* 4, no. 3 (1988): 31.

113. Hertl, interview.

114. There is no record of Hertl being monitored by the secret police.

115. Henry Jenkins, *Textual Poachers: Television Fans & Participatory Culture*, Studies in Culture and Communication (New York: Routledge, 1992); Ann Komaromi, “The Material Existence of Soviet Samizdat,” *Slavic Review* 63, no. 3 (2004): 597–618, https://doi.org/10.2307/1520346.

116. Ondřej Kafka, interview by Jaroslav Švelch, November 20, 2017.

117. Jonathan Bolton, *Worlds of Dissent: Charter 77, the Plastic People of the Universe, and Czech Culture under Communism* (Cambridge, MA: Harvard University Press, 2012), 37.

118. Ibid., 33.

119. Ibid., 99. Also see Jiří Suk, *Politika jako absurdní drama: Václav Havel v letech 1975 - 1989* (Prague: Nakladatelství Paseka, 2013).

120. Lašák, interview.

121. Knihovna Václava Havla, “Václav Havel a počítače,” Knihovna Václava Havla, November 19, 2012, http://www.vaclavhavel-library.org/cs/index/videa/105/vaclav-havel-a-pocitace-19-11-2012.

122. Suk, *Politika jako absurdní drama*, 218.

123. Václav Havel, “A letter to the Prime Minister Ladislav Adamec,” November 4, 1989, Record no. 873, Václav Havel Library, https://archive.vaclavhavel-library.org/viewArchive.php?itemDetail=873.

124. According to his collaborator's recollections, the machine was back in Havel's hands in August 1989. It is now exhibited in the Václav Havel Library. See Knihovna Václava Havla, "Václav Havel a počítače."

125. Lašák, interview.

126. Theodor H. Nelson, *Computer Lib: You Can and Must Understand Computers Now/Dream Machines: New Freedoms through Computer Screens—A Minority Report* (self-published, 1974).

127. Lašák bought his own, but Stárek's arrived from France, supplied by the exiled journalist and samizdat supporter Pavel Tigrid.

128. František Stárek, interview by Jaroslav Švelch, January 5, 2016.

129. Ibid.

130. Ibid.

131. Lašák, interview.

132. Daniel Jenne, *D-Text*, ZX Spectrum (Prague: Daje, 1986).

133. Albert Ball and Stuart C. Ball, *Jumping Jack*, ZX Spectrum (Imagine Software, 1983); Paco Suarez and Paco Portalo, *Bugaboo (The Flea)*, ZX Spectrum (Quicksilva, 1983).

134. Jiří Richter, interview by Jaroslav Švelch, January 8, 2016; Aleš Martiník, interview by Jaroslav Švelch, June 14, 2017.

135. Richta and collective of authors, *Civilizace na rozcestí: Společenské a lidské souvislosti vědeckotechnické revoluce*.

136. Ľudovít Barát, interview by Jaroslav Švelch, October 25, 2016.

137. Stachniak, "Red Clones," 15.

138. Based on my material, the only club that had stable international connections was the Atari club in Olomouc. Its cofounder Oldřich Burger used to travel to West Germany to visit his émigré mother, and he established contacts with the president of the German Atari Bit Byter User Club, Mr. Wolfgang Burger (no relationship). Oldřich Burger, "Mezinárodní klubová spolupráce," *Atari zpravodaj (Olomouc)*, nos. 4–5 (1986): 4–5; Burger and Poláček, interview.

4 Who's Afraid of Gameplay?

1. Mikrobáze, "Programová nabídka Mikrobáze (1/1985)," *Mikrobáze* 1, no. 1 (1985): 65.

2. Graeme Kirkpatrick, "Constitutive Tensions of Gaming's Field: UK gaming Magazines and the Formation of Gaming Culture, 1981–1995," *Game Studies* 12, no. 1 (2012), http://gamestudies.org/1201/articles/kirkpatrick.

3. Graeme Kirkpatrick, "Early Games Production, Gamer Subjectivation and the Containment of the Ludic Imagination," in *Fans and Videogames: Histories, Fandom, Archives*, ed. Melanie Swalwell, Helen Stuckey, and Angela Ndalianis, Routledge Advances in Game Studies 9 (New York: Routledge, 2017), 31.

4. Graeme Kirkpatrick, *The Formation of Gaming Culture: UK Gaming Magazines, 1981–1995* (London: Palgrave Pivot, 2015).

5. Gleb J. Albert, "Rezension zu: G. Kirkpatrick: The Formation of Gaming Culture," H-Soz-Kult: Kommunikation und Fachinformation für die Geschichtswissenschaften, published July 14, 2016, http://www.hsozkult.de/publicationreview/id/rezbuecher-24951.

6. Specifically, 18.6 percent of households owned a computer. John Schmitt and Jonathan Wadsworth, *Give PCs a Chance: Personal Computer Ownership and the Digital Divide in the United States and Great Britain* (London: Centre for Economic Performance, London School of Economics and Political Science, 2002).

7. Kirkpatrick, "Early Games Production, Gamer Subjectivation and the Containment of the Ludic Imagination," 33.

8. Claude Lévi-Strauss, *The Savage Mind* (London: Weidenfeld and Nicolson, 1966).

9. Donna Jeanne Haraway, *Simians, Cyborgs, and Women: The Reinvention of Nature* (New York: Routledge, 1991), 173.

10. Ibid., 177.

11. For examples of playful approaches among Soviet programmers, see: Ksenia Tatarchenko, "'A House with the Window to the West': The Akademgorodok Computer Center (1958–1993)" (PhD diss., Princeton University, 2013).

12. Nick Montfort, *Exploratory Programming for the Arts and Humanities* (Cambridge, MA: MIT Press, 2016).

13. Born in 1935, Bébr had his first job at the Regional Telecommunications Office in the Southern Bohemian city of České Budějovice. During the post-1968 screenings, he was fired because of his involvement in connecting international phone calls to the West during the Soviet invasion. However, he eventually found a job at the Prague Headquarters of the Czechoslovak Post, where he headed its computing division and had access to UNIVAC mainframes and HP minis. Richard Bébr, interview by Jaroslav Švelch, March 7, 2014.

14. Also corroborated by Ľudovít Barát's memories of computer fairs. Lay visitors enjoyed getting printouts of biorhythms and similar programs over other uses of the computer. Ľudovít Barát, interview by Jaroslav Švelch, October 25, 2016.

15. DECUS, *DECUS Program Library Catalog for PDP-8, FOCAL8* (Maynard, MA: Digital Equipment Computers Users Society, 1973); Frank Veraart, "Losing Meanings:

Computer Games in Dutch Domestic Use, 1975–2000," *IEEE Annals of the History of Computing* 33, no. 1 (January 2011): 52–65, https://doi.org/10.1109/MAHC.2009.66.

16. To describe human-computer interaction, he uses the expression—literally translated—"communication of computer with human."

17. Richard Bébr, "Hravé obvody," *Technický magazín* 24, no. 12 (1981): 20.

18. Redakce AR, "Programy pro praxi i zábavu," *Amatérské radio* 31, no. A1 (1982): 20.

19. A similar expression, "games with computer programs," was used by Ted Nelson in 1974. Theodor H. Nelson, *Computer Lib: You Can and Must Understand Computers Now/Dream Machines: New Freedoms through Computer Screens—A Minority Report* (self-published, 1974), 48.

20. Kirkpatrick, "Constitutive Tensions of Gaming's Field: UK Gaming Magazines and the Formation of Gaming Culture, 1981–1995."

21. Versions of these had been collected in David Ahl's best-selling book *BASIC Computer Games*. David H. Ahl, *BASIC Computer Games* (New York: Workman Publishing, 1978).

22. AR, "AR výpočetní technice '85," *Amatérské radio* 34, no. A2 (1985): 57–58; Alek Myslík, "mikro PF 86," *Amatérské radio* 35, no. A1 (1986): 17–18.

23. Matthew Smith, *Manic Miner*, ZX Spectrum (Software Projects, 1983); Ultimate Play the Game, *Atic Atac*, ZX Spectrum (ACG, 1983); Ultimate Play the Game, *Knight Lore*, ZX Spectrum (ACG, 1984).

24. Jaroslav Švelch, "Manic Miner," in *100 Greatest Video Game Franchises*, ed. Robert Mejia, Jaime Banks, and Aubrie Adams (Lanham, MD: Rowman & Littlefield Publishers, 2017), 112–113.

25. Lucie Skoumalová, "Vznik a vývoj českých kasin z pohledu krupiéra" (master's thesis, Charles University in Prague, 2012).

26. Kristen Haring, *Ham Radio's Technical Culture* (Cambridge, MA: MIT Press, 2007).

27. Gordon Calleja, *In-Game: From Immersion to Incorporation* (Cambridge, MA: MIT Press, 2011).

28. In addition, Swalwell's ethnographic research shows that games produce pleasure through intense sensory and kinesthetic engagement. She compares the visceral, bodily nature of video game play in the arcades and in social settings with the use of a home computer, which is perceived as more reserved and asensual. Matti Lahti, "As We Become Machines: Corporealized Pleasures in Video Games," in *The Video Game Theory Reader*, ed. Mark J. P. Wolf and Bernard Perron (New York: Routledge, 2003), 169; Melanie Swalwell, "Movement and Kinaesthetic Responsiveness: A Neglected Pleasure," in *The Pleasures of Computer Gaming: Essays on Cultural

History, Theory and Aesthetics, ed. Melanie Swalwell and Jason Wilson (Jefferson, NC: McFarland & Co, 2008), 72–93.

29. Sherry Turkle, *The Second Self: Computers and the Human Spirit*, 20th Anniversary Edition (Cambridge, MA: MIT Press, 2005), 84.

30. Jon Dovey and Helen W. Kennedy, *Game Cultures: Computer Games As New Media*, Issues in Cultural and Media Studies (Maidenhead, UK: Open University Press, 2006), 26.

31. Ibid.

32. Rob Cover, "Gaming (Ad)diction: Discourse, Identity, Time and Play in the Production of the Gamer Addiction Myth," *Game Studies* 6, no. 1 (2006), http://gamestudies.org/0601/articles/cover.

33. Frank Veraart describes a similar process that unfolded in the Dutch Hobby Computer Club. Veraart, "Losing Meanings."

34. Mikrobáze, "Malá úvaha nejen pro uživatele ZX Spectra," *Mikrobáze* 1, no. 1 (1985): 30.

35. kš [pseud.], "Program pro vnoučata," *Mikrobáze* 2, no. 4 (1986): 3.

36. Ibid.

37. Ladislav Zajíček, "Je libo pouček?," *Mikrobáze* 3, no. 5 (1987): 46.

38. Cover, "Gaming (Ad)diction."

39. Ibid.

40. Mikrobáze, "Malá úvaha nejen pro uživatele ZX Spectra," 29.

41. kš, "Program pro vnoučata," 3.

42. Myslík, "mikro PF 86," 17.

43. Tomáš Mastík, "Myš: Externí pohyblivý ovládač kurzoru pro mikropočítač ZX-Spectrum," *Amatérské radio* 35, no. A10 (1986): 377–379.

44. Petr Šimůnek and Daniel Jenne, "Univerzální interface Mirek," *Mikrobáze* 4, no. 7 (1988): 16–18.

45. A contemporary article in *Mikrobáze* suggests the same, and at least two interviewees agree on that. Ladislav Zajíček, "Hovory o programování: Je toho 80 kilo … můžu to tak nechat?," *Mikrobáze* 4, no. 10 (1988): 2–3; František Fuka, interview by Jaroslav Švelch, August 28, 2008, and February 2, 2017; David Hertl, interview by Jaroslav Švelch, February 10, 2013.

46. James Newman, *Playing with Videogames* (London: Routledge, 2008).

47. Understandable in the case of the ZX Spectrum, whose keyboard was notoriously fragile.

48. Antonín Hofmann and Zdeněk Stuchlík, “Kempston joystick,” *Amatérské radio* 36, no. A9 (1987): 155.

49. Karel Novotný, “Joystick,” *Amatérské radio* 37, no. A5 (1988): 177.

50. Sinclair 602, “Počítačové hry jako společenský fenomén,” *Sinclair 602* 2, no. 4 (1988): 4–5; Aleš Martiník, interview by Jaroslav Švelch, June 14, 2017.

51. Claudio Ciborra, *The Labyrinths of Information: Challenging the Wisdom of Systems* (Oxford: Oxford University Press, 2002); E. Gabriella Coleman, *Coding Freedom: The Ethics and Aesthetics of Hacking* (Princeton, NJ: Princeton University Press, 2013).

52. Montfort et al. provide intriguing discussions of poking on the Commodore 64. Nick Montfort et al., *10 PRINT CHR$(205.5+RND(1)); : GOTO 10* (Cambridge, MA: MIT Press, 2013).

53. Mikrobáze, “Úpravy her pro ZX Spectrum,” *Mikrobáze* 2, no. 3 (1986): 39.

54. Ibid., 44.

55. Mikrobáze, “Programová nabídka Mikrobáze (1/1985),” 65.

56. Zbyšek Bahenský, “Něco triků pro aktivní ‘spectristy’,” *Elektronika* 3, no. 10 (1989): 28.

57. Ladislav Zajíček, “V domovině Specter,” *Mikrobáze* 3, no. 7 (1987): 50.

58. Altar, “Informace o ALTARu,” Nakladatelství Altar, accessed October 28, 2017, http://www.altar.cz/altar/onas.html.

59. A popular horserace-themed *Monopoly* clone—called *Horserace Betting* (*Dostihy a sázky* in Czech)—was launched in 1984 and reportedly received negatively by the Communist Party’s press. ahr [pseud.], “‘Dostihy a sázky’ oslavily čtvrt století,” iDNES.cz, published January 14, 2010, http://sdeleni.idnes.cz/deskove-hry-dostihy-a-sazky-oslavily-ctvrt-stoleti-fh6-/kultura-sdeleni.aspx?c=A100108_103525_kultura-sdeleni_ahr.

60. Roberto Farné, “Pedagogy of Play,” *Topoi* 24, no. 2 (September 2005): 169–181, https://doi.org/10.1007/s11245-005-5053-5.

61. Miloš Zapletal, *Velká encyklopedie her: Hry v přírodě*, 2nd ed. (Prague: Olympia, 1987).

62. Vít Libovický and Daniel Dočekal, “Domácí počítače, s nimiž se (možná) setkáte,” in *Proč a nač je počítač: Kousněte si do jablka poznání (Magazín VTM pro příznivce informatiky a výpočetní techniky)* (Prague: Mladá fronta, 1987), 24–25.

63. Diary transcripts were kindly provided by the Blažek family in personal email communication.

64. Bohuslav Blažek, *Bludiště počítačových her* (Prague: Mladá fronta, 1990), 27.

65. Bohuslav Blažek, "Spor o počítačové hry," *Informace pro uživatele mikropočítačů* 1, no. 2 (Počítačové hry) (1989): 6.

66. A computer scientist, educationist and the co-inventor of the LOGO programming language, Papert intended to provide a playful environment in which children would learn to solve problems. Seymour Papert, *Mindstorms: Children, Computers, and Powerful Ideas* (New York: Basic Books, 1993). His work also inspired Soviet programming education initiatives. See Tatarchenko, "A House with the Window to the West."

67. Blažek, *Bludiště počítačových her*, 158.

68. Simon Egenfeldt-Nielsen, Jonas Heide Smith, and Susana Pajares Tosca, *Understanding Video Games: The Essential Introduction*, 2nd ed. (New York: Routledge, 2013).

69. Miroslav Háša, "Několik slov úvodem," in *Počítačové hry: Historie a současnost, 1. díl*, ed. František Fuka (Beroun: Zenitcentrum, 1988), 3.

70. Fuka intended the manuscript to be a single publication, and his suggested title was *What Mr. Sinclair Did Not Foresee*. The book was divided into two parts and released as teaching material under a different title without his knowledge. Fuka, interview.

71. We can see this history-in-the-making already in the very first 1985 *Mikrobáze* listings, where the 1983 platformer *Manic Miner*, one of the first British software blockbusters, is already labeled as "classic." Mikrobáze, "Programová nabídka Mikrobáze (1/1985)," 66.

72. "P" klub počítačových nadšenců, "Listárna," *Zápisník* 32, no. 22 (1988): 28–29.

73. Ibid., 29.

74. Dave Marshall, *Tomahawk*, ZX Spectrum (Digital Integration, 1985).

75. They were compiled from dozens of votes and thus not representative of overall preferences of Czechoslovak users. ZX Magazín, "Vašich 15 nej," *ZX Magazín* 2, no. 3 (1989): 2; ZX Magazín, "10x nej," *ZX Magazín* 2, no. 6 (1989): 1.

76. Kirkpatrick, *The Formation of Gaming Culture*, 51.

77. Egenfeldt-Nielsen, Smith, and Tosca, *Understanding Video Games*, 101.

78. Games were not contextualized in production narratives, but rather grouped together based on topic, genre, or simply convenience. The entries tended to describe what the player *does* in the game as well as the skills required to play it (such as dexterity, logical thinking, or orientation), an understandable inclusion for readers unfamiliar with game genres.

79. Mikrobáze, “Malá úvaha nejen pro uživatele ZX Spectra,” 30.

80. Paul Owens, Christian F. Urquhart, and F. David Thorpe, *Daley Thompson's Decathlon*, ZX Spectrum (Ocean Software, 1984); Mervyn J. Estcourt, *Deathchase*, ZX Spectrum (Micromega, 1983); Ultimate Play the Game, *Knight Lore*.

81. Mikrobáze, “Programová nabídka Mikrobáze (2/1986),” *Mikrobáze* 2, no. 2 (1986): 65.

82. Jan Flak, “Počítačová dilemata, II.,” *Počítačová dilemata* (Ostrava, Czech Republic: Československá televize Ostrava, September 14, 1988).

83. Mikrobáze, “Programová nabídka Mikrobáze (2/1986),” 67; Pete Cooke, Milos, and ASB, *Ski Star 2000*, ZX Spectrum (Richard Shepherd Software, 1985).

84. Mikrobáze, “Programová nabídka Mikrobáze (1/1985),” 65.

85. Blažek, *Bludiště počítačových her*, 54–55.

86. František Fuka, *Počítačové hry: Historie a současnost, 2. díl* (Beroun: Zenitcentrum, 1988), 46.

87. František Fuka, *Počítačové hry: Historie a současnost, 1. díl* (Beroun: Zenitcentrum, 1988), 34; Pete Cooke, *Academy*, ZX Spectrum (CRL Group, 1986); Geoff Crammond and Mike Follin, *The Sentinel*, ZX Spectrum (Firebird, 1986); Graftgold, *Quazatron*, ZX Spectrum (Hewson Consultants, 1986).

88. Fuka, *Počítačové hry: Historie a současnost, 1. díl*; Mikrobáze, “Programová nabídka Mikrobáze (2/1986).”

89. Andy Green et al., *Krakout*, Commodore 64 (Gremlin Graphics, 1987).

90. Blažek, *Bludiště počítačových her*, 60.

91. Ibid., 63.

92. Veraart, “Losing Meanings”; Kirkpatrick, “Constitutive Tensions of Gaming's Field: UK Gaming Magazines and the Formation of Gaming Culture, 1981–1995.”

93. Markku Reunanen, “How Those Crackers Became Us Demosceners,” *WiderScreen* 17, nos. 1–2 (April 15, 2014), http://widerscreen.fi/numerot/2014-1-2/crackers-became-us-demosceners/.

94. Kirkpatrick, *The Formation of Gaming Culture*.

95. Stanislav Hrda, interview by Jaroslav Švelch, October 4, 2016.

96. Tatarchenko, “A House with the Window to the West.”

97. Patryk Wasiak, “Playing and Copying: Social Practices of Home Computer Users in Poland during the 1980s,” in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. Gerd Alberts and Ruth Oldenziel (London: Springer, 2014), 136.

98. Tamás Beregi, “Hungary,” in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 227.

5 Lighting Up the Shadows

1. Michal Bechyně, “Ještě jednou Didaktik Gama,” *Mikrobáze* 5, no. 2 (1989): 27.

2. Petr Trojan, “Hráč (rozhovor s Františkem Fukou),” *Informace pro uživatele mikropočítačů* 1, no. 2 (Počítačové hry) (1989): 2.

3. Ramon Lobato, *Shadow Economies of Cinema: Mapping Informal Film Distribution*, Cultural Histories of Cinema (London: Palgrave Macmillan, 2012), 40.

4. Lobato, *Shadow Economies of Cinema*.

5. Ibid., 43.

6. Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network-Theory*, Clarendon Lectures in Management Studies (Oxford: Oxford University Press, 2005), 39.

7. Brian Larkin, *Signal and Noise: Media, Infrastructure, and Urban Culture in Nigeria* (Durham, NC: Duke University Press, 2008), 219.

8. Lobato, *Shadow Economies of Cinema*, 18.

9. I am using the word *modification* in a more general sense than the current meaning of the shortened terms “modding” and “mods.” See Hector Postigo, “Modification,” in *Debugging Game History: A Critical Lexicon*, ed. Henry Lowood and Raiford Guins, Game Histories (Cambridge, MA: MIT Press, 2016), 325–334.

10. Gérard Genette, *Paratexts: Thresholds of Interpretation*, trans. Jane E. Lewin, Literature, Culture, Theory 20 (Cambridge: Cambridge University Press, 1997).

11. Mia Consalvo, *Cheating: Gaining Advantage in Videogames* (Cambridge, MA: MIT Press, 2009); Todd Harper, *The Culture of Digital Fighting Games: Performance and Practice*, Routledge Studies in New Media and Cyberculture (New York: Routledge, Taylor & Francis Group, 2014); Annika Rockenberger, “Video Game Framings,” in *Examining Paratextual Theory and Its Applications in Digital Culture*, ed. Nadine Desrochers and Daniel Apollon (Hershey, PA: IGI Global, 2014), 252–286.

12. Jan Švelch, “‘Footage Not Representative’: Redefining Paratextuality for the Analysis of Official Communication in the Video Game Industry,” in *Contemporary Research on Intertextuality in Video Games*, ed. Christophe Duret and Christian-Marie Pons (Hershey, PA: IGI Global, 2016), 297–315.

13. Raffaele Cecco, *Exolon*, ZX Spectrum (Hewson Consultants, 1987).

14. Despite intermittent shortages of both tapes and tape decks, the fact that they were also used for audio made them less niche and more readily available than

floppy disks. See also: Alex Wade, *Playback: A Genealogy of 1980s British Videogames* (New York: Bloomsbury, 2016).

15. Crash, "Exolon (review)," *Crash* 4, no. 8 (1987): 14.

16. František Fuka, interview by Jaroslav Švelch, August 28, 2008, and February 2, 2017.

17. ZX Magazín, "Začali jsme …," *ZX Magazín* 3, no. 7 (1990): 27. The Polish connection is corroborated by Polish interview material collected by Wade. Wade, *Playback*.

18. Stanislav Hrda, interview by Jaroslav Švelch, October 4, 2016.

19. Patryk Wasiak, "Playing and Copying: Social Practices of Home Computer Users in Poland during the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. Gerd Alberts and Ruth Oldenziel (London: Springer, 2014), 129–150; Bruno Jakić, "Galaxy and the New Wave: Yugoslav Computer Culture in the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. G. Alberts and Ruth Oldenziel (London: Springer, 2014), 129–150.

20. Patryk Wasiak, "'Illegal Guys': A History of Digital Subcultures in Europe during the 1980s," *Zeithistorische Forschungen/Studies in Contemporary History, Online-Ausgabe* 9, no. H. 2 (2012), http://www.zeithistorische-forschungen.de/site/40209282/default.aspx#zitieren.

21. Franco Frey, "Multifaceted Device," *Crash* 3, no. 26 (1986): 86. Multiface saved the running code of the game, but not the loading screen. This might have motivated Eastern European pirates to commission their own loading screens, as seen in the case of *Exolon*.

22. Moj Mikro, "Mali oglasi," classified ad section, *Moj Mikro* 4, no. 11 (1987): 45–52.

23. MS-CID, "YS Games of the Year, 1989," *ZX Magazín* 3, no. 7 (1990).

24. On the Spectrum platform, most games traveled from their countries of origin to Poland or Yugoslavia, and then to other countries of the Soviet bloc, possibly via Czechoslovakia. There is little evidence of software making its way via the Soviet Union into the West. Full Tape Crack Pack, 2016, http://spectrum4ever.org.

25. Patrik Rak, Hana Gregorová, and Vítek Gregor, *Piškworks*, ZX Spectrum (Prague: Raxoft, 1990).

26. Jansoft, "Jansoft," classified ad, *Moj Mikro* 4, no. 5 (1987): 61; Future Soft, "Future Soft," classified ad, *Moj Mikro* 4, no. 5 (1987): 62.

27. For more about Stawicki, see fan resources such as the Russian-language SpeccyWiki ("Stawicki," *SpeccyWiki*, accessed April 6, 2018, http://speccy.info/M._Stawicki).

28. An analog version of the Stawicki illustration was published in *Bajtek*, year 1988, issue 3. It bears signs of being a redrawing or a mash-up of existing comics art. This

would fit with Stawicki's habit of redrawing images from magazines such as *Your Sinclair*.

29. Fuka, interview.

30. Some of his versions appear on the Full Tape Crack Pack website. Fuka, interview.

31. František Fuka, *Počítačové hry: Historie a současnost, 1. díl* (Beroun: Zenitcentrum, 1988).

32. Mikrobáze, "Úpravy her pro ZX Spectrum," *Mikrobáze* 2, no. 3 (1986): 44.

33. Markku Reunanen, Patryk Wasiak, and Daniel Botz, "Crack Intros: Piracy, Creativity and Communication," *International Journal of Communication* 9 (March 26, 2015): 20.

34. Jukka Vuorinen, "Ethical Codes in the Digital World: Comparisons of the Proprietary, the Open/Free and the Cracker System," *Ethics and Information Technology* 9, no. 1 (February 28, 2007): 27–38, https://doi.org/10.1007/s10676-006-9130-2.

35. Markku Reunanen, "How Those Crackers Became Us Demosceners," *WiderScreen* 17, nos. 1–2 (April 15, 2014), http://widerscreen.fi/numerot/2014-1-2/crackers-became-us-demosceners/.

36. Anders Carlsson, "The Forgotten Pioneers of Creative Hacking and Social Networking: Introducing the Demoscene," in *Re:live Media Art Histories 2009 Conference Proceedings*, ed. Sean Cubitt and Paul Thomas (Melbourne: University of Melbourne and Victorian College of the Arts and Music, 2009), 16–20; Jimmy Maher, *The Future Was Here: The Commodore Amiga*, Platform Studies (Cambridge, MA: MIT Press, 2012).

37. Maher, *The Future Was Here*; Wasiak, "Illegal Guys"; Reunanen, "How Those Crackers Became Us Demosceners"; Antti Silvast and Markku Reunanen, "Multiple Users, Diverse Users: Appropriation of Personal Computers by Demoscene Hackers," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. Gerd Alberts and Ruth Oldenziel (London: Springer, 2014), 151–163.

38. Jeff Smart, "The European Crackers Map, Anno 1988," *Illegal* 3, no. 26 (1988): not numbered.

39. Salve Big Ben/COSMOS, "The East is Coming!," *Illegal* 3, no. 31 (1988): not numbered.

40. In his account of the European computer underground, Wasiak claims that "fewer games were released for [the ZX Spectrum, Atari XE/XL, or Sharp] and their users did not develop sophisticated software exchange networks." As we have seen, this did not hold for the Czechoslovak Spectrum scene. In addition, some sources show that more titles were released for the Spectrum than for the Commodore 64. For a discussion of the number of releases, see: Mark Langshaw, "ZX Spectrum vs

Commodore 64: Gaming's Greatest Rivalry," Digital Spy, published April 22, 2012, http://www.digitalspy.com/gaming/news/a377768/zx-spectrum-vs-commodore-64-gamings-greatest-rivalry/.

41. Wade, *Playback*.

42. Lobato, *Shadow Economies of Cinema*, 40.

43. Josef Dubský, *Závěrečná zpráva z výzkumu č. 84-6: Názory občanů na vybrané problémy životní úrovně v ČSSR* (Ústav pro výzkum veřejného mínění při Federálním statistickém úřadu, 1986), http://dspace.soc.cas.cz:8080/xmlui/handle/123456789/3009.

44. Sbírka zákonů, "Zákon ze dne 28. března 1990, kterým se mění a doplňuje zákon č.35/1965 Sb., o dílech literárních, vědeckých a uměleckých (autorský zákon)," *Sbírka zákonů – Československá socialistická republika* 1990, no. 89 (1990): 379–381.

45. Oldřich Burger, "Vážení ATARI fans, …," in *Seznam programů ATARI XL/XE*, ed. Josef Kubelka (Rožnov pod Radhoštěm: FLOP—Vičar, 1990).

46. Tamás Beregi, "Hungary," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 219–234.

47. As an informant reminisced: "We believed that we were actually helping the friends that we would then give [the games] to."

48. Fuka, interview.

49. František Fuka, *Hry pro PC* (Brno, Czech Republic: Cybex, 1993).

50. Platinum Productions, *Raid over Moscow*, ZX Spectrum (US Gold, 1985); Jonathan F. Smith and F. David Thorpe, *Green Beret*, ZX Spectrum (Imagine Software, 1986).

51. Tero Pasanen, *Beyond the Pale: Gaming Controversies and Moral Panics as Rites of Passage*, Jyväskylä Studies in Humanities (Jyväskylä, Finland: University of Jyväskylä, 2017), https://jyx.jyu.fi/dspace/bitstream/handle/123456789/55170/978-951-39-7152-6_vaitos_02092017.pdf.

52. Fuka, interview.

53. Hrda, interview.

54. Miroslav Vaněk, *Byl to jenom rock'n'roll?: hudební alternativa v komunistickém Československu 1956–1989* (Prague: Academia, 2010); Kevin McDermott, *Communist Czechoslovakia, 1945–89: A Political and Social History*, European History in Perspective (New York: Palgrave Macmillan, 2015).

55. Eric S. Raymond, ed. *The On-line Hacker Jargon File, version 4.2.2* (self-published, 2000), http://manybooks.net/titles/anonetext02jarg422.html.

56. Bohuslav Blažek, *Bludiště počítačových her* (Prague: Mladá fronta, 1990), 136–137.

57. Trojan, "Hráč (rozhovor s Františkem Fukou)," 3.

58. František Fuka, "Hry pro ZX Spectrum," *Mikrobáze* 3, no. 6 (1987): 44–48; Vít Libovický, interview by Jaroslav Švelch, April 13, 2011.

59. Trojan, "Hráč (rozhovor s Františkem Fukou)," 3.

60. Petr Mihula, interview by Jaroslav Švelch, May 9, 2017; David Hertl, interview by Jaroslav Švelch, February 10, 2013.

61. Slavomír Labský, "ZX Spectrum kopiraky," *Busysoft*, 2001, http://busy.speccy.cz/tvorba/zxcopys.htm.

62. Mikrobáze, "Programová nabídka Mikrobáze (2/1986)," *Mikrobáze* 2, no. 2 (1986): 68.

63. Mikrobáze, "Výsledky ankety Mikrobáze z roku 1985," *Mikrobáze* 2, no. 6 (1986): 4–13.

64. Lewis Hyde, *The Gift: Imagination and the Erotic Life of Property* (New York: Vintage Books, 1983).

65. Given the hobbyist profile of the newsletter, we can expect a higher percentage and higher counts among the general population of micro owners.

66. Maher, *The Future Was Here*.

67. Hrda, interview; Martin Malý, interview by Jaroslav Švelch, August 12, 2014.

68. Although a part of "informal distribution," club distribution was partially formalized, especially when taken up by larger computer clubs. During its heyday in 1985–1986, all orders from the *Mikrobáze* collection had to be made by mail and were processed using a computerized database. At the 415th Svazarm club in Ostrava, the former wandering programmer Vlastimil Veselý was in charge of distribution in 1988–1989. He made copies and kept detailed records about all transactions in exchange for the privilege of borrowing one of the club's PMD 85 machines for home use. Vlastimil Veselý, interview by Jaroslav Švelch, February 11, 2013.

69. Burger, "Vážení ATARI fans, …"

70. Hertl, interview.

71. The price range is based on classified ads in *Amateur Radio* in 1989. The price of a cinema ticket was 6.90 crowns in 1989. Unie filmových distributorů, "Přehledy, statistiky," *Unie filmových distributorů*, 2016, http://www.ufd.cz/prehledy-statistiky.

72. Patryk Wasiak, "Computing behind the Iron Curtain: Social Impact of Home Computers in the Polish People's Republic," Tensions of Europe Working Paper 2010, no. 8 (2010): 1–17; Wasiak, "Playing and Copying: Social Practices of Home Computer Users in Poland during the 1980s."

73. Grant McCracken, "Culture and Consumption: A Theoretical Account of the Structure and Movement of the Cultural Meaning of Consumer Goods," *Journal of Consumer Research* 13, no. 1 (June 1986): 79.

74. Personalization of computer game cassette tapes shares many similarities with the practice of creating mixtapes. As Fenby-Hulse writes, "The personality of the receiver and the creator both imprinted on the mixtape." Kieran Fenby-Hulse, "Rethinking the Digital Playlist: Mixtapes, Nostalgia and Emotionally Durable Design," in *Networked Music Cultures: Contemporary Approaches, Emerging Issues*, ed. Raphaël Nowak and Andrew Whelan (London: Palgrave Macmillan UK, 2016), 176, http://dx.doi.org/10.1057/978-1-137-58290-4_11.

75. It would require research beyond the scope of this book to state it with certainty, but online discussions of cassette enthusiasts (or "tapeheads") suggest that the cassette was manufactured in South Korea by the Lucky-Goldstar conglomerate, later renamed to LG. "AudioStar HS-I, Type I Cassette/Tape, Anyone?," Tapeheads—Tape, Audio and Music Forums, published December 2012, http://www.tapeheads.net/showthread.php?t=22033.

76. Denton Designs, *The Great Escape*, ZX Spectrum (Ocean Software, 1986); David Braben, Ian C. Bell, and Torus, *Elite*, ZX Spectrum (Firebird, 1985).

77. Ultimate Play the Game, *Alien 8*, ZX Spectrum (ACG, 1984); Don Priestley, *Dictator*, ZX Spectrum (DK'Tronics, 1983).

78. Hertl, interview.

79. Trojan, "Hráč (rozhovor s Františkem Fukou)."

80. In contrast, Czechoslovak homebrew games often *did* include instructions, because their authors knew they would be distributed without a manual.

81. Fuka, "Hry pro ZX Spectrum," 44.

82. Incidentally, the instructions that followed turned out to be flawed. "P" klub počítačových nadšenců, "Jak se hraje … Pyjamarama," *Zápisník* 33, no. 20 (1989): 45; Chris Hinsley, *Pyjamarama*, ZX Spectrum (Mikro-Gen, 1984).

83. Beam Software, Philip Mitchell, and Veronika Megler, *The Hobbit*, ZX Spectrum (Melbourne House, 1982). It is unclear which version of *Defender* the list was referring to. The World of Spectrum archive list nine games with that title. World of Spectrum, "Archive," World of Spectrum, 2017, http://www.worldofspectrum.org/archive.html.

84. Consalvo, *Cheating*; Alexander R. Galloway, *Gaming: Essays on Algorithmic Culture* (Minneapolis: University of Minnesota Press, 2006); Stephanie Boluk and Patrick Lemieux, "Metagame," in *Debugging Game History: A Critical Lexicon*, ed. Henry Lowood and Raiford Guins, Game Histories (Cambridge, MA: MIT Press, 2016), 313–324; James Newman, "Playing (with) Videogames," *Convergence: The International*

Journal of Research into New Media Technologies 11, no. 1 (March 1, 2005): 48–67, https://doi.org/10.1177/135485650501100105.

85. Umberto Eco, "Towards a Semiotic Inquiry into the Television Message," trans. Paola Splendore, *Working Papers in Cultural Studies*, no. 3 (1972): 103–121.

86. Matthew Smith, *Jet Set Willy*, ZX Spectrum (Software Projects, 1984).

87. Fuka, "Hry pro ZX Spectrum"; Fuka, *Počítačové hry: Historie a současnost, 1. díl*; ZX Magazín, "POKE 3," *ZX Magazín* 2, no. 10 (1989): 3.

88. Mel Croucher and Andrew Stagg, *Deus Ex Machina*, ZX Spectrum (Automata UK, 1984); Jaroslav Švelch, "The Context of Innovation in Metaphorical Game Design: The Case of Deus Ex Machina," in *Exploring the Edges of Gaming: Proceedings of the Vienna Games Conference 2008–2009: Future and Reality of Gaming*, ed. Christoph Klimmt, Konstantin Mitgustsch, and Herbert Rosensting (Vienna: Braumüller, 2010), 303–313.

89. Bohuslav Blažek, "Metakomunikace (2): Kradení jako modus bytí," *Mikrobáze* 4, no. 8 (1988): 31.

90. Blažek, *Bludiště počítačových her*, 139.

91. Fuka, *Počítačové hry: Historie a současnost, 1. díl*, 95.

92. Michal Hlaváč, interview by Jaroslav Švelch, November 20, 2012.

93. Beam Software, Mitchell, and Megler, *The Hobbit*.

94. Hrda, interview.

95. Fuka, interview.

96. Clive Townsend, *Saboteur 2*, ZX Spectrum (Durell Software, 1987).

97. Newman, "Playing (with) Videogames," 93.

98. Club publications did from time to time print game instructions and hints, but could never cover the immense quantities of available games. Although it was technically possible to circulate walkthroughs or maps as text or graphics files in the informal distribution network, this practice seems to have been quite rare.

99. A methodological note: unlike computer clubs, funfairs and arcades have left next to no "paper trail." The following reconstruction is therefore disproportionately reliant on oral history.

100. Vlastimil Veselý, "Co se dělo v srpnu 1983," WEXova stránka, 2013, http://wexova.sweb.cz/historie/h8308.htm; Tomáš Smutný, interview by Jaroslav Švelch, January 23, 2013.

101. Their history dates back to at least the nineteenth century, and although they have had some ties to the Roma people, they tend to self-identify as a professional

group rather than an ethnic one. Martina Janečková, “Světští na Plzeňsku” (master’s thesis, Masarykova univerzita, 2014).

102. Ibid.

103. Česká televize, “Cirkusy a poutě,” *Retro* (Česká televize, 2010), http://www.ceskatelevize.cz/porady/10176269182-retro/210411000360016/.

104. Smutný, interview.

105. Martin Bach, “Muž, který bastlil v Československu herní automaty,” Games.cz, published September 2012, http://games.tiscali.cz/tema/muz-ktery-bastlil-v-ceskoslovensku-herni-automaty-60695.

106. Smutný, interview.

107. The full police file had been discarded, but the book of records, available at the Czech Security Services Archive, confirms that Smutný was held in custody between October 13 and December 22, 1987. (Archival item: Deník vyšetřovacích spisů OS SNB, odd. vyšetřování VB: 2.10.1978–13 6.1990, no. 5166-5169/2016, vol. 14.)

108. Personal communication with Vladimír Smejkal.

109. Mikrobáze, “Programová nabídka,” *Mikrobáze* 4, no. 1 (1988): 28–29; VD Program, “Reklama na objednávku - VD Program,” *ZX Magazín* 2, nos. 8–9 (1989): 13.

110. In Czechoslovakia, the *Computer Dilemmas* TV program included software audio signal, and the Czechoslovak Radio also reportedly experimented with this. Jan Flak, “Počítačová dilemata, I.,” *Počítačová dilemata* (Ostrava, Czech Republic: Československá televize Ostrava, September 7, 1988); Miroslav Háša, interview by Jaroslav Švelch, July 10, 2012. For more on software broadcasting in Yugoslavia, see: Jakić, “Galaxy and the New Wave: Yugoslav Computer Culture in the 1980s.” In Poland, the *Radiokomputer* program had a similar mission. See Tomasz Jordan, “Uwaga … Start!!! Radiokomputer,” *Bajtek* 25, no. 11 (1986): 28.

111. Crackers had little time to actually play games. Wasiak writes: “The time they had to invest into cracking and the time that went into cultivating trading partners and maintaining their social position within the scene, combined with the day-to-day pressures of ordinary teenage life, in fact left most with little time for playing games.” Wasiak, “Illegal Guys.”

112. James Newman, *Best Before: Videogames, Supersession and Obsolescence* (Abingdon, UK: Routledge, 2012), 124.

113. Petri Saarikoski and Jaakko Suominen, “Computer Hobbyists and the Gaming Industry in Finland,” *IEEE Annals of the History of Computing* 31, no. 3 (2009): 20–33, https://doi.org/10.1109/MAHC.2009.39.

6 Bastard Children of the West

1. Vít Libovický, interview by Jaroslav Švelch, April 13, 2011.

2. For a historical account that attempts to chart a history of "firsts," see: Simon Egenfeldt-Nielsen, Jonas Heide Smith, and Susana Pajares Tosca, *Understanding Video Games: The Essential Introduction*, 2nd ed. (New York: Routledge, 2013). For critiques from two different perspectives, see: Laine Nooney, "A Pedestal, a Table, a Love Letter: Archaeologies of Gender in Videogame History," *Game Studies* 13, no. 2 (December 2013), http://gamestudies.org/1302/articles/nooney; Carl Therrien, "Inspecting Video Game Historiography Through Critical Lens: Etymology of the First-Person Shooter Genre," *Game Studies* 15, no. 2 (December 2015), http://gamestudies.org/1502/articles/therrien.

3. For a classical account of the social and cultural value of imitation, see: Gabriel Tarde, *The Laws of Imitation* (New York: Henry Holt and Company, 1903).

4. Square, *Final Fantasy*, NES (Nintendo, 1987); id Software, *Doom*, PC (id Software, 1993); Mia Consalvo, *Atari to Zelda: Japan's Videogames in Global Contexts* (Cambridge, MA: MIT Press, 2015); Tristan Donovan, *Replay: The History of Video Games* (East Sussex, UK: Yellow Ant, 2010).

5. Christian Katzenbach, Sarah Herweg, and Lies van Roessel, "Copies, Clones, and Genre Building: Discourses on Imitation and Innovation in Digital Games," *International Journal of Communication* 10 (January 29, 2016): 838–859.

6. Ibid.

7. Note that many of the British-produced Spectrum games were in turn conversions of Japanese arcade games or American computer games. The influence of Spain during the "Golden Age of Spanish Software" has been elaborated on by Garin and Martínez as well as by Meda-Calvet. Lange and Liebe mention influential 1980s West German games such as *Giana Sisters* (1987) and *Turrican* (1989)—both heavily inspired by Japanese console titles—in their account of German game history, but mistakenly date the former to 1997. Manuel Garin and Victor Manuel Martínez, "Spain," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 521–534; Ignasi Meda-Calvet, "Bugaboo: A Spanish Case of Circulation and Co-production of Video Games," *Cogent Arts & Humanities* 3, no. 1 (December 31, 2016): 1190440, https://doi.org/10.1080/23311983.2016.1190440; Andreas Lange and Michael Liebe, "Germany," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 193–206.

8. Alexei Yurchak, *Everything Was Forever, until It Was No More: The Last Soviet Generation* (Princeton, NJ: Princeton University Press, 2006), 173.

9. John Urry, *Sociology beyond Societies: Mobilities for the Twenty-First Century*, International Library of Sociology (London: Routledge, 2000).

10. Claude Lévi-Strauss, *The Savage Mind* (London: Weidenfeld and Nicolson, 1966), 17.

11. The original quote: "[Readers] fragment texts and reassemble the broken shards according to their own blueprints, salvaging bits and pieces of the found material in making sense of their own social experience." Henry Jenkins, *Textual Poachers: Television Fans & Participatory Culture*, Studies in Culture and Communication (New York: Routledge, 1992), 27.

12. Manuela Perteghella, "Adaptation: 'Bastard Child' or Critique? Putting Terminology Centre Stage," *Journal of Romance Studies* 8, no. 3 (January 1, 2008): 55, https://doi.org/10.3828/jrs.8.3.51.

13. Jeff Minter, *Attack of the Mutant Camels*, Commodore 64 (Llamasoft, 1983); Riccardo Fassone, "Cammelli and Attack of the Mutant Camels: A Variantology of Italian Video Games of the 1980s," *Well Played* 6, no. 2 (2017): 68.

14. Unlike commentaries or reviews, the resulting texts are "properly literary" works, or in our case "proper" computer games, rather than reviews or walkthroughs.

15. In Genette's theory, which I will be using in this chapter, the term *transformation* is used in a different sense than in US copyright law. For Genette, transformation is akin to translation or localization, whereas US copyright law speaks of "transformative uses," which presuppose more substantial creative input by the user, and are therefore included under the fair use doctrine. See: Aaron Schwabach, *Fan Fiction and Copyright: Outsider Works and Intellectual Property Protection* (Farnham, UK: Ashgate Publishing, 2013).

16. Tom McLean, *Electronic Dreams: How 1980s Britain Learned to Love the Computer*, Bloomsbury Sigma 10 (London: Bloomsbury Sigma, 2016); Alex Wade, *Playback: A Genealogy of 1980s British Videogames* (New York: Bloomsbury, 2016).

17. František Fuka, *Počítačové hry: Historie a současnost, 2. díl* (Beroun: Zenitcentrum, 1988), 11–12.

18. Melanie Swalwell, *Homebrew Gaming and the Beginnings of Vernacular Digitality* (Cambridge, MA: MIT Press, forthcoming).

19. In today's vernacular, *homebrew* is a term mostly associated with amateur software written for proprietary video game platforms, not home computers. See: Casey O'Donnell, "Mixed Messages: The Ambiguity of the MOD Chip and Pirate Cultural Production for the Nintendo DS," *New Media & Society* 16, no. 5 (August 1, 2014): 737–752, https://doi.org/10.1177/1461444813489509.

20. Michaela Pfadenhauer, "Ethnography of Scenes: Towards a Sociological Life-World Analysis of (Post-traditional) Community-Building," *Forum Qualitative Sozialforschung/Forum: Qualitative Social Research* 6, no. 3 (September 30, 2005), http://www.qualitative-research.net/index.php/fqs/article/view/23.

21. Holly Kruse, "Subcultural Identity in Alternative Music Culture," *Popular Music* 12, no. 1 (January 1, 1993): 37.

22. Jukka Vuorinen, "Ethical Codes in the Digital World: Comparisons of the Proprietary, the Open/Free and the Cracker System," *Ethics and Information Technology* 9, no. 1 (February 28, 2007): 27–38, https://doi.org/10.1007/s10676-006-9130-2; Markku Reunanen, "How Those Crackers Became Us Demosceners," *WiderScreen* 17, nos. 1–2 (April 15, 2014), http://widerscreen.fi/numerot/2014-1-2/crackers-became-us-demosceners/.

23. These included variations on games such as *Hamurabi*, *Lunar Landing Game*, and *Snake*, some of which had been circulating in programmer circles since the 1970s, both in the West and—through imported magazines—in Czechoslovakia.

24. Jaroslav Švelch, "Say It with a Computer Game: Hobby Computer Culture and the Non-entertainment Uses of Homebrew Games in the 1980s Czechoslovakia," *Game Studies* 13, no. 2 (December 2013), http://gamestudies.org/1302/articles/svelch. Similar observations have been made about Finnish and Spanish communities. See Petri Saarikoski, Jaakko Suominen, and Markku Reunanen, "Pac-Man for the VIC-20: Game Clones and Program Listings in the Emerging Finnish Home Computer Market," *Well Played* 6, no. 2 (2017): 7–31; Meda-Calvet, "Bugaboo."

25. Fuka, *Počítačové hry: Historie a současnost, 2. díl*, 12.

26. Patrik Rak, Hana Gregorová, and Vítek Gregor, *Piškworks*, ZX Spectrum (Prague: Raxoft, 1990).

27. Petr Lášek, Alexandra Štorkánová, and Christo Bjalkovski, *Šest ran do klobouku*, ZX Spectrum (Prague: SPL, 1986); email correspondence with Alexandra Štorkánová.

28. In addition, the pornographic demo *Satan, or What Is Missing in the Bible* has been preserved. It features graphic sexual intercourse between a horned devil and the character of Eve, drawn in comics-style graphics. No details about this program (not even the year of release) are known.

29. Kruse, "Subcultural Identity in Alternative Music Culture."

30. Reunanen, "How Those Crackers Became Us Demosceners."

31. Pfadenhauer, "Ethnography of Scenes."

32. TRC's Tomáš Rylek remembers styling the name of his "company" after ACG, the business name of the British label Ultimate Play the Game; Fuxoft's rectangular logo was also inspired by Ultimate's. František Fuka, Miroslav Fídler, and Tomáš Rylek, "Golden Triangle po deseti letech," *Živel*, no. 6 (1997): 48–51.

33. The figures are based on the archives listed in the bibliography. Small numbers of titles have also survived for the IQ 151 and the Ondra. Almost no software has been preserved for the relatively popular ZX81, apart from the Manic Miner conversion

discussed later in the chapter. Games were also being made and converted for the Sord M5 and Amstrad CPC platforms, but these have not yet been made available. Homebrew production for the Commodore 64 only took off after 1989. As of 2018, the most extensive list of Czechoslovak games is maintained on the Game Archaeologist (Herní archeolog) blog (UgraUgra, "Master Game List," Herní archeolog, 2018, https://herniarcheolog.blogspot.com/p/master-game-list.html).

34. František Fuka, interview by Jaroslav Švelch, August 28, 2008, and February 2, 2017.

35. The Atari, for example, had eighteen screen modes as opposed to the Spectrum's one, and used two separate chips to operate sprite graphics. An Atari coder remarked that doing graphics on the Atari was "way complicated." Viktor Lošt'ák and Zdeněk Polách, interview by Jaroslav Švelch, January 26, 2016.

36. Karel Šuhajda, *Hlípa*, PMD 85 (Prague: 482. ZO Svazarmu, 1989); Ultimate Play the Game, *Knight Lore*, ZX Spectrum (ACG, 1984); Ultimate Play the Game, *Alien 8*, ZX Spectrum (ACG, 1984).

37. The figure would be higher if one also included Sharp MZ 800 ports of Spectrum titles. However, these often lack paratextual information, making it impossible to determine their release dates.

38. Martin Malý, interview by Jaroslav Švelch, August 12, 2014; Patrik Rak, interview by Jaroslav Švelch, December 8, 2014.

39. There were exceptions, especially among more experienced coders or university-trained computer engineers. An example would be Petr Mihula's *The Mystery of the Conundrum*, discussed in the next chapter. Petr Mihula, *Záhada hlavolamu*, ZX Spectrum (Brno, Czech Republic: MS-CID, 1988).

40. Even *Manic Miner*, the paradigmatic British bedroom game of the 1980s, was not coded directly on the Spectrum, but assembled on a TRS-80 computer connected to a Spectrum machine. Wade, *Playback*.

41. Stanislav Hrda, interview by Jaroslav Švelch, October 4, 2016; Tomáš Rylek, interview by Jaroslav Švelch, January 15, 2015.

42. Michal Hlaváč, interview by Jaroslav Švelch, November 20, 2012; Malý, interview.

43. Juul demonstrates this on the example of the Commodore 64 game *China Miner* in: Jesper Juul, "Paragaming: Good Fun with Bad Games," *The Ludologist*, published September 24, 2009, https://www.jesperjuul.net/ludologist/2009/09/24/paragaming-good-fun-with-bad-games/.

44. Alison Gazzard, "The Intertextual Arcade: Tracing Histories of Arcade Clones in 1980s Britain," *Reconstruction* 14, no. 1 (2014). For a similar study on Finland, see: Saarikoski, Suominen, and Reunanen, "Pac-Man for the VIC-20: Game Clones and Program Listings in the Emerging Finnish Home Computer Market."

45. *Retro Gamer* magazine's "Making Of" feature series, for instance, typically contains information about conversions. For an example, see: Craig Grannell, "The Making of Marble Madness," *Retro Gamer* 5, no. 53 (2008): 82–87.

46. Clara Fernández-Vara and Nick Montfort, *Videogame Editions for Play and Study*, TROPE 13-02 (Cambridge, MA: MIT Trope Tank, June 5, 2014), 6, http://dspace.mit.edu/handle/1721.1/87668.

47. Fernández-Vara and Montfort, *Videogame Editions for Play and Study*; Nick Montfort and Ian Bogost, *Racing the Beam: The Atari Video Computer System* (Cambridge, MA: MIT Press, 2009).

48. I have not collected enough occurrences of these terms to determine whether the distinction was emic in the 1980s Czechoslovak discourse. A diary entry from 1987 suggests that both ports and conversions might have been called "reworking" (*předělávka* in Czech). Vlastimil Veselý, "Co se dělo v prosinci 1987," WEXova stránka, 2013, http://wexova.sweb.cz/historie/h8712.htm.

49. Andrew S. Tanenbaum, Paul Klint, and Wim Bohm, "Guidelines for Software Portability," *Software: Practice and Experience* 8, no. 6 (November 1978): 682, https://doi.org/10.1002/spe.4380080604.

50. Although all of these machines used the same basic machine code instruction sets thanks to the Z80, many of the calls to hardware (such as display) were still platform-specific and had to be converted by hand.

51. Gazzard, "The Intertextual Arcade: Tracing Histories of Arcade Clones in 1980s Britain"; Bob Pape, *IT'S BEHIND YOU: The Making of a Computer Game* (Bizzley.com, 2013), http://www.bizzley.com/.

52. James Bagley, Charles Davies, and Keith Tinman, *Midnight Resistance*, ZX Spectrum (Ocean Software, 1990).

53. Aleš Martiník, interview by Jaroslav Švelch, June 14, 2017.

54. Frank Gasking, "Pushed to the Edge," Oldschool Gaming, published July 1, 2007, http://www.oldschool-gaming.com/view_article.php?art=pushed_to_the_edge.

55. The technique was adapted from foreign-language material that was circulating among ZX81 users at his university dormitory. However, its use in a fast-paced animated game was a feat in itself. Martiník, interview.

56. Vlastimil Veselý, *Flappy*, PMD 85 (Ostrava: VBG Sofware/415; ZO Svazarmu, 1987).

57. ssKO, *Flappy*, Sharp MZ 800 (dB-Soft, 1984).

58. Vlastimil Veselý, interview by Jaroslav Švelch, February 11, 2013.

59. Ibid.

60. Vlastimil Veselý, “Co se dělo v červnu 1987,” WEXova stránka, 2013, http://wexova.sweb.cz/historie/h8706.htm.

61. Ladislav Gavar, *Boulder Dash*, PMD 85 (Ostrava, Czech Republic: VBG Sofware/415. ZO Svazarmu, 1987); Ladislav Gavar, *Manic Miner*, PMD 85 (Ostrava, Czech Republic: VBG Sofware/415. ZO Svazarmu, 1987).

62. Tomáš Doležal, *Flappy*, Atari 8-bit (Doltari, 1988).

63. Karel Vlček, *Flappy*, ZX Spectrum (KVL, 1990); Michal Bačík, *Flappy*, Commodore 64 (59 Production, 1992).

64. Fernández-Vara and Montfort, *Videogame Editions for Play and Study*, 8.

65. For discussions on rules and fiction in games, see: Gonzalo Frasca, “Simulation versus Narrative: Introduction to Ludology,” in *The Video Game Theory Reader*, ed. Mark J. P. Wolf and Bernard Perron (New York: Routledge, 2003), 221–237; Jesper Juul, *Half-Real: Video Games between Real Rules and Fictional Worlds* (Cambridge, MA: MIT Press, 2005).

66. The Czechoslovak Atari conversion explicitly states that it is based on Sierra On-Line’s PC game. Peersoft, *Lazzy Larry*, Atari 8-bit (Peersoft, 1990); Sierra On-Line, *Leisure Suit Larry in the Land of the Lounge Lizards*, PC (Sierra On-Line, 1987).

67. Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network-Theory*, Clarendon Lectures in Management Studies (Oxford: Oxford University Press, 2005).

68. Rylek, interview. Homebrewer Martin Malý illustrates the slowness of BASIC with the following story: “When I discovered that the Spectrum BASIC has the CIRCLE command that draws circles, I set out to create a game called ping-pong, in which a ball would fly around the screen. I wrote the program in theory, at home on a piece of paper.…Then I realized it took two minutes before the ball moved from one side of the screen to the other, so I gave up on it.” Malý, interview.

69. Rylek, interview.

70. Jonathan F. Smith, *Terra Cresta*, ZX Spectrum (Imagine Software, 1986).

71. Montfort and Bogost, *Racing the Beam*.

72. Tomáš Rylek, *Star Fly*, ZX Spectrum (Prague: TRC, 1987); Tomáš Rylek, *Star Swallow*, ZX Spectrum (Prague: TRC, 1987).

73. Gérard Genette, *Palimpsests: Literature in the Second Degree*, trans. Channa Newman and Claude Doubinsky, vol. 8 of *Stages* (Lincoln: University of Nebraska Press, 1997), 87.

74. Gergo Vas, “The Funniest (and Worst) Engrish in Classic Japanese Video Games,” Kotaku, published February 2013, http://kotaku.com/5980508/the-funniest-and-worst-engrish-in-classic-japanese-video-games.

75. Rylek, interview.

76. Miroslav Fídler, interview by Jaroslav Švelch, November 18, 2014.

77. Miroslav Fídler, *Maglaxians*, ZX Spectrum (Prague: Cybexlab, 1985); Miroslav Fídler, *Itemiada*, ZX Spectrum (Prague: Cybexlab, 1985).

78. Miroslav Fídler and František Fuka, *Jet-Story*, ZX Spectrum (Prague: Cybexlab, 1988).

79. However, as Rylek is quick to point out, Fuka would in fact "not touch drugs or alcohol with a ten-foot pole." Rylek, interview.

80. He collected some of these tracks in the music demo series *Fuxoft Soundtracks 1–4*.

81. There were multiple reasons for these delays, including lower license fees for older films, inefficient mail communication, the need for localization, and manufacturing difficulties. Aleš Danielis, "Česká filmová distribuce po roce 1989," *Iluminace* 19, no. 1 (2007): 53–104; Luděk Havel, "Hollywood a normalizace: Distribuce amerických filmů v Československu 1970-1989" (master's thesis, Masaryk University, 2008).

82. Havel, "Hollywood a normalizace."

83. Ivo Slávik, "Rozhovor s autorom: František Fuka," *Bit* 1, no. 10 (1991): 38.

84. To remind the community of his programming skills, Fuka wrote 1988's shoot-'em-up *F.I.R.E.* "because a rumor has been spreading that Fuka cannot do machine code." Petr Trojan, "Hráč (rozhovor s Františkem Fukou)," *Informace pro uživatele mikropočítačů* 1, no. 2 (Počítačové hry) (1989), 3.

85. Nick Montfort, "Adventure," in *Debugging Game History: A Critical Lexicon*, ed. Henry Lowood and Raiford Guins, Game Histories (Cambridge, MA: MIT Press, 2016), 15.

86. William Crowther and Don Woods, *Adventure*, PDP-10 (Public domain, 1976).

87. František Fuka, *Indiana Jones a Chrám zkázy*, ZX Spectrum (Prague: Fuxoft, 1985).

88. Colin B. Harvey, "A Taxonomy of Transmedia Storytelling," in *Storyworlds across Media: Toward a Media-Conscious Narratology*, ed. Marie-Laure Ryan and Jan-Noël Thon, Frontiers of Narrative (Lincoln: University of Nebraska Press, 2014), 278–294.

89. Havel, "Hollywood a normalizace."

90. Angelsoft, *Indiana Jones in Revenge of the Ancients*, PC (Mindscape, 1987).

91. As of 2017, there are over twelve thousand entries in the World of Spectrum database, and over nine thousand entries in the Atari Mania's Atari 8-bit database. World of Spectrum, "Archive," World of Spectrum, 2017, http://www.worldofspectrum.org/archive.html; Atari Mania, "Atari 400 800 XL XE," Atari Mania, 2017, http://www.atarimania.com/atari-400-800-xl-xe.html.

92. Fuka, *Počítačové hry: Historie a současnost, 2. díl*, 12.

93. Mikrobáze, "Programová nabídka Mikrobáze (2/1986)," *Mikrobáze* 2, no. 2 (1986): 62.

94. Espen J. Aarseth, *Cybertext: Perspectives on Ergodic Literature* (Baltimore: Johns Hopkins University Press, 1997).

95. Beam Software, Philip Mitchell, and Veronika Megler, *The Hobbit*, ZX Spectrum (Melbourne House, 1982).

96. Lucasfilm, *Maniac Mansion*, PC (Lucasfilm, 1987).

97. The other title was *Fuksoft*, which will be discussed in chapter 7. Stanislav Hrda, Michal Hlaváč, and Martin Sústrik, *Fuksoft*, ZX Spectrum (Bratislava: Sybilasoft, 1987); František Fuka, *Indiana Jones 2*, ZX Spectrum (Prague: Fuxoft, 1987).

98. Miroslav Fídler, *Belegost*, ZX Spectrum (Prague: Cybexlab, 1989).

99. Consalvo, *Atari to Zelda*.

100. Don Priestley, *Dictator*, ZX Spectrum (DK'Tronics, 1983).

101. Steven Levy, *Hackers: Heroes of the Computer Revolution* (Sebastopol, CA: O'Reilly Media, 2010).

102. Lee Kristofferson, *System 15000*, ZX Spectrum (AVS, 1984); Steve Cartwright, *Hacker*, ZX Spectrum (Activision, 1985).

103. František Fuka, *Podraz 3*, ZX Spectrum (Prague: Fuxoft, 1986).

104. George Roy Hill, *The Sting* (Universal Pictures, 1973); Jeremy Kagan, *The Sting II* (Universal Studios, 1983); Havel, "Hollywood a normalizace."

105. The fact that Tim Coleman was "unemployed" was slightly subversive, because unemployment formally did not exist in socialist Czechoslovakia.

106. Timex 2097 is a reference to Timex 2048, a Spectrum-compatible computer designed by Timex North America. The RS-2368 modem refers to the RS-232 serial port standard.

107. Anton Tokár, *Podraz IV*, ZX Spectrum (Revúca, Slovakia: Antok Software, 1987); Patrik Rak, *Podraz 4*, ZX Spectrum (Praha: Raxoft, 1988); Anton Tokár and E. H. Becz, *Podraz 5*, ZX Spectrum (Revúca, Slovakia: Antok Software, 1988); Falcon Soft, *Podraz 5*, ZX Spectrum (Falcon Soft, n.d.); Anton Tokár and E. H. Becz, *Podraz 6*, ZX Spectrum (Revúca, Slovakia: Antok Software, 1988); O. Nechvátal, V. Baroš, and B. Jura, *Podraz 7*, ZX Spectrum (Pegas, 1991).

108. T.C.G., "Games," *T.C.G.—Total Computer Gang*, 2009, http://tcg.speccy.cz/index.php?pg=games-en.

109. Ondřej Kafka, *Expert*, ZX Spectrum (Prague: OKF, 1988); Marek Novotný and Pavel Maňas, *Cracker*, ZX Spectrum (Študák Software, 1989).

110. As one of the 1987 *Sting* clones put it in its introduction—inspired by the title of Levy's book on hackers, which was never published in Czech but whose title was already known—"We live among you, peacefully and unnoticed. You all know us, but everybody calls us a different name. For some, we are lunatics, for others gods, for some simply thieves. Only we ourselves know who we are: Hackers, heroes of the computer revolution." Pavel Kořenský, *Podraz na Indiana Jonese*, ZX Spectrum (PKCS, 1987).

111. Jenkins, *Textual Poachers*, 246.

112. Kenneth D. Pimple, "The Meme-ing of Folklore," *Journal of Folklore Research* 33, no. 3 (1996): 238.

113. There is a clear parallel to fan communities, in which authorship is likewise a powerful concept. Alexandra Elisabeth Herzog, "'But This Is My Story and This Is How I Wanted to Write It': Author's Notes as a Fannish Claim to Power in Fan Fiction Writing," *Transformative Works and Cultures* 11 (April 11, 2012), https://doi.org/10.3983/twc.2012.0406.

114. Rylek, interview.

115. Yuri Takhteyev, *Coding Places: Software Practice in a South American City*, Acting with Technology (Cambridge, MA: MIT Press, 2012), 208.

116. A popular title *Conquering the Castle*, for example, took place at the fictional medieval castle of Zvonkovice. František Brabec and Martin Dlouhý, *Dobývání hradu*, ZX Spectrum (Antic Software, 1986).

117. Perhaps the most notorious adaption of the former was *Sexeso*, which involved matching upper and lower halves of female nudes. It was a rare intergenerational collaboration between veteran hobbyist Jiří Pobříslo and 17-year-old Ondřej Kafka. Having seen Kafka's graphics for other projects, Pobříslo asked him to draw the erotic images without having met him in person. When Kafka appeared on Pobříslo's doorstep to deliver the data, the older man was shocked to discover that his collaborator was a teenager. Despite the overwhelmingly male and young demographics of the scene, *Sexeso* was a rare example of a male-targeted erotic game. Ondřej Kafka, interview by Jaroslav Švelch, November 20, 2017.

118. Andromeda Software, *Caesar the Cat*, Commodore 64 (Mirrorsoft, 1983); Andromeda Software, *Eureka!*, Commodore 64 (Domark, 1984); Andromeda Software, *Scarabaeus*, Commodore 64 (Ariolasoft, 1985).

119. Tamás Beregi, "Hungary," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 219–234.

120. Andromeda Software, *Tetris*, ZX Spectrum (Mirrorsoft, 1987).

121. Dan Ackerman, *The Tetris Effect: The Game that Hypnotized the World* (New York: PublicAffairs, 2016).

122. P. Konrad Budziszewski, "Poland," in *Video Games around the World*, ed. Mark J. P. Wolf (Cambridge, MA: MIT Press, 2015), 404; Borkowski, Marcin. *Pandora's Box*. ZX Spectrum. self-published, 1986.

123. P. Z. Karen Co. Development Group, *Blockout*, PC (California Dreams, 1989); P. Z. Karen Co. Development Group, *Street Rod*, PC (California Dreams, 1989); Budziszewski, "Poland."

124. Duke and Muraja, *Kung Fu*, ZX Spectrum (Bug-Byte Software, 1984); Duke, Mario, and F. David Thorpe, *Movie*, ZX Spectrum (Imagine Software, 1986); Duke, *Phantom Club*, ZX Spectrum (Ocean Software, 1988).

125. M. Rasulić, "Englezi uče karate," *Svet Kompjutera* 1, no. 11 (1984): 5.

126. Bruno Jakić, "Galaxy and the New Wave: Yugoslav Computer Culture in the 1980s," in *Hacking Europe: From Computer Cultures to Demoscenes*, ed. G. Alberts and Ruth Oldenziel (London: Springer, 2014), 129–150.

7 Empowered by Games

1. Bohuslav Blažek, "Počítače a kultura," *Informace pro uživatele mikropočítačů* 1, no. 1 (Počítač přítel člověka) (1989): 9.

2. František Fuka, Miroslav Fídler, and Tomáš Rylek, "Golden Triangle po deseti letech," *Živel*, no. 6 (1997): 48–51.

3. Ibid.

4. In her ethnographic work among early microcomputer users, Turkle also observed this kind of empowerment. One of her informants enjoyed "making things that nobody else has ever made." Sherry Turkle, *The Second Self: Computers and the Human Spirit*, 20th Anniversary Edition (Cambridge, MA: MIT Press, 2005), 144.

5. Theodor H. Nelson, *Computer Lib: You Can and Must Understand Computers Now/Dream Machines: New Freedoms through Computer Screens—A Minority Report* (self-published, 1974).

6. Richard Barbrook and Andy Cameron, "The Californian Ideology," *Science as Culture* 6, no. 1 (January 1, 1996): 44–72, https://doi.org/10.1080/09505439609526455.

7. Blažek, "Počítače a kultura," 9.

8. Henry Jenkins, *Textual Poachers: Television Fans & Participatory Culture*, Studies in Culture and Communication (New York: Routledge, 1992).

9. David Garcia and Geert Lovink, "The ABC of Tactical Media," 1997, http://subsol.c3.hu/subsol_2/contributors2/garcia-lovinktext.html.

10. Rita Raley, *Tactical Media*, Electronic Mediations 28 (Minneapolis: University of Minnesota Press, 2009).

11. Christine Harold, “Pranking Rhetoric: ‘Culture Jamming’ as Media Activism,” *Critical Studies in Media Communication* 21, no. 3 (September 1, 2004): 189–211, https://doi.org/10.1080/0739318042000212693.

12. Andrew Lawrence Roberts, *From Good King Wenceslas to the Good Soldier Švejk: A Dictionary of Czech Popular Culture* (Budapest: Central European University Press, 2005), 166–167.

13. Kevin McDermott, *Communist Czechoslovakia, 1945–89: A Political and Social History*, European History in Perspective (New York: Palgrave Macmillan, 2015), 19.

14. Jaroslav Major, “Hudba pro masy (konečně živě),” *Svět v obrazech* 44, no. 15 (1988): 22.

15. Federální statistický úřad—Ústav pro výzkum veřejného mínění, *Záverečná správa z výskumu č. 88-4: Názory našich občanov na vedúcu úlohu strany* (Prague: Ústav pro výzkum veřejného mínění, 1988), http://dspace.soc.cas.cz/bitstream/handle/123456789/3041/CSDA88-4.PDF?sequence=1; Federální statistický úřad—Ústav pro výzkum veřejného mínění, “Červen 1989” (Czech Social Science Data Archive, 2010), https://doi.org/10.14473/V8909.

16. Miroslav Vaněk and Pavel Mücke, *Velvet Revolutions: An Oral History of Czech Society*, Oxford Oral History Series (New York: Oxford University Press, 2016).

17. McDermott, *Communist Czechoslovakia, 1945–89*.

18. Vít Libovický, *Město robotů*, ZX Spectrum (Beroun: Zenitcentrum, 1989); Vít Libovický, interview by Jaroslav Švelch, April 13, 2011. The game was in fact a conversion and translation of the 1981 TRS-80 title *Forbidden City* (Demas, 1981).

19. For comparison, a movie ticket cost seven crowns and a copy of a Western game from a for-profit pirate cost two to fifteen crowns. See chapter 5 for more details.

20. The game was locked with a password that was to be broadcast on September 21, 1989. Due to an oversight during the preparation of the master copy, the password was accidentally written onto the tape, which made the game trivial to hack; and many players had already completed it before the password was broadcast. The contest was canceled, and winners were drawn in a lottery. Libovický, interview.

21. Žiga Turk and Matevž Kmet, “Tihotapci, pozor!,” *Moj Mikro* 1, nos. 7–8 (1984): 51; Mel Croucher, *Deus Ex Machina: The Best Game You Never Played in Your Life* (Luton, UK: Acorn Books, 2014).

22. Laine Nooney, “Easter Eggs,” in *The Johns Hopkins Guide to Digital Media*, ed. Marie-Laure Ryan, Lori Emerson, and Benjamin J. Robertson (Baltimore: Johns Hopkins University Press, 2014), 165.

23. Martin Malý, interview by Jaroslav Švelch, August 12, 2014.

24. Martin Malý, *Katanga*, ZX Spectrum (Chleby, Czech Republic: Demonsoft, 1988).

25. When describing Western "crack intros," Reunanen, Wasiak, and Botz use the term "scrollers," but I will stick to the expression "scrolling message," which was explicitly (in English) used by František Fuka in the 1988 game *F.I.R.E.* Markku Reunanen, Patryk Wasiak, and Daniel Botz, "Crack Intros: Piracy, Creativity and Communication," *International Journal of Communication* 9 (March 26, 2015): 20; František Fuka, *F.I.R.E.*, ZX Spectrum (Prague: Fuxoft, 1988).

26. Peter Harrap, *Monty on the Run*, ZX Spectrum (Gremlin Graphics, 1985).

27. Miroslav Fídler, *Galactic Gunners*, ZX Spectrum (Prague: Cybexlab, 1987).

28. In the game *F.I.R.E.*, the author—František Fuka—"greets" his favorite British coders, knowing it is highly unlikely they would ever read the message, but exhibiting his cultural capital by sharing his acquired taste and knowledge of the Western game industry. He writes (in English): "Hello to all programming legends: Jonathan Smith, Raffaele Cecco, Dominic Robinson, those guys at Level 9, and Peter 'Micronaut' Cooke." Fuka, *F.I.R.E.*

29. Patrik Rak, interview by Jaroslav Švelch, December 8, 2014; Malý, interview.

30. Malý, interview.

31. Michal Hlaváč, interview by Jaroslav Švelch, November 20, 2012; Stanislav Hrda, interview by Jaroslav Švelch, October 4, 2016.

32. František Fuka, *Indiana Jones 3*, ZX Spectrum (Prague: Fuxoft, 1990).

33. I have developed this argument in my previous work. Jaroslav Švelch, "What You Can't See Is What You Don't Get: Paradigms of Game World Visualization," in *Proceedings of the 2008 Conference on Future Play: Research, Play, Share*, presented at Future Play '08 (New York: ACM, 2008), 212–215, https://doi.org/10.1145/1496984.1497026.

34. František Fuka, *Počítačové hry: Historie a současnost, 2. díl* (Beroun: Zenitcentrum, 1988), 12.

35. Martin Sústrik, *Plutonia*, ZX Spectrum (Bratislava: Sybilasoft, 1987); Martin Sústrik, Stanislav Hrda, and Michal Hlaváč, *Tria*, ZX Spectrum (Bratislava: Sybilasoft, 1987); Martin Sústrik, Pavol Čejka, and Juraj Sústrik, *Stensontron*, ZX Spectrum (Bratislava: Sybilasoft, 1987).

36. Stanislav Hrda, Michal Hlaváč, and Martin Sústrik, *Fuksoft*, ZX Spectrum (Bratislava: Sybilasoft, 1987); Stanislav Hrda, Michal Hlaváč, and Sybilasoft, *Šatochin*, ZX Spectrum (Bratislava: Sybilasoft, 1988).

37. Juraj Hlaváč and Michal Hlaváč, *Sherlock Holmes*, ZX Spectrum (Bratislava: Sybilasoft, 1987); Michal Hlaváč, *Chrobák Truhlík*, ZX Spectrum (Bratislava: Sybilasoft, 1990). Like much genre fiction, Sherlock Holmes stories were framed in Czechoslovakia as literature for teens. The 1987 collection *The Man with a Pipe and a Violin*, for example, was explicitly targeted at a teen audience and recommended to

"twelve-year-old and older readers." The translations, however, were faithful to the original and unabridged. See Arthur Conan Doyle, *Muž s dýmkou a houslemi* (Prague: Albatros, 1987).

38. In addition, he wrote the story of *Super Discus*, a game that takes place in an insane asylum co-inhabited by fictionalized versions of Sybilasoft themselves. Martin Sústrik et al., *Super Discus*, ZX Spectrum (Bratislava: Sybilasoft, 1987).

39. Hlaváč, interview.

40. Sybilasoft teased, but never finished *Maryčka Magdónova*, an adaptation-cum-parody of an early twentieth-century Czech tragic, socially critical poem by Petr Bezruč about an orphan girl who looks after her siblings, but eventually commits suicide. The poem is considered a Czech classic and taught in high schools. In the adaptation, Maryčka was to rise from the grave into a dystopian world and eliminate her enemies.

41. Hrda, interview.

42. Ibid.

43. Interestingly, the authors of *Fuksoft*'s Atari conversion, who might not have been familiar with František Fuka's Spectrum work, changed the title to "Fuxosoft." They also removed some references to ZX Spectrum games. Stanislav Hrda himself became a main character in *Programmers' Haunted Castle*, on a mission to save one of the coauthors of that game. Syssoft and Titansoft, *Fuxosoft (ZX Fuk)*, Atari 8-bit (Syssoft, 1987); Tom & Jerry and Delphine Soft, *Zakliatý zámok programátorov*, ZX Spectrum (Bratislava: Tom & Jerry & Delphine Soft, 1988); Patrik Rak, *Fuksoft II*, ZX Spectrum (Prague: Raxoft, 1989).

44. Dan Dare, a heroic science fiction fighter pilot, first appeared in the British comics magazine *Eagle* in the late 1940s. New strips were being published throughout the 1980s. He was featured in three 1980s 8-bit games, starting with 1986's *Dan Dare: Pilot of the Future*. Malý had already used the image of Dan Dare from that title in his own game *Nick Carter*. Gang of Five, *Dan Dare: Pilot of the Future*, ZX Spectrum (Virgin Games, 1986); Martin Malý, *Nick Carter*, ZX Spectrum (Chleby, Czech Republic: Demonsoft, 1988).

45. Martin Malý, *Demon in Danger*, ZX Spectrum (Chleby, Czech Republic: Demonsoft, 1988).

46. Emily T. Metzgar, David D. Kurpius, and Karen M. Rowley, "Defining Hyperlocal Media: Proposing a Framework for Discussion," *New Media & Society* 13, no. 5 (August 2011): 772–787, doi:10.1177/1461444810385095.

47. Laine Nooney, "A Pedestal, a Table, a Love Letter: Archaeologies of Gender in Videogame History," *Game Studies* 13, no. 2 (December 2013), http://gamestudies.org/1302/articles/nooney.

48. Pandovisia Software, *XIV*, ZX Spectrum (Belgrade: Pandovisia Software, 1984).

49. Other possible candidates for hyperlocal games would be 1983's *Streets of London*, whose portrayal of the city is however too vague and nonspecific to be called truly hyperlocal, and 1984's *Hampstead*, which offers a satirical take on this posh London neighborhood, rather than presenting it as a place of the authors' everyday life. Grant Privett and Allan Webb, *Streets of London*, Commodore 64 (Supersoft, 1983); Trevor Lever and Peter Jones, *Hampstead*, ZX Spectrum (Supersoft, 1984).

50. Jaromír Nohavica and Mark Landry, *The Jarek Nohavica Songbook* (Ostrava, Czech Republic: Montanex, 2009), 166. Reprinted with permission from Jaromír Nohavica and Mark Landry.

51. Hrda, Hlaváč, and Sústrik, *Fuksoft*.

52. Alexei Yurchak, *Everything Was Forever, until It Was No More: The Last Soviet Generation* (Princeton, NJ: Princeton University Press, 2006), 104.

53. The standard length of military service was two years, reduced to one in the case of university graduates. See Eliška Veselovská, "Základní vojenská služba v letech 1945–2003 ve vzpomínkách pamětníků" (master's thesis, Masaryk University, 2011).

54. Vaněk and Mücke, *Velvet Revolutions*.

55. Jan Richter, "Jaroslav Foglar and his 'Rapid Arrows,'" Radio Prague, 2012, http://www.radio.cz/en/section/czech-history/jaroslav-foglar-and-his-rapid-arrows.

56. Petr Mihula, *Záhada hlavolamu*, ZX Spectrum (Brno, Czech Republic: MS-CID, 1988).

57. Yurchak, *Everything Was Forever, until It Was No More*.

58. Ibid., 73.

59. The background of Czechoslovak parades has been described in detail by Sobotková. Jitka Sobotková, "Komunistické slavnosti v Československu v letech 1948-1989" (PhD diss., Palacký University Olomouc, 2011).

60. Hrda, interview.

61. Ultravideo Software, *Space Saving Mission*, ZX Spectrum (Prague: Ultravideo Software, 1985).

62. Tomáš Rylek, interview by Jaroslav Švelch, January 15, 2015.

63. Dick Hebdige, *Subculture: The Meaning of Style* (Hoboken, NJ: Taylor & Francis, 2012), 104–105.

64. John Fiske, *Television Culture*, 2nd ed. (London: Routledge, 2007), 252.

65. Ibid., 149.

66. Hrda, Hlaváč, and Sybilasoft, *Šatochin*.

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Interviews

I conducted all the following interviews in person. Three of these were double interviews, to which the informant brought along a friend or a family member who was also active on the Czechoslovak hobby or gaming scene.

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Veselý, Vlastimil. Ostrava, Czech Republic, February 11, 2013. (I have also drawn from his diary from the 1980s, which is available online at http://wexova.sweb.cz.)

In addition, my material includes six 2010 interviews with computer game players who were children in the 1980s. Since publishing their names would bring no research benefit, I have chosen to anonymize them. Three of these interviews were conducted by my collaborator Petr Vnouček.

Hobbyist Newsletters and Fanzines

A portion of these is available in the Mikrobaze.szm.com, Speccy.cz, and Czech Atari Literature Preservation Project online fan archives. I obtained the rest from my informants' personal collections.

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Mikrobáze, newsletter published by the 602nd Basic Organization of Svazarm in collaboration with *Amateur Radio* magazine.

Sinclair 602, the 602's Sinclair club newsletter.

Spektrum (renamed to *ZX Magazín* in 1989), an independent Spectrum fanzine.

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Additional material was retrieved from *Illegal*, the international cracker and demoscene fanzine, available from the Scene.org file archive.

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ABC mladých techniků a přírodovědců [ABC of young technicians and scientists].

Annonce.

Bajt [Byte].

Bajtek, Polish hobby magazine.

Bit.

Computerworld.

Crash, British gaming magazine.

Excalibur.

Informace pro uživatele mikropočítačů [Information for microcomputer users].

Mladý svět [Young world].

Moj Mikro [My micro], Yugoslavian hobby magazine.

Pedagogika [Pedagogy].

Rudé právo [Red justice].

Svět v obrazech [Illustrated world].

Technický magazín [Technology magazine].

Učitelské noviny [Teachers' news].

Věda a technika mládeži [Science and technology for youth].

Your Sinclair, British gaming magazine.

Zahraniční obchod [Foreign trade].

Zápisník [Chronicle].

I have used the following TV programs, produced by Czechoslovak Television, available in the Archive and Program Repository of Czech Television and the Slovak Television Archive:

Jubilant: Tridsiate piate výročie Zväzarmu [Celebrating Svazarm's 35th anniversary].

Počítačová dilemata [Computer dilemmas].

Televizní klub mladých [Youth TV club], selected broadcasts.

Televizní noviny [Television news], selected broadcasts.

Period Statistics, Research, and Industry Reports

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I have conducted research in the following archives (full references in notes):

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Speccy Archiv (CS.Speccy.cz), ZX Spectrum Games—Svět osmibitové herní legendy (ZX-Spectrum.cz), Fly.Atari.org, and Archive.Free8bit.net. An extensive archive of Czechoslovak Sharp MZ 800 software is hosted by the Scav.cz website. All of these collections are linked from the book's website at ironcurtain.svelch.com. A growing list cataloging Czechoslovak games is maintained on the Herní archeolog (Game Archaeologist) blog (herniarcheolog.blogspot.com).

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